FHIR-STU3 — Diagnostic Medicine Module (sections 10 – 10.8)

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10 Diagnostic Medicine Module

URL: https://hl7.org/fhir/STU3/diagnostics-module.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Work Group Orders and Observations & Clinical Genomics | Ballot Status: Informative

10.0 Diagnostic Medicine Module

10.0.1 Introduction

The Diagnostics Module provides an overview and guide to the FHIR content that addresses ordering and reporting of clinical diagnostics including laboratory testing, imaging and genomics.

10.0.2 Index

The Diagnostics module covers the following resources:

Observation DiagnosticReport ProcedureRequest | ImagingStudy ImagingManifest Sequence | Specimen BodySite

The diagnostic resources and their relationships are shown below. The arrows represent the direction of the references between resources (for example, DiagnosticReport references ProcedureRequest). See the Workflow Module for information about the coordination of activities such as ordering and fulfilling of diagnostics.

Note: See the Genomics Implementation Guidance for additional information about how to use the Diagnostic resources for Clinical Genomic Reporting and Analysis.

Name | Aliases | Description

Observation | Vital Signs, Measurement, Results | Measurements and simple assertions made about a patient, device or other subject.

DiagnosticReport | Report, Test, Result, Results, Labs | The findings and interpretation of diagnostic tests performed on patients, groups of patients, devices, and locations, and/or specimens derived from these. The report includes clinical context such as requesting and provider information, and some mix of atomic results, images, textual and coded interpretations, and formatted representation of diagnostic reports.

ProcedureRequest | diagnostic request | A record of a request for diagnostic investigations, treatments, or operations to be performed.

ImagingStudy | | Representation of the content produced in a DICOM imaging study. A study comprises a set of series, each of which includes a set of Service-Object Pair Instances (SOP Instances - images or other data) acquired or produced in a common context. A series is of only one modality (e.g. X-ray, CT, MR, ultrasound), but a study may have multiple series of different modalities.

ImagingManifest | Manifest, XDS-I summary, Key Images | A text description of the DICOM SOP instances selected in the ImagingManifest; or the reason for, or significance of, the selection.

Sequence | | Raw data describing a biological sequence.

Specimen | | A sample to be used for analysis.

BodySite | anatomical location | Record details about the anatomical location of a specimen or body part. This resource may be used when a coded concept does not provide the necessary detail needed for the use case.

10.0.3 Security and Privacy

The diagnostic resources often represent patient-specific data, and as such are susceptible to data breaching. Necessary privacy and security provisions must be in place when searching and fetching this information. For more general considerations, see the Security and Privacy module.

10.0.4 Common Use Cases

Diagnostic resources are commonly used to plan, recommend, order and report clinical diagnostics:

* Planning a diagnostic test such as in this Careplan-example

* Recommending diagnostics such as is done by Clinical Decision Support tools. See the Clinical Reasoning module for more information.

* Ordering a Laboratory Test such as a lipid panel

* Reporting a Laboratory Test such as a lipid panel

* Ordering an imaging study such as a CT study

* Reporting an imaging study such as a CT study

* Ordering a genetic test such as a gene mutation analysis

* Reporting a genetic test such as HLA genotyping

There are many ways to use these resources independently as well. The Observation resource in particular is central to capturing many measurements and events in healthcare and is often used outside the context of diagnostic orders and reports.

* recording vital signs such as temperature

* recording other observations of a patient's social history such as tobacco use

* recording other characteristics of a patient such as pregnancy status (note: The Condition resource may be used to represent these as well)

* recording clinical assessment tool scores such as an Apgar score.

* identifying and describing a specimen or body site

10.0.5 Related Resources and Modules

The resources that represent the basic information about a patient and a clinical encounter can be found in the Administration Module. Other resources that represent core clinical information generated by healthcare providers during the course of a patient encounter are detailed in the Clinical Summary Module and the Medications Module.

10.0.6 Developmental Roadmap

* The Observation and Diagnostic Report resources have been tested and used in production tooling and as such have reached a maturity level FMM=4 where changes become less likely and require implementer community approval. Future work will focus on developing core profiles and clarifying use cases and boundaries and providing more domain specific examples. We anticipate Observation and Diagnostic Report reaching normative status by the end of this cycle.

* The ImagingStudy resource has had some limited testing and use. Further use of this resource in production tooling in order for it to reach a more stable FMM level.

* ImagingManifest: We expect to see more widespread use of this resource and future updates to its FMM level

* The ProcedureRequest resources has undergone substantial changes since DSTU2. Foremost of which is the merging of DSTU2 DiagnosticOrder and ProcedureRequestinto a single resources. In addition, as a result of the addition of FHIR workflow for STU3, its structure and content have been updated to align with the general FHIR workflow "request" pattern. Because of these, it remains a relatively immature resource. No more sweeping changes to its structure are expected. However, its scope and boundaries vis a vis other resources will be the topic of future workgroup efforts. We anticipate its broad use by implementers and that it will mature as a resource throught the upcoming cycle.

* The Specimen resource will be the focus of work group efforts for the upcoming cycle. A new SpecimenDefinition resource is planned for describing specimen types vs specimen instances and will take over some of the Specimen resourc'e's current scope. We expect to see more widespread use of this resource and future updates to its FMM level

* The Sequence resource remains draft and we expect to see more widespread use of this resource and future updates to its FMM level

* The BodySite resource is an immature resource with a FMM level of 1. To date there has been little implementation feedback.

* The Media resource is the newest member to the Diagnostics module having undergone substantial scope and structural changes to create an specialized observation resource focused on digital media such as images and audio files. Future work will focus on developing core profiles and providing more domain specific examples.

10.1 Resource Observation - Content

URL: https://hl7.org/fhir/STU3/observation.html

See the section in the Observation resource operations page for more information and examples

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Orders and Observations Work Group | Maturity Level: 5 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Measurements and simple assertions made about a patient, device or other subject.

10.1.1 Scope and Usage

This resource is an event resource from a FHIR workflow perspective - see Workflow.

Observations are a central element in healthcare, used to support diagnosis, monitor progress, determine baselines and patterns and even capture demographic characteristics. Most observations are simple name/value pair assertions with some metadata, but some observations group other observations together logically, or even are multi-component observations. Note that the DiagnosticReport resource provides a clinical or workflow context for a set of observations and the Observation resource is referenced by DiagnosticReport to represent lab, imaging, and other clinical and diagnostic data to form a complete report.

Uses for the Observation resource include:

* Vital signs such as body weight, blood pressure, and temperature

* Laboratory Data like blood glucose, or an estimated GFR

* Imaging results like bone density or fetal measurements

* Devices Measurements such as EKG data or Pulse Oximetry data

* Clinical assessment tools such as APGAR or a Glasgow Coma Score

* Personal characteristics: such as eye-color

* Social history like tobacco use, family support, or cognitive status

* Core characteristics like pregnancy status, or a death assertion

10.1.1.1 Core Profiles for Observation

The following core profiles for the Observation resource have been defined as well. If implementations use this Resource when expressing the profile-specific concepts as structured data, they SHALL conform to the following profiles:

Profile | Description
Vital signs | The FHIR Vital Signs profile sets a minimum expectations for the Observation Resource to record, search and fetch the vital signs (e.g. temperature, blood pressure, respiration rate, etc) associated with a patient

10.1.2 Boundaries and Relationships

In contrast to the Observation resource, the DiagnosticReport resource typically includes additional clinical context and some mix of atomic results, images, imaging reports, textual and coded interpretation, and formatted representations. Laboratory reports, pathology reports, and imaging reports should be represented using the DiagnosticReport resource. The Observation resource is referenced by the DiagnosticReport to provide the atomic results for a particular investigation.

"Laboratories routinely have a variable that is summative across a series of discrete variables - these are usually called 'impressions' or 'interpretations'. Sometimes they are algorithmically specified and sometimes they have the imprimatur of pathologists and they are conveyed in Observation or DiagnosticReport instead of the Clinical Impression resource. However, the Observation resources should not be used to record clinical diagnosis about a patient or subject that are typically captured in the Condition resource or the ClinicalImpression resource.

The Observation resource is often referenced by the Condition resource to provide specific subjective and objective data to support its assertions. There are other resources that can be considered "specializations" of the Observation resource and should be used for those specific contexts and use cases. They include AllergyIntolerance resource, FamilyMemberHistory resource, Procedure resource, and Questionnaire resource. In some cases, such as when source data is coming from an HL7 v2 feed, a system may not have information that allows it to distinguish diagnosis, allergy and other "specialized" types of observations from lab, vital sign and other observation types intended to be conveyed with this resource. In those circumstances, such specialized observations may also appear using this resource.

The Media resource captures a specific type of observation whose value is audio, video or image data. This resource is the preferred representation of such forms of information as it exposes the metadata relevant for interpreting the information. However, in some implementations, media information may appear in Observation susing the valueAttachment element and systems should be aware of this possibility.

This resource is referenced by AdverseEvent, ChargeItem, ClinicalImpression, Communication, CommunicationRequest, Condition, DiagnosticReport, FamilyMemberHistory, Goal, Immunization, ImmunizationRecommendation, MedicationAdministration, MedicationRequest, MedicationStatement, Procedure, ProcedureRequest, QuestionnaireResponse, ReferralRequest and Sequence

10.1.3 Resource Content

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Observation | I | | DomainResource | Measurements and simple assertions + If code is the same as a component code then the value element associated with the code SHALL NOT be present + dataAbsentReason SHALL only be present if Observation.value[x] is not present Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ProcedureRequest | ReferralRequest) | Fulfills plan, proposal or order
status | ?!? | 1..1 | code | registered | preliminary | final | amended + ObservationStatus (Required)
category | | 0..* | CodeableConcept | Classification of type of observation Observation Category Codes (Preferred)
code | ? | 1..1 | CodeableConcept | Type of observation (code / type) LOINC Codes (Example)
subject | ? | 0..1 | Reference(Patient | Group | Device | Location) | Who and/or what this is about
context | | 0..1 | Reference(Encounter | EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | ? | 0..1 | | Clinically relevant time/time-period for observation
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner | Organization | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | ?I | 0..1 | | Actual result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept

valueString | | | string
valueBoolean | | | boolean
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueAttachment | | | Attachment
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Observation Value Absent Reason (Extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Observation Interpretation Codes (Extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part SNOMED CT Body Structures (Example)
method | | 0..1 | CodeableConcept | How it was done Observation Methods (Example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation + Must have at least a low or a high or text
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Observation Reference Range Meaning Codes (Extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Observation Reference Range Applies To Codes (Example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | ? | 0..* | BackboneElement | Resource related to this observation
type | | 0..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by ObservationRelationshipType (Required)
target | | 1..1 | Reference(Observation | QuestionnaireResponse | Sequence) | Resource that is related to this one
component | ? | 0..* | BackboneElement | Component results
code | ? | 1..1 | CodeableConcept | Type of component observation (code / type) LOINC Codes (Example)
value[x] | ? | 0..1 | | Actual component result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueAttachment | | | Attachment
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Observation Value Absent Reason (Extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Observation Interpretation Codes (Extensible)
referenceRange | | 0..* | see referenceRange | Provides guide for interpretation of component result
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Observation xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for observation --></identifier>
 <basedOn><!-- 0..* Reference(CarePlan|DeviceRequest|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ProcedureRequest|ReferralRequest) Fulfills plan, proposal or order --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended + -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation --></category>
 <code><!-- 1..1 CodeableConcept Type of observation (code / type) --></code>
 <subject><!-- 0..1 Reference(Patient|Group|Device|Location) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Healthcare event during which this observation is made
    --></context>
 <effective[x]><!-- 0..1 dateTime|Period Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner|Organization|Patient|RelatedPerson) Who is responsible for the observation
    --></performer>
 <value[x]><!--  0..1 Quantity|CodeableConcept|string|boolean|Range|Ratio|
   SampledData|Attachment|time|dateTime|Period Actual result --></value[x]>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc. --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device|DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  <!-- 0..* Provides guide for interpretation -->
  <low><!--  0..1 Quantity(SimpleQuantity) Low Range, if relevant --></low>
  <high><!--  0..1 Quantity(SimpleQuantity) High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population --></appliesTo>
  <age><!-- 0..1 Range Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
```

```
 </referenceRange>
 <related>  <!-- 0..* Resource related to this observation -->
  <type value="[code]"/><!-- 0..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by -->
  <target><!-- 1..1 Reference(Observation|QuestionnaireResponse|Sequence) Resource that is related to this one
    --></target>
 </related>
 <component>  <!-- 0..* Component results -->
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type) --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|Range|Ratio|SampledData|
    Attachment|time|dateTime|Period Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc. --></interpretation>
  <referenceRange><!-- 0..* Content as for Observation.referenceRange Provides guide for interpretation of component
    result --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{
  "resourceType" : "Observation",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for observation
  "basedOn" : [{ Reference(CarePlan|DeviceRequest|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ProcedureRequest|ReferralRequest) }], // Fulfills plan, proposal or order
  "status" : "<code>", // R!  registered | preliminary | final | amended +
  "category" : [{ CodeableConcept }], // Classification of  type of observation
  "code" : { CodeableConcept }, // R!  Type of observation (code / type)
  "subject" : { Reference(Patient|Group|Device|Location) }, // Who and/or what this is about
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Healthcare event during which this observation is made
  // effective[x]: Clinically relevant time/time-period for observation. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "issued" : "<instant>", // Date/Time this was made available
  "performer" : [{ Reference(Practitioner|Organization|Patient|RelatedPerson) }], // Who is responsible for the
    observation
  // value[x]: Actual result. One of these 11:
  "valueQuantity" : { Quantity },
  "valueCodeableConcept" : { CodeableConcept },
  "valueString" : "<string>",
  "valueBoolean" : <boolean>,
  "valueRange" : { Range },
  "valueRatio" : { Ratio },
  "valueSampledData" : { SampledData },
  "valueAttachment" : { Attachment },
  "valueTime" : "<time>",
  "valueDateTime" : "<dateTime>",
  "valuePeriod" : { Period },
  "dataAbsentReason" : { CodeableConcept }, // C? Why the result is missing
  "interpretation" : { CodeableConcept }, // High, low, normal, etc.
  "comment" : "<string>", // Comments about result
  "bodySite" : { CodeableConcept }, // Observed body part
  "method" : { CodeableConcept }, // How it was done
  "specimen" : { Reference(Specimen) }, // Specimen used for this observation
  "device" : { Reference(Device|DeviceMetric) }, // (Measurement) Device
  "referenceRange" : [{ // Provides guide for interpretation
    "low" : { Quantity(SimpleQuantity) }, // C? Low Range, if relevant
    "high" : { Quantity(SimpleQuantity) }, // C? High Range, if relevant
    "type" : { CodeableConcept }, // Reference range qualifier
    "appliesTo" : [{ CodeableConcept }], // Reference range population
    "age" : { Range }, // Applicable age range, if relevant
    "text" : "<string>" // Text based reference range in an observation
  }],
  "related" : [{ // Resource related to this observation
    "type" : "<code>", // has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by
    "target" : { Reference(Observation|QuestionnaireResponse|Sequence) } // R!  Resource that is related to this one
  }],
  "component" : [{ // Component results
    "code" : { CodeableConcept }, // R!  Type of component observation (code / type)
    // value[x]: Actual component result. One of these 10:
    "valueQuantity" : { Quantity },
    "valueCodeableConcept" : { CodeableConcept },
    "valueString" : "<string>",
    "valueRange" : { Range },
    "valueRatio" : { Ratio },
    "valueSampledData" : { SampledData },
    "valueAttachment" : { Attachment },
    "valueTime" : "<time>",
    "valueDateTime" : "<dateTime>",
    "valuePeriod" : { Period },
    "dataAbsentReason" : { CodeableConcept }, // C? Why the component result is missing
    "interpretation" : { CodeableConcept }, // High, low, normal, etc.
    "referenceRange" : [{ Content as for Observation.referenceRange }] // Provides guide for interpretation of component
     result
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .
```

```
[ a fhir:Observation;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Observation.identifier [ Identifier ], ... ; # 0..* Business Identifier for observation
  fhir:Observation.basedOn [ Reference(CarePlan|DeviceRequest|ImmunizationRecommendation|MedicationRequest|
  NutritionOrder|ProcedureRequest|ReferralRequest) ], ... ; # 0..* Fulfills plan, proposal or order
  fhir:Observation.status [ code ]; # 1..1 registered | preliminary | final | amended +
  fhir:Observation.category [ CodeableConcept ], ... ; # 0..* Classification of  type of observation
  fhir:Observation.code [ CodeableConcept ]; # 1..1 Type of observation (code / type)
  fhir:Observation.subject [ Reference(Patient|Group|Device|Location) ]; # 0..1 Who and/or what this is about
  fhir:Observation.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Healthcare event during which this observation
     is made
  # Observation.effective[x] : 0..1 Clinically relevant time/time-period for observation. One of these 2
    fhir:Observation.effectiveDateTime [ dateTime ]
    fhir:Observation.effectivePeriod [ Period ]
  fhir:Observation.issued [ instant ]; # 0..1 Date/Time this was made available
  fhir:Observation.performer [ Reference(Practitioner|Organization|Patient|RelatedPerson) ], ... ; # 0..* Who is
    responsible for the observation
  # Observation.value[x] : 0..1 Actual result. One of these 11
    fhir:Observation.valueQuantity [ Quantity ]
    fhir:Observation.valueCodeableConcept [ CodeableConcept ]
    fhir:Observation.valueString [ string ]
    fhir:Observation.valueBoolean [ boolean ]
    fhir:Observation.valueRange [ Range ]
    fhir:Observation.valueRatio [ Ratio ]
    fhir:Observation.valueSampledData [ SampledData ]
    fhir:Observation.valueAttachment [ Attachment ]
    fhir:Observation.valueTime [ time ]
    fhir:Observation.valueDateTime [ dateTime ]
    fhir:Observation.valuePeriod [ Period ]
  fhir:Observation.dataAbsentReason [ CodeableConcept ]; # 0..1 Why the result is missing
  fhir:Observation.interpretation [ CodeableConcept ]; # 0..1 High, low, normal, etc.
  fhir:Observation.comment [ string ]; # 0..1 Comments about result
  fhir:Observation.bodySite [ CodeableConcept ]; # 0..1 Observed body part
  fhir:Observation.method [ CodeableConcept ]; # 0..1 How it was done
  fhir:Observation.specimen [ Reference(Specimen) ]; # 0..1 Specimen used for this observation
  fhir:Observation.device [ Reference(Device|DeviceMetric) ]; # 0..1 (Measurement) Device
  fhir:Observation.referenceRange [ # 0..* Provides guide for interpretation
    fhir:Observation.referenceRange.low [ Quantity(SimpleQuantity) ]; # 0..1 Low Range, if relevant
    fhir:Observation.referenceRange.high [ Quantity(SimpleQuantity) ]; # 0..1 High Range, if relevant
    fhir:Observation.referenceRange.type [ CodeableConcept ]; # 0..1 Reference range qualifier
    fhir:Observation.referenceRange.appliesTo [ CodeableConcept ], ... ; # 0..* Reference range population
    fhir:Observation.referenceRange.age [ Range ]; # 0..1 Applicable age range, if relevant
    fhir:Observation.referenceRange.text [ string ]; # 0..1 Text based reference range in an observation
  ], ...;
  fhir:Observation.related [ # 0..* Resource related to this observation
    fhir:Observation.related.type [ code ]; # 0..1 has-member | derived-from | sequel-to | replaces | qualified-by |
    interfered-by
    fhir:Observation.related.target [ Reference(Observation|QuestionnaireResponse|Sequence) ]; # 1..1 Resource that is
    related to this one
  ], ...;
  fhir:Observation.component [ # 0..* Component results
    fhir:Observation.component.code [ CodeableConcept ]; # 1..1 Type of component observation (code / type)
    # Observation.component.value[x] : 0..1 Actual component result. One of these 10
      fhir:Observation.component.valueQuantity [ Quantity ]
      fhir:Observation.component.valueCodeableConcept [ CodeableConcept ]
      fhir:Observation.component.valueString [ string ]
      fhir:Observation.component.valueRange [ Range ]
      fhir:Observation.component.valueRatio [ Ratio ]
      fhir:Observation.component.valueSampledData [ SampledData ]
      fhir:Observation.component.valueAttachment [ Attachment ]
      fhir:Observation.component.valueTime [ time ]
      fhir:Observation.component.valueDateTime [ dateTime ]
      fhir:Observation.component.valuePeriod [ Period ]
    fhir:Observation.component.dataAbsentReason [ CodeableConcept ]; # 0..1 Why the component result is missing
    fhir:Observation.component.interpretation [ CodeableConcept ]; # 0..1 High, low, normal, etc.
    fhir:Observation.component.referenceRange [ See Observation.referenceRange ], ... ; # 0..* Provides guide for
    interpretation of component result
  ], ...;
]
```

Changes since DSTU2

Observation

Observation.basedOn | Added Element

Observation.category | Max Cardinality changed from 1 to *

Observation.context | Renamed from encounter to context Add Reference(EpisodeOfCare)

Observation.value[x] | Add boolean

Observation.comment | Renamed from comments to comment

Observation.referenceRange.type | Added Element

Observation.referenceRange.appliesTo | Renamed from meaning to appliesTo Max Cardinality changed from 1 to *

Observation.related.target | Add Reference(Sequence)

Observation.component.interpretation | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 34 tests that all execute ok. All tests pass round-trip testing and 15 r3 resources are invalid (58 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints

Observation | I | | DomainResource | Measurements and simple assertions + If code is the same as a component code then the value element associated with the code SHALL NOT be present + dataAbsentReason SHALL only be present if Observation.value[x] is not present Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | Business Identifier for observation

basedOn | ? | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ProcedureRequest | ReferralRequest) | Fulfills plan, proposal or order

status | ?!? | 1..1 | code | registered | preliminary | final | amended + ObservationStatus (Required)

category | | 0..* | CodeableConcept | Classification of type of observation Observation Category Codes (Preferred)

code | ? | 1..1 | CodeableConcept | Type of observation (code / type) LOINC Codes (Example)

subject | ? | 0..1 | Reference(Patient | Group | Device | Location) | Who and/or what this is about

context | | 0..1 | Reference(Encounter | EpisodeOfCare) | Healthcare event during which this observation is made

effective[x] | ? | 0..1 | | Clinically relevant time/time-period for observation

effectiveDateTime | | | dateTime

effectivePeriod | | | Period

issued | ? | 0..1 | instant | Date/Time this was made available

performer | ? | 0..* | Reference(Practitioner | Organization | Patient | RelatedPerson) | Who is responsible for the observation

value[x] | ?I | 0..1 | | Actual result

valueQuantity | | | Quantity

valueCodeableConcept | | | CodeableConcept

valueString | | | string

valueBoolean | | | boolean

valueRange | | | Range

valueRatio | | | Ratio

valueSampledData | | | SampledData

valueAttachment | | | Attachment

valueTime | | | time

valueDateTime | | | dateTime

valuePeriod | | | Period

dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Observation Value Absent Reason (Extensible)

interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Observation Interpretation Codes (Extensible)

comment | | 0..1 | string | Comments about result

bodySite | | 0..1 | CodeableConcept | Observed body part SNOMED CT Body Structures (Example)

method | | 0..1 | CodeableConcept | How it was done Observation Methods (Example)

specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation

device | | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device

referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation + Must have at least a low or a high or text

low | I | 0..1 | SimpleQuantity | Low Range, if relevant

high | I | 0..1 | SimpleQuantity | High Range, if relevant

type | | 0..1 | CodeableConcept | Reference range qualifier Observation Reference Range Meaning Codes (Extensible)

appliesTo | | 0..* | CodeableConcept | Reference range population Observation Reference Range Applies To Codes (Example)

age | | 0..1 | Range | Applicable age range, if relevant

text | | 0..1 | string | Text based reference range in an observation

related | ? | 0..* | BackboneElement | Resource related to this observation

type | | 0..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by ObservationRelationshipType (Required)

target | | 1..1 | Reference(Observation | QuestionnaireResponse | Sequence) | Resource that is related to this one

component | ? | 0..* | BackboneElement | Component results

code | ? | 1..1 | CodeableConcept | Type of component observation (code / type) LOINC Codes (Example)

value[x] | ? | 0..1 | | Actual component result

valueQuantity | | | Quantity

valueCodeableConcept | | | CodeableConcept

valueString | | | string

valueRange | | | Range

valueRatio | | | Ratio

valueSampledData | | | SampledData

valueAttachment | | | Attachment

valueTime | | | time

valueDateTime | | | dateTime

valuePeriod | | | Period

dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Observation Value Absent Reason (Extensible)

interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Observation Interpretation Codes (Extensible)

referenceRange | | 0..* | see referenceRange | Provides guide for interpretation of component result

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Observation xmlns="http://hl7.org/fhir">
```

```
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for observation --></identifier>
 <basedOn><!-- 0..* Reference(CarePlan|DeviceRequest|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ProcedureRequest|ReferralRequest) Fulfills plan, proposal or order --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended + -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation --></category>
 <code><!-- 1..1 CodeableConcept Type of observation (code / type) --></code>
 <subject><!-- 0..1 Reference(Patient|Group|Device|Location) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Healthcare event during which this observation is made
    --></context>
 <effective[x]><!-- 0..1 dateTime|Period Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner|Organization|Patient|RelatedPerson) Who is responsible for the observation
    --></performer>
 <value[x]><!--  0..1 Quantity|CodeableConcept|string|boolean|Range|Ratio|
   SampledData|Attachment|time|dateTime|Period Actual result --></value[x]>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc. --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device|DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  <!-- 0..* Provides guide for interpretation -->
  <low><!--  0..1 Quantity(SimpleQuantity) Low Range, if relevant --></low>
  <high><!--  0..1 Quantity(SimpleQuantity) High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population --></appliesTo>
  <age><!-- 0..1 Range Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  <!-- 0..* Resource related to this observation -->
  <type value="[code]"/><!-- 0..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by -->
  <target><!-- 1..1 Reference(Observation|QuestionnaireResponse|Sequence) Resource that is related to this one
    --></target>
 </related>
 <component>  <!-- 0..* Component results -->
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type) --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|Range|Ratio|SampledData|
    Attachment|time|dateTime|Period Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc. --></interpretation>
  <referenceRange><!-- 0..* Content as for Observation.referenceRange Provides guide for interpretation of component
    result --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{
  "resourceType" : "Observation",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for observation
  "basedOn" : [{ Reference(CarePlan|DeviceRequest|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ProcedureRequest|ReferralRequest) }], // Fulfills plan, proposal or order
  "status" : "<code>", // R!  registered | preliminary | final | amended +
  "category" : [{ CodeableConcept }], // Classification of  type of observation
  "code" : { CodeableConcept }, // R!  Type of observation (code / type)
  "subject" : { Reference(Patient|Group|Device|Location) }, // Who and/or what this is about
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Healthcare event during which this observation is made
  // effective[x]: Clinically relevant time/time-period for observation. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "issued" : "<instant>", // Date/Time this was made available
  "performer" : [{ Reference(Practitioner|Organization|Patient|RelatedPerson) }], // Who is responsible for the
    observation
  // value[x]: Actual result. One of these 11:
  "valueQuantity" : { Quantity },
  "valueCodeableConcept" : { CodeableConcept },
  "valueString" : "<string>",
  "valueBoolean" : <boolean>,
  "valueRange" : { Range },
  "valueRatio" : { Ratio },
  "valueSampledData" : { SampledData },
  "valueAttachment" : { Attachment },
  "valueTime" : "<time>",
  "valueDateTime" : "<dateTime>",
  "valuePeriod" : { Period },
  "dataAbsentReason" : { CodeableConcept }, // C? Why the result is missing
  "interpretation" : { CodeableConcept }, // High, low, normal, etc.
  "comment" : "<string>", // Comments about result
  "bodySite" : { CodeableConcept }, // Observed body part
  "method" : { CodeableConcept }, // How it was done
  "specimen" : { Reference(Specimen) }, // Specimen used for this observation
  "device" : { Reference(Device|DeviceMetric) }, // (Measurement) Device
  "referenceRange" : [{ // Provides guide for interpretation
    "low" : { Quantity(SimpleQuantity) }, // C? Low Range, if relevant
    "high" : { Quantity(SimpleQuantity) }, // C? High Range, if relevant
```

```
    "type" : { CodeableConcept }, // Reference range qualifier
    "appliesTo" : [{ CodeableConcept }], // Reference range population
    "age" : { Range }, // Applicable age range, if relevant
    "text" : "<string>" // Text based reference range in an observation
  }],
  "related" : [{ // Resource related to this observation
    "type" : "<code>", // has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by
    "target" : { Reference(Observation|QuestionnaireResponse|Sequence) } // R!  Resource that is related to this one
  }],
  "component" : [{ // Component results
    "code" : { CodeableConcept }, // R!  Type of component observation (code / type)
    // value[x]: Actual component result. One of these 10:
    "valueQuantity" : { Quantity },
    "valueCodeableConcept" : { CodeableConcept },
    "valueString" : "<string>",
    "valueRange" : { Range },
    "valueRatio" : { Ratio },
    "valueSampledData" : { SampledData },
    "valueAttachment" : { Attachment },
    "valueTime" : "<time>",
    "valueDateTime" : "<dateTime>",
    "valuePeriod" : { Period },
    "dataAbsentReason" : { CodeableConcept }, // C? Why the component result is missing
    "interpretation" : { CodeableConcept }, // High, low, normal, etc.
    "referenceRange" : [{ Content as for Observation.referenceRange }] // Provides guide for interpretation of component
     result
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Observation;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Observation.identifier [ Identifier ], ... ; # 0..* Business Identifier for observation
  fhir:Observation.basedOn [ Reference(CarePlan|DeviceRequest|ImmunizationRecommendation|MedicationRequest|
  NutritionOrder|ProcedureRequest|ReferralRequest) ], ... ; # 0..* Fulfills plan, proposal or order
  fhir:Observation.status [ code ]; # 1..1 registered | preliminary | final | amended +
  fhir:Observation.category [ CodeableConcept ], ... ; # 0..* Classification of  type of observation
  fhir:Observation.code [ CodeableConcept ]; # 1..1 Type of observation (code / type)
  fhir:Observation.subject [ Reference(Patient|Group|Device|Location) ]; # 0..1 Who and/or what this is about
  fhir:Observation.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Healthcare event during which this observation
     is made
  # Observation.effective[x] : 0..1 Clinically relevant time/time-period for observation. One of these 2
    fhir:Observation.effectiveDateTime [ dateTime ]
    fhir:Observation.effectivePeriod [ Period ]
  fhir:Observation.issued [ instant ]; # 0..1 Date/Time this was made available
  fhir:Observation.performer [ Reference(Practitioner|Organization|Patient|RelatedPerson) ], ... ; # 0..* Who is
    responsible for the observation
  # Observation.value[x] : 0..1 Actual result. One of these 11
    fhir:Observation.valueQuantity [ Quantity ]
    fhir:Observation.valueCodeableConcept [ CodeableConcept ]
    fhir:Observation.valueString [ string ]
    fhir:Observation.valueBoolean [ boolean ]
    fhir:Observation.valueRange [ Range ]
    fhir:Observation.valueRatio [ Ratio ]
    fhir:Observation.valueSampledData [ SampledData ]
    fhir:Observation.valueAttachment [ Attachment ]
    fhir:Observation.valueTime [ time ]
    fhir:Observation.valueDateTime [ dateTime ]
    fhir:Observation.valuePeriod [ Period ]
  fhir:Observation.dataAbsentReason [ CodeableConcept ]; # 0..1 Why the result is missing
  fhir:Observation.interpretation [ CodeableConcept ]; # 0..1 High, low, normal, etc.
  fhir:Observation.comment [ string ]; # 0..1 Comments about result
  fhir:Observation.bodySite [ CodeableConcept ]; # 0..1 Observed body part
  fhir:Observation.method [ CodeableConcept ]; # 0..1 How it was done
  fhir:Observation.specimen [ Reference(Specimen) ]; # 0..1 Specimen used for this observation
  fhir:Observation.device [ Reference(Device|DeviceMetric) ]; # 0..1 (Measurement) Device
  fhir:Observation.referenceRange [ # 0..* Provides guide for interpretation
    fhir:Observation.referenceRange.low [ Quantity(SimpleQuantity) ]; # 0..1 Low Range, if relevant
    fhir:Observation.referenceRange.high [ Quantity(SimpleQuantity) ]; # 0..1 High Range, if relevant
    fhir:Observation.referenceRange.type [ CodeableConcept ]; # 0..1 Reference range qualifier
    fhir:Observation.referenceRange.appliesTo [ CodeableConcept ], ... ; # 0..* Reference range population
    fhir:Observation.referenceRange.age [ Range ]; # 0..1 Applicable age range, if relevant
    fhir:Observation.referenceRange.text [ string ]; # 0..1 Text based reference range in an observation
  ], ...;
  fhir:Observation.related [ # 0..* Resource related to this observation
    fhir:Observation.related.type [ code ]; # 0..1 has-member | derived-from | sequel-to | replaces | qualified-by |
    interfered-by
    fhir:Observation.related.target [ Reference(Observation|QuestionnaireResponse|Sequence) ]; # 1..1 Resource that is
    related to this one
  ], ...;
  fhir:Observation.component [ # 0..* Component results
    fhir:Observation.component.code [ CodeableConcept ]; # 1..1 Type of component observation (code / type)
    # Observation.component.value[x] : 0..1 Actual component result. One of these 10
```

```
      fhir:Observation.component.valueQuantity [ Quantity ]
      fhir:Observation.component.valueCodeableConcept [ CodeableConcept ]
      fhir:Observation.component.valueString [ string ]
      fhir:Observation.component.valueRange [ Range ]
      fhir:Observation.component.valueRatio [ Ratio ]
      fhir:Observation.component.valueSampledData [ SampledData ]
      fhir:Observation.component.valueAttachment [ Attachment ]
      fhir:Observation.component.valueTime [ time ]
      fhir:Observation.component.valueDateTime [ dateTime ]
      fhir:Observation.component.valuePeriod [ Period ]
    fhir:Observation.component.dataAbsentReason [ CodeableConcept ]; # 0..1 Why the component result is missing
    fhir:Observation.component.interpretation [ CodeableConcept ]; # 0..1 High, low, normal, etc.
    fhir:Observation.component.referenceRange [ See Observation.referenceRange ], ... ; # 0..* Provides guide for
    interpretation of component result
  ], ...;
]
```

Changes since DSTU2

Observation
Observation.basedOn | Added Element
Observation.category | Max Cardinality changed from 1 to *
Observation.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
Observation.value[x] | Add boolean
Observation.comment | Renamed from comments to comment
Observation.referenceRange.type | Added Element
Observation.referenceRange.appliesTo | Renamed from meaning to appliesTo Max Cardinality changed from 1 to *
Observation.related.target | Add Reference(Sequence)
Observation.component.interpretation | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 34 tests that all execute ok. All tests pass round-trip testing and 15 r3 resources are invalid (58 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

10.1.3.1 Terminology Bindings

Path | Definition | Type | Reference
Observation.status | Codes providing the status of an observation. | Required | ObservationStatus
Observation.category | Codes for high level observation categories. | Preferred | Observation Category Codes
Observation.code Observation.component.code | Codes identifying names of simple observations. | Example | LOINC Codes
Observation.dataAbsentReason Observation.component.dataAbsentReason | Codes specifying why the result (Observation.value[x]) is missing. | Extensible | Observation Value Absent Reason
Observation.interpretation Observation.component.interpretation | Codes identifying interpretations of observations. | Extensible | Observation Interpretation Codes
Observation.bodySite | Codes describing anatomical locations. May include laterality. | Example | SNOMED CT Body Structures
Observation.method | Methods for simple observations. | Example | Observation Methods
Observation.referenceRange.type | Code for the meaning of a reference range. | Extensible | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Codes identifying the population the reference range applies to. | Example | Observation Reference Range Applies To Codes
Observation.related.type | Codes specifying how two observations are related. | Required | ObservationRelationshipType

10.1.3.2 Constraints

* obs-3: On Observation.referenceRange: Must have at least a low or a high or text (expression on Observation.referenceRange: low.exists() or high.exists() or text.exists())

* obs-6: dataAbsentReason SHALL only be present if Observation.value[x] is not present (expression : dataAbsentReason.empty() or value.empty())

* obs-7: If code is the same as a component code then the value element associated with the code SHALL NOT be present (expression : value.empty() or component.code.where((coding.code = %resource.code.coding.code) and (coding.system = %resource.code.coding.system)).empty())

10.1.4 Notes:

At its simplest, a resource instances can consist of only a code and a value, and status flag. The relevance of other properties will vary based on the type of observation.

Profiles will be created to provide guidance on capturing certain types of simple observations. This resource focuses on the level of detail captured by most systems. However, any "simple" observation can easily be broken into numerous components and sub-components to provide additional information relevant in certain circumstances. As with other resources, extensions can be used to introduce this additional complexity.

10.1.4.1 Observation Grouping

Many observations have important relationships to other observations and need to be grouped together. Three structures have been defined to do this: DiagnosticReport and DiagnosticReport.result, Observation and Observation.component, and Observation and Observation.related. The table below provides guidance around which structure to use. Because the idea of

what to group together is often highly contextual and based upon the end user's point of view, the choice of which structure to use will be driven by jurisdiction, organizational practice and context. Profiling will normally be necessary for implementation.

DiagnosticReport and DiagnosticReport.result | Observation and Observation.component | Observation and Observation.related
DiagnosticReport relates directly to an order (ProcedureRequest). The DiagnosticReport.code names the panel and serves as the grouping element, which is traditionally referred to as a "panel" or "battery" by laboratories. The DiagnosticReport.result element references the individual observations. Several examples demonstrate observation grouping using DiagnosticReport as the grouping structure. | Observation.component is used for any supporting result that cannot reasonably be interpreted and used outside the scope of the Observation it is a component of. Components should only be used when there is only one method, one observation, one performer, one device, and one time. For example, systolic and diastolic blood pressure are represented as a single Observation (e.g. Blood pressure panel) because the two are almost always produced and interpreted together. Note that the component.code may in some cases only be able to be understood in relation to the Observation.code (for example, see the $stats operation). | Observation.related is used for any supporting result that can be interpreted and used on its own and has one or more different values for method, observation, performer, device, time, and/or error conditions. The top level observation specifies the grouping code in Observation.code, but typically does not have its own Observation.value, and the set of member observations are listed in the Observation.related element. This structure permits nested grouping when used with DiagnosticReport (e.g. complex micro isolate and sensitivities report).

10.1.4.2 Value[x] and Datatypes

* The element, Observation.value[x], has a variable name depending on the type as follows:

* valueQuantity

* valueCodeableConcept

* valueString

* valueBoolean

* valueRange

* valueRatio

* valueSampledData

* valueAttachment

* valueTime

* valueDateTime

* valuePeriod

Using codes for result values

When a result value is a represented as a predefined concept using a code, valueCodeableConcept is used. This element is bound to a value set comprised of a standard nomenclature such as SNOMED CT or a source system ("local") coded result values. Results may be coded in multiple value sets based on different code systems and these may be mapped using the ConceptMap resource and/or given as translations directly in the element as shown in the example below.

For example the LOINC 43304-5 Chlamydia trachomatis rRNA [Presence] in Unspecified specimen by Probe and target amplification method is typically associated with coded presence/absence concepts. Using the coded value for 'negative' with a standard code translation, valueCodeableConcept would be:

```
"valueCodeableConcept": {
"coding": [
  {
  "system": "http://snomed.info/sct",
  "code": "260385009",
  "display": "Negative"
  },
  {
  "https://acme.lab/resultcodes",
  "code": "NEG",
  "display": "Negative"
  }
],
"text": "Negative for Chlamydia Trachomatis rRNA"
  }
```

Text values for coded results:

When the data element is usually coded or the type associated with the code element defines a coded value, use valueCodeableConcept even if there is no appropriate code and only un-coded text is available. For example using text only, the valueCodeableConcept element would be:

```
"valueCodeableConcept": {
"text": "uncoded free text result"
  }
```

When a coded answer list includes a concept code for "other" and there is a free text description of the concept, the valueCodeableConcept.text element should be used to capture the full meaning of the source. In the example below, the answer code "Other" is provided in the valueCodeableConcept element and the text value supplied value in the CodeableConcept.text element.

```
  {
    "resourceType": "Observation",
    "code": {
      "coding": [{
        "system": "http://loinc.org",
        "code": "74076-1",
```

```
        "display": "Medication or substance involved"
      }]
    },
    "valueCodeableConcept": {
      "coding": [{
        "system": "http://loinc.org",
        "code": " LA20343-2",
        "display": "Other substance: PLEASE SPECIFY"
      }],
      "text": "Other: Blue pills I found under my couch"
    }
  }
```

* The Boolean data type is rarely used for value[x] because most observations result values are never truly Boolean due to exceptional values such as "unknown", therefore they should use the CodeableConcept data type instead.

* The special values "E" (error), "L" (below detection limit) and "U" (above detection limit) can be used are in the SampledData data type. However when using valueQuantity in an observation for above and below detection limit values, valueQuantity should be used by stating the limit along with the comparator. In addition, when there is an error the dataAbsentReason element should be used with the appropriate value ('error' or 'NaN'). For example if the value was below the lower limit of detection of <2.0 mmol/L the would be: If the value was "NaN" (i.e. an error) the element would be absent and element would be:

valueQuantity

```
    "valueQuantity": {
    "value": 2.0,
     "comparator" : "<",
    "unit": "mmol/l",
    "system": "http://unitsofmeasure.org",
    "code": "mmol/L"
     }
```

valueCodeableConcept

dataAbsentReason

```
"dataAbsentReason": {
    "coding": [
      {
        "system": "http://hl7.org/fhir/data-absent-reason",
        "code": "NaN",
        "display": "Not a Number"
      }
```

* Because there are multiple types allowed for the value element, multiple value search parameters are defined. There is no standard parameter for searching values of type Attachment, or Ratio

10.1.4.3 Physiologically Relevant Time of the Observation

The effectiveDateTime or effectivePeriod is the time that the observation is most relevant as an observation of the subject. For a biological subject (e.g. a human patient), this is the physiologically relevant time of the observation. In the case of an observation using a specimen, this represents the start and end of the specimen collection (e.g. 24 hour Urine Sodium), but if the collection time is sufficiently short, this is reported as a point in time value (e.g. normal venipuncture). In the case of an observation obtained directly from a subject (e.g. BP, Chest X-ray), this is the start and end time of the observation process, which again, is often reported as a single point in time.

10.1.4.4 Reference Range

Most common observations will only have one generic reference range. Reference ranges may be useful for lab tests and other measures like systolic blood pressure, but will have little relevance for something like "pregnancy status". Systems MAY choose to restrict to only supplying the relevant reference range based on knowledge about the patient (e.g. specific to the patient's age, gender, weight and other factors), but this may not be possible or appropriate. Whenever more than one reference range is supplied, the differences between them SHOULD be provided in the reference range and/or age properties.

10.1.4.5 Canceled or Aborted Observations

If a measurement or test could not be completed (for example if the specimen is unsatisfactory or the provider cancelled the order), then the status value should be updated to "cancelled" and the specific details given - preferably as coded values in the dataAbsentReason or valueCodeableConcept element. Additional information may be provided in comments as well. The specimen reject example demonstrates this using a coded value for unsatisfactory specimen in dataAbsentReason.

10.1.5 Operations defined for Observation

10.1.5.1 Searching for the Last N Observations

The lastn query operation meets the common need for searching for the most recent or "last known" Observations for a subject. Examples where this query could be used:

* Fetch the last 5 temperatures for a patient to view trends

* Get the most recent lab results for patient

* Fetch the last 3 results for all vitals for a patient

See the Last N Observations Query section in the Observation resource operations page for more information and examples

STU Note: This is the first draft of this operation and we are seeking input from the implementer community on it and whether it should be extended to other resources such as DiagnosticReport. This operation could be the subject of future connectathon testing.

Feedback is welcome here .

10.1.5.2 Retrieving Statistics for Laboratory Observations

The stats operation performs a set of statistical calculations on a set of clinical measurements such as a blood pressure as stored on the server. This operation is focused on Observation resources with valueQuantity elements that have UCUM unit codes. Examples where this operation could be used:

* Get the average, min, max and count of a series of BP measurements for a patient

* Determine 20th or 80th percentile on a set of measurements over a time period

Observation Statistics

10.1.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
based-on | reference | Reference to the test or procedure request. | Observation.basedOn (ReferralRequest, CarePlan, MedicationRequest, NutritionOrder, ProcedureRequest, DeviceRequest, ImmunizationRecommendation)
category | token | The classification of the type of observation | Observation.category
code | token | The code of the observation type | Observation.code | 8 Resources
code-value-concept | composite | Code and coded value parameter pair | Observation
code-value-date | composite | Code and date/time value parameter pair | Observation
code-value-quantity | composite | Code and quantity value parameter pair | Observation
code-value-string | composite | Code and string value parameter pair | Observation
combo-code | token | The code of the observation type or component type | Observation.code | Observation.component.code
combo-code-value-concept | composite | Code and coded value parameter pair, including in components | Observation | Observation.component
combo-code-value-quantity | composite | Code and quantity value parameter pair, including in components | Observation | Observation.component
combo-data-absent-reason | token | The reason why the expected value in the element Observation.value[x] or Observation.component.value[x] is missing. | Observation.dataAbsentReason | Observation.component.dataAbsentReason
combo-value-concept | token | The value or component value of the observation, if the value is a CodeableConcept | Observation.value.as(CodeableConcept) | Observation.component.value.as(CodeableConcept)
combo-value-quantity | quantity | The value or component value of the observation, if the value is a Quantity, or a SampledData (just search on the bounds of the values in sampled data) | Observation.value.as(Quantity) | Observation.component.value.as(Quantity)
component-code | token | The component code of the observation type | Observation.component.code
component-code-value-concept | composite | Component code and component coded value parameter pair | Observation.component
component-code-value-quantity | composite | Component code and component quantity value parameter pair | Observation.component
component-data-absent-reason | token | The reason why the expected value in the element Observation.component.value[x] is missing. | Observation.component.dataAbsentReason
component-value-concept | token | The value of the component observation, if the value is a CodeableConcept | Observation.component.value.as(CodeableConcept)
component-value-quantity | quantity | The value of the component observation, if the value is a Quantity, or a SampledData (just search on the bounds of the values in sampled data) | Observation.component.value.as(Quantity)
context | reference | Healthcare event (Episode-of-care or Encounter) related to the observation | Observation.context (EpisodeOfCare, Encounter)
data-absent-reason | token | The reason why the expected value in the element Observation.value[x] is missing. | Observation.dataAbsentReason
date | date | Obtained date/time. If the obtained element is a period, a date that falls in the period | Observation.effective | 18 Resources
device | reference | The Device that generated the observation data. | Observation.device (Device, DeviceMetric)
encounter | reference | Encounter related to the observation | Observation.context (Encounter) | 12 Resources
identifier | token | The unique id for a particular observation | Observation.identifier | 26 Resources
method | token | The method used for the observation | Observation.method
patient | reference | The subject that the observation is about (if patient) | Observation.subject (Patient) | 31 Resources
performer | reference | Who performed the observation | Observation.performer (Practitioner, Organization, Patient, RelatedPerson)
related | composite | Related Observations - search on related-type and related-target together | Observation.related
related-target | reference | Resource that is related to this one | Observation.related.target (Observation, Sequence, QuestionnaireResponse)
related-type | token | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by | Observation.related.type
specimen | reference | Specimen used for this observation | Observation.specimen (Specimen)
status | token | The status of the observation | Observation.status
subject | reference | The subject that the observation is about | Observation.subject (Group, Device, Patient, Location)
value-concept | token | The value of the observation, if the value is a CodeableConcept | Observation.value.as(CodeableConcept)
value-date | date | The value of the observation, if the value is a date or period of time | Observation.value.as(DateTime) | Observation.value.as(Period)
value-quantity | quantity | The value of the observation, if the value is a Quantity, or a SampledData (just search on the bounds of the values in sampled data) | Observation.value.as(Quantity)
value-string | string | The value of the observation, if the value is a string, and also searches in CodeableConcept.text | Observation.value.as(String)

10.1.7 Resource Observation - Examples

URL: https://hl7.org/fhir/STU3/observation-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.7 Resource Observation - Examples

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Example List:

Simple Weight Example | example | XML | JSON | Turtle
Simple Respiratory Rate Example | respiratory-rate | XML | JSON | Turtle
Simple Heart Rate Example | heart-rate | XML | JSON | Turtle
Simple Body Temperature Example | body-temperature | XML | JSON | Turtle
Simple Body Height Example | body-height | XML | JSON | Turtle
Simple Body Length Example | body-length | XML | JSON | Turtle
Simple Head Circumference Example | head-circumference | XML | JSON | Turtle
Simple BMI Example | bmi | XML | JSON | Turtle
Oxygen Saturation by Pulse Oximetry Example (DeviceMetricObservation profile) | satO2 | XML | JSON | Turtle
Blood Pressure Measurement | blood-pressure | XML | JSON | Turtle
Blood Pressure Measurement with an absent Diastolic measurement | blood-pressure-dar | XML | JSON | Turtle
Simple Mean Arterial Blood Pressure Example | mbp | XML | JSON | Turtle
Simple Panel of Heart Rate, Repiratory Rate. Blood Pressure and Body Temperature | vitals-panel | XML | JSON | Turtle
Blood Pressure Measurement cancelled | blood-pressure-cancel | XML | JSON | Turtle
Real-world patient - glucose | f001 | XML | JSON | Turtle
Specimen Reject Example | unsat | XML | JSON | Turtle
Real-world patient -base excess | f002 | XML | JSON | Turtle
Real-world patient - CO2 | f003 | XML | JSON | Turtle
Real-world patient - erythrocyte | f004 | XML | JSON | Turtle
Real-world patient - hemoglobin | f005 | XML | JSON | Turtle
An "ask at order" response using the datetime datetype for the result value | date-lastmp | XML | JSON | Turtle
Real-world patient - temperature | f202 | XML | JSON | Turtle
Real-world patient - CO2 | f203 | XML | JSON | Turtle
Real-world patient - creatinine | f204 | XML | JSON | Turtle
Real-world patient - estimated gfr | f205 | XML | JSON | Turtle
Real-world patient - staphylococcus | f206 | XML | JSON | Turtle
EKG Example using SampledData data type | ekg | XML | JSON | Turtle
Clinical assessment tool example - GCS using related Observations | glasgow | XML | JSON | Turtle
Clinical assessment tool example - GCS using related QuestionnaireAnswer | gcs-qa | XML | JSON | Turtle
Example of a 1 minute Apgar score including the individual score components | 1minute-apgar-score | XML | JSON | Turtle
Example of a 2 minute Apgar score including the individual score components | 2minute-apgar-score | XML | JSON | Turtle
Example of a 5 minute Apgar score including the individual score components | 5minute-apgar-score | XML | JSON | Turtle
Example of a 10 minute Apgar score including the individual score components | 10minute-apgar-score | XML | JSON | Turtle
Example of a 20 minute Apgar score including the individual score components | 20minute-apgar-score | XML | JSON | Turtle
Example of a very simple personal characteristic observation | eye-color | XML | JSON | Turtle
Example of an encoded BMD measurements of left femur | bmd | XML | JSON | Turtle
Genetics example 1 | example-genetics-1 | XML | JSON | Turtle | for Profile Observation-genetics
Genetics example 2 | example-genetics-2 | XML | JSON | Turtle | for Profile Observation-genetics
Genetics example 3 | example-genetics-3 | XML | JSON | Turtle | for Profile Observation-genetics
Genetics example 4 | example-genetics-4 | XML | JSON | Turtle | for Profile Observation-genetics
Genetics example 5 | example-genetics-5 | XML | JSON | Turtle | for Profile Observation-genetics
Example of diplotype data (with haplotypes observation as reference) | example-diplotype1 | XML | JSON | Turtle | for Profile Observation-genetics
Example of haplotype data that is basis of a diplotype data | example-haplotype1 | XML | JSON | Turtle | for Profile Observation-genetics
Example of another haplotype data that is basis of a diplotype data | example-haplotype2 | XML | JSON | Turtle | for Profile Observation-genetics
Example of phenotype data | example-phenotype | XML | JSON | Turtle | for Profile Observation-genetics
Example of a TPMT diplotype that link to two TPMT haplotype observations | example-TPMT-diplotype | XML | JSON | Turtle | for Profile Observation-genetics
Example of a TPMT haplotype observation | example-TPMT-haplotype-one | XML | JSON | Turtle | for Profile Observation-genetics
Example of another TPMT haplotype observation | example-TPMT-haplotype-two | XML | JSON | Turtle | for Profile Observation-genetics

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.1.8 Resource Observation - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/observation-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.8 Resource Observation - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 5 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Observation resource.

Observation

Definition | Measurements and simple assertions made about a patient, device or other subject.
Control | 1..1
Requirements | Observations are a key aspect of healthcare. This resource is used to capture those that do not require more sophisticated mechanisms.
Alternate Names | Vital Signs; Measurement; Results; Tests
Comments | Used for simple observations such as device measurements, laboratory atomic results, vital signs, height, weight, smoking status, comments, etc. Other resources are used to provide context for observations such as lab reports, etc.
Invariants | Defined on this element obs-6: dataAbsentReason SHALL only be present if Observation.value[x] is not present (expression : dataAbsentReason.empty() or value.empty(), xpath: not(exists(f:dataAbsentReason)) or (not(exists(*[starts-with(local-name(.), 'value')])))) obs-7: If code is the same as a component code then the value element associated with the code SHALL NOT be present (expression : value.empty() or component.code.where((coding.code = %resource.code.coding.code) and (coding.system = %resource.code.coding.system)).empty(), xpath: not(exists(f:*[starts-with(local-name(.), 'value')])) or not(count(for $coding in f:code/f:coding return parent::*/f:component/f:code/f:coding[f:code/@value=$coding/f:code/@value and f:system/@value=$coding/f:system/@value])=0))
Observation.identifier
Definition | A unique identifier assigned to this observation.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Allows observations to be distinguished and referenced.
Summary | true
Observation.basedOn
Definition | A plan, proposal or order that is fulfilled in whole or in part by this event.
Control | 0..*
Type | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ProcedureRequest | ReferralRequest)
Requirements | Allows tracing of authorization for the event and tracking whether proposals/recommendations were acted upon.
Alternate Names | Fulfills
Summary | true
Observation.status
Definition | The status of the result value.
Control | 1..1
Terminology Binding | ObservationStatus (Required)
Type | code
Is Modifier | true
Requirements | Need to track the status of individual results. Some results are finalized before the whole report is finalized.
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
Observation.category
Definition | A code that classifies the general type of observation being made.
Control | 0..*
Terminology Binding | Observation Category Codes (Preferred)
Type | CodeableConcept
Requirements | Used for filtering what observations are retrieved and displayed.
Comments | In addition to the required category valueset, this element allows various categorization schemes based on the owner’s definition of the category and effectively multiple categories can be used at once. The level of granularity is defined by the category concepts in the value set.
Observation.code
Definition | Describes what was observed. Sometimes this is called the observation "name".
Control | 1..1
Terminology Binding | LOINC Codes (Example)
Type | CodeableConcept
Requirements | Knowing what kind of observation is being made is essential to understanding the observation.
Alternate Names | Name
Summary | true
Observation.subject
Definition | The patient, or group of patients, location, or device whose characteristics (direct or indirect) are described by the observation and into whose record the observation is placed. Comments: Indirect characteristics may be those of a specimen, fetus, donor, other observer (for example a relative or EMT), or any observation made about the subject.
Control | 0..1
Type | Reference(Patient | Group | Device | Location)
Requirements | Observations have no value if you don't know who or what they're about.
Summary | true
Comments | One would expect this element to be a cardinality of 1..1. The only circumstance in which the subject can be missing is when the observation is made by a device that does not know the patient. In this case, the observation SHALL be matched to a patient through some context/channel matching technique, and at this point, the observation should be updated. If the target of the observation is different than the subject, the general extension observation-focal-subject. may be used. However, the distinction between the patient's own value for an observation versus that of the fetus, or the donor or blood product unit, etc., are often specified in the observation code.
Observation.context
Definition | The healthcare event (e.g. a patient and healthcare provider interaction) during which this observation is made.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Requirements | For some observations it may be important to know the link between an observation and a particular encounter.
Alternate Names | Encounter
Comments | This will typically be the encounter the event occurred within, but some events may be initiated prior to or after the official completion of an encounter or episode but still be tied to the context of the encounter or episode (e.g. pre-admission lab tests).

Observation.effective[x]
Definition | The time or time-period the observed value is asserted as being true. For biological subjects - e.g. human patients - this is usually called the "physiologically relevant time". This is usually either the time of the procedure or of specimen collection, but very often the source of the date/time is not known, only the date/time itself.
Control | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Knowing when an observation was deemed true is important to its relevance as well as determining trends.
Alternate Names | Occurrence
Summary | true
Comments | At least a date should be present unless this observation is a historical report.
Observation.issued
Definition | The date and time this observation was made available to providers, typically after the results have been reviewed and verified.
Control | 0..1
Type | instant
Summary | true
Comments | Updated when the result is updated.
Observation.performer
Definition | Who was responsible for asserting the observed value as "true".
Control | 0..*
Type | Reference(Practitioner | Organization | Patient | RelatedPerson)
Requirements | May give a degree of confidence in the observation and also indicates where follow-up questions should be directed.
Summary | true
Observation.value[x]
Definition | The information determined as a result of making the observation, if the information has a simple value.
Control | 0..1
Type | Quantity|CodeableConcept|string|boolean|Range|Ratio|SampledData|Attachment|time|dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | An observation exists to have a value, though it may not if it is in error, or if it represents a group of observations.
Summary | true
Comments | Normally, an observation will have either a single value or a set of related observations. A few observations (e.g. Apgar score) may have both a value and related observations (for an Apgar score, the observations from which the measure is derived). If a value is present, the datatype for this element should be determined by Observation.code. This element has a variable name depending on the type as follows: valueQuantity, valueCodeableConcept, valueString, valueBoolean, valueRange, valueRatio, valueSampledData, valueAttachment, valueTime, valueDateTime, or valuePeriod. (The name format is "'value' + the type name" with a capital on the first letter of the type). If the data element is usually coded or if the type associated with the Observation.value defines a coded value, use CodeableConcept instead of string datatype even if the value is uncoded text. A value set is bound to the ValueCodeableConcept element. For further discussion and examples see the notes section below.
Invariants | Affect this element obs-7: If code is the same as a component code then the value element associated with the code SHALL NOT be present (expression : value.empty() or component.code.where((coding.code = %resource.code.coding.code) and (coding.system = %resource.code.coding.system)).empty(), xpath: not(exists(f:*[starts-with(local-name(.), 'value')])) or not(count(for $coding in f:code/f:coding return parent::*/f:component/f:code/f:coding[f:code/@value=$coding/f:code/@value and f:system/@value=$coding/f:system/@value])=0))
Observation.dataAbsentReason
Definition | Provides a reason why the expected value in the element Observation.value[x] is missing.
Control | 0..1
Terminology Binding | Observation Value Absent Reason (Extensible)
Type | CodeableConcept
Requirements | For many results it is necessary to handle exceptional values in measurements.
Comments | Null or exceptional values can be represented two ways in FHIR Observations. One way is to simply include them in the value set and represent the exceptions in the value. For example, measurement values for a serology test could be "detected", "not detected", "inconclusive", or "specimen unsatisfactory". The alternate way is to use the value element for actual observations and use the explicit dataAbsentReason element to record exceptional values. For example, the dataAbsentReason code "error" could be used when the measurement was not completed. Note that an observation may only be reported if there are values to report. For example differential cell counts values may be reported only when > 0. Because of these options, use-case agreements are required to interpret general observations for null or exceptional values.
Invariants | Affect this element obs-6: dataAbsentReason SHALL only be present if Observation.value[x] is not present (expression : dataAbsentReason.empty() or value.empty(), xpath: not(exists(f:dataAbsentReason)) or (not(exists(*[starts-with(local-name(.), 'value')]))))
Observation.interpretation
Definition | The assessment made based on the result of the observation. Intended as a simple compact code often placed adjacent to the result value in reports and flow sheets to signal the meaning/normalcy status of the result. Otherwise known as abnormal flag.
Control | 0..1
Terminology Binding | Observation Interpretation Codes (Extensible)
Type | CodeableConcept
Requirements | For some results, particularly numeric results, an interpretation is necessary to fully understand the significance of a result.
Alternate Names | Abnormal Flag
Observation.comment
Definition | May include statements about significant, unexpected or unreliable values, or information about the source of the value where this may be relevant to the interpretation of the result.
Control | 0..1
Type | string
Requirements | Need to be able to provide free text additional information.
Observation.bodySite
Definition | Indicates the site on the subject's body where the observation was made (i.e. the target site).
Control | 0..1
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept

Comments | Only used if not implicit in code found in Observation.code. In many systems, this may be represented as a related observation instead of an inline component. If the use case requires BodySite to be handled as a separate resource (e.g. to identify and track separately) then use the standard extension body-site-instance.

Observation.method

Definition | Indicates the mechanism used to perform the observation.

Control | 0..1

Terminology Binding | Observation Methods (Example)

Type | CodeableConcept

Requirements | In some cases, method can impact results and is thus used for determining whether results can be compared or determining significance of results.

Comments | Only used if not implicit in code for Observation.code.

Observation.specimen

Definition | The specimen that was used when this observation was made.

Control | 0..1

Type | Reference(Specimen)

Comments | Should only be used if not implicit in code found in Observation.code. Observations are not made on specimens themselves; they are made on a subject, but in many cases by the means of a specimen. Note that although specimens are often involved, they are not always tracked and reported explicitly. Also note that observation resources may be used in contexts that track the specimen explicitly (e.g. Diagnostic Report).

Observation.device

Definition | The device used to generate the observation data.

Control | 0..1

Type | Reference(Device | DeviceMetric)

Comments | An extension should be used if further typing of the device is needed. Devices used to support obtaining an observation can be represented using either an extension or through the Observation.related element.

Observation.referenceRange

Definition | Guidance on how to interpret the value by comparison to a normal or recommended range.

Control | 0..*

Requirements | Knowing what values are considered "normal" can help evaluate the significance of a particular result. Need to be able to provide multiple reference ranges for different contexts.

Comments | Most observations only have one generic reference range. Systems MAY choose to restrict to only supplying the relevant reference range based on knowledge about the patient (e.g., specific to the patient's age, gender, weight and other factors), but this may not be possible or appropriate. Whenever more than one reference range is supplied, the differences between them SHOULD be provided in the reference range and/or age properties.

Invariants | Defined on this element obs-3: Must have at least a low or a high or text (expression : low.exists() or high.exists() or text.exists(), xpath: (exists(f:low) or exists(f:high)or exists(f:text)))

Observation.referenceRange.low

Definition | The value of the low bound of the reference range. The low bound of the reference range endpoint is inclusive of the value (e.g. reference range is >=5 - <=9). If the low bound is omitted, it is assumed to be meaningless (e.g. reference range is <=2.3).

Control | 0..1

Type | SimpleQuantity

Invariants | Affect this element obs-3: Must have at least a low or a high or text (expression : low.exists() or high.exists() or text.exists(), xpath: (exists(f:low) or exists(f:high)or exists(f:text)))

Observation.referenceRange.high

Definition | The value of the high bound of the reference range. The high bound of the reference range endpoint is inclusive of the value (e.g. reference range is >=5 - <=9). If the high bound is omitted, it is assumed to be meaningless (e.g. reference range is >= 2.3).

Control | 0..1

Type | SimpleQuantity

Invariants | Affect this element obs-3: Must have at least a low or a high or text (expression : low.exists() or high.exists() or text.exists(), xpath: (exists(f:low) or exists(f:high)or exists(f:text)))

Observation.referenceRange.type

Definition | Codes to indicate the what part of the targeted reference population it applies to. For example, the normal or therapeutic range.

Control | 0..1

Terminology Binding | Observation Reference Range Meaning Codes (Extensible)

Type | CodeableConcept

Requirements | Need to be able to say what kind of reference range this is - normal, recommended, therapeutic, etc, - for proper interpretation.

Comments | This SHOULD be populated if there is more than one range. If this element is not present then the normal range is assumed.

Observation.referenceRange.appliesTo

Definition | Codes to indicate the target population this reference range applies to. For example, a reference range may be based on the normal population or a particular sex or race.

Control | 0..*

Terminology Binding | Observation Reference Range Applies To Codes (Example)

Type | CodeableConcept

Requirements | Need to be able to identify the target population for proper interpretation.

Comments | This SHOULD be populated if there is more than one range. If this element is not present then the normal population is assumed.

Observation.referenceRange.age

Definition | The age at which this reference range is applicable. This is a neonatal age (e.g. number of weeks at term) if the meaning says so.

Control | 0..1

Type | Range

Requirements | Some analytes vary greatly over age.

Observation.referenceRange.text

Definition | Text based reference range in an observation which may be used when a quantitative range is not appropriate for an observation. An example would be a reference value of "Negative" or a list or table of 'normals'.

Control | 0..1

Type | string

Observation.related
Definition | A reference to another resource (usually another Observation) whose relationship is defined by the relationship type code.
Control | 0..*
Requirements | Normally, an observation will have either a value or a set of related observations. A few observations (e.g. Apgar score) may have both a value and a set of related observations or sometimes a QuestionnaireResponse from which the measure is derived.
Summary | true
Comments | For a discussion on the ways Observations can assembled in groups together see Notes below.
Observation.related.type
Definition | A code specifying the kind of relationship that exists with the target resource.
Control | 0..1
Terminology Binding | ObservationRelationshipType (Required)
Type | code
Requirements | A relationship type SHOULD be provided.
Comments | The "derived-from" type is the only logical choice when referencing the QuestionnaireAnswer resource.
Observation.related.target
Definition | A reference to the observation or QuestionnaireResponse resource that is related to this observation.
Control | 1..1
Type | Reference(Observation | QuestionnaireResponse | Sequence)
Observation.component
Definition | Some observations have multiple component observations. These component observations are expressed as separate code value pairs that share the same attributes. Examples include systolic and diastolic component observations for blood pressure measurement and multiple component observations for genetics observations.
Control | 0..*
Requirements | Component observations share the same attributes in the Observation resource as the primary observation and are always treated a part of a single observation (they are not separable). However, the reference range for the primary observation value is not inherited by the component values and is required when appropriate for each component observation.
Summary | true
Comments | For a discussion on the ways Observations can be assembled in groups together see Notes below.
Observation.component.code
Definition | Describes what was observed. Sometimes this is called the observation "code".
Control | 1..1
Terminology Binding | LOINC Codes (Example)
Type | CodeableConcept
Requirements | Knowing what kind of observation is being made is essential to understanding the observation.
Summary | true
Observation.component.value[x]
Definition | The information determined as a result of making the observation, if the information has a simple value.
Control | 0..1
Type | Quantity|CodeableConcept|string|Range|Ratio|SampledData|Attachment|time|dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | An observation exists to have a value, though it may not if it is in error, or if it represents a group of observations.
Summary | true
Comments | Normally, an observation will have either a single value or a set of related observations. A few observations (e.g. Apgar score) may have both a value and related observations (for an Apgar score, the observations from which the measure is derived). If a value is present, the datatype for this element should be determined by Observation.code. A CodeableConcept with just a text would be used instead of a string if the field was usually coded, or if the type associated with the Observation.code defines a coded value. For boolean values use valueCodeableConcept and select codes from <http://hl7.org/fhir/ValueSet/v2-0136> (these "yes/no" concepts can be mapped to the display name "true/false" or other mutually exclusive terms that may be needed"). The element, Observation.value[x], has a variable name depending on the type as follows: valueQuantity, valueCodeableConcept, valueRatio, valueChoice, valuePeriod, valueSampleData, or valueString (the name format is "'value' + the type name" with a capital on the first letter of the type).
Observation.component.dataAbsentReason
Definition | Provides a reason why the expected value in the element Observation.value[x] is missing.
Control | 0..1
Terminology Binding | Observation Value Absent Reason (Extensible)
Type | CodeableConcept
Requirements | For many results it is necessary to handle exceptional values in measurements.
Comments | "Null" or exceptional values can be represented two ways in FHIR Observations. One way is to simply include them in the value set and represent the exceptions in the value. For example, measurement values for a serology test could be "detected", "not detected", "inconclusive", or "test not done". The alternate way is to use the value element for actual observations and use the explicit dataAbsentReason element to record exceptional values. For example, the dataAbsentReason code "error" could be used when the measurement was not completed. Because of these options, use-case agreements are required to interpret general observations for exceptional values.
Invariants | Affect this element obs-6: dataAbsentReason SHALL only be present if Observation.value[x] is not present (expression : dataAbsentReason.empty() or value.empty(), xpath: not(exists(f:dataAbsentReason)) or (not(exists(*[starts-with(local-name(.), 'value')]))))
Observation.component.interpretation
Definition | The assessment made based on the result of the observation. Intended as a simple compact code often placed adjacent to the result value in reports and flow sheets to signal the meaning/normalcy status of the result. Otherwise known as abnormal flag.
Control | 0..1
Terminology Binding | Observation Interpretation Codes (Extensible)
Type | CodeableConcept
Requirements | For some results, particularly numeric results, an interpretation is necessary to fully understand the significance of a result.
Alternate Names | Abnormal Flag
Comments | The component interpretation applies only to the individual component value. For an overall interpretation all components together use thes Observation.interpretation element.
Observation.component.referenceRange

Definition | Guidance on how to interpret the value by comparison to a normal or recommended range.
Control | 0..*
Type | See Observation.referenceRange
Requirements | Knowing what values are considered "normal" can help evaluate the significance of a particular result. Need to be able to provide multiple reference ranges for different contexts.
Comments | Most observations only have one generic reference range. Systems MAY choose to restrict to only supplying the relevant reference range based on knowledge about the patient (e.g., specific to the patient's age, gender, weight and other factors), but this may not be possible or appropriate. Whenever more than one reference range is supplied, the differences between them SHOULD be provided in the reference range and/or age properties.

10.1.9 Resource Observation - Mappings

URL: https://hl7.org/fhir/STU3/observation-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.9 Resource Observation - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

SNOMED CT Attribute Binding

SNOMED CT Concept Domain Binding

HL7 v2 Mapping

Workflow Mapping

Mappings for the observation resource.

10.1.9.1 Mappings for RIM Mapping (http://hl7.org/v3)

Observation | Observation[classCode=OBS, moodCode=EVN]
identifier | id
basedOn | .outboundRelationship[typeCode=FLFS].target
status | status Amended & Final are differentiated by whether it is the subject of a ControlAct event with a type of "revise"
category | .outboundRelationship[typeCode="COMP].target[classCode="LIST", moodCode="EVN"].code
code | code
subject | participation[typeCode=SBJ]
context | inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
effective[x] | effectiveTime
issued | participation[typeCode=AUT].time
performer | participation[typeCode=PRF]
value[x] | value
dataAbsentReason | value.nullFlavor
interpretation | interpretationCode
comment | subjectOf.observationEvent[code="annotation"].value
bodySite | targetSiteCode
method | methodCode
specimen | participation[typeCode=SPC].specimen
device | participation[typeCode=DEV]
referenceRange | outboundRelationship[typeCode=REFV]/target[classCode=OBS, moodCode=EVN]
low | value:IVL_PQ.low
high | value:IVL_PQ.high
type | interpretationCode
appliesTo | interpretationCode
age | outboundRelationship[typeCode=PRCN].targetObservationCriterion[code="age"].value
text | value:ST
related | outBoundRelationship
type | .typeCode
target | .targetObservation
component | outBoundRelationship[typeCode=COMP]
code | code
value[x] | value
dataAbsentReason | value.nullFlavor
interpretation | interpretationCode
referenceRange | outboundRelationship[typeCode=REFV]/target[classCode=OBS, moodCode=EVN]

10.1.9.2 Mappings for SNOMED CT Attribute Binding (http://snomed.info/sct)

Attribute bindings link coded data elements in FHIR resources to a corresponding attribute in the SNOMED CT concept model. These bindings help to support:

* clarifying the intended meaning of the data element

* Quality checking the alignment between FHIR resource design and any coresponding SNOMED CT concept model

* Composition and decomposition of data instances by indicating the SNOMED CT concept model attribute whose value may be used to decompose a precoordinated concept into this data element

Observation
identifier
basedOn
status
category
code | 116680003 |Is a|
subject
context
effective[x]
issued
performer
value[x] | 363714003 |Interprets|
dataAbsentReason
interpretation | 363713009 |Has interpretation|
comment
bodySite | 718497002 |Inherent location|
method
specimen | 704319004 |Inherent in|
device | 424226004 |Using device|
referenceRange
low
high
type
appliesTo
age
text
related
type
target
component
code
value[x] | 363714003 |Interprets|
dataAbsentReason
interpretation | 363713009 |Has interpretation|
referenceRange

10.1.9.3 Mappings for SNOMED CT Concept Domain Binding (http://snomed.info/conceptdomain)

Concept domain bindings link a resource or an element to a set of SNOMED CT concepts that represent the intended semantics of the instances (whether or not SNOMED CT is used to encode that data element). This set of concepts is represented using a SNOMED CT expression constraint. Note that the 'Concept domain binding' may be a superset of the 'value set binding'. These bindings help to support:

* Quality checking FHIR resources by ensuring that (a) the intended semantics of the instances matches the valid range of the corresponding SNOMED CT attribute, and (b) the intended value set is appropriate for the intended semantics of the instances

* Semantic checking of data instances by helping to detect potential inconsistencies caused by overlap between the semantics incorporated in two concept domains

Observation | < 363787002 |Observable entity|
identifier
basedOn
status | < 445584004 |Report by finality status|
category
code | < 363787002 |Observable entity| OR < 386053000 |Evaluation procedure|
subject
context
effective[x]
issued
performer
value[x] | < 441742003 |Evaluation finding|
dataAbsentReason
interpretation | < 260245000 |Findings values|
comment
bodySite | < 123037004 |Body structure|
method
specimen | < 123038009 |Specimen|
device | < 49062001 |Device|
referenceRange

low
high
type | < 260245000 |Findings values| OR < 365860008 |General clinical state finding| OR < 250171008 |Clinical history or observation findings| OR < 415229000 |Racial group| OR < 365400002 |Finding of puberty stage| OR < 443938003 |Procedure carried out on subject|
appliesTo | < 260245000 |Findings values| OR < 365860008 |General clinical state finding| OR < 250171008 |Clinical history or observation findings| OR < 415229000 |Racial group| OR < 365400002 |Finding of puberty stage| OR < 443938003 |Procedure carried out on subject|
age
text
related
type
target
component
code | < 363787002 |Observable entity| OR < 386053000 |Evaluation procedure|
value[x] | 363714003 |Interprets| < 441742003 |Evaluation finding|
dataAbsentReason
interpretation | < 260245000 |Findings values|
referenceRange

10.1.9.4 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Observation | OBX
identifier | OBX.21 For OBX segments from systems without OBX-21 support a combination of ORC/OBR and OBX must be negotiated between trading partners to uniquely identify the OBX segment. Depending on how V2 has been implemented each of these may be an option: 1) OBR-3 + OBX-3 + OBX-4 or 2) OBR-3 + OBR-4 + OBX-3 + OBX-4 or 2) some other way to uniquely ID the OBR/ORC + OBX-3 + OBX-4.
basedOn | ORC
status | OBX-11
category
code | OBX-3
subject | PID-3
context | PV1
effective[x] | OBX-14, and/or OBX-19 after v2.4 (depends on who observation made)
issued | OBR.22 (or MSH.7), or perhaps OBX-19 (depends on who observation made)
performer | OBX.15 / (Practitioner) OBX-16, PRT-5:PRT-4='RO' / (Device) OBX-18 , PRT-10:PRT-4='EQUIP' / (Organization) OBX-23, PRT-8:PRT-4='PO'
value[x] | OBX.2, OBX.5, OBX.6
dataAbsentReason | N/A
interpretation | OBX-8
comment | NTE.3 (partner NTE to OBX, or sometimes another (child?) OBX)
bodySite | OBX-20
method | OBX-17
specimen | SPM segment
device | OBX-17 / PRT -10
referenceRange | OBX.7
low | OBX-7
high | OBX-7
type | OBX-10
appliesTo | OBX-10
age
text | OBX-7
related | Relationships established by OBX-4 usage
type | N/A
target
component | containment by OBX-4?
code | OBX-3
value[x] | OBX.2, OBX.5, OBX.6
dataAbsentReason | N/A
interpretation | OBX-8
referenceRange | OBX.7

10.1.9.5 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

Observation | Event
identifier | Event.identifier
basedOn | Event.basedOn
status | Event.status
category
code | Event.code
subject | Event.subject
context | Event.context
effective[x] | Event.occurrence[x]
issued
performer | Event.performer
value[x] | Event.notDone, Event.notDoneReason (maps to valeu[x] or dataAbsentReason)
dataAbsentReason | Event.notDone, Event.notDoneReason (maps to valeu[x] or dataAbsentReason)

interpretation
comment | Event.note
bodySite
method
specimen
device
referenceRange
low
high
type
appliesTo
age
text
related
type
target
component
code
value[x]
dataAbsentReason
interpretation
referenceRange

10.1.10 Resource Observation - Design Notes

URL: https://hl7.org/fhir/STU3/observation-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.10 Resource Observation - Design Notes

to do

10.1.11 Resource Observation - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/observation-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.11 Resource Observation - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Observation resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

10.1.11.1 Profiles

BMI | Defines constraints and extensions on the Observation resource for use in querying and retrieving Vital Signs information. | for Vital Signs Profile
BP | Defines constraints and extensions on the Observation resource for use in querying and retrieving Vital Signs information. | for Vital Signs Profile
BodyHeight | Defines constraints and extensions on the Observation resource for use in querying and retrieving Vital Signs information. | for Vital Signs Profile
BodyLength | Defines constraints and extensions on the Observation resource for use in querying and retrieving Vital Signs information. | for Vital Signs Profile
BodyTemp | Defines constraints and extensions on the Observation resource for use in querying and retrieving Vital Signs information. | for Vital Signs Profile
BodyWeight | Defines constraints and extensions on the Observation resource for use in querying and retrieving Vital Signs information. | for Vital Signs Profile
DeviceMetricObservation | This profile describes the direct or derived, qualitative or quantitative physiological measurement, setting, or calculation data produced by a medical device or a device component. | for Device Metric Observation Profile
HeadCircum | Defines constraints and extensions on the Observation resource for use in querying and retrieving Vital Signs information. | for Vital Signs Profile
HeartRate | Defines constraints and extensions on the Observation resource for use in querying and retrieving Vital Signs information. | for Vital Signs Profile
Observation-Genetics | Describes how the observation resource is used to report structured genetic test results | for Observation-genetics
OxygenSat | Defines constraints and extensions on the Observation resource for use in querying and retrieving Vital Signs information. | for Vital Signs Profile
RespRate | Defines constraints and extensions on the Observation resource for use in querying and retrieving Vital Signs information. | for Vital Signs Profile
VitalSigns | Defines constraints and extensions on the Observation resource for use in querying and retrieving Vital Signs information. | for Vital Signs Profile

VitalsPanel | Defines constraints and extensions on the Observation resource for use in querying and retrieving Vital Signs information. | for Vital Signs Profile

10.1.11.2 Extensions

event-OnBehalfOf | OnBehalfOf | for Event Pattern HL7 Extensions
event-definition | definition | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
event-performerRole | performerRole | for Event Pattern HL7 Extensions
event-reasonCode | reasonCode | for Event Pattern HL7 Extensions
event-reasonReference | reasonReference | for Event Pattern HL7 Extensions
observation-bodyPosition | bodyPosition | for Observation HL7 Extensions
observation-delta | delta | for Observation HL7 Extensions
observation-eventTiming | eventTiming | for Observation HL7 Extensions
observation-focal-subject | focal-subject | for Observation HL7 Extensions
observation-geneticsAlleleName | AlleleName | for Observation-genetics
observation-geneticsAllelicFrequency | AllelicFrequency | for Observation-genetics
observation-geneticsAllelicState | AllelicState | for Observation-genetics
observation-geneticsAminoAcidChangeName | AminoAcidChangeName | for Observation-genetics
observation-geneticsAminoAcidChangeType | AminoAcidChangeType | for Observation-genetics
observation-geneticsCopyNumberEvent | CopyNumberEvent | for Observation-genetics
observation-geneticsDNARegionName | DNARegionName | for Observation-genetics
observation-geneticsDNASequenceVariantName | DNASequenceVariantName | for Observation-genetics
observation-geneticsDNASequenceVariantType | DNASequenceVariantType | for Observation-genetics
observation-geneticsDNAVariantId | DNAVariantId | for Observation-genetics
observation-geneticsGene | Gene | for Observation-genetics
observation-geneticsGenomicSourceClass | GenomicSourceClass | for Observation-genetics
observation-geneticsInterpretation | Interpretation | for Observation-genetics
observation-geneticsPhaseSet | PhaseSet | for Observation-genetics
observation-geneticsSequence | Sequence | for Observation-genetics
observation-time-offset | time-offset | for Observation HL7 Extensions

Extensions that reference this resource:

condition-partOf | partOf | for Condition HL7 Extensions
event-reasonReference | reasonReference | for Event Pattern HL7 Extensions
family-member-history-genetics-observation | observation | for Family member history for genetics analysis
observation-geneticsInterpretation | Interpretation | for Observation-genetics
openEHR-test | test | for openEHR Archetype Profile
procedure-causedBy | causedBy | for Procedure HL7 Extensions
procedurerequest-geneticsItem | Item | for ProcedureRequest-Genetics

10.1.11.3 Search Extensions

No Search Extensions defined for this resource

10.1.12 Device Metric Observation Profile

URL: https://hl7.org/fhir/STU3/observation-devicemetricobservation.html

For the initial scope, this DeviceMetricObservation profile is only applicable to describe a measurement that is produced by any medical device that implements or derives from the ISO/IEEE 11073 standard.

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.12 Device Metric Observation Profile

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

10.1.12.1 Scope and Usage

The DeviceMetricObservation profile describes the direct or derived, qualitative or quantitative physiological measurement, setting, or calculation data produced by a medical device or a device component.

10.1.12.2 Content

Profiles:
DeviceMetricObservation | This profile describes the direct or derived, qualitative or quantitative physiological measurement, setting, or calculation data produced by a medical device or a device component.

10.1.12.3 StructureDefinition: DeviceMetricObservation

URL: https://hl7.org/fhir/STU3/devicemetricobservation.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.12.3 StructureDefinition: DeviceMetricObservation

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/devicemetricobservation`

This profile describes the direct or derived, qualitative or quantitative physiological measurement, setting, or calculation data produced by a medical device or a device component.

This profile was published on Mon, Nov 17, 2014 00:00+1100 as a draft by Health Level Seven International (Health Care Devices).

10.1.12.3.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from Observation.

Summary

Mandatory: 5 elements (+1 nested mandatory element) Must-Support: 13 elements Prohibited: 4 elements

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..*
identifier | S | 0..* | Identifier
status | S | 1..1 | code
code | S | 1..1 | CodeableConcept
subject | S | 1..1 | Reference(Patient), Reference(Device)
context | | 0..0
effectiveDateTime | S | 1..1 | dateTime
issued | | 0..0
value[x] | S | 0..1 | Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period
dataAbsentReason | | 0..0
interpretation | | 0..1 | CodeableConcept
comment | | 0..1 | string
bodySite | S | 0..1 | CodeableConcept
method | S | 0..1 | CodeableConcept
specimen | | 0..0
device | S | 1..1 | Reference(DeviceMetric)
referenceRange | S | 0..1 | BackboneElement
related | S | 0..1 | BackboneElement
type | S | 0..1 | code
target | S | 1..1 | Reference(Observation)
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | Measurements and simple assertions
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | S? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | S? | 1..1 | CodeableConcept | Type of observation (code / type) Binding: LOINC Codes (example)
subject | S? | 1..1 | Reference(Patient), Reference(Device) | Who and/or what this is about
effectiveDateTime | S? | 1..1 | dateTime | Clinically relevant time/time-period for observation
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
value[x] | S?I | 0..1 | Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual result
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result

bodySite | S | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | S | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
device | S | 1..1 | Reference(DeviceMetric) | (Measurement) Device
referenceRange | SI | 0..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..1 | BackboneElement | Resource related to this observation
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 0..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required)
target | S | 1..1 | Reference(Observation) | Resource that is related to this one
component | ?I | 0..* | BackboneElement | Component results
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | ? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ? | 0..1 | Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual component result
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Device Metric Observation Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation  --></category>
 <code><!-- 1..1 CodeableConcept Type of observation (code / type)  --></code>
 <subject><!-- 1..1 Reference(Patient)|Reference(Device)
     Who and/or what this is about --></subject>
 <context><!-- 0..0 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effectiveDateTime value="[dateTime]"/><!-- 1..1 Clinically relevant time/time-period for observation -->
 <issued value="[instant]"/><!-- 0..0 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <value[x]><!-- 0..1 Quantity|CodeableConcept|string|Range|Ratio|
   SampledData|Attachment|time|dateTime|Period Actual result --></value[x]>
 <dataAbsentReason><!-- 0..0 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..0 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 1..1 Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..1 BackboneElement  <!--  0..1 Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..1 BackboneElement  <!--  0..1 Resource related to this observation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
```

```
  <type value="[code]"/><!-- 0..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Observation)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|
    Range|Ratio|SampledData|Attachment|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Device Metric Observation Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], //Classification of  type of observation
    "code" : { CodeableConcept }, // R! Type of observation (code / type)
// value[x]: Who and/or what this is about. One of these 2:
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
    "subject" : { Reference(Device) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "effectiveDateTime" : "<dateTime>", // R! Clinically relevant time/time-period for observation
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
// value[x]: Actual result. One of these 10:
    "valueQuantity" : { Quantity }, //Actual result
    "valueCodeableConcept" : { CodeableConcept }, //Actual result
    "valueString" : "<string>", //Actual result
    "valueRange" : { Range }, //Actual result
    "valueRatio" : { Ratio }, //Actual result
    "valueSampledData" : { SampledData }, //Actual result
    "valueAttachment" : { Attachment }, //Actual result
    "valueTime" : "<time>", //Actual result
    "valueDateTime" : "<dateTime>", //Actual result
    "valuePeriod" : { Period }, //Actual result
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "device" : { Reference(DeviceMetric) }, // R! (Measurement) Device
    "referenceRange" : { BackboneElement }, // C?Provides guide for interpretation
    "related" : { BackboneElement }, // C?Resource related to this observation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

This structure is derived from Observation.

Summary

Mandatory: 5 elements (+1 nested mandatory element) Must-Support: 13 elements Prohibited: 4 elements

Differential View

This structure is derived from Observation.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		
identifier	S	0..*	Identifier	
status	S	1..1	code	
code	S	1..1	CodeableConcept	
subject	S	1..1	Reference(Patient), Reference(Device)	
context		0..0		
effectiveDateTime	S	1..1	dateTime	
issued		0..0		
value[x]	S	0..1	Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period	
dataAbsentReason		0..0		

interpretation | | 0..1 | CodeableConcept
comment | | 0..1 | string
bodySite | S | 0..1 | CodeableConcept
method | S | 0..1 | CodeableConcept
specimen | | 0..0
device | S | 1..1 | Reference(DeviceMetric)
referenceRange | S | 0..1 | BackboneElement
related | S | 0..1 | BackboneElement
type | S | 0..1 | code
target | S | 1..1 | Reference(Observation)
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | Measurements and simple assertions
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | S? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | S? | 1..1 | CodeableConcept | Type of observation (code / type) Binding: LOINC Codes (example)
subject | S? | 1..1 | Reference(Patient), Reference(Device) | Who and/or what this is about
effectiveDateTime | S? | 1..1 | dateTime | Clinically relevant time/time-period for observation
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
value[x] | S?I | 0..1 | Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual result
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | S | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | S | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
device | S | 1..1 | Reference(DeviceMetric) | (Measurement) Device
referenceRange | SI | 0..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..1 | BackboneElement | Resource related to this observation
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 0..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required)
target | S | 1..1 | Reference(Observation) | Resource that is related to this one
component | ?I | 0..* | BackboneElement | Component results
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | ? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ? | 0..1 | Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual component result
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Device Metric Observation Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
```

```
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation  --></category>
 <code><!-- 1..1 CodeableConcept Type of observation (code / type)  --></code>
 <subject><!-- 1..1 Reference(Patient)|Reference(Device)
     Who and/or what this is about --></subject>
 <context><!-- 0..0 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effectiveDateTime value="[dateTime]"/><!-- 1..1 Clinically relevant time/time-period for observation -->
 <issued value="[instant]"/><!-- 0..0 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <value[x]><!-- 0..1 Quantity|CodeableConcept|string|Range|Ratio|
   SampledData|Attachment|time|dateTime|Period Actual result --></value[x]>
 <dataAbsentReason><!-- 0..0 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..0 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 1..1 Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..1 BackboneElement  <!--  0..1 Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..1 BackboneElement  <!--  0..1 Resource related to this observation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 0..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Observation)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|
    Range|Ratio|SampledData|Attachment|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Device Metric Observation Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], //Classification of  type of observation
    "code" : { CodeableConcept }, // R! Type of observation (code / type)
// value[x]: Who and/or what this is about. One of these 2:
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
    "subject" : { Reference(Device) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "effectiveDateTime" : "<dateTime>", // R! Clinically relevant time/time-period for observation
```

```
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
// value[x]: Actual result. One of these 10:
    "valueQuantity" : { Quantity }, //Actual result
    "valueCodeableConcept" : { CodeableConcept }, //Actual result
    "valueString" : "<string>", //Actual result
    "valueRange" : { Range }, //Actual result
    "valueRatio" : { Ratio }, //Actual result
    "valueSampledData" : { SampledData }, //Actual result
    "valueAttachment" : { Attachment }, //Actual result
    "valueTime" : "<time>", //Actual result
    "valueDateTime" : "<dateTime>", //Actual result
    "valuePeriod" : { Period }, //Actual result
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "device" : { Reference(DeviceMetric) }, // R! (Measurement) Device
    "referenceRange" : { BackboneElement }, // C?Provides guide for interpretation
    "related" : { BackboneElement }, // C?Resource related to this observation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

Other representations of profile: Schematron

10.1.12.3.2 Terminology Bindings

Path | Name | Conformance | ValueSet

Observation.language | Common Languages | extensible | Common Languages

Observation.status | ObservationStatus | required | ObservationStatus

Observation.category | Observation Category Codes | preferred | Observation Category Codes

Observation.code | LOINC Codes | example | LOINC Codes

Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures

Observation.method | Observation Methods | example | Observation Methods

Observation.referenceRange.type | Observation Reference Range Meaning Codes | extensible | Observation Reference Range Meaning Codes

Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes

Observation.related.type | ObservationRelationshipType | required | ObservationRelationshipType

Observation.component.code | LOINC Codes | example | LOINC Codes

Observation.component.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason

Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.13 Observation HL7 Extensions

URL: https://hl7.org/fhir/STU3/observation-extensions.html

Defines common extensions used with or related to the Observation resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.13 Observation HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

10.1.13.1 Content

Extensions:

observation-bodyPosition | bodyPosition : The position of the body when the observation was done, e.g. standing, sitting. To be used only when the body position in not precoordinated in the observation code.

observation-delta | delta : The qualitative change in the value relative to the previous measurement. Usually only recorded if the change is clinically significant.

observation-focal-subject | focal-subject : The focus of an observation when not the patient of record (i.e. different from Observation.subject). For example, the focal subject may be a fetus, or a donor.

observation-time-offset | time-offset : A specific offset time in milliseconds from the stated time in the Observation.appliesDateTime to allow for representation of sequential recording of sampled data from the same lead or data stream. For example, an ECG recorder may record sequentially 3 leads four time to receive 12-lead ECG, see ISO 22077 .

observation-eventTiming | eventTiming : Regular life events the observation is tied to. This is useful for recording imprecise or "fuzzy" times when the observation occurred. For example, a blood glucose measurement "after breakfast". It is interpreted as relative to the effectiveTime element or the issued element if effectiveTime is absent. For example the blood glucose measurement could be reported that same day ("today after breakfast") or some other day ("yesterday after breakfast").

10.1.14 Observation-genetics

URL: https://hl7.org/fhir/STU3/observation-genetic.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.14 Observation-genetics

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

10.1.14.1 Overview

Observation-genetics-profile extends Observation resource to enable reporting of structured genetic test results. In addition, the genetics profile contextualizes well established standards from the field of clinical genetics into the standards of healthcare (e.g. HGNC - HUGO Gene Nomenclature Committee's international standard for gene names, symbols, and identifiers).

For use cases, details on how this resource interact with other Clinical Genomics resources or profiles, please refer to implementation guidance document here .

10.1.14.1.1 Genetic Standards and Resources include:

* Variant Databases: dbSNP , ClinVar , and COSMIC

* Reference Sequences: RefSeq and ENSEMBL

* Gene Symbols and Identifiers: HGNC - Human Genee Nomenclature Committee

* Variant Nomenclature: HGVS nomenclature from the Human Genome Variantion Society

* Locus: Gene

10.1.14.2 Scope and Usage

The Observation-genetics profile supports reporting of a DNA variant at the genomic, cDNA, and protein change level. In addition, a condition context may be provided, as AssessedCondition. For large genomic tests, a condition may be used as an input into the analytic pipeline to aid in the identification of clinically relevant variants related to the test order. It is strongly encouraged to provide all available information in this profile for any reported variants, because receiving systems (e.g. discovery research, outcomes analysis, and public health reporting) may use this information to normalize variants over time or across sources. However, these data should not be used to dynamically correct/change variant representations for clinical use outside of the laboratory, due to insufficient information.

Implementers should be aware that semantic equivalency of results of genetic variants cannot be guaranteed unless there is an agreed upon standard between sending and receiving systems.

This FHIR genomics work is based on work of the HL7 Clinical Genomics Workgroup and modeled based on the Domain Analysis Model and SMART on FHIR Genomics as published in JAMIA 2015 (http://jamia.oxfordjournals.org/content/early/2015/07/21/jamia.ocv045.long).

The HL7 Clinical Genomics Work Group emphasizes the importance of transmitting structured genetic findings within the clinical, translational, and research environments fully integrated with other clinical data, in order to drive outcomes analysis, operational decision making, discovery research, and public health reporting. The standard doesn't currently cover the reporting of clinically relevant negative or wild type results within genetic data portion of the message.

Here is the document of HL7 Version 3 Domain Analysis Model where the examples used in genetics profile are from (Page 5).

10.1.14.3 Interpretation

Extension - geneticsInterpretation points to an Observation entity. In this Observation entity, Observation.component is used for recording clinical interpretations for variation.

Here is a LOINC panel that could be supported by Observation.component:

LOINC # | Component | Description/Comments

51963-7 | Medication Assessed | A coded medication accessed in a pharmacogenetic test (recommend RxNorm).

51964-5 | Drug Efficacy Analysis Overall Interpretation | Overall predicted phenotype for drug efficacy for all DNA Sequence Variations identified in a single case. LOINC Answer List values can be seen in table below.

51967-8 | Genetic disease assessed | A coded disease which is associated with the region of DNA covered by the genetic test (recommend SNOMED).

51969-4 | Genetic analysis summary report | Narrative report in disease diagnostic-based format, which is used for pharmacogenomic reporting as well and disease risk or diagnosis. These reports currently follow the same formatting recommendations.

51971-0 | Drug metabolism analysis overall interpretation | Overall predicted phenotype for drug metabolism for all DNA Sequence Variations identified in a single case. LOINC Answer List values can be seen in table below.

53039-4 | Genetic Disease Analysis Overall Carrier Interpretation | Carrier Identification interpretation of all identified DNA Sequence Variations along with any known clinical information for the benefit of aiding clinicians in understanding the results overall. LOINC Answer List values can be seen in table below.

LOINC Code | Sequence | Answer text | LOINC answer code

51964-5 | 1 | Responsive | LA6677-4

2 | Resistant | LA6676-6

3 | Negative | LA6577-6

4 | Inconclusive | LA9663-1

5 | Failure | LA9664-9

51971-0 | 1 | Ultrarapid metabolizer | LA10315-2

2 | Extensive metabolizer | LA10316-0

3 | Intermediate metabolizer | LA10317-8

4 | Poor metabolizer | LA9657-3

5 | Inconclusive | LA9663-1
53039-4 | 1 | Carrier | LA10314-5
2 | Negative | LA6577-6
3 | Inconclusive | LA9663-1
4 | Failure | LA9664-9

In addition, in this Observation entity, Observation.related is used to link itself with Observation-genetics profile. Observation.related.type is 'derived-from' as it provides clinical interpretation for variation recoreded in Observation-genetics entity.

10.1.14.4 Content

Profiles:
Observation-Genetics | Describes how the observation resource is used to report structured genetic test results
Extensions:
observation-geneticsDNASequenceVariantName | DNASequenceVariantName : Human Genome Variation Society (HGVS) nomenclature for a single or set of DNA Sequence Variation(s) identified in testing. The use of the nomenclature is also used to describe non-variations (aka. wild types). LOINC Code: (48004-6).
observation-geneticsDNAVariantId | DNAVariantId : Identifier for DNA sequence variant. If a germline variant, ClinVar or dbSNP identifier should be used. If a somatic variant, COSMIC identifier should be used, unless in ClinVar then either maybe used. Need to provide the code system used (ClinVar, dbSNP, COSMIC) LOINC Code: (48003-8).
observation-geneticsDNASequenceVariantType | DNASequenceVariantType : Codified type for associated DNA sequence variant. DNA sequence variants use the HGVS notation, which implies the DNA sequence variant type. LOINC Code: (48019-4).
observation-geneticsAminoAcidChangeName | AminoAcidChangeName : Human Genome Variation Society (HGVS) nomenclature for an amino acid change. Reference sequence ID used for HGVS naming must be annotated. An amino acid is a sequence feature that corresponds to a single amino acid residue in a polypeptide (SO:0001237). LOINC Code: (48005-3).
observation-geneticsAminoAcidChangeType | AminoAcidChangeType : Codified type for associated Amino Acid Change. LOINC Code: (48006-1).
observation-geneticsGene | Gene : A region (or regions) that includes all of the sequence elements necessary to encode a functional transcript. A gene may include regulatory regions, transcribed regions and/or other functional sequence regions (SO:0000704). This element is the official gene symbol approved by the HGNC, which is a short abbreviated form of the gene name (HGNC). LOINC Code: (48018-6).
observation-geneticsDNARegionName | DNARegionName : A human readable name for the region of interest. Typically Exon #, Intron # or other. NOTE: This is not standardized and is mainly for convenience and display purposes. LOINC Code: (47999-8).
observation-geneticsAlleleName | AlleleName : An allele is one of a set of coexisting sequence variants of a gene (SO:0001023). This element is the common name for the allele. LOINC Code: (48008-7).
observation-geneticsAllelicState | AllelicState : The level of occurrence of a single DNA Sequence Variant within a set of chromosomes. Heterozygous indicates the DNA sequence variant is only present in one of the two genes contained in homologous chromosomes. Homozygous indicates the DNA Sequence Variant is present in both genes contained in homologous chromosomes. Hemizygous indicates the DNA Sequence Variant exists in the only single copy of a gene in a non-homologous chromosome (the male X and Y chromosome are non-homologous). Hemiplasmic indicates that the DNA Sequence Variant is present in some but not all of the copies of mitochondrial DNA. Homoplasmic indicates that the DNA Sequence Variant is present in all of the copies of mitochondrial DNA. LOINC Code: (53034-5).
observation-geneticsAllelicFrequency | AllelicFrequency : A physical quality which inheres to the allele by virtue of the number instances of the allele within a population. LOINC Code: (81258-6).
observation-geneticsCopyNumberEvent | CopyNumberEvent : A variation that increases or decreases the copy number of a given region (SO:0001019). Values: amplification/deletion/LOH.
observation-geneticsGenomicSourceClass | GenomicSourceClass : Source of sample used to determine the sequence in sequencing lab -- germline, somatic, prenatal. LOINC Code: (48002-0).
observation-geneticsPhaseSet | PhaseSet : Chromosomal phase set identifier (UUID/OID). All sequences with the same PhaseSet identifier are asserted to be on the same chromosome (cis).
observation-geneticsSequence | Sequence : Refers to the Sequence resource, representing raw genetics data.
observation-geneticsInterpretation | Interpretation : Clinical Interpretations for variant. It's a reference to an Observation resource.
Examples:
Genetics 1 | Genetics example 1
Genetics 2 | Genetics example 2
Genetics 3 | Genetics example 3
Genetics 4 | Genetics example 4
Genetics 5 | Genetics example 5
Diplotype(with Haplotypes Ref) | Example of diplotype data (with haplotypes observation as reference)
Haplotype | Example of haplotype data that is basis of a diplotype data
Haplotype-2 | Example of another haplotype data that is basis of a diplotype data
Phenotype | Example of phenotype data
TPMT-diplotype | Example of a TPMT diplotype that link to two TPMT haplotype observations
TPMT-haplotype-one | Example of a TPMT haplotype observation
TPMT-haplotype-two | Example of another TPMT haplotype observation

10.1.14.5 Search Parameters

Search parameters defined by this package. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths | Source
amino-acid-change | string | HGVS Protein Change | | XML / JSON
dna-variant | string | HGVS DNA variant | | XML / JSON
gene-amino-acid-change | string | HGNC gene symbol and HGVS Protein change | | XML / JSON
gene-dnavariant | string | HGNC gene symbol and HGVS DNA Variant | | XML / JSON
gene-identifier | token | HGNC gene symbol and identifier | | XML / JSON

10.1.14.6 StructureDefinition: Observation-Genetics

URL: https://hl7.org/fhir/STU3/observation-genetics.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.14.6 StructureDefinition: Observation-Genetics

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/observation-genetics`

Describes how the observation resource is used to report structured genetic test results

This profile was published on Mon, Mar 14, 2016 00:00+1100 as a draft by Health Level Seven International (Clinical Genomics).

10.1.14.6.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from Observation.

Summary

Extensions

This structure refers to these extensions:

* http://hl7.org/fhir/StructureDefinition/observation-geneticsDNASequenceVariantName

* http://hl7.org/fhir/StructureDefinition/observation-geneticsDNAVariantId

* http://hl7.org/fhir/StructureDefinition/observation-geneticsDNASequenceVariantType

* http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeName

* http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeType

* http://hl7.org/fhir/StructureDefinition/observation-geneticsGene

* http://hl7.org/fhir/StructureDefinition/observation-geneticsDNARegionName

* http://hl7.org/fhir/StructureDefinition/observation-geneticsAlleleName

* http://hl7.org/fhir/StructureDefinition/observation-geneticsAllelicState

* http://hl7.org/fhir/StructureDefinition/observation-geneticsAllelicFrequency

* http://hl7.org/fhir/StructureDefinition/observation-geneticsCopyNumberEvent

* http://hl7.org/fhir/StructureDefinition/observation-geneticsGenomicSourceClass

* http://hl7.org/fhir/StructureDefinition/observation-geneticsPhaseSet

* http://hl7.org/fhir/StructureDefinition/observation-geneticsSequence

* http://hl7.org/fhir/StructureDefinition/observation-geneticsInterpretation

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..*
observation-geneticsDNASequenceVariantName | | 0..1 | CodeableConcept | DNA sequence variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNASequenceVariantName Binding: ???? (preferred)
observation-geneticsDNAVariantId | | 0..1 | CodeableConcept | DNA sequence variant ID URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNAVariantId
observation-geneticsDNASequenceVariantType | | 0..1 | CodeableConcept | DNA sequence variant type URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNASequenceVariantType
observation-geneticsAminoAcidChangeName | | 0..1 | CodeableConcept | Amino acid change URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeName Binding: ???? (preferred)
observation-geneticsAminoAcidChangeType | | 0..1 | CodeableConcept | Amino Acid Change Type URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeType
observation-geneticsGene | | 0..1 | CodeableConcept | HGNC gene symbol URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGene Binding: ???? (preferred)
observation-geneticsDNARegionName | | 0..1 | string | DNA region name URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNARegionName
observation-geneticsAlleleName | | 0..1 | CodeableConcept | Name of allele URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAlleleName Binding: ???? (preferred)

observation-geneticsAllelicState | | 0..1 | CodeableConcept | The level of occurrence of a single DNA sequence variant within a set of chromosomes: Heteroplasmic / Homoplasmic / Homozygous / Heterozygous / Hemizygous URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAllelicState
observation-geneticsAllelicFrequency | | 0..1 | decimal | Allele frequency URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAllelicFrequency
observation-geneticsCopyNumberEvent | | 0..1 | CodeableConcept | Copy number variation URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsCopyNumberEvent
observation-geneticsGenomicSourceClass | | 0..1 | CodeableConcept | Genomic source class URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGenomicSourceClass
observation-geneticsPhaseSet | | 0..1 | uri | Phase set identifier (UUID/OID) URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsPhaseSet
observation-geneticsSequence | | 0..* | Reference(Sequence) | Sequence URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsSequence
observation-geneticsInterpretation | | 0..1 | Reference(Observation) | Clinical interpretations for variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsInterpretation

Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | Measurements and simple assertions
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
observation-geneticsDNASequenceVariantName | I | 0..1 | CodeableConcept | DNA sequence variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNASequenceVariantName Binding: ???? (preferred)
observation-geneticsDNAVariantId | I | 0..1 | CodeableConcept | DNA sequence variant ID URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNAVariantId
observation-geneticsDNASequenceVariantType | I | 0..1 | CodeableConcept | DNA sequence variant type URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNASequenceVariantType
observation-geneticsAminoAcidChangeName | I | 0..1 | CodeableConcept | Amino acid change URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeName Binding: ???? (preferred)
observation-geneticsAminoAcidChangeType | I | 0..1 | CodeableConcept | Amino Acid Change Type URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeType
observation-geneticsGene | I | 0..1 | CodeableConcept | HGNC gene symbol URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGene Binding: ???? (preferred)
observation-geneticsDNARegionName | I | 0..1 | string | DNA region name URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNARegionName
observation-geneticsAlleleName | I | 0..1 | CodeableConcept | Name of allele URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAlleleName Binding: ???? (preferred)
observation-geneticsAllelicState | I | 0..1 | CodeableConcept | The level of occurrence of a single DNA sequence variant within a set of chromosomes: Heteroplasmic / Homoplasmic / Homozygous / Heterozygous / Hemizygous URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAllelicState
observation-geneticsAllelicFrequency | I | 0..1 | decimal | Allele frequency URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAllelicFrequency
observation-geneticsCopyNumberEvent | I | 0..1 | CodeableConcept | Copy number variation URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsCopyNumberEvent
observation-geneticsGenomicSourceClass | I | 0..1 | CodeableConcept | Genomic source class URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGenomicSourceClass
observation-geneticsPhaseSet | I | 0..1 | uri | Phase set identifier (UUID/OID) URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsPhaseSet
observation-geneticsSequence | I | 0..* | Reference(Sequence) | Sequence URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsSequence
observation-geneticsInterpretation | I | 0..1 | Reference(Observation) | Clinical interpretations for variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsInterpretation
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | ? | 1..1 | CodeableConcept | Type of observation (code / type) Binding: LOINC Codes (example)
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for observation
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
value[x] | ?I | 0..1 | Quantity, CodeableConcept, string, boolean, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual result
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text

id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | ?I | 0..* | BackboneElement | Resource related to this observation
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | | 0..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required)
target | | 1..1 | Reference(Observation), Reference(QuestionnaireResponse), Reference(Sequence) | Resource that is related to this one
component | ?I | 0..* | BackboneElement | Component results
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | ? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ? | 0..1 | Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual component result
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Observation-genetics -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation  --></category>
 <code><!-- 1..1 CodeableConcept Type of observation (code / type)  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!-- 0..1 dateTime|Period
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <value[x]><!-- 0..1 Quantity|CodeableConcept|string|boolean|Range|
   Ratio|SampledData|Attachment|time|dateTime|Period Actual result --></value[x]>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Resource related to this observation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 0..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Observation)|Reference(QuestionnaireResponse)|Reference(Sequence)
      Resource that is related to this one --></target>
 </related>
```

```
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|
    Range|Ratio|SampledData|Attachment|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Observation-genetics
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified orderOpen
      { // DNA sequence variant // C?
        // from Element: extension
      },
      { // DNA sequence variant ID // C?
        // from Element: extension
      },
      { // DNA sequence variant type // C?
        // from Element: extension
      },
      { // Amino acid change // C?
        // from Element: extension
      },
      { // Amino Acid Change Type // C?
        // from Element: extension
      },
      { // HGNC gene symbol // C?
        // from Element: extension
      },
      { // DNA region name // C?
        // from Element: extension
      },
      { // Name of allele // C?
        // from Element: extension
      },
      { // The level of occurrence of a single DNA sequence variant within a set of chromosomes: Heteroplasmic /
    Homoplasmic / Homozygous / Heterozygous / Hemizygous // C?
        // from Element: extension
      },
      { // Allele frequency // C?
        // from Element: extension
      },
      { // Copy number variation // C?
        // from Element: extension
      },
      { // Genomic source class // C?
        // from Element: extension
      },
      { // Phase set identifier (UUID/OID) // C?
        // from Element: extension
      },
      { // Sequence // C?
        // from Element: extension
      },
      { // Clinical interpretations for variant // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], //Classification of  type of observation
    "code" : { CodeableConcept }, // R! Type of observation (code / type)
// value[x]: Who and/or what this is about. One of these 4:
    "subject" : { Reference(Patient) }, //Who and/or what this is about
    "subject" : { Reference(Group) }, //Who and/or what this is about
    "subject" : { Reference(Device) }, //Who and/or what this is about
    "subject" : { Reference(Location) }, //Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 2:
```

```
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for observation
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }], //Who is responsible for the observation
// value[x]: Actual result. One of these 11:
    "valueQuantity" : { Quantity }, //Actual result
    "valueCodeableConcept" : { CodeableConcept }, //Actual result
    "valueString" : "<string>", //Actual result
    "valueBoolean" : <boolean>, //Actual result
    "valueRange" : { Range }, //Actual result
    "valueRatio" : { Ratio }, //Actual result
    "valueSampledData" : { SampledData }, //Actual result
    "valueAttachment" : { Attachment }, //Actual result
    "valueTime" : "<time>", //Actual result
    "valueDateTime" : "<dateTime>", //Actual result
    "valuePeriod" : { Period }, //Actual result
    "dataAbsentReason" : { CodeableConcept } //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Resource related to this observation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

This structure is derived from Observation.

Summary

Extensions

This structure refers to these extensions:

* http://hl7.org/fhir/StructureDefinition/observation-geneticsDNASequenceVariantName
* http://hl7.org/fhir/StructureDefinition/observation-geneticsDNAVariantId
* http://hl7.org/fhir/StructureDefinition/observation-geneticsDNASequenceVariantType
* http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeName
* http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeType
* http://hl7.org/fhir/StructureDefinition/observation-geneticsGene
* http://hl7.org/fhir/StructureDefinition/observation-geneticsDNARegionName
* http://hl7.org/fhir/StructureDefinition/observation-geneticsAlleleName
* http://hl7.org/fhir/StructureDefinition/observation-geneticsAllelicState
* http://hl7.org/fhir/StructureDefinition/observation-geneticsAllelicFrequency
* http://hl7.org/fhir/StructureDefinition/observation-geneticsCopyNumberEvent
* http://hl7.org/fhir/StructureDefinition/observation-geneticsGenomicSourceClass
* http://hl7.org/fhir/StructureDefinition/observation-geneticsPhaseSet
* http://hl7.org/fhir/StructureDefinition/observation-geneticsSequence
* http://hl7.org/fhir/StructureDefinition/observation-geneticsInterpretation

Differential View

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints

Observation | | 0..*

observation-geneticsDNASequenceVariantName | | 0..1 | CodeableConcept | DNA sequence variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNASequenceVariantName Binding: ???? (preferred)

observation-geneticsDNAVariantId | | 0..1 | CodeableConcept | DNA sequence variant ID URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNAVariantId

observation-geneticsDNASequenceVariantType | | 0..1 | CodeableConcept | DNA sequence variant type URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNASequenceVariantType

observation-geneticsAminoAcidChangeName | | 0..1 | CodeableConcept | Amino acid change URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeName Binding: ???? (preferred)

observation-geneticsAminoAcidChangeType | | 0..1 | CodeableConcept | Amino Acid Change Type URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeType

observation-geneticsGene | | 0..1 | CodeableConcept | HGNC gene symbol URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGene Binding: ???? (preferred)

observation-geneticsDNARegionName | | 0..1 | string | DNA region name URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNARegionName

observation-geneticsAlleleName | | 0..1 | CodeableConcept | Name of allele URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAlleleName Binding: ???? (preferred)
observation-geneticsAllelicState | | 0..1 | CodeableConcept | The level of occurrence of a single DNA sequence variant within a set of chromosomes: Heteroplasmic / Homoplasmic / Homozygous / Heterozygous / Hemizygous URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAllelicState
observation-geneticsAllelicFrequency | | 0..1 | decimal | Allele frequency URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAllelicFrequency
observation-geneticsCopyNumberEvent | | 0..1 | CodeableConcept | Copy number variation URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsCopyNumberEvent
observation-geneticsGenomicSourceClass | | 0..1 | CodeableConcept | Genomic source class URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGenomicSourceClass
observation-geneticsPhaseSet | | 0..1 | uri | Phase set identifier (UUID/OID) URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsPhaseSet
observation-geneticsSequence | | 0..* | Reference(Sequence) | Sequence URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsSequence
observation-geneticsInterpretation | | 0..1 | Reference(Observation) | Clinical interpretations for variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsInterpretation
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | Measurements and simple assertions
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
observation-geneticsDNASequenceVariantName | I | 0..1 | CodeableConcept | DNA sequence variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNASequenceVariantName Binding: ???? (preferred)
observation-geneticsDNAVariantId | I | 0..1 | CodeableConcept | DNA sequence variant ID URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNAVariantId
observation-geneticsDNASequenceVariantType | I | 0..1 | CodeableConcept | DNA sequence variant type URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNASequenceVariantType
observation-geneticsAminoAcidChangeName | I | 0..1 | CodeableConcept | Amino acid change URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeName Binding: ???? (preferred)
observation-geneticsAminoAcidChangeType | I | 0..1 | CodeableConcept | Amino Acid Change Type URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeType
observation-geneticsGene | I | 0..1 | CodeableConcept | HGNC gene symbol URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGene Binding: ???? (preferred)
observation-geneticsDNARegionName | I | 0..1 | string | DNA region name URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNARegionName
observation-geneticsAlleleName | I | 0..1 | CodeableConcept | Name of allele URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAlleleName Binding: ???? (preferred)
observation-geneticsAllelicState | I | 0..1 | CodeableConcept | The level of occurrence of a single DNA sequence variant within a set of chromosomes: Heteroplasmic / Homoplasmic / Homozygous / Heterozygous / Hemizygous URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAllelicState
observation-geneticsAllelicFrequency | I | 0..1 | decimal | Allele frequency URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAllelicFrequency
observation-geneticsCopyNumberEvent | I | 0..1 | CodeableConcept | Copy number variation URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsCopyNumberEvent
observation-geneticsGenomicSourceClass | I | 0..1 | CodeableConcept | Genomic source class URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGenomicSourceClass
observation-geneticsPhaseSet | I | 0..1 | uri | Phase set identifier (UUID/OID) URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsPhaseSet
observation-geneticsSequence | I | 0..* | Reference(Sequence) | Sequence URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsSequence
observation-geneticsInterpretation | I | 0..1 | Reference(Observation) | Clinical interpretations for variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsInterpretation
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | ? | 1..1 | CodeableConcept | Type of observation (code / type) Binding: LOINC Codes (example)
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for observation
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
value[x] | ?I | 0..1 | Quantity, CodeableConcept, string, boolean, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual result
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)

specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | ?I | 0..* | BackboneElement | Resource related to this observation
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | | 0..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required)
target | | 1..1 | Reference(Observation), Reference(QuestionnaireResponse), Reference(Sequence) | Resource that is related to this one
component | ?I | 0..* | BackboneElement | Component results
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | ? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ? | 0..1 | Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual component result
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Observation-genetics -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation  --></category>
 <code><!-- 1..1 CodeableConcept Type of observation (code / type)  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!-- 0..1 dateTime|Period
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <value[x]><!-- 0..1 Quantity|CodeableConcept|string|boolean|Range|
   Ratio|SampledData|Attachment|time|dateTime|Period Actual result --></value[x]>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
```

```
 <related>  0..* BackboneElement  <!--  0..* Resource related to this observation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 0..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Observation)|Reference(QuestionnaireResponse)|Reference(Sequence)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|
    Range|Ratio|SampledData|Attachment|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Observation-genetics
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified orderOpen
      { // DNA sequence variant // C?
        // from Element: extension
      },
      { // DNA sequence variant ID // C?
        // from Element: extension
      },
      { // DNA sequence variant type // C?
        // from Element: extension
      },
      { // Amino acid change // C?
        // from Element: extension
      },
      { // Amino Acid Change Type // C?
        // from Element: extension
      },
      { // HGNC gene symbol // C?
        // from Element: extension
      },
      { // DNA region name // C?
        // from Element: extension
      },
      { // Name of allele // C?
        // from Element: extension
      },
      { // The level of occurrence of a single DNA sequence variant within a set of chromosomes: Heteroplasmic /
    Homoplasmic / Homozygous / Heterozygous / Hemizygous // C?
        // from Element: extension
      },
      { // Allele frequency // C?
        // from Element: extension
      },
      { // Copy number variation // C?
        // from Element: extension
      },
      { // Genomic source class // C?
        // from Element: extension
      },
      { // Phase set identifier (UUID/OID) // C?
        // from Element: extension
      },
      { // Sequence // C?
        // from Element: extension
      },
      { // Clinical interpretations for variant // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], //Classification of  type of observation
    "code" : { CodeableConcept }, // R! Type of observation (code / type)
// value[x]: Who and/or what this is about. One of these 4:
```

```
    "subject" : { Reference(Patient) }, //Who and/or what this is about
    "subject" : { Reference(Group) }, //Who and/or what this is about
    "subject" : { Reference(Device) }, //Who and/or what this is about
    "subject" : { Reference(Location) }, //Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 2:
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for observation
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }], //Who is responsible for the observation
// value[x]: Actual result. One of these 11:
    "valueQuantity" : { Quantity }, //Actual result
    "valueCodeableConcept" : { CodeableConcept }, //Actual result
    "valueString" : "<string>", //Actual result
    "valueBoolean" : <boolean>, //Actual result
    "valueRange" : { Range }, //Actual result
    "valueRatio" : { Ratio }, //Actual result
    "valueSampledData" : { SampledData }, //Actual result
    "valueAttachment" : { Attachment }, //Actual result
    "valueTime" : "<time>", //Actual result
    "valueDateTime" : "<dateTime>", //Actual result
    "valuePeriod" : { Period }, //Actual result
    "dataAbsentReason" : { CodeableConcept } //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Resource related to this observation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

Other representations of profile: Schematron

10.1.14.6.2 Terminology Bindings

Path	Name	Conformance	ValueSet
Observation.language	Common Languages	extensible	Common Languages
Observation.status	ObservationStatus	required	ObservationStatus
Observation.category	Observation Category Codes	preferred	Observation Category Codes
Observation.code	LOINC Codes	example	LOINC Codes
Observation.dataAbsentReason	Observation Value Absent Reason	extensible	Observation Value Absent Reason
Observation.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes
Observation.bodySite	SNOMED CT Body Structures	example	SNOMED CT Body Structures
Observation.method	Observation Methods	example	Observation Methods
Observation.referenceRange.type	Observation Reference Range Meaning Codes	extensible	Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo	Observation Reference Range Applies To Codes	example	Observation Reference Range Applies To Codes
Observation.related.type	ObservationRelationshipType	required	ObservationRelationshipType
Observation.component.code	LOINC Codes	example	LOINC Codes
Observation.component.dataAbsentReason	Observation Value Absent Reason	extensible	Observation Value Absent Reason
Observation.component.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes

10.1.14.6.3 Constraints

Id	Path	Details	Requirements
obs-3	Observation.referenceRange	Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()	

10.1.14.13 Observation-genetic-Observation-gene-identifier

URL: https://hl7.org/fhir/STU3/observation-genetic-Observation-gene-identifier.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.14.13 Observation-genetic-Observation-gene-identifier

Search Parameter gene-identifier (type: token) defined as part of the observation-genetic Profile.

HGNC gene symbol and identifier

Other Details:

* Full URL : http://hl7.org/fhir/SearchParameter/observation-genetic-Observation-gene-identifier

* Paths :

* Targets :

10.1.14.14 Observation-genetic-Observation-dna-variant

URL: https://hl7.org/fhir/STU3/observation-genetic-Observation-dna-variant.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.14.14 Observation-genetic-Observation-dna-variant

Search Parameter dna-variant (type: string) defined as part of the observation-genetic Profile.

HGVS DNA variant

Other Details:

* Full URL : http://hl7.org/fhir/SearchParameter/observation-genetic-Observation-dna-variant

* Paths :

* Targets :

10.1.14.15 Observation-genetic-Observation-amino-acid-change

URL: https://hl7.org/fhir/STU3/observation-genetic-Observation-amino-acid-change.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.14.15 Observation-genetic-Observation-amino-acid-change

Search Parameter amino-acid-change (type: string) defined as part of the observation-genetic Profile.

HGVS Protein Change

Other Details:

* Full URL : http://hl7.org/fhir/SearchParameter/observation-genetic-Observation-amino-acid-change

* Paths :

* Targets :

10.1.14.16 Observation-genetic-Observation-gene-dnavariant

URL: https://hl7.org/fhir/STU3/observation-genetic-Observation-gene-dnavariant.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.14.16 Observation-genetic-Observation-gene-dnavariant

Search Parameter gene-dnavariant (type: string) defined as part of the observation-genetic Profile.

HGNC gene symbol and HGVS DNA Variant

Other Details:

* Full URL : http://hl7.org/fhir/SearchParameter/observation-genetic-Observation-gene-dnavariant

* Paths :

* Targets :

10.1.14.17 Observation-genetic-Observation-gene-amino-acid-change

URL: https://hl7.org/fhir/STU3/observation-genetic-Observation-gene-amino-acid-change.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.14.17 Observation-genetic-Observation-gene-amino-acid-change

Search Parameter gene-amino-acid-change (type: string) defined as part of the observation-genetic Profile.

HGNC gene symbol and HGVS Protein change

Other Details:

* Full URL : http://hl7.org/fhir/SearchParameter/observation-genetic-Observation-gene-amino-acid-change

* Paths :

* Targets :

10.1.15 Vital Signs Profile

URL: https://hl7.org/fhir/STU3/observation-vitalsigns.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.15 Vital Signs Profile

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

10.1.16 Introduction

Vital signs will be one of the first areas where there is a need for a single, global vocabulary to allow for ubiquitous access and re-use. Particularly with the use of wearables by patients where they want to/need to share information from those devices. To meet this need there must be a consistent vocabulary and a common syntax to achieve semantic interoperability. The FHIR Vital Signs profile sets minimum expectations for the Observation resource to record, search and fetch the vital signs associated with a patient that include the primary vital signs plus additional measurements such as height, weight and BMI. Support for basic mandatory searching of resources is defined below in the Quick Start section. When a FHIR implementation supports any of the vital signs listed below, the implementation SHALL conform to this profile for the vital sign observation.

These requirements were originally developed, balloted, and published in FHIR DSTU2 as part of the ONC sponsored Data Access Framework (DAF) project and were subsequently updated to define the minimum mandatory conformance requirements needed for accessing patient data as defined by the Argonaut pilot implementations.

10.1.17 Scope and Usage

Example Usage Scenarios:

The following are example usage scenarios for this profile:

* Query for vital signs of a particular patient

10.1.17.0.0.1 Mandatory Data Elements and Terminology

The following data-elements are mandatory (i.e. data SHALL be present). These are presented below in a simple human-readable explanation. Profile-specific guidance and valid examples are provided as well. Note that many of the examples capture more than the minimum required. The links to the Profile Definitions provide the formal views of the profile content, descriptions, mappings and the StructureDefinitions in JSON and XML.

Each Observation must have:

* a status

* a category code of 'vital-signs'

* a "magic value" which tells you what is being measured

* LOINC was chosen for the "magic values" because this aligns with the most countries, but it can be treated as simply a fixed core set of common codes to communicate basic vital signs. Implementers that need to use a different code system can still map accordingly.

* a patient

* a time indicating when the measurement was taken

* a numeric result value and standard UCUM unit which is taken from the Unit Code column in the table below.

* note: if there is no numeric result then you have to supply a reason

10.1.18 Formal View of Profile Content

Vital Signs Profile : Link to the formal definition views for the vital signs listed in this table.

* The table below represents a minimum set of vital sign concepts, the required codes ("magic values"), and UCUM units of measure codes used for representing vital signs observations. These are extensible bindings and require that when a system supports any of these vital signs concepts, they must represent them using these codes. In addition, if you have a blood pressure observation, you must have both a systolic and a diastolic component, though one or both may have dataAbsentReason instead of a value.

* The first column of this table links to the formal views of the individual profile for each vital sign.

* If a more specific code or another code system is recorded or required, implementers must support both the values (LOINC) listed below and the translated code - e.g. method specific LOINC codes, SNOMED CT concepts, system specific (local) codes.

* In addition the implementer may choose to provide alternate codes in addition to the standard codes defined here. The examples illustrate using other codes as translations.

* Other profiles may make rules about which vital sign must be present or must be present as part of a panel or expand the list to include other vital signs. For implementers using LOINC, optional qualifier codes are provided in the notes below.

Profile Name | "Magic Value" (LOINC) | LOINC Name and Comments | UCUM Unit Code | Examples

Vital Signs Panel | 85353-1 | Vital signs, weight, height, head circumference, oxygen saturation and BMI panel - It represent a panel of vital signs listed in this table. All members of the panel are optional and note that querying for the panel may miss individual results that are not part of the actual panel. When used, Observation.valueQuantity is not present; instead, related links (with type=has-member) reference the vital signs observations (e.g. respiratory rate, heart rate, BP, etc.). This code replaces the deprecated code 8716-3 - Vital signs which is used in the Argonaut Data Query Implementation Guide. | - | Vital Signs Panel Example

Respiratory Rate | 9279-1 | Respiratory Rate | /min | Respiratory Rate Example

Heart rate | 8867-4 | Heart rate - To supplement this vital sign observation, 8887-2 - Heart rate device type MAY be used as an additional observation. | /min | Heart Rate Example

Oxygen saturation | 59408-5 | Oxygen saturation in Arterial blood by Pulse oximetry - This code replaces the deprecated code 2710-2 - Deprecated Oxygen saturation in Capillary blood by Oximetry which had been listed in C-CDA. | % | Oxygen Saturation Example

Body temperature | 8310-5 | Body temperature - o supplement this vital sign observation, 8327-9 - Body temperature measurement site (oral, forehead, rectal, etc.) and 8326-1 -Type of body temperature device MAY be used as additional observations | Cel, [degF] | Body Temperature Example

Body height | 8302-2 | Body height | cm, [in_i] | Body height Example

Body length | 8306-3 | Body height --lying - Like height, but lying down, typically this is used for infants | cm, [in_i] | Body Length Example

Head circumference | 8287-5 | Head Occipital-frontal circumference by Tape measure | cm, [in_i] | Head Circumference Example

Body weight | 29463-7 | Body weight - To supplement this vital sign observation, 8352-7 - Clothing worn during measure and 8361-8 - Body position with respect to gravity MAY be used as additional observations. | g, kg,[lb_av] | Body Weight Example

Body mass index | 39156-5 | Body mass index (BMI) [Ratio] | kg/m2 | Body Mass Example

Blood pressure systolic and diastolic | 85354-9 | Blood pressure panel with all children optional - This is a component observation. It has no value in Observation.valueQuantity and contains at least one component (systolic and/or diastolic). To supplement this vital sign observation, 8478-0 - Mean blood pressure, 8357-6 - Blood pressure method, 41904-4 - Blood pressure measurement site, 8358-4 - Blood pressure device cuff size, 41901-0 - Type of blood pressure device MAY be used as additional observations. | - | Blood Pressure Example, Blood Pressure Example with missing Diastolic measurement

Systolic blood pressure | 8480-6 | Systolic blood pressure - Observation.component code for a blood pressure Observation | mm[Hg] | Blood Pressure Example

Diastolic blood pressure | 8462-4 | Diastolic blood pressure - Observation.component code for a blood pressure Observation | mm[Hg] | Blood Pressure Example

10.1.19 Quick Start

Below is an overview of required search and read operations

Clients

* A client has connected to a server and fetched all of a patient's vital signs by searching by category using GET [base]/Observation?patient=[id]&category=vital-signs.

* A client has connected to a server and fetched all of a patient's vital signs searching by category code and date range using GET [base]/Observation?patient=[id]&category=vital-signs&date=[date]{&date=[date]}.

* A client has connected to a server and fetched any of a patient's vital signs by searching by one or more of the codes listed above using GET [base]/Observation?patient=[id]&code[vital sign LOINC{,LOINC2,LOINC3,...}].

* A client SHOULD be capable of connecting to a server and fetching any of a patient's vital signs searching by one or more of the codes listed above and date range using GET [base]/Observation?patient=[id]&code=[LOINC{,LOINC2...}]vital-signs&date=[date]{&date=[date]}.

Servers

* A server is capable of returning all of a patient's vital signs that it supports using GET [base]/Observation?patient=[id]&category=vital-signs.

* A server is capable of returning all of a patient's vital signs queried by date range using GET [base]/Observation?patient=[id]&category=vital-signs&date=[date]{&date=[date]}.

* A server is capable of returning any of a patient's vital signs queried by one or more of the codes listed above using GET [base]/Observation?patient=[id]&code[vital sign LOINC{,LOINC2,LOINC3,...}].

* A server SHOULD be capable of returning any of a patient's vital signs queried by one or more of the codes listed above and date range using GET [base]/Observation?patient=[id]&code=[LOINC{,LOINC2...}]vital-signs&date=[date]{&date=[date]}.

* A server has ensured that every API request includes a valid Authorization token, supplied via:Authorization: Bearer {server-specific-token-here}

* A server has rejected any unauthorized requests by returning an HTTP 401 Unauthorized response code.

10.1.19.0.1 GET [base]/Observation?patient=[id]&category=vital-signs

Example: Search for all Vital Signs measurements for a patient

GET [base]/Observation?patient=1186747&category=vital-signs

Support: Mandatory to support search by category code.

Implementation Notes: Search based on vital sign category code. This search fetches a bundle of all Observation resources with category 'vital-signs' for the specified patient (how to search by reference) and (how to search by token). The table above is the minimum set, additional vital signs are allowed.

Response Class:

* (Status 200): successful operation

* (Status 400): invalid parameter

* (Status 401/4xx): unauthorized request

* (Status 403): insufficient scope

10.1.19.0.2 GET [base]/Observation?patient=[id]&code=[vital sign LOINC{,LOINC2,LOINC3,...}]

Example: Search for all heart rate observations for a patient:

GET [base]/Observation?patient=1186747&code=8867-4

Example: Search for all heart rate, respiratory rate and blood pressure observations for a patient:

GET [base]/Observation?patient=1186747&code=8867-4,9279-1,85354-9

Support: Mandatory to support search by vital sign LOINC(s) listed above.

Implementation Notes: 1)Search based on vital sign LOINC code(s). This fetches a bundle of all Observation resources for specific vital sign(s) listed in the table above for the specified patient (how to search by reference) and [how to search by token)]. 2) The "code" parameter searches only Observation.code. For example when fetching blood pressures the resource will be only be returned when the search is based on 55284-4(Systolic and Diastolic BP). Using the component codes 8480-6(Systolic BP) or 8462-4 (Diastolic BP) will not return the resource . In order to search both Observation.code and Observation.component.code in a single query, use the "combo-code" search parameter.

Response Class:

* (Status 200): successful operation

* (Status 400): invalid parameter

* (Status 401/4xx): unauthorized request

* (Status 403): insufficient scope

10.1.19.0.3 GET [base]/Observation?patient=[id]&category=vital-signs&date=[date]{&date=[date]}

Example: Find all the blood pressures after 2015-01-14

GET [base]/Observation?patient=555580&code=85354-9&date=ge2015-01-14

Support: Mandatory to support search by category code and date

Implementation Notes: Search based on vital sign category code and date. This fetches a bundle of all Observation resources with category 'vital-signs' for the specified patient for a specified time period (how to search by reference) and (how to search by token).

Response Class:

* (Status 200): successful operation

* (Status 400): invalid parameter

* (Status 401/4xx): unauthorized request

* (Status 403): insufficient scope

10.1.19.1 Content

Profiles:
VitalSigns | FHIR Vital Signs Profile
BodyWeight | FHIR Body Weight Profile
RespRate | FHIR Respiratory Rate Profile
HeartRate | FHIR Heart Rate Profile
BodyTemp | FHIR Body Temperature Profile
BodyHeight | FHIR Body Height Profile
BodyLength | FHIR Body Length Profile
HeadCircum | FHIR Head Circumference Profile
OxygenSat | FHIR Oxygen Saturation Profile
BMI | FHIR Body Mass Index (BMI) Profile
BP | FHIR Blood Pressure Profile
VitalsPanel | FHIR Vital Signs Panel Profile

10.1.19.2 StructureDefinition: VitalSigns

URL: https://hl7.org/fhir/STU3/vitalsigns.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.19.2 StructureDefinition: VitalSigns

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/vitalsigns`

FHIR Vital Signs Profile

This profile was published on Fri, Mar 25, 2016 00:00+1100 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.19.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table
* Snapshot Table
* XML Template
* JSON Template
* All

This structure is derived from Observation.

Summary

10.1.19.3 Complete Summary of the Mandatory Requirements

One status in Observation.statuswhich has a required binding to:

* ObservationStatus value set.

A category in Observation.category which must have:

* a fixed Observation.category.coding.system="http://hl7.org/fhir/observation-category"
* a fixed Observation.category.coding.code= "vital-signs"

A code in Observation.code

* a fixed Observation.code.coding.system= "http://loinc.org"
* a LOINC code in which has an binding to:

Observation.code.coding.code

extensible

* Vital Signs value set.

One patient in Observation.subject

A date and time in effectiveDateTime or effectivePeriod

Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason

* Each Observation.valueQuantity must have:
* One numeric value in Observation.valueQuantity.value
* a fixed Observation.valueQuantity.system="http://unitsofmeasure"
* a UCUM unit code in which has an binding to:

Observation.valueQuantity.code

required

* Vital Signs Units value set.
* Observation.DataAbsentReason is bound to Observation Value Absent Reason value set.

When using a panel code to group component observations (Note: See the comments regarding blood pressure in the table above), one or more Observation.component.code each of which must have:

* a fixed ="<http://loinc.org>"

Observation.component.code.coding.system

* a LOINC code in which has an binding to:

Observation.code.coding.code

extensible

* Vital Signs Units value set.

Either one Observation.component.valueQuantity or, if there is no value, one code in Observation.component.DataAbsentReason

* Each Observation.component.valueQuantity must have:
* One numeric value in Observation.component.valueQuantity.value
* a fixed Observation.component.valueQuantity.system= "<http://unitsofmeasure.org>"
* a UCUM unit code in which has an binding to:

Observation.component.valueQuantity.code

required

* Vital Signs Units value set.
* Observation.component.DataAbsentReason is bound to Observation Value Absent Reason value set.

When using a panel code to group observations, one or more reference to Vitals Signs Observations in Observation.related.target

* a fixed Observation.related.type= "has-member"

Mandatory: 5 elements (+11 nested mandatory elements) Must-Support: 22 elements Fixed Value: 3 elements

Structures

This structure refers to these other structures:

* http://hl7.org/fhir/StructureDefinition/vitalsigns

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Vital Signs Profile vs-2: If there is no component or related element then either a value[x] or a data absent reason must be present
status | S | 1..1 | code | Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S | 1..1 | CodeableConcept | Coded Responses from C-CDA Vital Sign Results Binding: Vital Signs (extensible)
subject | S | 1..1 | Reference(Patient)
effective[x] | SI | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
valueQuantity | SI | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Binding: Observation Value Absent Reason (extensible)
related | S | 0..* | BackboneElement | Used when reporting vital signs panel components
type | S | 1..1 | code | Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns)
component | SI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
code | S | 1..1 | CodeableConcept | Binding: Vital Signs (extensible)
valueQuantity | SI | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Binding: Observation Value Absent Reason (extensible)
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Vital Signs Profile vs-2: If there is no component or related element then either a value[x] or a data absent reason must be present
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Coded Responses from C-CDA Vital Sign Results Binding: Vital Signs (extensible)
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations

modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code><!-- 1..1 CodeableConcept Coded Responses from C-CDA Vital Sign Results  --></code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1
      Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
```

```
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Coded Responses from C-CDA Vital Sign Results
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from Observation.

Summary

10.1.19.4 Complete Summary of the Mandatory Requirements

One status in Observation.statuswhich has a required binding to:

* ObservationStatus value set.

A category in Observation.category which must have:

* a fixed Observation.category.coding.system="http://hl7.org/fhir/observation-category"

* a fixed Observation.category.coding.code= "vital-signs"

A code in Observation.code

* a fixed Observation.code.coding.system= "http://loinc.org"

* a LOINC code in which has an binding to:

Observation.code.coding.code

extensible

* Vital Signs value set.

One patient in Observation.subject

A date and time in effectiveDateTime or effectivePeriod

Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason

* Each Observation.valueQuantity must have:

* One numeric value in Observation.valueQuantity.value

* a fixed Observation.valueQuantity.system="http://unitsofmeasure"

* a UCUM unit code in which has an binding to:

Observation.valueQuantity.code

required

* Vital Signs Units value set.

* Observation.DataAbsentReason is bound to Observation Value Absent Reason value set.

When using a panel code to group component observations (Note: See the comments regarding blood pressure in the table above), one or more Observation.component.code each of which must have:

* a fixed ="<http://loinc.org>"

Observation.component.code.coding.system

* a LOINC code in which has an binding to:

Observation.code.coding.code

extensible

* Vital Signs Units value set.

Either one Observation.component.valueQuantity or, if there is no value, one code in Observation.component.DataAbsentReason

* Each Observation.component.valueQuantity must have:

* One numeric value in Observation.component.valueQuantity.value

* a fixed Observation.component.valueQuantity.system= "<http://unitsofmeasure.org>"

* a UCUM unit code in which has an binding to:

Observation.component.valueQuantity.code

required

* Vital Signs Units value set.

* Observation.component.DataAbsentReason is bound to Observation Value Absent Reason value set.

When using a panel code to group observations, one or more reference to Vitals Signs Observations in Observation.related.target

* a fixed Observation.related.type= "has-member"

Mandatory: 5 elements (+11 nested mandatory elements) Must-Support: 22 elements Fixed Value: 3 elements

Structures

This structure refers to these other structures:

* http://hl7.org/fhir/StructureDefinition/vitalsigns

Differential View

This structure is derived from Observation.

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		FHIR Vital Signs Profile vs-2: If there is no component or related element then either a value[x] or a data absent reason must be present
status	S	1..1	code	Binding: ObservationStatus (required)
category	SI	1..*	CodeableConcept	vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code	S	1..1	CodeableConcept	Coded Responses from C-CDA Vital Sign Results Binding: Vital Signs (extensible)
subject	S	1..1	Reference(Patient)	
effective[x]	SI	1..1	dateTime, Period	Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
valueQuantity	SI	0..1	Quantity	Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
value	S	1..1	decimal	
unit	S	1..1	string	

system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Binding: Observation Value Absent Reason (extensible)
related | S | 0..* | BackboneElement | Used when reporting vital signs panel components
type | S | 1..1 | code | Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns)
component | SI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
code | S | 1..1 | CodeableConcept | Binding: Vital Signs (extensible)
valueQuantity | SI | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Binding: Observation Value Absent Reason (extensible)
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Vital Signs Profile vs-2: If there is no component or related element then either a value[x] or a data absent reason must be present
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Coded Responses from C-CDA Vital Sign Results Binding: Vital Signs (extensible)
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one

component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code><!-- 1..1 CodeableConcept Coded Responses from C-CDA Vital Sign Results  --></code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1
      Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
```

```
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Coded Responses from C-CDA Vital Sign Results
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Other representations of profile: Schematron

10.1.19.4.1 Terminology Bindings

Path	Name	Conformance	ValueSet
Observation.language	Common Languages	extensible	Common Languages
Observation.status	ObservationStatus	required	ObservationStatus
Observation.category	Observation Category Codes	preferred	Observation Category Codes
Observation.code	Vital Signs	extensible	Vital Signs
Observation.valueQuantity	Vital Signs Units	required	Vital Signs Units
Observation.valueQuantity.comparator	QuantityComparator	required	QuantityComparator
Observation.dataAbsentReason	Observation Value Absent Reason	extensible	Observation Value Absent Reason
Observation.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes
Observation.bodySite	SNOMED CT Body Structures	example	SNOMED CT Body Structures
Observation.method	Observation Methods	example	Observation Methods
Observation.referenceRange.type	Observation Reference Range Meaning Codes	extensible	Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo	Observation Reference Range Applies To Codes	example	Observation Reference Range Applies To Codes
Observation.related.type	ObservationRelationshipType	required	ObservationRelationshipType
Observation.component.code	Vital Signs	extensible	Vital Signs
Observation.component.valueQuantity	Vital Signs Units	required	Vital Signs Units
Observation.component.valueQuantity.comparator	QuantityComparator	required	QuantityComparator

Observation.component.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.19.4.2 Constraints

Id | Path | Details | Requirements
vs-2 | Observation | If there is no component or related element then either a value[x] or a data absent reason must be present Expression : (component.empty() and related.empty()) implies (dataAbsentReason.exists() or value.exists())
vs-4 | Observation.category | Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category' Expression : where(coding.system='http://hl7.org/fhir/observation-category' and coding.code='vital-signs').exists()
vs-1 | Observation.effective[x] | Datetime must be at least to day. Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.19.11 StructureDefinition: BodyWeight

URL: https://hl7.org/fhir/STU3/bodyweight.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.19.11 StructureDefinition: BodyWeight

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/bodyweight`

FHIR Body Weight Profile

This profile was published on Fri, Mar 25, 2016 00:00+1100 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.19.11.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from Vital Signs Profile.

Summary

10.1.19.11.2 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '29463-7'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* Observation.valueQuantity.code = 'kg', 'g', or '[lb_av]'

Fixed Value: 2 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

This structure is derived from Vital Signs Profile.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Body Weight Profile
code | | 1..1 | CodeableConcept | Body Weight
coding | | | Coding | Slice: Unordered, Open by value:code
coding | | 0..* | Coding
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 29463-7
valueQuantity | | 0..1 | Quantity
code | | 1..1 | code | Binding: Body Weight Units (required)
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints

Observation | I | 0..* | | FHIR Body Weight Profile
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Body Weight Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 29463-7
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Weight Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)

unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Body Weight -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1 Coded responses from the common UCUM units for vital signs value set.  -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
```

```
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Body Weight
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from Vital Signs Profile.

Summary

10.1.19.11.3 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '29463-7'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* Observation.valueQuantity.code = 'kg', 'g', or '[lb_av]'

Fixed Value: 2 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

Differential View

This structure is derived from Vital Signs Profile.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Body Weight Profile
code		1..1	CodeableConcept	Body Weight
coding			Coding	Slice: Unordered, Open by value:code

coding | | 0..* | Coding
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 29463-7
valueQuantity | | 0..1 | Quantity
code | | 1..1 | code | Binding: Body Weight Units (required)
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Body Weight Profile
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Body Weight Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 29463-7
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Weight Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one

component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Body Weight -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1 Coded responses from the common UCUM units for vital signs value set.  -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
```

```
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Body Weight
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Other representations of profile: Schematron

10.1.19.11.4 Terminology Bindings

Path	Name	Conformance	ValueSet
Observation.language	Common Languages	extensible	Common Languages
Observation.status	ObservationStatus	required	ObservationStatus
Observation.category	Observation Category Codes	preferred	Observation Category Codes
Observation.code	Vital Signs	extensible	Vital Signs
Observation.valueQuantity	Vital Signs Units	required	Vital Signs Units
Observation.valueQuantity.comparator	QuantityComparator	required	QuantityComparator
Observation.valueQuantity.code	Body Weight Units	required	Body Weight Units
Observation.dataAbsentReason	Observation Value Absent Reason	extensible	Observation Value Absent Reason
Observation.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes

Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | extensible | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.related.type | ObservationRelationshipType | required | ObservationRelationshipType
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.valueQuantity | Vital Signs Units | required | Vital Signs Units
Observation.component.valueQuantity.comparator | QuantityComparator | required | QuantityComparator
Observation.component.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.19.11.5 Constraints

Id | Path | Details | Requirements
vs-4 | Observation.category | Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category' Expression : where(coding.system='http://hl7.org/fhir/observation-category' and coding.code='vital-signs').exists()
vs-1 | Observation.effective[x] | Datetime must be at least to day. Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.19.18 StructureDefinition: RespRate

URL: https://hl7.org/fhir/STU3/resprate.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.19.18 StructureDefinition: RespRate

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/resprate`

FHIR Respiratory Rate Profile

This profile was published on Fri, Mar 25, 2016 00:00+1100 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.19.18.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from Vital Signs Profile.

Summary

10.1.19.18.2 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '9279-1'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* a fixedObservation.valueQuantity.code = '/min'

Fixed Value: 3 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

This structure is derived from Vital Signs Profile.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Respiratory Rate Profile
code | | 1..1 | CodeableConcept | Respiratory Rate
coding | | | Coding | Slice: Unordered, Open by value:code

coding | | 0..* | Coding
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 9279-1
valueQuantity | | 0..1 | Quantity
code | | 1..1 | code | Fixed Value: /min
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Respiratory Rate Profile
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Respiratory Rate Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 9279-1
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: /min
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present

id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Respiratory Rate -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1
      Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
```

```
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Respiratory Rate
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from Vital Signs Profile.

Summary

10.1.19.18.3 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '9279-1'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* a fixedObservation.valueQuantity.code = '/min'

Fixed Value: 3 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

Differential View

This structure is derived from Vital Signs Profile.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Respiratory Rate Profile
code | | 1..1 | CodeableConcept | Respiratory Rate
coding | | | Coding | Slice: Unordered, Open by value:code
coding | | 0..* | Coding
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 9279-1
valueQuantity | | 0..1 | Quantity
code | | 1..1 | code | Fixed Value: /min
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Respiratory Rate Profile
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Respiratory Rate Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 9279-1
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: /min
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)

age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Respiratory Rate -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1
      Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
```

```
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Respiratory Rate
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Other representations of profile: Schematron

10.1.19.18.4 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | extensible | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | Vital Signs | extensible | Vital Signs
Observation.valueQuantity | Vital Signs Units | required | Vital Signs Units
Observation.valueQuantity.comparator | QuantityComparator | required | QuantityComparator
Observation.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | extensible | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.related.type | ObservationRelationshipType | required | ObservationRelationshipType
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.valueQuantity | Vital Signs Units | required | Vital Signs Units
Observation.component.valueQuantity.comparator | QuantityComparator | required | QuantityComparator
Observation.component.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.19.18.5 Constraints

Id | Path | Details | Requirements
vs-4 | Observation.category | Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category' Expression : where(coding.system='http://hl7.org/fhir/observation-category' and coding.code='vital-signs').exists()
vs-1 | Observation.effective[x] | Datetime must be at least to day. Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.19.25 StructureDefinition: HeartRate

URL: https://hl7.org/fhir/STU3/heartrate.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.19.25 StructureDefinition: HeartRate

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/heartrate`

FHIR Heart Rate Profile

This profile was published on Fri, Mar 25, 2016 00:00+1100 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.19.25.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from Vital Signs Profile.

Summary

10.1.19.25.2 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '8867-4'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* a fixedObservation.valueQuantity.code = '/min'

Fixed Value: 3 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

This structure is derived from Vital Signs Profile.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Heart Rate Profile
code | | 1..1 | CodeableConcept | Heart Rate
coding | | | Coding | Slice: Unordered, Open by value:code
coding | | 0..* | Coding
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 8867-4
valueQuantity | | 0..1 | Quantity
code | | 1..1 | code | Fixed Value: /min
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Heart Rate Profile
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Heart Rate Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 8867-4
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: /min
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components

id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required)
Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Heart Rate -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1
      Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
```

```
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Heart Rate
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from Vital Signs Profile.

Summary

10.1.19.25.3 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '8867-4'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* a fixedObservation.valueQuantity.code = '/min'

Fixed Value: 3 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

Differential View

This structure is derived from Vital Signs Profile.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Heart Rate Profile
code | | 1..1 | CodeableConcept | Heart Rate
coding | | | Coding | Slice: Unordered, Open by value:code
coding | | 0..* | Coding
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 8867-4
valueQuantity | | 0..1 | Quantity
code | | 1..1 | code | Fixed Value: /min
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Heart Rate Profile
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Heart Rate Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 8867-4
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: /min
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)

extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Heart Rate -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1
      Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
```

```
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Heart Rate
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
```

```
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Other representations of profile: Schematron

10.1.19.25.4 Terminology Bindings

Path	Name	Conformance	ValueSet
Observation.language	Common Languages	extensible	Common Languages
Observation.status	ObservationStatus	required	ObservationStatus
Observation.category	Observation Category Codes	preferred	Observation Category Codes
Observation.code	Vital Signs	extensible	Vital Signs
Observation.valueQuantity	Vital Signs Units	required	Vital Signs Units
Observation.valueQuantity.comparator	QuantityComparator	required	QuantityComparator
Observation.dataAbsentReason	Observation Value Absent Reason	extensible	Observation Value Absent Reason
Observation.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes
Observation.bodySite	SNOMED CT Body Structures	example	SNOMED CT Body Structures
Observation.method	Observation Methods	example	Observation Methods
Observation.referenceRange.type	Observation Reference Range Meaning Codes	extensible	Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo	Observation Reference Range Applies To Codes	example	Observation Reference Range Applies To Codes
Observation.related.type	ObservationRelationshipType	required	ObservationRelationshipType
Observation.component.code	Vital Signs	extensible	Vital Signs
Observation.component.valueQuantity	Vital Signs Units	required	Vital Signs Units
Observation.component.valueQuantity.comparator	QuantityComparator	required	QuantityComparator
Observation.component.dataAbsentReason	Observation Value Absent Reason	extensible	Observation Value Absent Reason
Observation.component.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes

10.1.19.25.5 Constraints

Id	Path	Details	Requirements
vs-4	Observation.category	Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category' Expression : where(coding.system='http://hl7.org/fhir/observation-category' and coding.code='vital-signs').exists()	
vs-1	Observation.effective[x]	Datetime must be at least to day. Expression : ($this as dateTime).toString().length() >= 8	
obs-3	Observation.referenceRange	Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()	
vs-3	Observation.component	If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()	

10.1.19.32 StructureDefinition: BodyTemp

URL: https://hl7.org/fhir/STU3/bodytemp.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.19.32 StructureDefinition: BodyTemp

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/bodytemp`

FHIR Body Temperature Profile

This profile was published on Fri, Mar 25, 2016 00:00+1100 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.19.32.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from Vital Signs Profile.

Summary

10.1.19.32.2 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '8310-5'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* a fixedObservation.valueQuantity.code = 'Cel', or '[degF]'

Fixed Value: 2 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

This structure is derived from Vital Signs Profile.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Body Temperature Profile
code | | 1..1 | CodeableConcept | Body Temperature
coding | | | Coding | Slice: Unordered, Open by value:code
coding | | 0..* | Coding
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 8310-5
valueQuantity | | 0..1 | Quantity
code | | 1..1 | code | Binding: Body Temperature Units (required)
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Body Temperature Profile
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Body Temperature Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 8310-5
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Temperature Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored

low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Body Temperature -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1 Coded responses from the common UCUM units for vital signs value set.  -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
```

```
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Body Temperature
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from Vital Signs Profile.

Summary

10.1.19.32.3 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '8310-5'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* a fixedObservation.valueQuantity.code = 'Cel', or '[degF]'

Fixed Value: 2 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

Differential View

This structure is derived from Vital Signs Profile.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Body Temperature Profile
code | | 1..1 | CodeableConcept | Body Temperature
coding | | | Coding | Slice: Unordered, Open by value:code
coding | | 0..* | Coding
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 8310-5
valueQuantity | | 0..1 | Quantity
code | | 1..1 | code | Binding: Body Temperature Units (required)
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Body Temperature Profile
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Body Temperature Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 8310-5
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Temperature Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)

comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Body Temperature -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
```

```
    Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1 Coded responses from the common UCUM units for vital signs value set.  -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Body Temperature
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
```

```
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Other representations of profile: Schematron

10.1.19.32.4 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | extensible | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | Vital Signs | extensible | Vital Signs
Observation.valueQuantity | Vital Signs Units | required | Vital Signs Units
Observation.valueQuantity.comparator | QuantityComparator | required | QuantityComparator
Observation.valueQuantity.code | Body Temperature Units | required | Body Temperature Units
Observation.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | extensible | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.related.type | ObservationRelationshipType | required | ObservationRelationshipType
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.valueQuantity | Vital Signs Units | required | Vital Signs Units
Observation.component.valueQuantity.comparator | QuantityComparator | required | QuantityComparator
Observation.component.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.19.32.5 Constraints

Id | Path | Details | Requirements
vs-4 | Observation.category | Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category' Expression : where(coding.system='http://hl7.org/fhir/observation-category' and coding.code='vital-signs').exists()
vs-1 | Observation.effective[x] | Datetime must be at least to day. Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.19.39 StructureDefinition: BodyHeight

URL: https://hl7.org/fhir/STU3/bodyheight.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.19.39 StructureDefinition: BodyHeight

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/bodyheight`

FHIR Body Height Profile

This profile was published on Fri, Mar 25, 2016 00:00+1100 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.19.39.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from Vital Signs Profile.

Summary

10.1.19.39.2 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '8302-2'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* a fixedObservation.valueQuantity.code = 'cm', or '[in_i]'

Fixed Value: 2 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

This structure is derived from Vital Signs Profile.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Body Height Profile
code | | 1..1 | CodeableConcept | Body Height
coding | | | Coding | Slice: Unordered, Open by value:code
coding | | 0..* | Coding
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 8302-2
valueQuantity | | 0..1 | Quantity
code | | 1..1 | code | Binding: Body Length Units (required)
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Body Height Profile
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Body Height Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 8302-2
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Length Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)

method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Body Height -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
```

```
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1 Coded responses from the common UCUM units for vital signs value set.  -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Body Height
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
```

```
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from Vital Signs Profile.

Summary

10.1.19.39.3 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '8302-2'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* a fixedObservation.valueQuantity.code = 'cm', or '[in_i]'

Fixed Value: 2 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

Differential View

This structure is derived from Vital Signs Profile.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Body Height Profile
code | | 1..1 | CodeableConcept | Body Height
coding | | | Coding | Slice: Unordered, Open by value:code
coding | | 0..* | Coding
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 8302-2
valueQuantity | | 0..1 | Quantity
code | | 1..1 | code | Binding: Body Length Units (required)
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Body Height Profile
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Body Height Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 8302-2
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)

value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Length Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Body Height -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
```

```
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1 Coded responses from the common UCUM units for vital signs value set.  -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Body Height
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
```

```
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Other representations of profile: Schematron

10.1.19.39.4 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | extensible | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | Vital Signs | extensible | Vital Signs
Observation.valueQuantity | Vital Signs Units | required | Vital Signs Units
Observation.valueQuantity.comparator | QuantityComparator | required | QuantityComparator
Observation.valueQuantity.code | Body Length Units | required | Body Length Units
Observation.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | extensible | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.related.type | ObservationRelationshipType | required | ObservationRelationshipType
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.valueQuantity | Vital Signs Units | required | Vital Signs Units
Observation.component.valueQuantity.comparator | QuantityComparator | required | QuantityComparator
Observation.component.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.19.39.5 Constraints

Id | Path | Details | Requirements
vs-4 | Observation.category | Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category' Expression : where(coding.system='http://hl7.org/fhir/observation-category' and coding.code='vital-signs').exists()
vs-1 | Observation.effective[x] | Datetime must be at least to day. Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.19.46 StructureDefinition: BodyLength

URL: https://hl7.org/fhir/STU3/bodylength.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.19.46 StructureDefinition: BodyLength

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/bodylength`

FHIR Body Length Profile

This profile was published on Fri, Mar 25, 2016 00:00+1100 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.19.46.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from Vital Signs Profile.

Summary

10.1.19.46.2 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '8306-3'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* a fixedObservation.valueQuantity.code = 'cm', or '[in_i]'

Fixed Value: 2 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

This structure is derived from Vital Signs Profile.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Body Length Profile
code		1..1	CodeableConcept	Body Length
coding			Coding	Slice: Unordered, Open by value:code
coding		0..*	Coding	
system		0..1	uri	Fixed Value: http://loinc.org
code		0..1	code	Fixed Value: 8306-3
valueQuantity		0..1	Quantity	
code		1..1	code	Binding: Body Length Units (required)

Documentation for this format

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		FHIR Body Length Profile
id	?	0..1	id	Logical id of this artifact
meta	?	0..1	Meta	Metadata about the resource
implicitRules	?!?	0..1	uri	A set of rules under which this content was created
language		0..1	code	Language of the resource content Binding: Common Languages (extensible)
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension		0..*	Extension	Additional Content defined by implementations
modifierExtension	?!	0..*	Extension	Extensions that cannot be ignored
identifier	?	0..*	Identifier	Business Identifier for observation
basedOn	?	0..*	Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest)	Fulfills plan, proposal or order
status	?!S?	1..1	code	registered | preliminary | final | amended + Binding: ObservationStatus (required)
category	SI	1..*	CodeableConcept	Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code	S?	1..1	CodeableConcept	Body Length Binding: Vital Signs (extensible)
id		0..1	string	xml:id (or equivalent in JSON)
coding	?		Coding	Code defined by a terminology system Slice: Unordered, Open by value:code
coding	?	0..*	Coding	Code defined by a terminology system
id		0..1	string	xml:id (or equivalent in JSON)
system	?	0..1	uri	Identity of the terminology system Fixed Value: http://loinc.org
version	?	0..1	string	Version of the system - if relevant
code	?	0..1	code	Symbol in syntax defined by the system Fixed Value: 8306-3
display	?	0..1	string	Representation defined by the system
userSelected	?	0..1	boolean	If this coding was chosen directly by the user
text	?	0..1	string	Plain text representation of the concept
subject	S?	1..1	Reference(Patient)	Who and/or what this is about
context		0..1	Reference(Encounter), Reference(EpisodeOfCare)	Healthcare event during which this observation is made
effective[x]	S?I	1..1	dateTime, Period	Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued	?	0..1	instant	Date/Time this was made available
performer	?	0..*	Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson)	Who is responsible for the observation
valueQuantity	S?I	0..1	Quantity	Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id		0..1	string	xml:id (or equivalent in JSON)
value	S?	1..1	decimal	Numerical value (with implicit precision)
comparator	?!?	0..1	code	< | <= | >= | > - how to understand the value Binding: QuantityComparator (required)

unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Length Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Body Length -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
```

```
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1 Coded responses from the common UCUM units for vital signs value set.  -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
     Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
     High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
     Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
     Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
      Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Body Length
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
```

```
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from Vital Signs Profile.

Summary

10.1.19.46.3 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '8306-3'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* a fixedObservation.valueQuantity.code = 'cm', or '[in_i]'

Fixed Value: 2 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

Differential View

This structure is derived from Vital Signs Profile.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Body Length Profile
code		1..1	CodeableConcept	Body Length
coding			Coding	Slice: Unordered, Open by value:code
coding		0..*	Coding	
system		0..1	uri	Fixed Value: http://loinc.org
code		0..1	code	Fixed Value: 8306-3
valueQuantity		0..1	Quantity	
code		1..1	code	Binding: Body Length Units (required)

Documentation for this format

Snapshot View

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		FHIR Body Length Profile
id	?	0..1	id	Logical id of this artifact
meta	?	0..1	Meta	Metadata about the resource
implicitRules	?!?	0..1	uri	A set of rules under which this content was created
language		0..1	code	Language of the resource content Binding: Common Languages (extensible)
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension		0..*	Extension	Additional Content defined by implementations
modifierExtension	?!	0..*	Extension	Extensions that cannot be ignored
identifier	?	0..*	Identifier	Business Identifier for observation
basedOn	?	0..*	Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest)	Fulfills plan, proposal or order
status	?!S?	1..1	code	registered | preliminary | final | amended + Binding: ObservationStatus (required)
category	SI	1..*	CodeableConcept	Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code	S?	1..1	CodeableConcept	Body Length Binding: Vital Signs (extensible)
id		0..1	string	xml:id (or equivalent in JSON)
coding	?		Coding	Code defined by a terminology system Slice: Unordered, Open by value:code
coding	?	0..*	Coding	Code defined by a terminology system
id		0..1	string	xml:id (or equivalent in JSON)
system	?	0..1	uri	Identity of the terminology system Fixed Value: http://loinc.org
version	?	0..1	string	Version of the system - if relevant
code	?	0..1	code	Symbol in syntax defined by the system Fixed Value: 8306-3
display	?	0..1	string	Representation defined by the system
userSelected	?	0..1	boolean	If this coding was chosen directly by the user
text	?	0..1	string	Plain text representation of the concept
subject	S?	1..1	Reference(Patient)	Who and/or what this is about

context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Length Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Body Length -->
```

```
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1 Coded responses from the common UCUM units for vital signs value set.  -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Body Length
```

```
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Other representations of profile: Schematron

10.1.19.46.4 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | extensible | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | Vital Signs | extensible | Vital Signs
Observation.valueQuantity | Vital Signs Units | required | Vital Signs Units
Observation.valueQuantity.comparator | QuantityComparator | required | QuantityComparator
Observation.valueQuantity.code | Body Length Units | required | Body Length Units
Observation.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | extensible | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.related.type | ObservationRelationshipType | required | ObservationRelationshipType
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.valueQuantity | Vital Signs Units | required | Vital Signs Units
Observation.component.valueQuantity.comparator | QuantityComparator | required | QuantityComparator
Observation.component.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.19.46.5 Constraints

Id | Path | Details | Requirements
vs-4 | Observation.category | Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category' Expression : where(coding.system='http://hl7.org/fhir/observation-category' and coding.code='vital-signs').exists()
vs-1 | Observation.effective[x] | Datetime must be at least to day. Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.19.53 StructureDefinition: HeadCircum

URL: https://hl7.org/fhir/STU3/headcircum.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.19.53 StructureDefinition: HeadCircum

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/headcircum`

FHIR Head Circumference Profile

This profile was published on Fri, Mar 25, 2016 00:00+1100 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.19.53.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary
* Differential Table
* Snapshot Table
* XML Template
* JSON Template
* All

This structure is derived from Vital Signs Profile.

Summary

10.1.19.53.2 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'
* a fixed Observation.code.coding.code= '8287-5'
* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value
* a fixedObservation.valueQuantity.code = 'cm', or '[in_i]'

Fixed Value: 2 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

This structure is derived from Vital Signs Profile.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Head Circumference Profile
code		1..1	CodeableConcept	Head Circumference
coding			Coding	Slice: Unordered, Open by value:code
coding		0..*	Coding	
system		0..1	uri	Fixed Value: http://loinc.org
code		0..1	code	Fixed Value: 8287-5
valueQuantity		0..1	Quantity	
code		1..1	code	Binding: Body Length Units (required)

Documentation for this format

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		FHIR Head Circumference Profile
id	?	0..1	id	Logical id of this artifact
meta	?	0..1	Meta	Metadata about the resource
implicitRules	?!?	0..1	uri	A set of rules under which this content was created
language		0..1	code	Language of the resource content Binding: Common Languages (extensible)
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension		0..*	Extension	Additional Content defined by implementations
modifierExtension	?!	0..*	Extension	Extensions that cannot be ignored
identifier	?	0..*	Identifier	Business Identifier for observation
basedOn	?	0..*	Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest)	Fulfills plan, proposal or order
status	?!S?	1..1	code	registered | preliminary | final | amended + Binding: ObservationStatus (required)
category	SI	1..*	CodeableConcept	Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code	S?	1..1	CodeableConcept	Head Circumference Binding: Vital Signs (extensible)
id		0..1	string	xml:id (or equivalent in JSON)
coding	?		Coding	Code defined by a terminology system Slice: Unordered, Open by value:code
coding	?	0..*	Coding	Code defined by a terminology system
id		0..1	string	xml:id (or equivalent in JSON)
system	?	0..1	uri	Identity of the terminology system Fixed Value: http://loinc.org
version	?	0..1	string	Version of the system - if relevant
code	?	0..1	code	Symbol in syntax defined by the system Fixed Value: 8287-5
display	?	0..1	string	Representation defined by the system
userSelected	?	0..1	boolean	If this coding was chosen directly by the user
text	?	0..1	string	Plain text representation of the concept
subject	S?	1..1	Reference(Patient)	Who and/or what this is about
context		0..1	Reference(Encounter), Reference(EpisodeOfCare)	Healthcare event during which this observation is made
effective[x]	S?I	1..1	dateTime, Period	Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued	?	0..1	instant	Date/Time this was made available

performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Length Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Head Circumference -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
```

```
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1 Coded responses from the common UCUM units for vital signs value set.  -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Head Circumference
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
```

```
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from Vital Signs Profile.

Summary

10.1.19.53.3 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '8287-5'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* a fixedObservation.valueQuantity.code = 'cm', or '[in_i]'

Fixed Value: 2 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

Differential View

This structure is derived from Vital Signs Profile.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Head Circumference Profile
code		1..1	CodeableConcept	Head Circumference
coding			Coding	Slice: Unordered, Open by value:code
coding		0..*	Coding	
system		0..1	uri	Fixed Value: http://loinc.org
code		0..1	code	Fixed Value: 8287-5
valueQuantity		0..1	Quantity	
code		1..1	code	Binding: Body Length Units (required)
Documentation for this format				

Snapshot View

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		FHIR Head Circumference Profile
id	?	0..1	id	Logical id of this artifact
meta	?	0..1	Meta	Metadata about the resource
implicitRules	?!?	0..1	uri	A set of rules under which this content was created
language		0..1	code	Language of the resource content Binding: Common Languages (extensible)
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension		0..*	Extension	Additional Content defined by implementations
modifierExtension	?!	0..*	Extension	Extensions that cannot be ignored
identifier	?	0..*	Identifier	Business Identifier for observation
basedOn	?	0..*	Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest)	Fulfills plan, proposal or order
status	?!S?	1..1	code	registered | preliminary | final | amended + Binding: ObservationStatus (required)
category	SI	1..*	CodeableConcept	Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code	S?	1..1	CodeableConcept	Head Circumference Binding: Vital Signs (extensible)
id		0..1	string	xml:id (or equivalent in JSON)
coding	?		Coding	Code defined by a terminology system Slice: Unordered, Open by value:code
coding	?	0..*	Coding	Code defined by a terminology system
id		0..1	string	xml:id (or equivalent in JSON)
system	?	0..1	uri	Identity of the terminology system Fixed Value: http://loinc.org

version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 8287-5
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Length Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
```

```
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Head Circumference -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1 Coded responses from the common UCUM units for vital signs value set.  -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
```

```
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Head Circumference
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Other representations of profile: Schematron

10.1.19.53.4 Terminology Bindings

Path	Name	Conformance	ValueSet
Observation.language	Common Languages	extensible	Common Languages
Observation.status	ObservationStatus	required	ObservationStatus
Observation.category	Observation Category Codes	preferred	Observation Category Codes
Observation.code	Vital Signs	extensible	Vital Signs
Observation.valueQuantity	Vital Signs Units	required	Vital Signs Units
Observation.valueQuantity.comparator	QuantityComparator	required	QuantityComparator
Observation.valueQuantity.code	Body Length Units	required	Body Length Units
Observation.dataAbsentReason	Observation Value Absent Reason	extensible	Observation Value Absent Reason
Observation.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes
Observation.bodySite	SNOMED CT Body Structures	example	SNOMED CT Body Structures
Observation.method	Observation Methods	example	Observation Methods
Observation.referenceRange.type	Observation Reference Range Meaning Codes	extensible	Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo	Observation Reference Range Applies To Codes	example	Observation Reference Range Applies To Codes
Observation.related.type	ObservationRelationshipType	required	ObservationRelationshipType
Observation.component.code	Vital Signs	extensible	Vital Signs
Observation.component.valueQuantity	Vital Signs Units	required	Vital Signs Units
Observation.component.valueQuantity.comparator	QuantityComparator	required	QuantityComparator
Observation.component.dataAbsentReason	Observation Value Absent Reason	extensible	Observation Value Absent Reason
Observation.component.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes

10.1.19.53.5 Constraints

Id	Path	Details	Requirements
vs-4	Observation.category	Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category' Expression : where(coding.system='http://hl7.org/fhir/observation-category' and coding.code='vital-signs').exists()	
vs-1	Observation.effective[x]	Datetime must be at least to day. Expression : ($this as dateTime).toString().length() >= 8	
obs-3	Observation.referenceRange	Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()	
vs-3	Observation.component	If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()	

10.1.19.60 StructureDefinition: OxygenSat

URL: https://hl7.org/fhir/STU3/oxygensat.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.19.60 StructureDefinition: OxygenSat

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/oxygensat`

FHIR Oxygen Saturation Profile

This profile was published on Fri, Mar 25, 2016 00:00+1100 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.19.60.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from Vital Signs Profile.

Summary

"####Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '59408-5'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* a fixedObservation.valueQuantity.code = '%' "

Fixed Value: 3 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

This structure is derived from Vital Signs Profile.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Oxygen Saturation Profile
code		1..1	CodeableConcept	Oxygen Saturation
coding			Coding	Slice: Unordered, Open by value:code
coding		0..*	Coding	
system		0..1	uri	Fixed Value: http://loinc.org
code		0..1	code	Fixed Value: 59408-5
valueQuantity		0..1	Quantity	
code		1..1	code	Fixed Value: %

Documentation for this format

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		FHIR Oxygen Saturation Profile
id	?	0..1	id	Logical id of this artifact
meta	?	0..1	Meta	Metadata about the resource
implicitRules	?!?	0..1	uri	A set of rules under which this content was created
language		0..1	code	Language of the resource content Binding: Common Languages (extensible)
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension		0..*	Extension	Additional Content defined by implementations
modifierExtension	?!	0..*	Extension	Extensions that cannot be ignored
identifier	?	0..*	Identifier	Business Identifier for observation
basedOn	?	0..*	Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest)	Fulfills plan, proposal or order
status	?!S?	1..1	code	registered | preliminary | final | amended + Binding: ObservationStatus (required)
category	SI	1..*	CodeableConcept	Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code	S?	1..1	CodeableConcept	Oxygen Saturation Binding: Vital Signs (extensible)
id		0..1	string	xml:id (or equivalent in JSON)
coding	?		Coding	Code defined by a terminology system Slice: Unordered, Open by value:code
coding	?	0..*	Coding	Code defined by a terminology system
id		0..1	string	xml:id (or equivalent in JSON)
system	?	0..1	uri	Identity of the terminology system Fixed Value: http://loinc.org
version	?	0..1	string	Version of the system - if relevant
code	?	0..1	code	Symbol in syntax defined by the system Fixed Value: 59408-5
display	?	0..1	string	Representation defined by the system
userSelected	?	0..1	boolean	If this coding was chosen directly by the user

text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: %
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
```

```
 <code> 1..1 CodeableConcept  <!-- 1..1 Oxygen Saturation -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1
      Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Oxygen Saturation
```

```
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from Vital Signs Profile.

Summary

"####Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '59408-5'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* a fixedObservation.valueQuantity.code = '%' "

Fixed Value: 3 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

Differential View

This structure is derived from Vital Signs Profile.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Oxygen Saturation Profile
code		1..1	CodeableConcept	Oxygen Saturation
coding			Coding	Slice: Unordered, Open by value:code
coding		0..*	Coding	
system		0..1	uri	Fixed Value: http://loinc.org
code		0..1	code	Fixed Value: 59408-5
valueQuantity		0..1	Quantity	
code		1..1	code	Fixed Value: %
Documentation for this format				

Snapshot View

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		FHIR Oxygen Saturation Profile
id	?	0..1	id	Logical id of this artifact
meta	?	0..1	Meta	Metadata about the resource
implicitRules	?!?	0..1	uri	A set of rules under which this content was created
language		0..1	code	Language of the resource content Binding: Common Languages (extensible)
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension		0..*	Extension	Additional Content defined by implementations
modifierExtension	?!	0..*	Extension	Extensions that cannot be ignored
identifier	?	0..*	Identifier	Business Identifier for observation
basedOn	?	0..*	Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest)	Fulfills plan, proposal or order
status	?!S?	1..1	code	registered | preliminary | final | amended + Binding: ObservationStatus (required)
category	SI	1..*	CodeableConcept	Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code	S?	1..1	CodeableConcept	Oxygen Saturation Binding: Vital Signs (extensible)

id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 59408-5
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: %
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
```

```
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Oxygen Saturation -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1
      Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
```

```
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Oxygen Saturation
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Other representations of profile: Schematron

10.1.19.60.2 Terminology Bindings

Path | Name | Conformance | ValueSet

Observation.language | Common Languages | extensible | Common Languages

Observation.status | ObservationStatus | required | ObservationStatus

Observation.category | Observation Category Codes | preferred | Observation Category Codes

Observation.code | Vital Signs | extensible | Vital Signs

Observation.valueQuantity | Vital Signs Units | required | Vital Signs Units

Observation.valueQuantity.comparator | QuantityComparator | required | QuantityComparator

Observation.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason

Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures

Observation.method | Observation Methods | example | Observation Methods

Observation.referenceRange.type | Observation Reference Range Meaning Codes | extensible | Observation Reference Range Meaning Codes

Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes

Observation.related.type | ObservationRelationshipType | required | ObservationRelationshipType

Observation.component.code | Vital Signs | extensible | Vital Signs

Observation.component.valueQuantity | Vital Signs Units | required | Vital Signs Units

Observation.component.valueQuantity.comparator | QuantityComparator | required | QuantityComparator

Observation.component.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason

Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.19.60.3 Constraints

Id | Path | Details | Requirements

vs-4 | Observation.category | Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category' Expression : where(coding.system='http://hl7.org/fhir/observation-category' and coding.code='vital-signs').exists()

vs-1 | Observation.effective[x] | Datetime must be at least to day. Expression : ($this as dateTime).toString().length() >= 8

obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()

vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.19.67 StructureDefinition: BMI

URL: https://hl7.org/fhir/STU3/bmi.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.19.67 StructureDefinition: BMI

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/bmi`

FHIR Body Mass Index (BMI) Profile

This profile was published on Fri, Mar 25, 2016 00:00+1100 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.19.67.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from Vital Signs Profile.

Summary

10.1.19.67.2 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '39156-5'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* a fixedObservation.valueQuantity.code = 'kg/m2'

Fixed Value: 3 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

This structure is derived from Vital Signs Profile.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Body Mass Index (BMI) Profile
code | | 1..1 | CodeableConcept | Body Mass Index (BMI)
coding | | | Coding | Slice: Unordered, Open by value:code
coding | | 0..* | Coding
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 39156-5
valueQuantity | | 0..1 | Quantity
code | | 1..1 | code | Fixed Value: kg/m2
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Body Mass Index (BMI) Profile
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Body Mass Index (BMI) Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)

system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 39156-5
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: kg/m2
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
```

```
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Body Mass Index (BMI) -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1
      Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
```

```
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Body Mass Index (BMI)
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from Vital Signs Profile.

Summary

10.1.19.67.3 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '39156-5'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* a fixedObservation.valueQuantity.code = 'kg/m2'

Fixed Value: 3 elements

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

Differential View

This structure is derived from Vital Signs Profile.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Body Mass Index (BMI) Profile
code		1..1	CodeableConcept	Body Mass Index (BMI)
coding			Coding	Slice: Unordered, Open by value:code
coding		0..*	Coding	
system		0..1	uri	Fixed Value: http://loinc.org
code		0..1	code	Fixed Value: 39156-5
valueQuantity		0..1	Quantity	
code		1..1	code	Fixed Value: kg/m2

Documentation for this format

Snapshot View

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		FHIR Body Mass Index (BMI) Profile
id	?	0..1	id	Logical id of this artifact
meta	?	0..1	Meta	Metadata about the resource
implicitRules	?!?	0..1	uri	A set of rules under which this content was created
language		0..1	code	Language of the resource content Binding: Common Languages (extensible)
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension		0..*	Extension	Additional Content defined by implementations
modifierExtension	?!	0..*	Extension	Extensions that cannot be ignored
identifier	?	0..*	Identifier	Business Identifier for observation

basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Body Mass Index (BMI) Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 39156-5
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: kg/m2
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Vital Signs Profile -->
```

```
<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Body Mass Index (BMI) -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1
      Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Body Mass Index (BMI)
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Other representations of profile: Schematron

10.1.19.67.4 Terminology Bindings

Path | Name | Conformance | ValueSet

Observation.language | Common Languages | extensible | Common Languages

Observation.status | ObservationStatus | required | ObservationStatus

Observation.category | Observation Category Codes | preferred | Observation Category Codes

Observation.code | Vital Signs | extensible | Vital Signs

Observation.valueQuantity | Vital Signs Units | required | Vital Signs Units

Observation.valueQuantity.comparator | QuantityComparator | required | QuantityComparator

Observation.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason

Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures

Observation.method | Observation Methods | example | Observation Methods

Observation.referenceRange.type | Observation Reference Range Meaning Codes | extensible | Observation Reference Range Meaning Codes

Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes

Observation.related.type | ObservationRelationshipType | required | ObservationRelationshipType

Observation.component.code | Vital Signs | extensible | Vital Signs

Observation.component.valueQuantity | Vital Signs Units | required | Vital Signs Units

Observation.component.valueQuantity.comparator | QuantityComparator | required | QuantityComparator

Observation.component.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason

Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.19.67.5 Constraints

Id | Path | Details | Requirements

vs-4 | Observation.category | Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category' Expression : where(coding.system='http://hl7.org/fhir/observation-category' and coding.code='vital-signs').exists()

vs-1 | Observation.effective[x] | Datetime must be at least to day. Expression : ($this as dateTime).toString().length() >= 8

obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()

vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.19.74 StructureDefinition: BP

URL: https://hl7.org/fhir/STU3/bp.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.19.74 StructureDefinition: BP

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/bp`

FHIR Blood Pressure Profile

This profile was published on Fri, Mar 25, 2016 00:00+1100 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.19.74.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from Vital Signs Profile.

Summary

10.1.19.74.2 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= 85354-9'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* One which must have

Observation.component.code

* a fixed Observation.component.code.coding.system='http://loinc.org'

* fixed Observation.component.code.coding.code= '8480-6'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* One which must have

Observation.component.code

* a fixed Observation.component.code.coding.system='http://loinc.org'

* fixed Observation.component.code.coding.code= '8462-4'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* fixedObservation.component.valueQuantity.code = 'mm[Hg]'

Fixed Value: 8 elements Prohibited: 1 element

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

* The element Observation.component is sliced based on the value of value:code.coding.code

* The element Observation.component.code.coding is sliced based on the value of value:code

This structure is derived from Vital Signs Profile.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Blood Pressure Profile
code | | 1..1 | CodeableConcept | Blood Pressure
coding | | | Coding | Slice: Unordered, Open by value:code
coding | | 0..* | Coding
system | | 0..1 | uri | Fixed Value: http://loinc.org

code | | 0..1 | code | Fixed Value: 85354-9
valueQuantity | | 0..0
component | | | BackboneElement | Slice: Unordered, Open by value:code.coding.code
component | | 0..* | BackboneElement
code | | 1..1 | CodeableConcept
coding | | | Coding | Slice: Unordered, Open by value:code
coding | | 0..* | Coding | Systolic Blood Pressure
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 8480-6
valueQuantity | | 0..1 | Quantity
code | | 1..1 | code | Fixed Value: mm[Hg]
component | | 0..* | BackboneElement
code | | 1..1 | CodeableConcept
coding | | | Coding | Slice: Unordered, Open by value:code
coding | | 0..* | Coding | Diastolic Blood Pressure
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 8462-4
valueQuantity | | 0..1 | Quantity
code | | 1..1 | code | Fixed Value: mm[Hg]
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Blood Pressure Profile
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Blood Pressure Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 85354-9
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)

extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | | BackboneElement | Used when reporting systolic and diastolic blood pressure. Slice: Unordered, Open by value:code.coding.code
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure.
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Systolic Blood Pressure
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 8480-6
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: mm[Hg]
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure.
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Diastolic Blood Pressure
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 8462-4
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: mm[Hg]
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result

Documentation for this format

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
```

```
  Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
<status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
<category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
<code> 1..1 CodeableConcept  <!-- 1..1 Blood Pressure -->
 <-- coding sliced by value:code  in the specified orderOpen-->
 <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
  <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
  <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
  <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
  <display value="[string]"/><!-- 0..1 Representation defined by the system -->
  <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
 </coding>
 <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
</code>
<subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
<context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
    Healthcare event during which this observation is made --></context>
<effective[x]><!--  1..1 dateTime|Period
    Often just a dateTime for Vital Signs --></effective[x]>
<issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
<performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
    Who is responsible for the observation --></performer>
<valueQuantity> 0..0 Quantity  <!-- 0..0 Vital Sign Value recorded with UCUM -->
</valueQuantity>
<dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
<interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
<comment value="[string]"/><!-- 0..1 Comments about result -->
<bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
<method><!-- 0..1 CodeableConcept How it was done  --></method>
<specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
<device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
<referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <low><!-- 0..1 Quantity
     Low Range, if relevant --></low>
 <high><!-- 0..1 Quantity
     High Range, if relevant --></high>
 <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
 <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
 <age><!-- 0..1 Range
     Applicable age range, if relevant --></age>
 <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
</referenceRange>
<related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
 <target><!-- 1..1 Reference(Vital Signs Profile)
     Resource that is related to this one --></target>
</related>
<-- component sliced by value:code.coding.code  in the specified orderOpen-->
<component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <code> 1..1 CodeableConcept  <!-- 1..1 Type of component observation (code / type) -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Systolic Blood Pressure -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1
      Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
</component>
<component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <code> 1..1 CodeableConcept  <!-- 1..1 Type of component observation (code / type) -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Diastolic Blood Pressure -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
```

```
  </code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>

 </component>
</Observation>
```

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Blood Pressure
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }], //Who is responsible for the observation
    "dataAbsentReason" : { CodeableConcept } //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [ //  sliced by value:code.coding.code  in the specified orderOpen
      { // Used when reporting systolic and diastolic blood pressure. // C?
        // from Element: extension
        "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
        "code" : { CodeableConcept }, // R! Type of component observation (code / type)
        "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
        "dataAbsentReason" : { CodeableConcept }, //Why the component result is missing
        "interpretation" : { CodeableConcept }, //High, low, normal, etc.
// value[x]: Provides guide for interpretation of component result. One of these 0:
      },
      { // Used when reporting systolic and diastolic blood pressure. // C?
        // from Element: extension
        "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
        "code" : { CodeableConcept }, // R! Type of component observation (code / type)
        "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
        "dataAbsentReason" : { CodeableConcept }, //Why the component result is missing
        "interpretation" : { CodeableConcept }, //High, low, normal, etc.
// value[x]: Provides guide for interpretation of component result. One of these 0:
      }
    ],
  }
```

This structure is derived from Vital Signs Profile.

Summary

10.1.19.74.3 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= 85354-9'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* One which must have

Observation.component.code

* a fixed Observation.component.code.coding.system='http://loinc.org'

* fixed Observation.component.code.coding.code= '8480-6'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* One which must have

Observation.component.code

* a fixed Observation.component.code.coding.system='http://loinc.org'

* fixed Observation.component.code.coding.code= '8462-4'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* fixedObservation.component.valueQuantity.code = 'mm[Hg]'

Fixed Value: 8 elements Prohibited: 1 element

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

* The element Observation.component is sliced based on the value of value:code.coding.code

* The element Observation.component.code.coding is sliced based on the value of value:code

Differential View

This structure is derived from Vital Signs Profile.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Blood Pressure Profile
code | | 1..1 | CodeableConcept | Blood Pressure
coding | | | Coding | Slice: Unordered, Open by value:code
coding | | 0..* | Coding
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 85354-9
valueQuantity | | 0..0
component | | | BackboneElement | Slice: Unordered, Open by value:code.coding.code
component | | 0..* | BackboneElement
code | | 1..1 | CodeableConcept
coding | | | Coding | Slice: Unordered, Open by value:code
coding | | 0..* | Coding | Systolic Blood Pressure
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 8480-6
valueQuantity | | 0..1 | Quantity
code | | 1..1 | code | Fixed Value: mm[Hg]
component | | 0..* | BackboneElement
code | | 1..1 | CodeableConcept
coding | | | Coding | Slice: Unordered, Open by value:code
coding | | 0..* | Coding | Diastolic Blood Pressure
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 8462-4
valueQuantity | | 0..1 | Quantity
code | | 1..1 | code | Fixed Value: mm[Hg]
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Blood Pressure Profile
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations

modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Blood Pressure Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 85354-9
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 0..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | | BackboneElement | Used when reporting systolic and diastolic blood pressure. Slice: Unordered, Open by value:code.coding.code
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure.
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Systolic Blood Pressure
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 8480-6
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: mm[Hg]
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)

interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure.
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Diastolic Blood Pressure
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 8462-4
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: mm[Hg]
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Blood Pressure -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..0 Quantity  <!-- 0..0 Vital Sign Value recorded with UCUM -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
```

```
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..* BackboneElement  <!--  0..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <-- component sliced by value:code.coding.code  in the specified orderOpen-->
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code> 1..1 CodeableConcept  <!-- 1..1 Type of component observation (code / type) -->
   <-- coding sliced by value:code  in the specified orderOpen-->
   <coding> 0..* Coding  <!-- 0..* Systolic Blood Pressure -->
    <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
    <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
    <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
    <display value="[string]"/><!-- 0..1 Representation defined by the system -->
    <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
   </coding>
   <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
  </code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code> 1..1 CodeableConcept  <!-- 1..1 Type of component observation (code / type) -->
   <-- coding sliced by value:code  in the specified orderOpen-->
   <coding> 0..* Coding  <!-- 0..* Diastolic Blood Pressure -->
    <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
    <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
    <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
    <display value="[string]"/><!-- 0..1 Representation defined by the system -->
    <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
   </coding>
   <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
  </code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>

 </component>
</Observation>
```

JSON Template

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
```

```
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Blood Pressure
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }], //Who is responsible for the observation
    "dataAbsentReason" : { CodeableConcept } //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C?Used when reporting vital signs panel components
    "component" : [ //  sliced by value:code.coding.code  in the specified orderOpen
      { // Used when reporting systolic and diastolic blood pressure. // C?
        // from Element: extension
        "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
        "code" : { CodeableConcept }, // R! Type of component observation (code / type)
        "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
        "dataAbsentReason" : { CodeableConcept }, //Why the component result is missing
        "interpretation" : { CodeableConcept }, //High, low, normal, etc.
// value[x]: Provides guide for interpretation of component result. One of these 0:
      },
      { // Used when reporting systolic and diastolic blood pressure. // C?
        // from Element: extension
        "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
        "code" : { CodeableConcept }, // R! Type of component observation (code / type)
        "valueQuantity" : { Quantity }, //Vital Sign Value recorded with UCUM
        "dataAbsentReason" : { CodeableConcept }, //Why the component result is missing
        "interpretation" : { CodeableConcept }, //High, low, normal, etc.
// value[x]: Provides guide for interpretation of component result. One of these 0:
      }
    ],
  }
```

Other representations of profile: Schematron

10.1.19.74.4 Terminology Bindings

Path | Name | Conformance | ValueSet

Observation.language | Common Languages | extensible | Common Languages

Observation.status | ObservationStatus | required | ObservationStatus

Observation.category | Observation Category Codes | preferred | Observation Category Codes

Observation.code | Vital Signs | extensible | Vital Signs

Observation.valueQuantity.comparator | QuantityComparator | required | QuantityComparator

Observation.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason

Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures

Observation.method | Observation Methods | example | Observation Methods

Observation.referenceRange.type | Observation Reference Range Meaning Codes | extensible | Observation Reference Range Meaning Codes

Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes

Observation.related.type | ObservationRelationshipType | required | ObservationRelationshipType

Observation.component.code | Vital Signs | extensible | Vital Signs

Observation.component.valueQuantity | Vital Signs Units | required | Vital Signs Units

Observation.component.valueQuantity.comparator | QuantityComparator | required | QuantityComparator

Observation.component.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason

Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

Observation.component.code | Vital Signs | extensible | Vital Signs

Observation.component.valueQuantity | Vital Signs Units | required | Vital Signs Units

Observation.component.valueQuantity.comparator | QuantityComparator | required | QuantityComparator

Observation.component.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason

Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.19.74.5 Constraints

Id | Path | Details | Requirements

vs-4 | Observation.category | Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category' Expression : where(coding.system='http://hl7.org/fhir/observation-category' and coding.code='vital-signs').exists()

vs-1 | Observation.effective[x] | Datetime must be at least to day. Expression : ($this as dateTime).toString().length() >= 8

obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()

10.1.19.81 StructureDefinition: VitalsPanel

URL: https://hl7.org/fhir/STU3/vitalspanel.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.19.81 StructureDefinition: VitalsPanel

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/vitalspanel`

FHIR Vital Signs Panel Profile

This profile was published on Fri, Mar 25, 2016 00:00+1100 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.19.81.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from Vital Signs Profile.

Summary

10.1.19.81.2 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '85353-1'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* No value in Observation.value[x]

* At Least one Vitals Sign Observation reference in Observation.related

Mandatory: 1 element Fixed Value: 2 elements Prohibited: 1 element

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

This structure is derived from Vital Signs Profile.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Vital Signs Panel Profile
code		1..1	CodeableConcept	Vital Signs Panel
coding			Coding	Slice: Unordered, Open by value:code
coding		0..*	Coding	
system		0..1	uri	Fixed Value: http://loinc.org
code		0..1	code	Fixed Value: 85353-1
valueQuantity		0..0		
related		1..*	BackboneElement	

Documentation for this format

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		FHIR Vital Signs Panel Profile
id	?	0..1	id	Logical id of this artifact
meta	?	0..1	Meta	Metadata about the resource
implicitRules	?!?	0..1	uri	A set of rules under which this content was created
language		0..1	code	Language of the resource content Binding: Common Languages (extensible)
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension		0..*	Extension	Additional Content defined by implementations
modifierExtension	?!	0..*	Extension	Extensions that cannot be ignored
identifier	?	0..*	Identifier	Business Identifier for observation

basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Vital Signs Panel Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 85353-1
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 1..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
```

```
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Vital Signs Panel -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..0 Quantity  <!-- 0..0 Vital Sign Value recorded with UCUM -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  1..* BackboneElement  <!--  1..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
```

```
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Vital Signs Panel
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C? R! Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from Vital Signs Profile.

Summary

10.1.19.81.3 Complete Summary of the Mandatory Requirements

* One code in which must have

Observation.code

* a fixed Observation.code.coding.system='http ://loinc.org'

* a fixed Observation.code.coding.code= '85353-1'

* Other additional codes are allowed - e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

* No value in Observation.value[x]

* At Least one Vitals Sign Observation reference in Observation.related

Mandatory: 1 element Fixed Value: 2 elements Prohibited: 1 element

Slices

This structure defines the following Slices:

* The element Observation.code.coding is sliced based on the value of value:code

Differential View

This structure is derived from Vital Signs Profile.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Vital Signs Panel Profile
code		1..1	CodeableConcept	Vital Signs Panel
coding			Coding	Slice: Unordered, Open by value:code
coding		0..*	Coding	
system		0..1	uri	Fixed Value: http://loinc.org
code		0..1	code	Fixed Value: 85353-1
valueQuantity		0..0		
related		1..*	BackboneElement	
Documentation for this format				

Snapshot View

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		FHIR Vital Signs Panel Profile
id	?	0..1	id	Logical id of this artifact
meta	?	0..1	Meta	Metadata about the resource
implicitRules	?!?	0..1	uri	A set of rules under which this content was created
language		0..1	code	Language of the resource content Binding: Common Languages (extensible)
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension		0..*	Extension	Additional Content defined by implementations
modifierExtension	?!	0..*	Extension	Extensions that cannot be ignored
identifier	?	0..*	Identifier	Business Identifier for observation
basedOn	?	0..*	Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest)	Fulfills plan, proposal or order

status | ?!S? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | 1..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred) vs-4: Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category'
code | S? | 1..1 | CodeableConcept | Vital Signs Panel Binding: Vital Signs (extensible)
id | | 0..1 | string | xml:id (or equivalent in JSON)
coding | ? | | Coding | Code defined by a terminology system Slice: Unordered, Open by value:code
coding | ? | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | xml:id (or equivalent in JSON)
system | ? | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 85353-1
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
text | ? | 0..1 | string | Plain text representation of the concept
subject | S? | 1..1 | Reference(Patient) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | S?I | 1..1 | dateTime, Period | Often just a dateTime for Vital Signs vs-1: Datetime must be at least to day.
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
comment | | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (extensible)
appliesTo | | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
related | S?I | 1..* | BackboneElement | Used when reporting vital signs panel components
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
type | S | 1..1 | code | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by Binding: ObservationRelationshipType (required) Fixed Value: has-member
target | S | 1..1 | Reference(VitalSigns) | Resource that is related to this one
component | S?I | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
valueQuantity | S? | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | S? | 1..1 | string | Unit representation
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded responses from the common UCUM units for vital signs value set.
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Vital Signs Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
```

```
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!--  1..* CodeableConcept Classification of  type of observation  --></category>
 <code> 1..1 CodeableConcept  <!-- 1..1 Vital Signs Panel -->
  <-- coding sliced by value:code  in the specified orderOpen-->
  <coding> 0..* Coding  <!-- 0..* Code defined by a terminology system -->
   <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
   <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
   <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!-- 0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
 </code>
 <subject><!-- 1..1 Reference(Patient) Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..0 Quantity  <!-- 0..0 Vital Sign Value recorded with UCUM -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..1 Quantity
      Low Range, if relevant --></low>
  <high><!-- 0..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  1..* BackboneElement  <!--  1..* Used when reporting vital signs panel components -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <type value="[code]"/><!-- 1..1 has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by  -->
  <target><!-- 1..1 Reference(Vital Signs Profile)
      Resource that is related to this one --></target>
 </related>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <valueQuantity> 0..1 Quantity  <!-- 0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!-- 1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value  -->
   <unit value="[string]"/><!-- 1..1 Unit representation -->
   <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
   <code value="[code]"/><!-- 1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Vital Signs Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
```

```
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C? R! Classification of  type of observation
    "code" : { CodeableConcept }, // R! Vital Signs Panel
    "subject" : { Reference(Patient) }, // R! Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }], //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }] //Who is responsible for the observation
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //High, low, normal, etc.
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "related" : [{ BackboneElement }], // C? R! Used when reporting vital signs panel components
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Other representations of profile: Schematron

10.1.19.81.4 Terminology Bindings

Path	Name	Conformance	ValueSet
Observation.language	Common Languages	extensible	Common Languages
Observation.status	ObservationStatus	required	ObservationStatus
Observation.category	Observation Category Codes	preferred	Observation Category Codes
Observation.code	Vital Signs	extensible	Vital Signs
Observation.valueQuantity.comparator	QuantityComparator	required	QuantityComparator
Observation.dataAbsentReason	Observation Value Absent Reason	extensible	Observation Value Absent Reason
Observation.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes
Observation.bodySite	SNOMED CT Body Structures	example	SNOMED CT Body Structures
Observation.method	Observation Methods	example	Observation Methods
Observation.referenceRange.type	Observation Reference Range Meaning Codes	extensible	Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo	Observation Reference Range Applies To Codes	example	Observation Reference Range Applies To Codes
Observation.related.type	ObservationRelationshipType	required	ObservationRelationshipType
Observation.component.code	Vital Signs	extensible	Vital Signs
Observation.component.valueQuantity	Vital Signs Units	required	Vital Signs Units
Observation.component.valueQuantity.comparator	QuantityComparator	required	QuantityComparator
Observation.component.dataAbsentReason	Observation Value Absent Reason	extensible	Observation Value Absent Reason
Observation.component.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes

10.1.19.81.5 Constraints

Id	Path	Details	Requirements
vs-4	Observation.category	Must have a category of 'vital-signs' and a code system 'http://hl7.org/fhir/observation-category' Expression : where(coding.system='http://hl7.org/fhir/observation-category' and coding.code='vital-signs').exists()	
vs-1	Observation.effective[x]	Datetime must be at least to day. Expression : ($this as dateTime).toString().length() >= 8	
obs-3	Observation.referenceRange	Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()	
vs-3	Observation.component	If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()	

10.1.20 Resource Observation - Operations

URL: https://hl7.org/fhir/STU3/observation-operations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.20 Resource Observation - Operations

Orders and Observations Work Group | Maturity Level: 5 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This resource has 2 operations associated with it:

$stats	Observation Statistics
$lastn	Last N Observations Query

For more information about operations, including how they are invoked, see Operations.

10.1.20.1 Observation Statistics

The Statistics operation performs a set of statistical calculations on a set of clinical measurements such as a blood pressure as stored on the server. This operation is focused on Observation resources with valueQuantity elements that have UCUM unit codes.

The set of Observations is defined by 4 parameters:

* the subject of the observations for which the statistics are being generated (subject)

* which observations to generate statistics for (code and system, or coding)

* the time period over which to generate statistics 'duration or period)

* the set of statistical analyses to return (statistic)

Possible statistical analyses (see StatisticsCode):

* average ("Average"): The mean of N measurements over the stated period

* maximum ("Maximum"): The maximum value of N measurements over the stated period

* minimum ("Minimum"): The minimum value of N measurements over the stated period

* count ("Count"): The [number] of valid measurements over the stated period that contributed to the other statistical outputs

* totalcount ("Total Count"): The total [number] of valid measurements over the stated period, including observations that were ignored because they did not contain valid result values

* median ("Median"): The median of N measurements over the stated period

* std-dev ("Standard Deviation"): The standard deviation of N measurements over the stated period

* sum ("Sum"): The sum of N measurements over the stated period

* variance ("Variance"): The variance of N measurements over the stated period

* 20-percent ("20th Percentile"): The 20th Percentile of N measurements over the stated period

* 80-percent ("80th Percentile"): The 80th Percentile of N measurements over the stated period

* 4-lower ("Lower Quartile"): The lower Quartile Boundary of N measurements over the stated period

* 4-upper ("Upper Quartile"): The upper Quartile Boundary of N measurements over the stated period

* 4-dev ("Quartile Deviation"): The difference between the upper and lower Quartiles is called the Interquartile range. (IQR = Q3-Q1) Quartile deviation or Semi-interquartile range is one-half the difference between the first and the third quartiles.

* 5-1 ("1st Quintile"): The lowest of four values that divide the N measurements into a frequency distribution of five classes with each containing one fifth of the total population

* 5-2 ("2nd Quintile"): The second of four values that divide the N measurements into a frequency distribution of five classes with each containing one fifth of the total population

* 5-3 ("3rd Quintile"): The third of four values that divide the N measurements into a frequency distribution of five classes with each containing one fifth of the total population

* 5-4 ("4th Quintile"): The fourth of four values that divide the N measurements into a frequency distribution of five classes with each containing one fifth of the total population

* skew ("Skew"): Skewness is a measure of the asymmetry of the probability distribution of a real-valued random variable about its mean. The skewness value can be positive or negative, or even undefined. Source: Wikipedia

* kurtosis ("Kurtosis"): Kurtosis is a measure of the "tailedness" of the probability distribution of a real-valued random variable. Source: Wikipedia

* regression ("Regression"): Linear regression is an approach for modeling two-dimensional sample points with one independent variable and one dependent variable (conventionally, the x and y coordinates in a Cartesian coordinate system) and finds a linear function (a non-vertical straight line) that, as accurately as possible, predicts the dependent variable values as a function of the independent variables. Source: Wikipedia This Statistic code will return both a gradient and an intercept value.

If successful, the operation returns an Observation resource for each code with the results of the statistical calculations as component value pairs where the component code = the statistical code. The Observation also contains the input parameters patient,code and duration parameters. If unsuccessful, an OperationOutcome with an error message will be returned.

The client can request that all the observations on which the statistics are based be returned as well, using the include parameter. If an include parameter is specified, a limit may also be specified; the sources observations are subsetted at the server's discretion if count > limit. This functionality is included with the intent of supporting graphical presentation

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Observation-stats
```

Formal Definition (as a OperationDefinition).

URL: [base]/Observation/$stats

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

subject | 1..1 | uri | | | The subject of the relevant Observations, which has the value of the Observation.subject.reference. E.g. 'Patient/123'. Reference can be to an absolute URL, but servers only perform stats on their own observations

code | 0..* | string | | | The test code(s) upon which the statistics are being performed. Provide along with a system, or as a coding. For example, the code = "45667-2" will evaluate all relevant Observations with this code in Observation.code. For LOINC codes that are panels, (e.g. 55284-4) this means to include all the individual codes that are part of a panel

system | 0..1 | uri | | | The system for the code(s). Or provide a coding instead

coding | 0..* | Coding | | | The test code upon which the statistics are being performed, as a Coding

duration | 0..1 | decimal | | | The time period of interest given as hours. For example, the duration = "1" represents the last hour - the time period from on hour ago to now

period | 0..1 | Period | | | The time period over which the calculations to be performed, if a duration is not provided

statistic | 1..* | code | | | average|max|min|count The statistical operations to be performed on the relevant operations. Multiple statistics operations can be specified. These codes are defined here

include | 0..1 | boolean | | | Whether to return the observations on which the statistics are based

limit | 0..1 | positiveInt | | | If an include parameter is specified, a limit may also be specified to limit the number of source Observations returned. If the include paramter is absent or equal to "false" the limit parameter SHALL be ignored by the server

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..* | Observation | | | A set of observations, one observation for each code, each containing one component for each statistic. The Observation.component.code contains the statistic, and is relative to the Observation.code and cannot be interpreted independently. The Observation will also contain a subject, effectivePeriod, and code reflecting the input parameters. The status is fixed to final.

source | 0..* | Observation | | | Source observations on which the statistics are based

This operation cannot be performed on observations that the user is not authorized to see. It is assumed that the server has identified and secured the context appropriately, and can either associate the authorization context with a single patient, or determine whether the context has the rights to the nominated patient, if there is one. If there is no nominated patient (e.g. the operation is invoked at the system level) and the context is not associated with a single patient record, then the server should return an error. Specifying the relationship between the context, a user and patient records is outside the scope of this specification.

10.1.20.1.1 Examples

Get the average, min, max and count of a series of BP measurements for a patient (Request):

```
GET [base]/Observation/$stats?subject=Patient/123&code=55284-
    4&system=http://loinc.org&duration=1&statistic=average&statistic=min&statistic=max&statistic=count
```

Response:

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
  The outcome for this particular operation is 2 Observation resources - one for the diastolic
  and one for the systolic components, because the requested LOINC code is a panel code
-->
<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="return"/>
    <resource>
      <Observation>
        <status value="final"/>
        <code>
          <coding>
            <system value="http://loinc.org"/>
            <code value="8462-4"/>
            <display value="Diastolic blood pressure"/>
          </coding>
        </code>
        <subject>
           <reference value="Patient/123"/>
        </subject>
        <effectivePeriod>
           <start value="2016-05-25T19:26:16Z"/>
           <end value="2016-05-25T20:26:16Z"/>
        </effectivePeriod>
        <!--The Observation.component.code is relative to the Observation.code and cannot be interpreted
    independently.-->
        <component>
           <code>
              <!--   the average parameter code -->
              <coding>
                 <system value="http://hl7.org/fhir/observation-paramcode"/>
                 <code value="average"/>
                 <display value="Average"/>
              </coding>
           </code>
           <valueQuantity>
              <value value="79.7"/>
              <unit value="mmHg"/>
              <system value="http://unitsofmeasure.org"/>
              <code value="mm[Hg]"/>
           </valueQuantity>
        </component>
        <component>
           <!--  the max parameter code -->
           <code>
              <coding>
```

```
               <system value="http://hl7.org/fhir/observation-paramcode"/>
               <code value="max"/>
               <display value="Maximum"/>
            </coding>
         </code>
         <valueQuantity>
            <value value="110"/>
            <unit value="mmHg"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="mm[Hg]"/>
         </valueQuantity>
      </component>
      <component>
         <!--  the min parameter code -->
         <code>
            <coding>
               <system value="http://hl7.org/fhir/observation-paramcode"/>
               <code value="min"/>
               <display value="Minimum"/>
            </coding>
         </code>
         <valueQuantity>
            <value value="56"/>
            <unit value="mmHg"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="mm[Hg]"/>
         </valueQuantity>
      </component>
      <component>
         <!--  the count parameter code -->
         <code>
            <coding>
               <system value="http://hl7.org/fhir/observation-paramcode"/>
               <code value="min"/>
               <display value="Minimum"/>
            </coding>
         </code>
         <valueQuantity>
            <value value="7"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="{observations}"/>
         </valueQuantity>
      </component>
   </Observation>
  </resource>
</parameter>
<parameter>
  <name value="return"/>
  <resource>
    <Observation>
      <status value="final"/>
      <code>
         <coding>
            <system value="http://loinc.org"/>
            <code value="8462-4"/>
            <display value="Diastolic blood pressure"/>
         </coding>
      </code>
      <subject>
         <reference value="Patient/123"/>
      </subject>
      <effectivePeriod>
         <start value="2016-05-25T19:26:16Z"/>
         <end value="2016-05-25T20:26:16Z"/>
      </effectivePeriod>
      <!--The Observation.component.code is relative to the Observation.code and cannot be interpreted
  independently.-->
      <component>
         <code>
            <!--   the average parameter code -->
            <coding>
               <system value="http://hl7.org/fhir/observation-paramcode"/>
               <code value="average"/>
               <display value="Average"/>
            </coding>
         </code>
         <valueQuantity>
            <value value="79.7"/>
            <unit value="mmHg"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="mm[Hg]"/>
         </valueQuantity>
      </component>
      <component>
         <!--  the max parameter code -->
         <code>
            <coding>
               <system value="http://hl7.org/fhir/observation-paramcode"/>
               <code value="max"/>
```

```
                <display value="Maximum"/>
             </coding>
          </code>
          <valueQuantity>
             <value value="110"/>
             <unit value="mmHg"/>
             <system value="http://unitsofmeasure.org"/>
             <code value="mm[Hg]"/>
          </valueQuantity>
       </component>
       <component>
          <!--  the min parameter code -->
          <code>
             <coding>
                <system value="http://hl7.org/fhir/observation-paramcode"/>
                <code value="min"/>
                <display value="Minimum"/>
             </coding>
          </code>
          <valueQuantity>
             <value value="56"/>
             <unit value="mmHg"/>
             <system value="http://unitsofmeasure.org"/>
             <code value="mm[Hg]"/>
          </valueQuantity>
       </component>
       <component>
          <!--  the count parameter code -->
          <code>
             <coding>
                <system value="http://hl7.org/fhir/observation-paramcode"/>
                <code value="count"/>
                <display value="Count"/>
             </coding>
          </code>
          <valueQuantity>
             <value value="7"/>
             <system value="http://unitsofmeasure.org"/>
             <code value="{observations}"/>
          </valueQuantity>
       </component>
     </Observation>
   </resource>
  </parameter>
</Parameters>
```

10.1.20.2 Last N Observations Query

The lastn query meets the common need for searching for the most recent or last n=number of observations for a subject. For example, retrieving the last 5 temperatures for a patient to view trends or fetching the most recent laboratory results or vitals signs. To ask a server to return the last n=number of observations, the lastn queryuses the normal search parameters defined for the Observation resource. However, rather than their normal use, they are interpreted as inputs - i.e.. instead of requiring that the resources literally contain the search parameters, they are passed to a server algorithm of some kind that uses them to determine the most appropriate matches.

The response from an lastn query is a set of observations:

* filtered by additional parameters

* 'GROUP BY' Observation.code

* sorted from most recent to the oldest

* limited to the number of requested responses per group specified by the optional max query parameter

* If no maximum number is given then only the most recent Observation in each group is returned.

The set of returned observations should represent distinct real world observations and not the same observation with changes in status or versions. If there are no matches, the lastn query SHALL return an empty search set with no error, but may include an operation outcome with further advice.

The official URL for this operation definition is

`http://hl7.org/fhir/OperationDefinition/Observation-lastn`

Formal Definition (as a OperationDefinition).

URL: [base]/Observation/$lastn

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

max | 0..1 | positiveInt | | | max is an optional input parameter to the lastn query operation. It is used to specify the maximum number of Observations to return from each group. For example for the query "Fetch the last 3 results for all vitals for a patient" max = 3.

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Bundle | | | The set of most recent N Observations that match the lastn query search criteria. Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The key differences between this query operation and simply searching Observation using the combination of _count and _sort parameters are:

* The lastn query returns only the last N resource grouped by code. Using the _count query method doesn't restrict the total mathches so you may need to page through several "A" Observations before getting to Observation "B".

* The server is responsible for grouping the observations by codes. This frees the client from needing to determine which codes she should ask for.

This operation cannot be performed on observations that the user is not authorized to see. It is assumed that the server has identified and secured the context appropriately, and can either associate the authorization context with a single patient, or determine whether the context has the rights to the nominated patient, if there is one. If there is no nominated patient (e.g. the operation is invoked at the system level) and the context is not associated with a single patient record, then the server should return an error. Specifying the relationship between the context, a user and patient records is outside the scope of this specification.

10.1.20.2.1 Examples

1): Fetch the last 3 results for all vitals for a patient 2): Fetch the last lab results for a patient 3): Fetch the last 3 blood pressures, respiratory rate and heart rates for a patient (Request):

```
1) GET [base]/Observation$lastn?max=3&patient=Patient/123&category=vital-signs

-------------------------------------------

2) GET [base]/Observation$lastn?patient=Patient/123&category=laboratory

-------------------------------------------

3) GET [base]/Observation$lastn?max=3&patient=Patient/123&category=vital-signs&code=9279-1,8867-4,85354-9
```

1): A Server returns from an imaginary Last N algorithm the last 3 of 9 different vital signs observations on a patient. e.g. body temperature, body weight, body height... (truncated for readability) 2): A Server returns from an imaginary Last N algorithm the last laboratory observations for a patient. e.g. CBC Panel, UA, Chem Panel, Lipid Panel components and HbA1c, (truncated for readability) 3): A Server returns from an imaginary Last N algorithm the last 3 blood pressures, respiratory rate and heart rates for a patient as specified by code from the request. (truncated for readability) (Response):

```
============================ Example 1 ============================

HTTP/1.1 200 OK
[other headers]

{
  "resourceType": "Bundle",
  "id": "26419249-18b3-45de-b10e-dca0b2e72b",
  "meta": {
    "lastUpdated": "2017-02-18T03:28:49Z"
  },
  "type": "searchset",
  "total": 27,
  // the patient temperature was being taken several times a day
  "entry": [{
    "fullUrl": "http://server/path/Observation/20170219-06temp",
    "resource": {
      "resourceType": "Observation",
      "id": "20170219temp",
      .. snip ...
  ,{
   "fullUrl": "http://server/path/Observation/20170219-05temp",
       "resource": {
         "resourceType": "Observation",
         "id": "20170218temp",
      .. snip ...
    }
      ,{
       "fullUrl": "http://server/path/Observation/20170219-04temp",
           "resource": {
             "resourceType": "Observation",
             "id": "20170217temp",
          .. snip ...
    },
    // the patient was being weighed daily
    {
    "fullUrl": "http://server/path/Observation/20170219weight",
    "resource": {
      "resourceType": "Observation",
      "id": "20170219weight",
      .. snip ...
  ,{
   "fullUrl": "http://server/path/Observation/20170218weight",
       "resource": {
         "resourceType": "Observation",
         "id": "20170218weight",
      .. snip ...
    }
      ,{
       "fullUrl": "http://server/path/Observation/20170217weight",
```

```
          "resource": {
            "resourceType": "Observation",
            "id": "20170217weight",
          .. snip ...
    },
    // height was recorded occasionally over the years
        {
        "fullUrl": "http://server/path/Observation/20170101height",
        "resource": {
          "resourceType": "Observation",
          "id": "20170219height",
          .. snip ...
      ,{
       "fullUrl": "http://server/path/Observation/20160101height",
           "resource": {
             "resourceType": "Observation",
             "id": "20160101height",
          .. snip ...
        }
          ,{
           "fullUrl": "http://server/path/Observation/20131001height",
               "resource": {
                 "resourceType": "Observation",
                 "id": "20131001height",
              .. snip ...
    }
  ]
}

---------------------------------

========================== Example 2 ============================


HTTP/1.1 200 OK
[other headers]

{
  "resourceType": "Bundle",
  "id": "26419249-18b3-45de-b10e-dca0b2e72b",
  "meta": {
    "lastUpdated": "2017-02-18T03:28:49Z"
  },
  "type": "searchset",
  "total": 45,
  // the patient's last CBC Panel component observations
  "entry": [{
    "fullUrl": "http://server/path/Observation/20170219-CBCPanel-RBCs",
    "resource": {
      "resourceType": "Observation",
      "id": "0170219-CBCPanel-RBCs",
      ... snip ...
      "code" : {
    "coding" : [
      {
        "system" : "http://loinc.org",
        "code" : "789-8",
        "display" : "RBC # Bld Auto"
      }
    ],
    "text" : "RBC # Bld Auto"
  },
  ...snip....
  },
  // the rest of the patient CBC Panel components observations
  { ...snip...
  },
  // the patient's last UA Panel component observations
    {
    "fullUrl": "http://server/path/Observation/20160610-UA-Color",
    "resource": {
      "resourceType": "Observation",
      "id": "20160610-UA-Color",
      .. snip ...
      "code" : {
   "coding" : [
     {
       "system" : "http://loinc.org",
       "code" : "5778-6",
       "display" : "Color of Urine"
     }
   ],
   "text" : "COLOR"
 },
    ...snip....
   },
   // the rest of the patient UA Panel component observations
   { ...snip...
   },
```

```
   // the patient's last Metabolic Panel component observations
  {
   "fullUrl": "http://server/path/Observation/20170219MetPanel-Sodium",
       "resource": {
         "resourceType": "Observation",
         "id": "20170219MetPanel-Sodium",
      ... snip ...
        "code" : {
          "coding" : [
          {
            "system" : "http://loinc.org",
            "code" : "2951-2",
            "display" : "Sodium SerPl-sCnc"
            }
          ],
          "text" : "Sodium SerPl-sCnc"
          },
      ...snip....
      },
      // the rest of the patient Metabolic Panel component observations
      { ...snip...
      },
    // the patient Lipase Panel components observations
        {
        "fullUrl": "http://server/path/Observation/20170219LipidPanel-Chol",
        "resource": {
          "resourceType": "Observation",
          "id": "20170219LipidPanel-Chol",
          ... snip ...
          "code": {
                 "coding": [
                   {
                     "system": "http://loinc.org",
                     "code": "14647-2",
                     "display": "Cholesterol [Moles/volume] in Serum or Plasma"
                   }
                 ],
                 "text": "Cholesterol"
          },
          ...snip....
          },
          // the patient HbA1c observation
          {
          "fullUrl": "http://server/path/Observation/20170219HbA1c",
          "resource": {
            "resourceType": "Observation",
            "id": "20170219HbA1c",
            ... snip ...
            "code": {
                   "coding": [
                     {
                       "system": "http://loinc.org",
                       "code": "4548-4",
                       "display": "Hemoglobin A1c/Hemoglobin.total in Blood"
                     }
                   ],
                   "text": "HbA1c"
            },
            ...snip....
    }
  ]
}

---------------------------------


========================== Example 3 ============================


HTTP/1.1 200 OK
[other headers]

{
  "resourceType": "Bundle",
  "id": "26419249-18b3-45de-b10e-dca0b2e72b",
  "meta": {
    "lastUpdated": "2017-02-18T03:28:49Z"
  },
  "type": "searchset",
  "total": 9,
  // Returns the patient's last 3 code = 9279-1 Respiratory Rate observations All from 20170219
  "entry": [{
    "fullUrl": "http://server/path/Observation/20170219-06RespRate",
    "resource": {
      "resourceType": "Observation",
      "id": "20170219-06RespRate",
      .. snip ...
     "code": {
       "coding": [
```

```
        {
          "system": "http://loinc.org",
          "code": "9279-1",
          "display": "Respiratory rate"
        }
      ],
      "text": "Respiratory rate"
},
    .. snip ...
 ,{
  "fullUrl": "http://server/path/Observation/20170219-05RespRate",
      "resource": {
        "resourceType": "Observation",
        "id": "20170219-05RespRate",
     .. snip ...
   }
     ,{
      "fullUrl": "http://server/path/Observation/20170219-04RespRate",
          "resource": {
            "resourceType": "Observation",
            "id": "20170217-04RespRate",
         .. snip ...
   },
   // Returns the patient's last 3 code =  8867-4 Heart Rate observations All from 20170219
   // the patient was being weighed daily
   {
   "fullUrl": "http://server/path/Observation/20170219-06HeartRate",
   "resource": {
     "resourceType": "Observation",
     "id": "20170219-06HeartRate",
     .. snip ...
     ],
   "code": {
     "coding": [
       {
         "system": "http://loinc.org",
         "code": "8867-4",
         "display": "Heart rate"
       }
     ],
     "text": "Heart rate"
 ...snip...
 ,{
  "fullUrl": "http://server/path/Observation/20170219-05HeartRate",
      "resource": {
        "resourceType": "Observation",
        "id": "20170219-05HeartRate",
     .. snip ...
   }
     ,{
      "fullUrl": "http://server/path/Observation/20170219-04HeartRate",
          "resource": {
            "resourceType": "Observation",
            "id": "20170219-04HeartRate",
         .. snip ...
   },
   // Returns the patient's last 3 code =  85354-9 Blood pressure systolic and diastolic All from 20170219
       {
       "fullUrl": "http://server/path/Observation/20170219-06BloodPressure",
       "resource": {
         "resourceType": "Observation",
         "id": "20170219-06BloodPressure",
         .. snip ...
         "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "85354-9",
              "display": "Bood pressure panel with all children optional"
            }
          ],
          "text": "Blood pressure systolic & diastolic"
        },
        ...snip...
               "component": [
        {
          "code": {
            "coding": [
              {
                "system": "http://loinc.org",
                "code": "8480-6",
                "display": "Systolic blood pressure"
              },
              {
           ...snip...
                      {
             "code": {
               "coding": [
                 {
```

```
                  "system": "http://loinc.org",
                  "code": "8462-4",
                  "display": "Diastolic blood pressure"
                }
              ]
            },
            ...snip....
     ,{
      "fullUrl": "http://server/path/Observation/20170219-05BloodPressure",
          "resource": {
            "resourceType": "Observation",
            "id": "20170219-05BloodPressure",
          .. snip ...
       }
         ,{
          "fullUrl": "http://server/path/Observation/20170219-04BloodPressure",
              "resource": {
                "resourceType": "Observation",
                "id": "20170219-04BloodPressure",
              .. snip ...
   }
  ]
}
```

10.1.21 Resource Observation - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/observation-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.1.21 Resource Observation - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for Observation.

Functional status for this map: 34 tests that all execute ok. All tests pass round-trip testing and 15 r3 resources are invalid (58 errors). (see documentation)

10.1.21.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Observation2to3" = "R2 to R3 Conversions for Observation"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Observation" alias ObservationR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Observation" alias Observation as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Observation extends DomainResource
  input src : ObservationR2 as source
  input tgt : Observation as target

  "Observation-identifier" : for src.identifier make tgt.identifier
  "Observation-status" : for src.status make tgt.status
  "Observation-category" : for src.category make tgt.category
  "Observation-code" : for src.code make tgt.code
  "Observation-subject" : for src.subject make tgt.subject
  "Observation-encounter" : for src.encounter make tgt.context
  "Observation-effective" : for src.effective make tgt.effective
  "Observation-issued" : for src.issued make tgt.issued
  "Observation-performer" : for src.performer make tgt.performer
  "Observation-value" : for src.value make tgt.value
  "Observation-dataAbsentReason" : for src.dataAbsentReason make tgt.dataAbsentReason
  "Observation-interpretation" : for src.interpretation make tgt.interpretation
  "Observation-comments" : for src.comments make tgt.comment
  "Observation-bodySite" : for src.bodySite make tgt.bodySite
  "Observation-method" : for src.method make tgt.method
  "Observation-specimen" : for src.specimen make tgt.specimen
  "Observation-device" : for src.device make tgt.device
  "Observation-referenceRange" : for src.referenceRange as vs0 make tgt.referenceRange as vt0 then
    obsReferenceRange(vs0, vt0)
  "Observation-related" : for src.related as vs0 make tgt.related as vt0 then obsRelated(vs0, vt0)
  "Observation-component" : for src.component as vs0 make tgt.component as vt0 then obsComponent(vs0, vt0)
endgroup

group obsReferenceRange extends BackboneElement
  input src as source
  input tgt as target

  "Observation.referenceRange-low" : for src.low make tgt.low
  "Observation.referenceRange-high" : for src.high make tgt.high
  "Observation.referenceRange-meaning" : for src.meaning make tgt.appliesTo
  "Observation.referenceRange-age" : for src.age make tgt.age
  "Observation.referenceRange-text" : for src.text make tgt.text
endgroup

group obsRelated extends BackboneElement
```

```
  input src as source
  input tgt as target

  "Observation.related-type" : for src.type make tgt.type
  "Observation.related-target" : for src.target make tgt.target
endgroup

group obsComponent extends BackboneElement
  input src as source
  input tgt as target

  "Observation.component-code" : for src.code make tgt.code
  "Observation.component-value" : for src.value make tgt.value
  "Observation.component-dataAbsentReason" : for src.dataAbsentReason make tgt.dataAbsentReason
  "Observation.component-referenceRange" : for src.referenceRange as vs0 make tgt.referenceRange as vt0 then
    obsReferenceRange(vs0, vt0)
endgroup
```

10.1.21.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Observation3to2" = "R3 to R2 Conversion for Observation"

uses "http://hl7.org/fhir/StructureDefinition/Observation" alias Observation as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Observation" alias ObservationR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Observation extends DomainResource
  input src : Observation as source
  input tgt : ObservationR2 as target

  "Observation-identifier" : for src.identifier make tgt.identifier
  "Observation-status" : for src.status make tgt.status
  "Observation-category" : for src.category make tgt.category
  "Observation-code" : for src.code make tgt.code
  "Observation-subject" : for src.subject make tgt.subject
  "Observation-encounter" : for src.context make tgt.encounter
  "Observation-effective" : for src.effective make tgt.effective
  "Observation-issued" : for src.issued make tgt.issued
  "Observation-performer" : for src.performer make tgt.performer
  "Observation-value" : for src.value make tgt.value
  "Observation-dataAbsentReason" : for src.dataAbsentReason make tgt.dataAbsentReason
  "Observation-interpretation" : for src.interpretation make tgt.interpretation
  "Observation-comments" : for src.comment make tgt.comments
  "Observation-bodySite" : for src.bodySite make tgt.bodySite
  "Observation-method" : for src.method make tgt.method
  "Observation-specimen" : for src.specimen make tgt.specimen
  "Observation-device" : for src.device make tgt.device
  "Observation-referenceRange" : for src.referenceRange as vs0 make tgt.referenceRange as vt0 then
    obsReferenceRange(vs0, vt0)
  "Observation-related" : for src.related as vs0 make tgt.related as vt0 then obsRelated(vs0, vt0)
  "Observation-component" : for src.component as vs0 make tgt.component as vt0 then obsComponent(vs0, vt0)
endgroup

group obsReferenceRange extends BackboneElement
  input src as source
  input tgt as target

  "Observation.referenceRange-low" : for src.low make tgt.low
  "Observation.referenceRange-high" : for src.high make tgt.high
  "Observation.referenceRange-meaning" : for src.appliesTo make tgt.meaning
  "Observation.referenceRange-age" : for src.age make tgt.age
  "Observation.referenceRange-text" : for src.text make tgt.text
endgroup

group obsRelated extends BackboneElement
  input src as source
  input tgt as target

  "Observation.related-type" : for src.type make tgt.type
  "Observation.related-target" : for src.target make tgt.target
endgroup

group obsComponent extends BackboneElement
  input src as source
  input tgt as target

  "Observation.component-code" : for src.code make tgt.code
  "Observation.component-value" : for src.value make tgt.value
  "Observation.component-dataAbsentReason" : for src.dataAbsentReason make tgt.dataAbsentReason
  "Observation.component-referenceRange" : for src.referenceRange as vs0 make tgt.referenceRange as vt0 then
    obsReferenceRange(vs0, vt0)
endgroup
```

10.1.21.3 R3 Validation Errors

genetics-example1-somatic | ERROR @ Observation.contained: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator) ERROR @ Observation.extension[1]: The extension http://hl7.org/fhir/StructureDefinition/geneticsGenomeBuild is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[2]: The extension http://hl7.org/fhir/StructureDefinition/geneticsChromosome is unknown, and not allowed here (src =

InstanceValidator) ERROR @ Observation.extension[3]: The extension http://hl7.org/fhir/StructureDefinition/geneticsGenomicStart is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[4]: The extension http://hl7.org/fhir/StructureDefinition/geneticsGenomicStop is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[5]: The extension http://hl7.org/fhir/StructureDefinition/geneticsSpecies is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[6]: The extension http://hl7.org/fhir/StructureDefinition/geneticsCIGAR is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[7]: The extension http://hl7.org/fhir/StructureDefinition/geneticsAssessedCondition is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[8]: The extension http://hl7.org/fhir/StructureDefinition/geneticsReferenceAllele is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[9]: The extension http://hl7.org/fhir/StructureDefinition/geneticsObservedAllele is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[10]: The extension http://hl7.org/fhir/StructureDefinition/geneticsGene is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[11]: The extension http://hl7.org/fhir/StructureDefinition/geneticsTranscriptReferenceSequenceId is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[12]: The extension http://hl7.org/fhir/StructureDefinition/geneticsDNASequenceVariation is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[13]: The extension http://hl7.org/fhir/StructureDefinition/geneticsDNASequenceVariationType is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[14]: The extension http://hl7.org/fhir/StructureDefinition/geneticsAminoAcidChange is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[15]: The extension http://hl7.org/fhir/StructureDefinition/geneticsAlleleName is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[16]: The extension http://hl7.org/fhir/StructureDefinition/geneticsGenomicSourceClass is unknown, and not allowed here (src = InstanceValidator)

genetics-example2-germline | ERROR @ Observation.extension[1]: The extension http://hl7.org/fhir/StructureDefinition/geneticsGenomeBuild is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[2]: The extension http://hl7.org/fhir/StructureDefinition/geneticsChromosome is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[3]: The extension http://hl7.org/fhir/StructureDefinition/geneticsGenomicStart is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[4]: The extension http://hl7.org/fhir/StructureDefinition/geneticsGenomicStop is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[5]: The extension http://hl7.org/fhir/StructureDefinition/geneticsReferenceAllele is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[6]: The extension http://hl7.org/fhir/StructureDefinition/geneticsObservedAllele is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[7]: The extension http://hl7.org/fhir/StructureDefinition/geneticsGene is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[8]: The extension http://hl7.org/fhir/StructureDefinition/geneticsDNASequenceVariation is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[9]: The extension http://hl7.org/fhir/StructureDefinition/geneticsVariationId is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[10]: The extension http://hl7.org/fhir/StructureDefinition/geneticsDNARegionName is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[11]: The extension http://hl7.org/fhir/StructureDefinition/geneticsDNASequenceVariationType is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[12]: The extension http://hl7.org/fhir/StructureDefinition/geneticsAminoAcidChange is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[13]: The extension http://hl7.org/fhir/StructureDefinition/geneticsGenomicSourceClass is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[14]: The extension http://hl7.org/fhir/StructureDefinition/geneticsSpecies is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[15]: The extension http://hl7.org/fhir/StructureDefinition/geneticsCIGAR is unknown, and not allowed here (src = InstanceValidator)

genetics-example3-mutationlist | ERROR @ Observation.extension[1]: The extension http://hl7.org/fhir/StructureDefinition/geneticsGenomeBuild is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[2]: The extension http://hl7.org/fhir/StructureDefinition/geneticsGene is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[3]: The extension http://hl7.org/fhir/StructureDefinition/geneticsSpecies is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[4]: The extension http://hl7.org/fhir/StructureDefinition/geneticsResult is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[5]: The extension http://hl7.org/fhir/StructureDefinition/geneticsResult is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[6]: The extension http://hl7.org/fhir/StructureDefinition/geneticsResult is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[7]: The extension http://hl7.org/fhir/StructureDefinition/geneticsResult is unknown, and not allowed here (src = InstanceValidator)

genetics-example3-mutationlist-1 | ERROR @ Observation.extension[1]: The extension http://hl7.org/fhir/StructureDefinition/geneticsDNASequenceVariation is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[2]: The extension http://hl7.org/fhir/StructureDefinition/geneticsDNARegionName is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[3]: The extension http://hl7.org/fhir/StructureDefinition/geneticsAminoAcidChange is unknown, and not allowed here (src = InstanceValidator)

genetics-example3-mutationlist-2 | ERROR @ Observation.extension[1]: The extension http://hl7.org/fhir/StructureDefinition/geneticsDNARegionName is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[2]: The extension http://hl7.org/fhir/StructureDefinition/geneticsAminoAcidChange is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[3]: The extension http://hl7.org/fhir/StructureDefinition/geneticsVariationId is unknown, and not allowed here (src = InstanceValidator)

genetics-example3-mutationlist-3 | ERROR @ Observation.extension[1]: The extension http://hl7.org/fhir/StructureDefinition/geneticsDNARegionName is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[2]: The extension http://hl7.org/fhir/StructureDefinition/geneticsVariationId is unknown, and not allowed here (src = InstanceValidator)

genetics-example3-mutationlist-4 | ERROR @ Observation.extension[1]: The extension http://hl7.org/fhir/StructureDefinition/geneticsDNARegionName is unknown, and not allowed here (src = InstanceValidator) ERROR @ Observation.extension[2]: The extension http://hl7.org/fhir/StructureDefinition/geneticsVariationId is unknown, and not allowed here (src = InstanceValidator)

obs-uslab-example1 | ERROR @ Observation.extension: The extension http://hl7.org/fhir/StructureDefinition/uslab-observationkind is unknown, and not allowed here (src = InstanceValidator)

obs-uslab-example2 | ERROR @ Observation.extension: The extension http://hl7.org/fhir/StructureDefinition/uslab-observationkind is unknown, and not allowed here (src = InstanceValidator)

obs-uslab-example3 | ERROR @ Observation.extension: The extension http://hl7.org/fhir/StructureDefinition/uslab-observationkind is unknown, and not allowed here (src = InstanceValidator)

obs-uslab-example4 | ERROR @ Observation.extension: The extension http://hl7.org/fhir/StructureDefinition/uslab-observationkind is unknown, and not allowed here (src = InstanceValidator)

obs-uslab-example5 | ERROR @ Observation.extension: The extension http://hl7.org/fhir/StructureDefinition/uslab-observationkind is unknown, and not allowed here (src = InstanceValidator)

obs-uslab-example6 | ERROR @ Observation.extension: The extension http://hl7.org/fhir/StructureDefinition/uslab-observationkind is unknown, and not allowed here (src = InstanceValidator)

obs-uslab-example7 | ERROR @ Observation.extension: The extension http://hl7.org/fhir/StructureDefinition/uslab-observationkind is unknown, and not allowed here (src = InstanceValidator)

obs-uslab-example8 | ERROR @ Observation.extension: The extension http://hl7.org/fhir/StructureDefinition/uslab-observationkind is unknown, and not allowed here (src = InstanceValidator) WARNING @ Observation.code.coding[2].display: value should not start or finish with whitespace (src = InstanceValidator)

10.2 Resource DiagnosticReport - Content

URL: https://hl7.org/fhir/STU3/diagnosticreport.html

The datatype has a element that may be used to distinguish the identifiers assigned by the requester and the performer of the request (known as the 'Placer' and 'Filler' in the HL7 Version 2 Messaging Standard). Use the identifier type code "PLAC" for the Placer Identifier and "FILL" for the Filler identifier as is shown in the example below:

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Orders and Observations Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner

The findings and interpretation of diagnostic tests performed on patients, groups of patients, devices, and locations, and/or specimens derived from these. The report includes clinical context such as requesting and provider information, and some mix of atomic results, images, textual and coded interpretations, and formatted representation of diagnostic reports.

10.2.1 Scope and Usage

This resource is an event resource from a FHIR workflow perspective - see Workflow. It is the intent of the Orders and Observation Workgroup to align this resource with the workflow pattern for event resources.

A diagnostic report is the set of information that is typically provided by a diagnostic service when investigations are complete. The information includes a mix of atomic results, text reports, images, and codes. The mix varies depending on the nature of the diagnostic procedure, and sometimes on the nature of the outcomes for a particular investigation. In FHIR, the report can be conveyed in a variety of ways including a Document, RESTful API, or Messaging framework. Included within each of these, would be the DiagnosticReport resource itself.

The DiagnosticReport resource has information about the diagnostic report itself, and about the subject and, in the case of lab tests, the specimen of the report. It can also refer to the request details and atomic observations details or image instances. Report conclusions can be expressed as a simple text blob, structured coded data or as an attached fully formatted report such as a PDF.

The DiagnosticReport resource is suitable for the following kinds of diagnostic reports:

* Laboratory (Clinical Chemistry, Hematology, Microbiology, etc.)

* Pathology / Histopathology / related disciplines

* Imaging Investigations (x-ray, CT, MRI etc.)

* Other diagnostics - Cardiology, Gastroenterology etc.

The DiagnosticReport resource is not intended to support cumulative result presentation (tabular presentation of past and present results in the resource). The DiagnosticReport resource does not yet provide full support for detailed structured reports of sequencing; this is planned for a future release.

10.2.2 Background and Context

Diagnostic Report Names

The words "tests", "results", "observations", "panels" and "batteries" are often used interchangeably when describing the various parts of a diagnostic report. This leads to much confusion. The naming confusion is worsened because of the wide variety of forms that the result of a diagnostic investigation can take, as described above. Languages other than English have their own variations on this theme.

This resource uses one particular set of terms. A practitioner "requests" a set of "tests". The diagnostic service returns a "report" which may contain a "narrative" - a written summary of the outcomes, and/or "results" - the individual pieces of atomic data which each are "observations". The results are assembled in "groups" which are nested structures of Observations (traditionally referred to as "panels" or " batteries" by laboratories) that can be used to represent relationships between the individual data items.

10.2.3 Boundaries and Relationships

Note that many diagnostic processes are procedures that generate observations and diagnostic reports. In many cases, such an observation does not require an explicit representation of the procedure used to create the observation, but where there are details of interest about how the diagnostic procedure was performed, the Procedure resource is used to describe the activity.

In contrast to the Observation resource, the DiagnosticReport resource typically includes additional clinical context and some mix of atomic results, images, imaging reports, textual and coded interpretation, and formatted representations. Laboratory reports, pathology reports, and imaging reports should be represented using the DiagnosticReport resource. The Observation

resource is referenced by the DiagnosticReport to provide the atomic results for a particular investigation.

If you have a highly structured report, then use DiagnosticReport - it has data and workflow support. Details about the request for a diagnostic investigation are captured in the various "request" resources (e.g., the ProcedureRequest) resource and allow the report to connect to clinical workflows. For more narrative driven reports with less work flow (histology/mortuary, etc.), the Composition resource would be more appropriate.

Image and media representations of the report and supporting images are referenced in the DiagnosticReport resource. The details and actual image instances can be referenced directly in Diagnostic report using the "imaging" element or by indirect reference through the ImagingManifest or ImagingStudy resources which represent the content produced in a DICOM imaging study or set of DICOM Instances of a patient.

This resource is referenced by ChargeItem, ClinicalImpression, Condition and Procedure

10.2.4 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | | | DomainResource | A Diagnostic report - a combination of request information, atomic results, images, interpretation, as well as formatted reports Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Business identifier for report
basedOn | | 0..* | Reference(CarePlan | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ProcedureRequest | ReferralRequest) | What was requested
status | ?!? | 1..1 | code | registered | partial | preliminary | final + DiagnosticReportStatus (Required)
category | ? | 0..1 | CodeableConcept | Service category Diagnostic Service Section Codes (Example)
code | ? | 1..1 | CodeableConcept | Name/Code for this diagnostic report LOINC Diagnostic Report Codes (Preferred)
subject | ? | 0..1 | Reference(Patient | Group | Device | Location) | The subject of the report - usually, but not always, the patient
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Health care event when test ordered
effective[x] | ? | 0..1 | | Clinically relevant time/time-period for report
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ? | 0..1 | instant | DateTime this version was released
performer | ? | 0..* | BackboneElement | Participants in producing the report
role | ? | 0..1 | CodeableConcept | Type of performer Procedure Performer Role Codes (Example)
actor | ? | 1..1 | Reference(Practitioner | Organization) | Practitioner or Organization participant
specimen | | 0..* | Reference(Specimen) | Specimens this report is based on
result | | 0..* | Reference(Observation) | Observations - simple, or complex nested groups
imagingStudy | | 0..* | Reference(ImagingStudy | ImagingManifest) | Reference to full details of imaging associated with the diagnostic report
image | ? | 0..* | BackboneElement | Key images associated with this report
comment | | 0..1 | string | Comment about the image (e.g. explanation)
link | ? | 1..1 | Reference(Media) | Reference to the image source
conclusion | | 0..1 | string | Clinical Interpretation of test results
codedDiagnosis | | 0..* | CodeableConcept | Codes for the conclusion SNOMED CT Clinical Findings (Example)
presentedForm | | 0..* | Attachment | Entire report as issued
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DiagnosticReport xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier for report --></identifier>
 <basedOn><!-- 0..* Reference(CarePlan|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ProcedureRequest|ReferralRequest) What was requested --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | partial | preliminary | final + -->
 <category><!-- 0..1 CodeableConcept Service category --></category>
 <code><!-- 1..1 CodeableConcept Name/Code for this diagnostic report --></code>
 <subject><!-- 0..1 Reference(Patient|Group|Device|Location) The subject of the report - usually, but not always, the
    patient --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Health care event when test ordered --></context>
 <effective[x]><!-- 0..1 dateTime|Period Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 DateTime this version was released -->
 <performer>  <!-- 0..* Participants in producing the report -->
  <role><!-- 0..1 CodeableConcept Type of performer --></role>
  <actor><!-- 1..1 Reference(Practitioner|Organization) Practitioner or Organization  participant --></actor>
 </performer>
 <specimen><!-- 0..* Reference(Specimen) Specimens this report is based on --></specimen>
 <result><!-- 0..* Reference(Observation) Observations - simple, or complex nested groups --></result>
```

```
 <imagingStudy><!-- 0..* Reference(ImagingStudy|ImagingManifest) Reference to full details of imaging associated with
    the diagnostic report --></imagingStudy>
 <image>  <!-- 0..* Key images associated with this report -->
  <comment value="[string]"/><!-- 0..1 Comment about the image (e.g. explanation) -->
  <link><!-- 1..1 Reference(Media) Reference to the image source --></link>
 </image>
 <conclusion value="[string]"/><!-- 0..1 Clinical Interpretation of test results -->
 <codedDiagnosis><!-- 0..* CodeableConcept Codes for the conclusion --></codedDiagnosis>
 <presentedForm><!-- 0..* Attachment Entire report as issued --></presentedForm>
</DiagnosticReport>
```

JSON Template

```
{
  "resourceType" : "DiagnosticReport",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier for report
  "basedOn" : [{ Reference(CarePlan|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ProcedureRequest|ReferralRequest) }], // What was requested
  "status" : "<code>", // R!  registered | partial | preliminary | final +
  "category" : { CodeableConcept }, // Service category
  "code" : { CodeableConcept }, // R!  Name/Code for this diagnostic report
  "subject" : { Reference(Patient|Group|Device|Location) }, // The subject of the report - usually, but not always, the
    patient
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Health care event when test ordered
  // effective[x]: Clinically relevant time/time-period for report. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "issued" : "<instant>", // DateTime this version was released
  "performer" : [{ // Participants in producing the report
    "role" : { CodeableConcept }, // Type of performer
    "actor" : { Reference(Practitioner|Organization) } // R!  Practitioner or Organization  participant
  }],
  "specimen" : [{ Reference(Specimen) }], // Specimens this report is based on
  "result" : [{ Reference(Observation) }], // Observations - simple, or complex nested groups
  "imagingStudy" : [{ Reference(ImagingStudy|ImagingManifest) }], // Reference to full details of imaging associated
    with the diagnostic report
  "image" : [{ // Key images associated with this report
    "comment" : "<string>", // Comment about the image (e.g. explanation)
    "link" : { Reference(Media) } // R!  Reference to the image source
  }],
  "conclusion" : "<string>", // Clinical Interpretation of test results
  "codedDiagnosis" : [{ CodeableConcept }], // Codes for the conclusion
  "presentedForm" : [{ Attachment }] // Entire report as issued
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DiagnosticReport;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DiagnosticReport.identifier [ Identifier ], ... ; # 0..* Business identifier for report
  fhir:DiagnosticReport.basedOn [ Reference(CarePlan|ImmunizationRecommendation|MedicationRequest|NutritionOrder|
  ProcedureRequest|ReferralRequest) ], ... ; # 0..* What was requested
  fhir:DiagnosticReport.status [ code ]; # 1..1 registered | partial | preliminary | final +
  fhir:DiagnosticReport.category [ CodeableConcept ]; # 0..1 Service category
  fhir:DiagnosticReport.code [ CodeableConcept ]; # 1..1 Name/Code for this diagnostic report
  fhir:DiagnosticReport.subject [ Reference(Patient|Group|Device|Location) ]; # 0..1 The subject of the report -
    usually, but not always, the patient
  fhir:DiagnosticReport.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Health care event when test ordered
  # DiagnosticReport.effective[x] : 0..1 Clinically relevant time/time-period for report. One of these 2
    fhir:DiagnosticReport.effectiveDateTime [ dateTime ]
    fhir:DiagnosticReport.effectivePeriod [ Period ]
  fhir:DiagnosticReport.issued [ instant ]; # 0..1 DateTime this version was released
  fhir:DiagnosticReport.performer [ # 0..* Participants in producing the report
    fhir:DiagnosticReport.performer.role [ CodeableConcept ]; # 0..1 Type of performer
    fhir:DiagnosticReport.performer.actor [ Reference(Practitioner|Organization) ]; # 1..1 Practitioner or Organization
      participant
  ], ...;
  fhir:DiagnosticReport.specimen [ Reference(Specimen) ], ... ; # 0..* Specimens this report is based on
  fhir:DiagnosticReport.result [ Reference(Observation) ], ... ; # 0..* Observations - simple, or complex nested groups
  fhir:DiagnosticReport.imagingStudy [ Reference(ImagingStudy|ImagingManifest) ], ... ; # 0..* Reference to full details
     of imaging associated with the diagnostic report
  fhir:DiagnosticReport.image [ # 0..* Key images associated with this report
    fhir:DiagnosticReport.image.comment [ string ]; # 0..1 Comment about the image (e.g. explanation)
    fhir:DiagnosticReport.image.link [ Reference(Media) ]; # 1..1 Reference to the image source
  ], ...;
  fhir:DiagnosticReport.conclusion [ string ]; # 0..1 Clinical Interpretation of test results
  fhir:DiagnosticReport.codedDiagnosis [ CodeableConcept ], ... ; # 0..* Codes for the conclusion
  fhir:DiagnosticReport.presentedForm [ Attachment ], ... ; # 0..* Entire report as issued
]
```

Changes since DSTU2

DiagnosticReport

DiagnosticReport.basedOn | Renamed from request to basedOn Remove Reference(DiagnosticOrder), Add Reference(CarePlan), Add Reference(ImmunizationRecommendation), Add Reference(MedicationRequest), Add Reference(NutritionOrder)
DiagnosticReport.subject | Min Cardinality changed from 1 to 0
DiagnosticReport.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
DiagnosticReport.effective[x] | Min Cardinality changed from 1 to 0
DiagnosticReport.issued | Min Cardinality changed from 1 to 0
DiagnosticReport.performer | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to * Remove Reference(Practitioner), Remove Reference(Organization)
DiagnosticReport.performer.role | Added Element
DiagnosticReport.performer.actor | Added Element
DiagnosticReport.imagingStudy | Remove Reference(ImagingObjectSelection), Add Reference(ImagingManifest)

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 11 tests of which 3 fail to execute. 8 fail round-trip testing and 3 r3 resources are invalid (22 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | | | DomainResource | A Diagnostic report - a combination of request information, atomic results, images, interpretation, as well as formatted reports Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Business identifier for report
basedOn | | 0..* | Reference(CarePlan | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ProcedureRequest | ReferralRequest) | What was requested
status | ?!? | 1..1 | code | registered | partial | preliminary | final + DiagnosticReportStatus (Required)
category | ? | 0..1 | CodeableConcept | Service category Diagnostic Service Section Codes (Example)
code | ? | 1..1 | CodeableConcept | Name/Code for this diagnostic report LOINC Diagnostic Report Codes (Preferred)
subject | ? | 0..1 | Reference(Patient | Group | Device | Location) | The subject of the report - usually, but not always, the patient
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Health care event when test ordered
effective[x] | ? | 0..1 | | Clinically relevant time/time-period for report
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ? | 0..1 | instant | DateTime this version was released
performer | ? | 0..* | BackboneElement | Participants in producing the report
role | ? | 0..1 | CodeableConcept | Type of performer Procedure Performer Role Codes (Example)
actor | ? | 1..1 | Reference(Practitioner | Organization) | Practitioner or Organization participant
specimen | | 0..* | Reference(Specimen) | Specimens this report is based on
result | | 0..* | Reference(Observation) | Observations - simple, or complex nested groups
imagingStudy | | 0..* | Reference(ImagingStudy | ImagingManifest) | Reference to full details of imaging associated with the diagnostic report
image | ? | 0..* | BackboneElement | Key images associated with this report
comment | | 0..1 | string | Comment about the image (e.g. explanation)
link | ? | 1..1 | Reference(Media) | Reference to the image source
conclusion | | 0..1 | string | Clinical Interpretation of test results
codedDiagnosis | | 0..* | CodeableConcept | Codes for the conclusion SNOMED CT Clinical Findings (Example)
presentedForm | | 0..* | Attachment | Entire report as issued
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DiagnosticReport xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier for report --></identifier>
 <basedOn><!-- 0..* Reference(CarePlan|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ProcedureRequest|ReferralRequest) What was requested --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | partial | preliminary | final + -->
 <category><!-- 0..1 CodeableConcept Service category --></category>
 <code><!-- 1..1 CodeableConcept Name/Code for this diagnostic report --></code>
 <subject><!-- 0..1 Reference(Patient|Group|Device|Location) The subject of the report - usually, but not always, the
    patient --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Health care event when test ordered --></context>
 <effective[x]><!-- 0..1 dateTime|Period Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 DateTime this version was released -->
 <performer>  <!-- 0..* Participants in producing the report -->
  <role><!-- 0..1 CodeableConcept Type of performer --></role>
  <actor><!-- 1..1 Reference(Practitioner|Organization) Practitioner or Organization  participant --></actor>
 </performer>
 <specimen><!-- 0..* Reference(Specimen) Specimens this report is based on --></specimen>
 <result><!-- 0..* Reference(Observation) Observations - simple, or complex nested groups --></result>
 <imagingStudy><!-- 0..* Reference(ImagingStudy|ImagingManifest) Reference to full details of imaging associated with
    the diagnostic report --></imagingStudy>
 <image>  <!-- 0..* Key images associated with this report -->
  <comment value="[string]"/><!-- 0..1 Comment about the image (e.g. explanation) -->
  <link><!-- 1..1 Reference(Media) Reference to the image source --></link>
 </image>
 <conclusion value="[string]"/><!-- 0..1 Clinical Interpretation of test results -->
 <codedDiagnosis><!-- 0..* CodeableConcept Codes for the conclusion --></codedDiagnosis>
 <presentedForm><!-- 0..* Attachment Entire report as issued --></presentedForm>
```

```
</DiagnosticReport>
```

JSON Template

```
{
  "resourceType" : "DiagnosticReport",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier for report
  "basedOn" : [{ Reference(CarePlan|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ProcedureRequest|ReferralRequest) }], // What was requested
  "status" : "<code>", // R!  registered | partial | preliminary | final +
  "category" : { CodeableConcept }, // Service category
  "code" : { CodeableConcept }, // R!  Name/Code for this diagnostic report
  "subject" : { Reference(Patient|Group|Device|Location) }, // The subject of the report - usually, but not always, the
    patient
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Health care event when test ordered
  // effective[x]: Clinically relevant time/time-period for report. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "issued" : "<instant>", // DateTime this version was released
  "performer" : [{ // Participants in producing the report
    "role" : { CodeableConcept }, // Type of performer
    "actor" : { Reference(Practitioner|Organization) } // R!  Practitioner or Organization  participant
  }],
  "specimen" : [{ Reference(Specimen) }], // Specimens this report is based on
  "result" : [{ Reference(Observation) }], // Observations - simple, or complex nested groups
  "imagingStudy" : [{ Reference(ImagingStudy|ImagingManifest) }], // Reference to full details of imaging associated
    with the diagnostic report
  "image" : [{ // Key images associated with this report
    "comment" : "<string>", // Comment about the image (e.g. explanation)
    "link" : { Reference(Media) } // R!  Reference to the image source
  }],
  "conclusion" : "<string>", // Clinical Interpretation of test results
  "codedDiagnosis" : [{ CodeableConcept }], // Codes for the conclusion
  "presentedForm" : [{ Attachment }] // Entire report as issued
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DiagnosticReport;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DiagnosticReport.identifier [ Identifier ], ... ; # 0..* Business identifier for report
  fhir:DiagnosticReport.basedOn [ Reference(CarePlan|ImmunizationRecommendation|MedicationRequest|NutritionOrder|
  ProcedureRequest|ReferralRequest) ], ... ; # 0..* What was requested
  fhir:DiagnosticReport.status [ code ]; # 1..1 registered | partial | preliminary | final +
  fhir:DiagnosticReport.category [ CodeableConcept ]; # 0..1 Service category
  fhir:DiagnosticReport.code [ CodeableConcept ]; # 1..1 Name/Code for this diagnostic report
  fhir:DiagnosticReport.subject [ Reference(Patient|Group|Device|Location) ]; # 0..1 The subject of the report -
    usually, but not always, the patient
  fhir:DiagnosticReport.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Health care event when test ordered
  # DiagnosticReport.effective[x] : 0..1 Clinically relevant time/time-period for report. One of these 2
    fhir:DiagnosticReport.effectiveDateTime [ dateTime ]
    fhir:DiagnosticReport.effectivePeriod [ Period ]
  fhir:DiagnosticReport.issued [ instant ]; # 0..1 DateTime this version was released
  fhir:DiagnosticReport.performer [ # 0..* Participants in producing the report
    fhir:DiagnosticReport.performer.role [ CodeableConcept ]; # 0..1 Type of performer
    fhir:DiagnosticReport.performer.actor [ Reference(Practitioner|Organization) ]; # 1..1 Practitioner or Organization
      participant
  ], ...;
  fhir:DiagnosticReport.specimen [ Reference(Specimen) ], ... ; # 0..* Specimens this report is based on
  fhir:DiagnosticReport.result [ Reference(Observation) ], ... ; # 0..* Observations - simple, or complex nested groups
  fhir:DiagnosticReport.imagingStudy [ Reference(ImagingStudy|ImagingManifest) ], ... ; # 0..* Reference to full details
     of imaging associated with the diagnostic report
  fhir:DiagnosticReport.image [ # 0..* Key images associated with this report
    fhir:DiagnosticReport.image.comment [ string ]; # 0..1 Comment about the image (e.g. explanation)
    fhir:DiagnosticReport.image.link [ Reference(Media) ]; # 1..1 Reference to the image source
  ], ...;
  fhir:DiagnosticReport.conclusion [ string ]; # 0..1 Clinical Interpretation of test results
  fhir:DiagnosticReport.codedDiagnosis [ CodeableConcept ], ... ; # 0..* Codes for the conclusion
  fhir:DiagnosticReport.presentedForm [ Attachment ], ... ; # 0..* Entire report as issued
]
```

Changes since DSTU2

DiagnosticReport

DiagnosticReport.basedOn | Renamed from request to basedOn Remove Reference(DiagnosticOrder), Add Reference(CarePlan), Add Reference(ImmunizationRecommendation), Add Reference(MedicationRequest), Add Reference(NutritionOrder)

DiagnosticReport.subject | Min Cardinality changed from 1 to 0

DiagnosticReport.context | Renamed from encounter to context Add Reference(EpisodeOfCare)

DiagnosticReport.effective[x] | Min Cardinality changed from 1 to 0

DiagnosticReport.issued | Min Cardinality changed from 1 to 0

DiagnosticReport.performer | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to * Remove Reference(Practitioner), Remove Reference(Organization)

DiagnosticReport.performer.role | Added Element
DiagnosticReport.performer.actor | Added Element
DiagnosticReport.imagingStudy | Remove Reference(ImagingObjectSelection), Add Reference(ImagingManifest)

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 11 tests of which 3 fail to execute. 8 fail round-trip testing and 3 r3 resources are invalid (22 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

10.2.4.1 Terminology Bindings

Path | Definition | Type | Reference
DiagnosticReport.status | The status of the diagnostic report as a whole. | Required | DiagnosticReportStatus
DiagnosticReport.category | Codes for diagnostic service sections. | Example | Diagnostic Service Section Codes
DiagnosticReport.code | Codes that describe Diagnostic Reports. | Preferred | LOINC Diagnostic Report Codes
DiagnosticReport.performer.role | Indicate a role of diagnostic report performer | Example | Procedure Performer Role Codes
DiagnosticReport.codedDiagnosis | Diagnosis codes provided as adjuncts to the report. | Example | SNOMED CT Clinical Findings

10.2.5 Notes:

10.2.5.1 Identifiers

identifier

type

```
<!-- Placer identifier-->
<identifier>
<type>
<coding>
  <system value="http://hl7.org/fhir/identifier-type"/>
  <code value="PLAC"/>
</coding>
<text value="Placer"/>
</type>
<system value="urn:oid:1.3.4.5.6.7"/>
<value value="2345234234234"/>
</identifier>
<!-- Filler identifier-->
<identifier>
<type>
<coding>
  <system value="http://hl7.org/fhir/identifier-type"/>
  <code value="PLAC"/>
</coding>
<text value="Placer"/>
</type>
<system value=" http://hl7.org/fhir/identifier-type"/>
<value value="567890"/>
</identifier>
```

10.2.5.2 Clinically Relevant Time

If the diagnostic procedure was performed on the patient directly, the effective[x] element is a dateTime, the time it was performed. If specimens were taken, the clinically relevant time of the report can be derived from the specimen collection times, but since detailed specimen information is not always available, and nor is the clinically relevant time always exactly the specimen collection time (e.g. complex timed tests), the reports SHALL always include a effective[x] element. Note that HL7 v2 messages often carry a diagnostically relevant time without carrying any specimen information.

10.2.5.3 Associated Observations

* The DiagnosticReport.code always contains the name of the report itself. The report can also contain a set of Observations in the DiagnosticReport.result element. These Observations can be simple observations (e.g. atomic results) or groups/panels of other observations. The Observation.code indicates the nature of the observation or panel (e.g. individual measure, organism isolate/sensitivity or antibody functional testing). When relevant, the observation can specify a particular specimen from which the result comes.

* Examples of nesting groups/panels within an observation include reporting a "profile" consisting of several panels as is shown in this example, a group of antibiotic isolate/sensitivities for a bacterial culture, or a set of perinatal measurements on a single fetus.

* There is rarely a need for more than two levels of nesting in the Observation tree. One known use is for organism sensitivities - an example of a complex Micro Isolate and Sensitivities is provided.

10.2.5.4 Associated Images

ImagingStudy and ImageObjectStudy and the DiagnosticReport.image element are somewhat overlapping - typically, the list of image references in the image element will also be found in one of the imaging study resources. However each caters to different types of displays for different types of purposes. Neither, either, or both may be provided.

10.2.5.5 Diagnostic Report Status

* Applications consuming diagnostic reports must take careful note of updated (revised) reports, and ensure that retracted reports are appropriately handled.

* For applications providing diagnostic reports, a report should not be final until all the individual data items reported with it are final or appended.

* If the report has been withdrawn following a previous final release, the DiagnosticReport and associated Observations should be retracted by replacing the status codes with the concept "entered-in-error", and setting the conclusion/comment (if provided) and the text narrative to some text like "This report has been withdrawn" in the appropriate language. A reason for retraction may be provided in the narrative.

10.2.5.6 Report Content

Typically, a report is either:all data, no narrative (e.g. Core lab) or a mix of data with some concluding narrative (e.g. Structured Pathology Report, Bone Density), or all narrative (for example a typical imaging report, histopathology). This resource provides for these 3 different presentations:

* As atomic data items: a hierarchical set of nested references to Observation resources often including pathologist/radiologist interpretation(s), one or more images, and possibly with a conclusion and/or one or more coded diagnoses

* As narrative: an XHTML presentation in the standard resource narrative

* As a "presented form": A rich text representation of the report - typically a PDF

Note that the conclusion and the coded diagnoses are part of the atomic data, and SHOULD be duplicated in the narrative and in the presented form if the latter is present. The narrative and the presented form serve the same function: a representation of the report for a human. The presented form is included since diagnostic service reports often contain presentation features that are not easy to reproduce in the HTML narrative. Whether or not the presented form is included, the narrative must be a clinically safe view of the diagnostic report; at a minimum, this could be fulfilled by a note indicating that the narrative is not proper representation of the report, and that the presented form must be used, or a generated view from the atomic data. However consumers of the report will best be served if the narrative contains clinically relevant data from the form. Commonly, the following patterns are used:

* Simple Laboratory Reports: A single set of atomic observations, and a tabular presentation in narrative. Typically this is encountered in high volume areas such as Biochemistry and Hematology

* Histopathology Report: A document report in a presented form and the narrative. Possibly a few key images, and some coded diagnoses for registries. If the service is creating a structured report, some atomic data may be included

* imaging Report; A document report in a presented form and the narrative, with an imaging study reference and possibly some key images. Some imaging reports such as a Bone Density Scan may include some atomic data

Note that the nature of reports from the various disciplines that provide diagnostic reports are changing quickly, as expert systems provide improved narrative reporting in high volume reports, structured reporting brings additional data to areas that have classically been narrative based, and the nature of the imaging and laboratory procedures are merging. As a consequence the patterns described above are only examples of how a diagnostic report can be used.

10.2.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
based-on | reference | Reference to the procedure request. | DiagnosticReport.basedOn (ReferralRequest, CarePlan, MedicationRequest, NutritionOrder, ProcedureRequest, ImmunizationRecommendation)
category | token | Which diagnostic discipline/department created the report | DiagnosticReport.category
code | token | The code for the report as a whole, as opposed to codes for the atomic results, which are the names on the observation resource referred to from the result | DiagnosticReport.code | 8 Resources
context | reference | Healthcare event (Episode of Care or Encounter) related to the report | DiagnosticReport.context (EpisodeOfCare, Encounter)
date | date | The clinically relevant time of the report | DiagnosticReport.effective | 18 Resources
diagnosis | token | A coded diagnosis on the report | DiagnosticReport.codedDiagnosis
encounter | reference | The Encounter when the order was made | DiagnosticReport.context (Encounter) | 12 Resources
identifier | token | An identifier for the report | DiagnosticReport.identifier | 26 Resources
image | reference | A reference to the image source. | DiagnosticReport.image.link (Media)
issued | date | When the report was issued | DiagnosticReport.issued
patient | reference | The subject of the report if a patient | DiagnosticReport.subject (Patient) | 31 Resources
performer | reference | Who was the source of the report (organization) | DiagnosticReport.performer.actor (Practitioner, Organization)
result | reference | Link to an atomic result (observation resource) | DiagnosticReport.result (Observation)
specimen | reference | The specimen details | DiagnosticReport.specimen (Specimen)
status | token | The status of the report | DiagnosticReport.status
subject | reference | The subject of the report | DiagnosticReport.subject (Group, Device, Patient, Location)

10.2.7 Resource DiagnosticReport - Examples

URL: https://hl7.org/fhir/STU3/diagnosticreport-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.7 Resource DiagnosticReport - Examples

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner

Example List:

General Lab Report Example | 101 | XML | JSON | Turtle
Various reports gathered from v2 messages | 72ac8493-52ac-41bd-8d5d-7258c289b5ea | XML | JSON | Turtle
Real-world patient example | f001 | XML | JSON | Turtle
Real-world patient example | f201 | XML | JSON | Turtle
Real-world patient example | f202 | XML | JSON | Turtle
Imaging Example (Bone Density DXA) | 102 | XML | JSON | Turtle
An example of a complex Micro Isolate and Sensitivities | micro | XML | JSON | Turtle
Abdominal Ultrasound example | ultrasound | XML | JSON | Turtle
An Example of a report with multiple specimen and panels | ghp | XML | JSON | Turtle
An example of a principally narrative anatomic pathology report | pap | XML | JSON | Turtle
An example of a presented form anatomic pathology report | gingival-mass | XML | JSON | Turtle
Lipid profile | lipids | XML | JSON | Turtle | for Profile Example Lipid Profile
An example of a genetics test report with Family member history | dg2 | XML | JSON | Turtle | for Profile DiagnosticReport-Genetics
An example of a comprehensive bone marrow report | report | XML | JSON | Turtle | for Profile DiagnosticReport-Genetics
An example of a PGx haplotype report | example-pgx | XML | JSON | Turtle | for Profile DiagnosticReport-Genetics
An example of a HLA genotyping results report | hla-1 | XML | JSON | Turtle | for Profile Profile for HLA Genotyping Results

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.2.8 Resource DiagnosticReport - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/diagnosticreport-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.8 Resource DiagnosticReport - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner

Detailed Descriptions for the elements in the DiagnosticReport resource.

DiagnosticReport
Definition | The findings and interpretation of diagnostic tests performed on patients, groups of patients, devices, and locations, and/or specimens derived from these. The report includes clinical context such as requesting and provider information, and some mix of atomic results, images, textual and coded interpretations, and formatted representation of diagnostic reports.
Control | 1..1
Requirements | To support reporting for any diagnostic report into a clinical data repository.
Alternate Names | Report; Test; Result; Results; Labs; Laboratory
Comments | This is intended to capture a single report, and is not suitable for use in displaying summary information that covers multiple reports. For example, this resource has not been designed for laboratory cumulative reporting formats nor detailed structured reports for sequencing.
DiagnosticReport.identifier
Definition | Identifiers assigned to this report by the performer or other systems.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Need to know what identifier to use when making queries about this report from the source laboratory, and for linking to the report outside FHIR context.
Alternate Names | ReportID; Filler ID; Placer ID
Summary | true
Comments | Usually assigned by the Information System of the diagnostic service provider (filler id).
DiagnosticReport.basedOn
Definition | Details concerning a test or procedure requested.
Control | 0..*
Type | Reference(CarePlan | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ProcedureRequest | ReferralRequest)
Requirements | This allows tracing of authorization for the report and tracking whether proposals/recommendations were acted upon.
Alternate Names | Request
Comments | Note: Usually there is one test request for each result, however in some circumstances multiple test requests may be represented using a single test result resource. Note that there are also cases where one request leads to multiple reports.
DiagnosticReport.status
Definition | The status of the diagnostic report as a whole.
Control | 1..1
Terminology Binding | DiagnosticReportStatus (Required)
Type | code
Is Modifier | true
Requirements | Diagnostic services routinely issue provisional/incomplete reports, and sometimes withdraw previously released reports.
Summary | true
Comments | This is labeled as "Is Modifier" because applications need to take appropriate action if a report is withdrawn.
DiagnosticReport.category

Definition | A code that classifies the clinical discipline, department or diagnostic service that created the report (e.g. cardiology, biochemistry, hematology, MRI). This is used for searching, sorting and display purposes.
Control | 0..1
Terminology Binding | Diagnostic Service Section Codes (Example)
Type | CodeableConcept
Alternate Names | Department; Sub-department; Service; Discipline
Summary | true
Comments | The level of granularity is defined by the category concepts in the value set. More fine-grained filtering can be performed using the metadata and/or terminology hierarchy in DiagnosticReport.code.
DiagnosticReport.code
Definition | A code or name that describes this diagnostic report.
Control | 1..1
Terminology Binding | LOINC Diagnostic Report Codes (Preferred)
Type | CodeableConcept
Alternate Names | Type
Summary | true
DiagnosticReport.subject
Definition | The subject of the report. Usually, but not always, this is a patient. However diagnostic services also perform analyses on specimens collected from a variety of other sources.
Control | 0..1
Type | Reference(Patient | Group | Device | Location)
Requirements | SHALL know the subject context.
Alternate Names | Patient
Summary | true
DiagnosticReport.context
Definition | The healthcare event (e.g. a patient and healthcare provider interaction) which this DiagnosticReport per is about.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Requirements | Links the request to the Encounter context.
Alternate Names | Encounter
Summary | true
Comments | This will typically be the encounter the event occurred within, but some events may be initiated prior to or after the official completion of an encounter or episode but still be tied to the context of the encounter or episode (e.g. pre-admission lab tests).
DiagnosticReport.effective[x]
Definition | The time or time-period the observed values are related to. When the subject of the report is a patient, this is usually either the time of the procedure or of specimen collection(s), but very often the source of the date/time is not known, only the date/time itself.
Control | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Need to know where in the patient history to file/present this report.
Alternate Names | Observation time; Effective Time; Occurrence
Summary | true
Comments | If the diagnostic procedure was performed on the patient, this is the time it was performed. If there are specimens, the diagnostically relevant time can be derived from the specimen collection times, but the specimen information is not always available, and the exact relationship between the specimens and the diagnostically relevant time is not always automatic.
DiagnosticReport.issued
Definition | The date and time that this version of the report was released from the source diagnostic service.
Control | 0..1
Type | instant
Requirements | Clinicians need to be able to check the date that the report was released.
Alternate Names | Date Created; Date published; Date Issued
Summary | true
Comments | May be different from the update time of the resource itself, because that is the status of the record (potentially a secondary copy), not the actual release time of the report.
DiagnosticReport.performer
Definition | Indicates who or what participated in producing the report.
Control | 0..*
Alternate Names | Laboratory; Service; Practitioner; Department; Company
Summary | true
DiagnosticReport.performer.role
Definition | Describes the type of participation (e.g. a responsible party, author, or verifier).
Control | 0..1
Terminology Binding | Procedure Performer Role Codes (Example)
Type | CodeableConcept
Summary | true
DiagnosticReport.performer.actor
Definition | The reference to the practitioner or organization involved in producing the report. For example, the diagnostic service that is responsible for issuing the report.
Control | 1..1
Type | Reference(Practitioner | Organization)
Requirements | Need to know whom to contact if there are queries about the results. Also may need to track the source of reports for secondary data analysis.
Summary | true

Comments | This is not necessarily the source of the atomic data items. It is the entity that takes responsibility for the clinical report.
DiagnosticReport.specimen
Definition | Details about the specimens on which this diagnostic report is based.
Control | 0..*
Type | Reference(Specimen)
Requirements | Need to be able to report information about the collected specimens on which the report is based.
Comments | If the specimen is sufficiently specified with a code in the test result name, then this additional data may be redundant. If there are multiple specimens, these may be represented per observation or group.
DiagnosticReport.result
Definition | Observations that are part of this diagnostic report. Observations can be simple name/value pairs (e.g. "atomic" results), or they can be grouping observations that include references to other members of the group (e.g. "panels").
Control | 0..*
Type | Reference(Observation)
Requirements | Need to support individual results, or report groups of results, where the result grouping is arbitrary, but meaningful. This structure is recursive - observations can contain observations.
Alternate Names | Data; Atomic Value; Result; Atomic result; Data; Test; Analyte; Battery; Organizer
DiagnosticReport.imagingStudy
Definition | One or more links to full details of any imaging performed during the diagnostic investigation. Typically, this is imaging performed by DICOM enabled modalities, but this is not required. A fully enabled PACS viewer can use this information to provide views of the source images.
Control | 0..*
Type | Reference(ImagingStudy | ImagingManifest)
Comments | ImagingStudy and ImageManifest and the image element are somewhat overlapping - typically, the list of image references in the image element will also be found in one of the imaging study resources. However each caters to different types of displays for different types of purposes. Neither, either, or both may be provided.
DiagnosticReport.image
Definition | A list of key images associated with this report. The images are generally created during the diagnostic process, and may be directly of the patient, or of treated specimens (i.e. slides of interest).
Control | 0..*
Requirements | Many diagnostic services include images in the report as part of their service.
Alternate Names | DICOM; Slides; Scans
Summary | true
DiagnosticReport.image.comment
Definition | A comment about the image. Typically, this is used to provide an explanation for why the image is included, or to draw the viewer's attention to important features.
Control | 0..1
Type | string
Requirements | The provider of the report should make a comment about each image included in the report.
Comments | The comment should be displayed with the image. It would be common for the report to include additional discussion of the image contents in other sections such as the conclusion.
DiagnosticReport.image.link
Definition | Reference to the image source.
Control | 1..1
Type | Reference(Media)
Summary | true
DiagnosticReport.conclusion
Definition | Concise and clinically contextualized impression / summary of the diagnostic report.
Control | 0..1
Type | string
Requirements | Need to be able to provide a conclusion that is not lost among the basic result data.
Alternate Names | Report
DiagnosticReport.codedDiagnosis
Definition | Codes for the conclusion.
Control | 0..*
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
DiagnosticReport.presentedForm
Definition | Rich text representation of the entire result as issued by the diagnostic service. Multiple formats are allowed but they SHALL be semantically equivalent.
Control | 0..*
Type | Attachment
Requirements | Gives laboratory the ability to provide its own fully formatted report for clinical fidelity.
Comments | "application/pdf" is recommended as the most reliable and interoperable in this context.

10.2.9 Resource DiagnosticReport - Mappings

URL: https://hl7.org/fhir/STU3/diagnosticreport-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.9 Resource DiagnosticReport - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner

Mappings:

RIM Mapping

HL7 v2 Mapping

Workflow Mapping

Mappings for the diagnosticreport resource.

10.2.9.1 Mappings for RIM Mapping (http://hl7.org/v3)

DiagnosticReport | Observation[classCode=OBS, moodCode=EVN]
identifier | id
basedOn | outboundRelationship[typeCode=FLFS].target
status | statusCode Note: final and amended are distinguished by whether observation is the subject of a ControlAct event of type "revise"
category | inboundRelationship[typeCode=COMP].source[classCode=LIST, moodCode=EVN, code < LabService].code
code | code
subject | participation[typeCode=SBJ]
context | inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
effective[x] | effectiveTime
issued | participation[typeCode=VRF or AUT].time
performer | .participation[typeCode=PRF]
role | .functionCode
actor | .role
specimen | participation[typeCode=SBJ]
result | outboundRelationship[typeCode=COMP].target
imagingStudy | outboundRelationship[typeCode=COMP].target[classsCode=DGIMG, moodCode=EVN]
image | outboundRelationship[typeCode=COMP].target
comment | .inboundRelationship[typeCode=COMP].source[classCode=OBS, moodCode=EVN, code="annotation"].value
link | .value.reference
conclusion | inboundRelationship[typeCode="SPRT"].source[classCode=OBS, moodCode=EVN, code=LOINC:48767-8].value (type=ST)
codedDiagnosis | inboundRelationship[typeCode=SPRT].source[classCode=OBS, moodCode=EVN, code=LOINC:54531-9].value (type=CD)
presentedForm | text (type=ED)

10.2.9.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

DiagnosticReport | ORU -> OBR
identifier | OBR-51
basedOn | ORC? OBR-2/3?
status | OBR-25 (not 1:1 mapping)
category | OBR-24
code | OBR-4 (HL7 v2 doesn't provide an easy way to indicate both the ordered test and the performed panel)
subject | PID-3 (no HL7 v2 mapping for Group or Device)
context | PV1-19
effective[x] | OBR-7
issued | OBR-22
performer | PRT-8 (where this PRT-4-Participation = "PO")
role | PRT-8 (where this PRT-4-Participation = "PO")
actor | PRT-8 (where this PRT-4-Participation = "PO")
specimen | SPM
result | OBXs
imagingStudy
image | OBX?
comment
link
conclusion | OBX
codedDiagnosis | OBX
presentedForm | OBX

10.2.9.3 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

DiagnosticReport | Event
identifier | Event.identifier
basedOn | Event.basedOn
status | Event.status
category
code | Event.code
subject | Event.subject
context | Event.context
effective[x] | Event.occurrence[x]
issued
performer | Event.performer
role | Event.performer.role
actor | Event.performer.actor
specimen
result
imagingStudy

image
comment
link
conclusion
codedDiagnosis
presentedForm

10.2.10 Resource DiagnosticReport - Design Notes

URL: https://hl7.org/fhir/STU3/diagnosticreport-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.10 Resource DiagnosticReport - Design Notes

to do

10.2.11 Resource DiagnosticReport - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/diagnosticreport-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.11 Resource DiagnosticReport - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the DiagnosticReport resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

10.2.11.1 Profiles

DiagnosticReport-Genetics | Describes how the DiagnosticReport resource is used to report structured genetic test results | for DiagnosticReport-Genetics
HLAResult | Describes how the HLA genotyping results | for Profile for HLA Genotyping Results
LipidProfile | Describes how the lab report is used for a standard Lipid Profile - Cholesterol, Triglyceride and Cholesterol fractions. Uses LOINC codes | for Example Lipid Profile

10.2.11.2 Extensions

DiagnosticReport-geneticsAnalysis | Analysis | for DiagnosticReport-Genetics
DiagnosticReport-geneticsAssessedCondition | AssessedCondition | for DiagnosticReport-Genetics
DiagnosticReport-geneticsFamilyMemberHistory | FamilyMemberHistory | for DiagnosticReport-Genetics
diagnosticReport-addendumOf | addendumOf | for DiagnosticReport HL7 Extensions
diagnosticReport-extends | extends | for DiagnosticReport HL7 Extensions
diagnosticReport-locationPerformed | locationPerformed | for DiagnosticReport HL7 Extensions
diagnosticReport-replaces | replaces | for DiagnosticReport HL7 Extensions
diagnosticReport-summaryOf | summaryOf | for DiagnosticReport HL7 Extensions
event-OnBehalfOf | OnBehalfOf | for Event Pattern HL7 Extensions
event-definition | definition | for Event Pattern HL7 Extensions
event-notDone | notDone | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
event-reasonCode | reasonCode | for Event Pattern HL7 Extensions
event-reasonReference | reasonReference | for Event Pattern HL7 Extensions
hla-genotyping-results-allele-database | allele-database | for Profile for HLA Genotyping Results
hla-genotyping-results-glstring | glstring | for Profile for HLA Genotyping Results
hla-genotyping-results-haploid | haploid | for Profile for HLA Genotyping Results
hla-genotyping-results-method | method | for Profile for HLA Genotyping Results
procedurerequest-geneticsItem | Item | for ProcedureRequest-Genetics

Extensions that reference this resource:

diagnosticReport-addendumOf | addendumOf | for DiagnosticReport HL7 Extensions
diagnosticReport-extends | extends | for DiagnosticReport HL7 Extensions
diagnosticReport-replaces | replaces | for DiagnosticReport HL7 Extensions
diagnosticReport-summaryOf | summaryOf | for DiagnosticReport HL7 Extensions
openEHR-test | test | for openEHR Archetype Profile
procedure-causedBy | causedBy | for Procedure HL7 Extensions

10.2.11.3 Search Extensions

No Search Extensions defined for this resource

10.2.12 Example Lipid Profile

URL: https://hl7.org/fhir/STU3/lipid-report.html

Describes how the lab report is used for a standard Lipid Profile - Cholesterol, Triglyceride and Cholesterol fractions. Uses LOINC codes

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.12 Example Lipid Profile

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative

10.2.12.1 Content

Profiles:
LipidProfile | Lipid Lab Report
Cholesterol | Describes how the lab report is used for a standard Lipid Profile - Cholesterol, Triglyceride and Cholesterol fractions. Uses LOINC codes
Triglyceride | Triglyceride Result
HDLCholesterol | HDL Cholesterol Result
LDLCholesterol | LDL Cholesterol Result
Examples:
Lipid profile | Lipid profile

10.2.12.2 StructureDefinition: LipidProfile

URL: https://hl7.org/fhir/STU3/lipidprofile.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.12.2 StructureDefinition: LipidProfile

Orders and Observations Work Group | Maturity Level: 1 | Informative | Use Context: Not Intended for Production use

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/lipidprofile`

Lipid Lab Report

This profile was published on Sat, May 12, 2012 00:00+1000 as a draft by Grahame Grieve.

10.2.12.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary
* Differential Table
* Snapshot Table
* XML Template
* JSON Template
* All

This structure is derived from DiagnosticReport.

Summary

This is a test summary lipids

Mandatory: 4 elements Must-Support: 5 elements Fixed Value: 1 element Prohibited: 1 element

Structures

This structure refers to these other structures:

* http://hl7.org/fhir/StructureDefinition/cholesterol
* http://hl7.org/fhir/StructureDefinition/triglyceride
* http://hl7.org/fhir/StructureDefinition/hdlcholesterol
* http://hl7.org/fhir/StructureDefinition/ldlcholesterol

Slices

This structure defines the following Slices:

* The element DiagnosticReport.result is sliced based on the value of value:reference.code (ordered, Closed)

This structure is derived from DiagnosticReport.

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | | 1..1 | | Lipid Lab Report
code | | 1..1 | CodeableConcept | LOINC Code for Lipid Panel with LDL Fixed Value:
{"coding":[{"system":"http://loinc.org","code":"57698-3","display":"Lipid panel with direct LDL - Serum or Plasma"}]}
result | | | Reference(Observation) | Slice: Ordered, Closed by value:reference.code
result | S | 1..1 | Reference(Example Lipid Profile) | Cholesterol Result

result | S | 1..1 | Reference(Example Lipid Profile) | Triglyceride Result
result | S | 1..1 | Reference(Example Lipid Profile) | HDL Cholesterol Result
result | S | 0..1 | Reference(Example Lipid Profile) | LDL Cholesterol result, if reported
conclusion | S | 0..1 | string | Clinical Interpretation of Lipid Panel
codedDiagnosis | | 0..0
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | I | 1..1 | | Lipid Lab Report
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business identifier for report
basedOn | | 0..* | Reference(CarePlan), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | What was requested
status | ?!? | 1..1 | code | registered | partial | preliminary | final + Binding: DiagnosticReportStatus (required)
category | ? | 0..1 | CodeableConcept | Service category Binding: Diagnostic Service Section Codes (example)
code | ? | 1..1 | CodeableConcept | LOINC Code for Lipid Panel with LDL Binding: LOINC Diagnostic Report Codes (preferred) Fixed Value: {"coding":[{"system":"http://loinc.org","code":"57698-3","display":"Lipid panel with direct LDL - Serum or Plasma"}]}
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | The subject of the report - usually, but not always, the patient
context | ? | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Health care event when test ordered
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for report
issued | ? | 0..1 | instant | DateTime this version was released
performer | ?I | 0..* | BackboneElement | Participants in producing the report
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
role | ? | 0..1 | CodeableConcept | Type of performer Binding: Procedure Performer Role Codes (example)
actor | ? | 1..1 | Reference(Practitioner), Reference(Organization) | Practitioner or Organization participant
specimen | | 0..* | Reference(Specimen) | Specimens this report is based on
result | | | Reference(Observation) | Observations - simple, or complex nested groups Slice: Ordered, Closed by value:reference.code
result | S | 1..1 | Reference(Example Lipid Profile) | Cholesterol Result
result | S | 1..1 | Reference(Example Lipid Profile) | Triglyceride Result
result | S | 1..1 | Reference(Example Lipid Profile) | HDL Cholesterol Result
result | S | 0..1 | Reference(Example Lipid Profile) | LDL Cholesterol result, if reported
imagingStudy | | 0..* | Reference(ImagingStudy), Reference(ImagingManifest) | Reference to full details of imaging associated with the diagnostic report
image | ?I | 0..* | BackboneElement | Key images associated with this report
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
comment | | 0..1 | string | Comment about the image (e.g. explanation)
link | ? | 1..1 | Reference(Media) | Reference to the image source
conclusion | S | 0..1 | string | Clinical Interpretation of Lipid Panel
presentedForm | | 0..* | Attachment | Entire report as issued
Documentation for this format

```
<!-- Example Lipid Profile -->

<DiagnosticReport xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business identifier for report --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|Reference(NutritionOrder)|
   Reference(ProcedureRequest)|Reference(ReferralRequest) What was requested --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | partial | preliminary | final +  -->
 <category><!-- 0..1 CodeableConcept Service category  --></category>
 <code><!-- 1..1 CodeableConcept LOINC Code for Lipid Panel with LDL  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     The subject of the report - usually, but not always, the patient --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Health care event when test ordered --></context>
```

```
 <effective[x]><!-- 0..1 dateTime|Period
     Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 DateTime this version was released -->
 <performer>  0..* BackboneElement  <!--  0..* Participants in producing the report -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <role><!-- 0..1 CodeableConcept Type of performer  --></role>
  <actor><!-- 1..1 Reference(Practitioner)|Reference(Organization)
      Practitioner or Organization  participant --></actor>
 </performer>
 <specimen><!-- 0..* Reference(Specimen)
     Specimens this report is based on --></specimen>
 <-- result sliced by value:reference.code  in any order-->
 <result><!-- 1..1 Reference(Example Lipid Profile) Cholesterol Result --></result>
 <result><!-- 1..1 Reference(Example Lipid Profile) Triglyceride Result --></result>
 <result><!-- 1..1 Reference(Example Lipid Profile) HDL Cholesterol Result --></result>
 <result><!-- 0..1 Reference(Example Lipid Profile)
     LDL Cholesterol result, if reported --></result>
 <imagingStudy><!-- 0..* Reference(ImagingStudy)|Reference(ImagingManifest)
     Reference to full details of imaging associated with the diagnostic report --></imagingStudy>
 <image>  0..* BackboneElement  <!--  0..* Key images associated with this report -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <comment value="[string]"/><!-- 0..1 Comment about the image (e.g. explanation) -->
  <link><!-- 1..1 Reference(Media)
      Reference to the image source --></link>
 </image>
 <conclusion value="[string]"/><!-- 0..1 Clinical Interpretation of Lipid Panel -->
 <codedDiagnosis><!-- 0..0 CodeableConcept No codes for a lipid panel  --></codedDiagnosis>
 <presentedForm><!-- 0..* Attachment
     Entire report as issued --></presentedForm>
</DiagnosticReport>
```

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business identifier for report
// value[x]: What was requested. One of these 6:
    "basedOn" : [{ Reference(CarePlan) }], //What was requested
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //What was requested
    "basedOn" : [{ Reference(MedicationRequest) }], //What was requested
    "basedOn" : [{ Reference(NutritionOrder) }], //What was requested
    "basedOn" : [{ Reference(ProcedureRequest) }], //What was requested
    "basedOn" : [{ Reference(ReferralRequest) }], //What was requested
    "status" : "<code>", // R! registered | partial | preliminary | final +
    "category" : { CodeableConcept }, //Service category
    "code" : { CodeableConcept }, // R!
// value[x]: The subject of the report - usually, but not always, the patient. One of these 4:
    "subject" : { Reference(Patient) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Group) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Device) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Location) }, //The subject of the report - usually, but not always, the patient
// value[x]: Health care event when test ordered. One of these 2:
    "context" : { Reference(Encounter) }, //Health care event when test ordered
    "context" : { Reference(EpisodeOfCare) }, //Health care event when test ordered
// value[x]: Clinically relevant time/time-period for report. One of these 2:
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for report
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for report
    "issued" : "<instant>", //DateTime this version was released
    "performer" : [{ BackboneElement }], // C?Participants in producing the report
    "specimen" : [{ Reference(Specimen) }] //Specimens this report is based on
    "result" : [ //  sliced by value:reference.code  in any order
      { // Cholesterol Result // R!
        // from Element: extension
      },
      { // Triglyceride Result // R!
        // from Element: extension
      },
      { // HDL Cholesterol Result // R!
        // from Element: extension
      },
      { // LDL Cholesterol result, if reported //
        // from Element: extension
      }
    ],
// value[x]: Reference to full details of imaging associated with the diagnostic report. One of these 2:
    "imagingStudy" : [{ Reference(ImagingStudy) }], //Reference to full details of imaging associated with the
    diagnostic report
    "imagingStudy" : [{ Reference(ImagingManifest) }], //Reference to full details of imaging associated with the
    diagnostic report
    "image" : [{ BackboneElement }], // C?Key images associated with this report
    "conclusion" : "<string>", //Clinical Interpretation of Lipid Panel
    "presentedForm" : [{ Attachment }], //Entire report as issued
  }
```

This structure is derived from DiagnosticReport.

Summary

This is a test summary lipids

Mandatory: 4 elements Must-Support: 5 elements Fixed Value: 1 element Prohibited: 1 element

Structures

This structure refers to these other structures:

* http://hl7.org/fhir/StructureDefinition/cholesterol

* http://hl7.org/fhir/StructureDefinition/triglyceride

* http://hl7.org/fhir/StructureDefinition/hdlcholesterol

* http://hl7.org/fhir/StructureDefinition/ldlcholesterol

Slices

This structure defines the following Slices:

* The element DiagnosticReport.result is sliced based on the value of value:reference.code (ordered, Closed)

Differential View

This structure is derived from DiagnosticReport.

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | | 1..1 | | Lipid Lab Report
code | | 1..1 | CodeableConcept | LOINC Code for Lipid Panel with LDL Fixed Value: {"coding":[{"system":"http://loinc.org","code":"57698-3","display":"Lipid panel with direct LDL - Serum or Plasma"}]}
result | | | Reference(Observation) | Slice: Ordered, Closed by value:reference.code
result | S | 1..1 | Reference(Example Lipid Profile) | Cholesterol Result
result | S | 1..1 | Reference(Example Lipid Profile) | Triglyceride Result
result | S | 1..1 | Reference(Example Lipid Profile) | HDL Cholesterol Result
result | S | 0..1 | Reference(Example Lipid Profile) | LDL Cholesterol result, if reported
conclusion | S | 0..1 | string | Clinical Interpretation of Lipid Panel
codedDiagnosis | | 0..0
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | I | 1..1 | | Lipid Lab Report
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business identifier for report
basedOn | | 0..* | Reference(CarePlan), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | What was requested
status | ?!? | 1..1 | code | registered | partial | preliminary | final + Binding: DiagnosticReportStatus (required)
category | ? | 0..1 | CodeableConcept | Service category Binding: Diagnostic Service Section Codes (example)
code | ? | 1..1 | CodeableConcept | LOINC Code for Lipid Panel with LDL Binding: LOINC Diagnostic Report Codes (preferred) Fixed Value: {"coding":[{"system":"http://loinc.org","code":"57698-3","display":"Lipid panel with direct LDL - Serum or Plasma"}]}
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | The subject of the report - usually, but not always, the patient
context | ? | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Health care event when test ordered
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for report
issued | ? | 0..1 | instant | DateTime this version was released
performer | ?I | 0..* | BackboneElement | Participants in producing the report
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
role | ? | 0..1 | CodeableConcept | Type of performer Binding: Procedure Performer Role Codes (example)
actor | ? | 1..1 | Reference(Practitioner), Reference(Organization) | Practitioner or Organization participant
specimen | | 0..* | Reference(Specimen) | Specimens this report is based on
result | | | Reference(Observation) | Observations - simple, or complex nested groups Slice: Ordered, Closed by value:reference.code
result | S | 1..1 | Reference(Example Lipid Profile) | Cholesterol Result
result | S | 1..1 | Reference(Example Lipid Profile) | Triglyceride Result
result | S | 1..1 | Reference(Example Lipid Profile) | HDL Cholesterol Result
result | S | 0..1 | Reference(Example Lipid Profile) | LDL Cholesterol result, if reported
imagingStudy | | 0..* | Reference(ImagingStudy), Reference(ImagingManifest) | Reference to full details of imaging associated with the diagnostic report
image | ?I | 0..* | BackboneElement | Key images associated with this report
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored

comment | | 0..1 | string | Comment about the image (e.g. explanation)
link | ? | 1..1 | Reference(Media) | Reference to the image source
conclusion | S | 0..1 | string | Clinical Interpretation of Lipid Panel
presentedForm | | 0..* | Attachment | Entire report as issued
Documentation for this format

XML Template

```
<!-- Example Lipid Profile -->

<DiagnosticReport xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business identifier for report --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|Reference(NutritionOrder)|
   Reference(ProcedureRequest)|Reference(ReferralRequest) What was requested --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | partial | preliminary | final +  -->
 <category><!-- 0..1 CodeableConcept Service category  --></category>
 <code><!-- 1..1 CodeableConcept LOINC Code for Lipid Panel with LDL  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     The subject of the report - usually, but not always, the patient --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Health care event when test ordered --></context>
 <effective[x]><!-- 0..1 dateTime|Period
     Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 DateTime this version was released -->
 <performer>  0..* BackboneElement  <!--  0..* Participants in producing the report -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <role><!-- 0..1 CodeableConcept Type of performer  --></role>
  <actor><!-- 1..1 Reference(Practitioner)|Reference(Organization)
      Practitioner or Organization  participant --></actor>
 </performer>
 <specimen><!-- 0..* Reference(Specimen)
     Specimens this report is based on --></specimen>
 <-- result sliced by value:reference.code  in any order-->
 <result><!-- 1..1 Reference(Example Lipid Profile) Cholesterol Result --></result>
 <result><!-- 1..1 Reference(Example Lipid Profile) Triglyceride Result --></result>
 <result><!-- 1..1 Reference(Example Lipid Profile) HDL Cholesterol Result --></result>
 <result><!-- 0..1 Reference(Example Lipid Profile)
     LDL Cholesterol result, if reported --></result>
 <imagingStudy><!-- 0..* Reference(ImagingStudy)|Reference(ImagingManifest)
     Reference to full details of imaging associated with the diagnostic report --></imagingStudy>
 <image>  0..* BackboneElement  <!--  0..* Key images associated with this report -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <comment value="[string]"/><!-- 0..1 Comment about the image (e.g. explanation) -->
  <link><!-- 1..1 Reference(Media)
      Reference to the image source --></link>
 </image>
 <conclusion value="[string]"/><!-- 0..1 Clinical Interpretation of Lipid Panel -->
 <codedDiagnosis><!-- 0..0 CodeableConcept No codes for a lipid panel  --></codedDiagnosis>
 <presentedForm><!-- 0..* Attachment
     Entire report as issued --></presentedForm>
</DiagnosticReport>
```

JSON Template

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business identifier for report
// value[x]: What was requested. One of these 6:
    "basedOn" : [{ Reference(CarePlan) }], //What was requested
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //What was requested
    "basedOn" : [{ Reference(MedicationRequest) }], //What was requested
    "basedOn" : [{ Reference(NutritionOrder) }], //What was requested
    "basedOn" : [{ Reference(ProcedureRequest) }], //What was requested
    "basedOn" : [{ Reference(ReferralRequest) }], //What was requested
    "status" : "<code>", // R! registered | partial | preliminary | final +
    "category" : { CodeableConcept }, //Service category
    "code" : { CodeableConcept }, // R!
// value[x]: The subject of the report - usually, but not always, the patient. One of these 4:
    "subject" : { Reference(Patient) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Group) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Device) }, //The subject of the report - usually, but not always, the patient
```

```
    "subject" : { Reference(Location) }, //The subject of the report - usually, but not always, the patient
// value[x]: Health care event when test ordered. One of these 2:
    "context" : { Reference(Encounter) }, //Health care event when test ordered
    "context" : { Reference(EpisodeOfCare) }, //Health care event when test ordered
// value[x]: Clinically relevant time/time-period for report. One of these 2:
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for report
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for report
    "issued" : "<instant>", //DateTime this version was released
    "performer" : [{ BackboneElement }], // C?Participants in producing the report
    "specimen" : [{ Reference(Specimen) }] //Specimens this report is based on
    "result" : [ //  sliced by value:reference.code  in any order
      { // Cholesterol Result // R!
        // from Element: extension
      },
      { // Triglyceride Result // R!
        // from Element: extension
      },
      { // HDL Cholesterol Result // R!
        // from Element: extension
      },
      { // LDL Cholesterol result, if reported //
        // from Element: extension
      }
    ],
// value[x]: Reference to full details of imaging associated with the diagnostic report. One of these 2:
    "imagingStudy" : [{ Reference(ImagingStudy) }], //Reference to full details of imaging associated with the
    diagnostic report
    "imagingStudy" : [{ Reference(ImagingManifest) }], //Reference to full details of imaging associated with the
    diagnostic report
    "image" : [{ BackboneElement }], // C?Key images associated with this report
    "conclusion" : "<string>", //Clinical Interpretation of Lipid Panel
    "presentedForm" : [{ Attachment }], //Entire report as issued
  }
```

Other representations of profile: Schematron

10.2.12.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
DiagnosticReport.language	Common Languages	extensible	Common Languages
DiagnosticReport.status	DiagnosticReportStatus	required	DiagnosticReportStatus
DiagnosticReport.category	Diagnostic Service Section Codes	example	Diagnostic Service Section Codes
DiagnosticReport.code	LOINC Diagnostic Report Codes	preferred	LOINC Diagnostic Report Codes
DiagnosticReport.performer.role	Procedure Performer Role Codes	example	Procedure Performer Role Codes

10.2.12.7 StructureDefinition: Cholesterol

URL: https://hl7.org/fhir/STU3/cholesterol.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.12.7 StructureDefinition: Cholesterol

Orders and Observations Work Group | Maturity Level: 1 | Informative | Use Context: Not Intended for Production use

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/cholesterol`

Describes how the lab report is used for a standard Lipid Profile - Cholesterol, Triglyceride and Cholesterol fractions. Uses LOINC codes

This profile was published on Sat, May 12, 2012 00:00+1000 as a draft by Grahame Grieve.

10.2.12.7.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from Observation.

Summary

Mandatory: 3 elements (+3 nested mandatory elements) Must-Support: 9 elements Fixed Value: 5 elements Prohibited: 6 elements

This structure is derived from Observation.

Name	Flags	Card.	Type	Description & Constraints

Observation | | 1..1
code | S | 1..1 | CodeableConcept | Cholesterol Fixed Value: {"coding":[{"system":"http://loinc.org","code":"35200-5","display":"Cholesterol"}]}
valueQuantity | S | 0..1 | Quantity | Cholesterol value
value | S | 0..1 | decimal | Actual numerical value
comparator | | 0..0
unit | S | 1..1 | string | Fixed Value: mmol/L
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Fixed Value: mmol/L
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | ---
comment | S | 0..1 | string | Comments about result
referenceRange | S | 1..1 | BackboneElement
low | | 0..0
high | | 1..1 | SimpleQuantity | Fixed Value: {"value":4.5}
type | | 0..0
appliesTo | | 0..0
age | | 0..0
related | | 0..0
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 1..1 | | Measurements and simple assertions
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | S? | 1..1 | CodeableConcept | Cholesterol Binding: LOINC Codes (example) Fixed Value: {"coding":[{"system":"http://loinc.org","code":"35200-5","display":"Cholesterol"}]}
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for observation
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Cholesterol value
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 0..1 | decimal | Actual numerical value
unit | S? | 1..1 | string | Unit representation Fixed Value: mmol/L
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded form of the unit Fixed Value: mmol/L
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
comment | S | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
high | I | 1..1 | SimpleQuantity | A fixed quantity (no comparator) Fixed Value: {"value":4.5}
text | | 0..1 | string | Text based reference range in an observation
component | ?I | 0..* | BackboneElement | Component results
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | ? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ? | 0..1 | Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual component result
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Example Lipid Profile -->
<Observation xmlns="http://hl7.org/fhir"
```

```
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
     --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation  --></category>
 <code><!-- 1..1 CodeableConcept Cholesterol  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!-- 0..1 dateTime|Period
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Cholesterol value -->
  <value value="[decimal]"/><!-- 0..1 Actual numerical value -->
  <comparator value="[code]"/><!-- 0..0 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1 Coded form of the unit -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept + | ++ | +++ | - | -- | ---  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..0 Quantity
      Low Range, if relevant --></low>
  <high><!--  1..1 Quantity
      A fixed quantity (no comparator) --></high>
  <type><!-- 0..0 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..0 CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..0 BackboneElement  <!--  0..0 Resource related to this observation -->
 </related>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|
    Range|Ratio|SampledData|Attachment|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], //Classification of  type of observation
    "code" : { CodeableConcept }, // R!
// value[x]: Who and/or what this is about. One of these 4:
```

```
    "subject" : { Reference(Patient) }, //Who and/or what this is about
    "subject" : { Reference(Group) }, //Who and/or what this is about
    "subject" : { Reference(Device) }, //Who and/or what this is about
    "subject" : { Reference(Location) }, //Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 2:
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for observation
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }] //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }], //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Cholesterol value
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //+ | ++ | +++ | - | -- | ---
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

This structure is derived from Observation.

Summary

Mandatory: 3 elements (+3 nested mandatory elements) Must-Support: 9 elements Fixed Value: 5 elements Prohibited: 6 elements

Differential View

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 1..1
code | S | 1..1 | CodeableConcept | Cholesterol Fixed Value: {"coding":[{"system":"http://loinc.org","code":"35200-5","display":"Cholesterol"}]}
valueQuantity | S | 0..1 | Quantity | Cholesterol value
value | S | 0..1 | decimal | Actual numerical value
comparator | | 0..0
unit | S | 1..1 | string | Fixed Value: mmol/L
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Fixed Value: mmol/L
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | ---
comment | S | 0..1 | string | Comments about result
referenceRange | S | 1..1 | BackboneElement
low | | 0..0
high | | 1..1 | SimpleQuantity | Fixed Value: {"value":4.5}
type | | 0..0
appliesTo | | 0..0
age | | 0..0
related | | 0..0
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 1..1 | | Measurements and simple assertions
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | S? | 1..1 | CodeableConcept | Cholesterol Binding: LOINC Codes (example) Fixed Value: {"coding":[{"system":"http://loinc.org","code":"35200-5","display":"Cholesterol"}]}
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for observation

issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Cholesterol value
id | | 0..1 | string | xml:id (or equivalent in JSON)
value | S? | 0..1 | decimal | Actual numerical value
unit | S? | 1..1 | string | Unit representation Fixed Value: mmol/L
system | S?I | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | S? | 1..1 | code | Coded form of the unit Fixed Value: mmol/L
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
comment | S | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
high | I | 1..1 | SimpleQuantity | A fixed quantity (no comparator) Fixed Value: {"value":4.5}
text | | 0..1 | string | Text based reference range in an observation
component | ?I | 0..* | BackboneElement | Component results
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | ? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ? | 0..1 | Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual component result
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Example Lipid Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation  --></category>
 <code><!-- 1..1 CodeableConcept Cholesterol  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!-- 0..1 dateTime|Period
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity> 0..1 Quantity  <!-- 0..1 Cholesterol value -->
  <value value="[decimal]"/><!-- 0..1 Actual numerical value -->
  <comparator value="[code]"/><!-- 0..0 < | <= | >= | > - how to understand the value  -->
  <unit value="[string]"/><!-- 1..1 Unit representation -->
  <system value="[uri]"/><!-- 1..1 System that defines coded unit form -->
  <code value="[code]"/><!-- 1..1 Coded form of the unit -->
 </valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept + | ++ | +++ | - | -- | ---  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..0 Quantity
```

```
      Low Range, if relevant --></low>
  <high><!--  1..1 Quantity
      A fixed quantity (no comparator) --></high>
  <type><!-- 0..0 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..0 CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..0 BackboneElement  <!--  0..0 Resource related to this observation -->
 </related>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|
    Range|Ratio|SampledData|Attachment|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], //Classification of  type of observation
    "code" : { CodeableConcept }, // R!
// value[x]: Who and/or what this is about. One of these 4:
    "subject" : { Reference(Patient) }, //Who and/or what this is about
    "subject" : { Reference(Group) }, //Who and/or what this is about
    "subject" : { Reference(Device) }, //Who and/or what this is about
    "subject" : { Reference(Location) }, //Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 2:
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for observation
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }] //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }], //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Cholesterol value
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //+ | ++ | +++ | - | -- | ---
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

Other representations of profile: Schematron

10.2.12.7.2 Terminology Bindings

Path	Name	Conformance	ValueSet
Observation.language	Common Languages	extensible	Common Languages
Observation.status	ObservationStatus	required	ObservationStatus
Observation.category	Observation Category Codes	preferred	Observation Category Codes
Observation.code	LOINC Codes	example	LOINC Codes
Observation.dataAbsentReason	Observation Value Absent Reason	extensible	Observation Value Absent Reason
Observation.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes
Observation.bodySite	SNOMED CT Body Structures	example	SNOMED CT Body Structures

Observation.method | Observation Methods | example | Observation Methods
Observation.related.type | ObservationRelationshipType | required | ObservationRelationshipType
Observation.component.code | LOINC Codes | example | LOINC Codes
Observation.component.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.2.12.14 StructureDefinition: Triglyceride

URL: https://hl7.org/fhir/STU3/triglyceride.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.12.14 StructureDefinition: Triglyceride

Orders and Observations Work Group | Maturity Level: 1 | Informative | Use Context: Not Intended for Production use

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/triglyceride`

Triglyceride Result

This profile was published on Sat, May 12, 2012 00:00+1000 as a draft by Grahame Grieve.

10.2.12.14.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from Observation.

Summary

Mandatory: 3 elements Must-Support: 5 elements Prohibited: 5 elements

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 1..1 | | Triglyceride Result
code | S | 1..1 | CodeableConcept | Triglyceride Required Pattern: {"coding":[{"system":"http://loinc.org","code":"35217-9"}]}
valueQuantity | S | 0..1 | Quantity | Triglyceride value
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | ---
comment | S | 0..1 | string | Comments about result
referenceRange | S | 1..1 | BackboneElement
low | | 0..0
high | | 1..1 | SimpleQuantity
type | | 0..0
appliesTo | | 0..0
age | | 0..0
related | | 0..0

Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 1..1 | | Triglyceride Result
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | S? | 1..1 | CodeableConcept | Triglyceride Binding: LOINC Codes (example) Required Pattern: {"coding":[{"system":"http://loinc.org","code":"35217-9"}]}
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for observation
issued | ? | 0..1 | instant | Date/Time this was made available

performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Triglyceride value
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
comment | S | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
high | I | 1..1 | SimpleQuantity | High Range, if relevant
text | | 0..1 | string | Text based reference range in an observation
component | ?I | 0..* | BackboneElement | Component results
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | ? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ? | 0..1 | Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual component result
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Example Lipid Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation  --></category>
 <code><!-- 1..1 CodeableConcept Triglyceride  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!-- 0..1 dateTime|Period
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity><!-- 0..1 Quantity Triglyceride value --></valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept + | ++ | +++ | - | -- | ---  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..0 Quantity
      Low Range, if relevant --></low>
  <high><!-- 1..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..0 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..0 CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..0 BackboneElement  <!--  0..0 Resource related to this observation -->
 </related>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|
```

```
    Range|Ratio|SampledData|Attachment|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], //Classification of  type of observation
    "code" : { CodeableConcept }, // R! Triglyceride
// value[x]: Who and/or what this is about. One of these 4:
    "subject" : { Reference(Patient) }, //Who and/or what this is about
    "subject" : { Reference(Group) }, //Who and/or what this is about
    "subject" : { Reference(Device) }, //Who and/or what this is about
    "subject" : { Reference(Location) }, //Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 2:
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for observation
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }] //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }], //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Triglyceride value
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //+ | ++ | +++ | - | -- | ---
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

This structure is derived from Observation.

Summary

Mandatory: 3 elements Must-Support: 5 elements Prohibited: 5 elements

Differential View

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 1..1 | | Triglyceride Result
code | S | 1..1 | CodeableConcept | Triglyceride Required Pattern: {"coding":[{"system":"http://loinc.org","code":"35217-9"}]}
valueQuantity | S | 0..1 | Quantity | Triglyceride value
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | ---
comment | S | 0..1 | string | Comments about result
referenceRange | S | 1..1 | BackboneElement
low | | 0..0
high | | 1..1 | SimpleQuantity
type | | 0..0
appliesTo | | 0..0
age | | 0..0
related | | 0..0
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 1..1 | | Triglyceride Result

id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | S? | 1..1 | CodeableConcept | Triglyceride Binding: LOINC Codes (example) Required Pattern: {"coding":[{"system":"http://loinc.org","code":"35217-9"}]}
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for observation
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | Triglyceride value
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
comment | S | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
high | I | 1..1 | SimpleQuantity | High Range, if relevant
text | | 0..1 | string | Text based reference range in an observation
component | ?I | 0..* | BackboneElement | Component results
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | ? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ? | 0..1 | Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual component result
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Example Lipid Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation  --></category>
 <code><!-- 1..1 CodeableConcept Triglyceride  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!-- 0..1 dateTime|Period
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity><!-- 0..1 Quantity Triglyceride value --></valueQuantity>
```

```
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept + | ++ | +++ | - | -- | ---  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..0 Quantity
      Low Range, if relevant --></low>
  <high><!-- 1..1 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..0 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..0 CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..0 BackboneElement  <!--  0..0 Resource related to this observation -->
 </related>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|
    Range|Ratio|SampledData|Attachment|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], //Classification of  type of observation
    "code" : { CodeableConcept }, // R! Triglyceride
// value[x]: Who and/or what this is about. One of these 4:
    "subject" : { Reference(Patient) }, //Who and/or what this is about
    "subject" : { Reference(Group) }, //Who and/or what this is about
    "subject" : { Reference(Device) }, //Who and/or what this is about
    "subject" : { Reference(Location) }, //Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 2:
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for observation
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }] //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }], //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //Triglyceride value
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //+ | ++ | +++ | - | -- | ---
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

Other representations of profile: Schematron

10.2.12.14.2 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | extensible | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | LOINC Codes | example | LOINC Codes
Observation.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.related.type | ObservationRelationshipType | required | ObservationRelationshipType
Observation.component.code | LOINC Codes | example | LOINC Codes
Observation.component.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.2.12.21 StructureDefinition: HDLCholesterol

URL: https://hl7.org/fhir/STU3/hdlcholesterol.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.12.21 StructureDefinition: HDLCholesterol

Orders and Observations Work Group | Maturity Level: 1 | Informative | Use Context: Not Intended for Production use

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/hdlcholesterol`

HDL Cholesterol Result

This profile was published on Sat, May 12, 2012 00:00+1000 as a draft by Grahame Grieve.

10.2.12.21.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from Observation.

Summary

Mandatory: 3 elements Must-Support: 5 elements Fixed Value: 2 elements Prohibited: 5 elements

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 1..1 | | HDL Cholesterol Result
code | S | 1..1 | CodeableConcept | HDL Cholesterol Fixed Value: {"coding":[{"system":"http://loinc.org","code":"2085-9","display":"HDL Cholesterol"}]}
valueQuantity | S | 0..1 | Quantity | HDL Cholesterol value
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | ---
comment | S | 0..1 | string | Comments about result
referenceRange | S | 1..1 | BackboneElement
low | | 1..1 | SimpleQuantity | Fixed Value: {"value":1.5}
high | | 0..0
type | | 0..0
appliesTo | | 0..0
age | | 0..0
related | | 0..0

Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 1..1 | | HDL Cholesterol Result
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation

basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | S? | 1..1 | CodeableConcept | HDL Cholesterol Binding: LOINC Codes (example) Fixed Value: {"coding":[{"system":"http://loinc.org","code":"2085-9","display":"HDL Cholesterol"}]}
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for observation
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | HDL Cholesterol value
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
comment | S | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 1..1 | SimpleQuantity | A fixed quantity (no comparator) Fixed Value: {"value":1.5}
text | | 0..1 | string | Text based reference range in an observation
component | ?I | 0..* | BackboneElement | Component results
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | ? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ? | 0..1 | Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual component result
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Example Lipid Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation  --></category>
 <code><!-- 1..1 CodeableConcept HDL Cholesterol  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!-- 0..1 dateTime|Period
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity><!-- 0..1 Quantity HDL Cholesterol value --></valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept + | ++ | +++ | - | -- | ---  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!--  1..1 Quantity
      A fixed quantity (no comparator) --></low>
  <high><!-- 0..0 Quantity
      High Range, if relevant --></high>
```

```
  <type><!-- 0..0 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..0 CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..0 BackboneElement  <!--  0..0 Resource related to this observation -->
 </related>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|
    Range|Ratio|SampledData|Attachment|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], //Classification of  type of observation
    "code" : { CodeableConcept }, // R!
// value[x]: Who and/or what this is about. One of these 4:
    "subject" : { Reference(Patient) }, //Who and/or what this is about
    "subject" : { Reference(Group) }, //Who and/or what this is about
    "subject" : { Reference(Device) }, //Who and/or what this is about
    "subject" : { Reference(Location) }, //Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 2:
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for observation
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }] //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }], //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //HDL Cholesterol value
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //+ | ++ | +++ | - | -- | ---
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

This structure is derived from Observation.

Summary

Mandatory: 3 elements Must-Support: 5 elements Fixed Value: 2 elements Prohibited: 5 elements

Differential View

This structure is derived from Observation.

Name	Flags	Card.	Type	Description & Constraints
Observation		1..1		HDL Cholesterol Result
code	S	1..1	CodeableConcept	HDL Cholesterol Fixed Value: {"coding":[{"system":"http://loinc.org","code":"2085-9","display":"HDL Cholesterol"}]}
valueQuantity	S	0..1	Quantity	HDL Cholesterol value
interpretation	S	0..1	CodeableConcept	+ | ++ | +++ | - | -- | ---
comment	S	0..1	string	Comments about result
referenceRange	S	1..1	BackboneElement	

low | | 1..1 | SimpleQuantity | Fixed Value: {"value":1.5}
high | | 0..0
type | | 0..0
appliesTo | | 0..0
age | | 0..0
related | | 0..0
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 1..1 | | HDL Cholesterol Result
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | S? | 1..1 | CodeableConcept | HDL Cholesterol Binding: LOINC Codes (example) Fixed Value: {"coding":[{"system":"http://loinc.org","code":"2085-9","display":"HDL Cholesterol"}]}
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for observation
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | HDL Cholesterol value
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
comment | S | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
low | I | 1..1 | SimpleQuantity | A fixed quantity (no comparator) Fixed Value: {"value":1.5}
text | | 0..1 | string | Text based reference range in an observation
component | ?I | 0..* | BackboneElement | Component results
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | ? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ? | 0..1 | Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual component result
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Example Lipid Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
```

```
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation  --></category>
 <code><!-- 1..1 CodeableConcept HDL Cholesterol  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!-- 0..1 dateTime|Period
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity><!-- 0..1 Quantity HDL Cholesterol value --></valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept + | ++ | +++ | - | -- | ---  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!--  1..1 Quantity
      A fixed quantity (no comparator) --></low>
  <high><!-- 0..0 Quantity
      High Range, if relevant --></high>
  <type><!-- 0..0 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..0 CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..0 BackboneElement  <!--  0..0 Resource related to this observation -->
 </related>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|
    Range|Ratio|SampledData|Attachment|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], //Classification of  type of observation
    "code" : { CodeableConcept }, // R!
// value[x]: Who and/or what this is about. One of these 4:
    "subject" : { Reference(Patient) }, //Who and/or what this is about
    "subject" : { Reference(Group) }, //Who and/or what this is about
    "subject" : { Reference(Device) }, //Who and/or what this is about
    "subject" : { Reference(Location) }, //Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 2:
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for observation
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }] //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }], //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //HDL Cholesterol value
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //+ | ++ | +++ | - | -- | ---
    "comment" : "<string>", //Comments about result
```

```
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

Other representations of profile: Schematron

10.2.12.21.2 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | extensible | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | LOINC Codes | example | LOINC Codes
Observation.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.related.type | ObservationRelationshipType | required | ObservationRelationshipType
Observation.component.code | LOINC Codes | example | LOINC Codes
Observation.component.dataAbsentReason | Observation Value Absent Reason | extensible | Observation Value Absent Reason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.2.12.28 StructureDefinition: LDLCholesterol

URL: https://hl7.org/fhir/STU3/ldlcholesterol.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.12.28 StructureDefinition: LDLCholesterol

Orders and Observations Work Group | Maturity Level: 1 | Informative | Use Context: Not Intended for Production use

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/ldlcholesterol`

LDL Cholesterol Result

This profile was published on Sat, May 12, 2012 00:00+1000 as a draft by Grahame Grieve.

10.2.12.28.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary
* Differential Table
* Snapshot Table
* XML Template
* JSON Template
* All

This structure is derived from Observation.

Summary

Mandatory: 3 elements Must-Support: 5 elements Fixed Value: 1 element Prohibited: 5 elements

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 1..1 | | LDL Cholesterol Result
code | S | 1..1 | CodeableConcept | LDL Cholesterol -measured or calculated per code Binding: LDLCodes (required)
valueQuantity | S | 0..1 | Quantity | LDL Cholesterol value
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | ---
comment | S | 0..1 | string | Comments about result
referenceRange | S | 1..1 | BackboneElement
low | | 0..0
high | | 1..1 | SimpleQuantity | Fixed Value: {"value":3.0}
type | | 0..0
appliesTo | | 0..0
age | | 0..0
related | | 0..0
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 1..1 | | LDL Cholesterol Result

id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | S? | 1..1 | CodeableConcept | LDL Cholesterol -measured or calculated per code Binding: LDLCodes (required)
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for observation
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | LDL Cholesterol value
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
comment | S | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
high | I | 1..1 | SimpleQuantity | A fixed quantity (no comparator) Fixed Value: {"value":3.0}
text | | 0..1 | string | Text based reference range in an observation
component | ?I | 0..* | BackboneElement | Component results
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | ? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ? | 0..1 | Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual component result
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Example Lipid Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation  --></category>
 <code><!-- 1..1 CodeableConcept LDL Cholesterol -measured or calculated per code  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!-- 0..1 dateTime|Period
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity><!-- 0..1 Quantity LDL Cholesterol value --></valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept + | ++ | +++ | - | -- | ---  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
```

```
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..0 Quantity
      Low Range, if relevant --></low>
  <high><!--  1..1 Quantity
      A fixed quantity (no comparator) --></high>
  <type><!-- 0..0 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..0 CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..0 BackboneElement  <!--  0..0 Resource related to this observation -->
 </related>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|
    Range|Ratio|SampledData|Attachment|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], //Classification of  type of observation
    "code" : { CodeableConcept }, // R! LDL Cholesterol -measured or calculated per code
// value[x]: Who and/or what this is about. One of these 4:
    "subject" : { Reference(Patient) }, //Who and/or what this is about
    "subject" : { Reference(Group) }, //Who and/or what this is about
    "subject" : { Reference(Device) }, //Who and/or what this is about
    "subject" : { Reference(Location) }, //Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 2:
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for observation
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
    "performer" : [{ Reference(Practitioner) }] //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }], //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //LDL Cholesterol value
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //+ | ++ | +++ | - | -- | ---
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

This structure is derived from Observation.

Summary

Mandatory: 3 elements Must-Support: 5 elements Fixed Value: 1 element Prohibited: 5 elements

Differential View

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 1..1 | | LDL Cholesterol Result
code | S | 1..1 | CodeableConcept | LDL Cholesterol -measured or calculated per code Binding: LDLCodes (required)
valueQuantity | S | 0..1 | Quantity | LDL Cholesterol value
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | ---
comment | S | 0..1 | string | Comments about result
referenceRange | S | 1..1 | BackboneElement
low | | 0..0
high | | 1..1 | SimpleQuantity | Fixed Value: {"value":3.0}
type | | 0..0
appliesTo | | 0..0
age | | 0..0
related | | 0..0
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 1..1 | | LDL Cholesterol Result
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business Identifier for observation
basedOn | ? | 0..* | Reference(CarePlan), Reference(DeviceRequest), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | Fulfills plan, proposal or order
status | ?!? | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | S? | 1..1 | CodeableConcept | LDL Cholesterol -measured or calculated per code Binding: LDLCodes (required)
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | Who and/or what this is about
context | | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Healthcare event during which this observation is made
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for observation
issued | ? | 0..1 | instant | Date/Time this was made available
performer | ? | 0..* | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(RelatedPerson) | Who is responsible for the observation
valueQuantity | S?I | 0..1 | Quantity | LDL Cholesterol value
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
comment | S | 0..1 | string | Comments about result
bodySite | | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device), Reference(DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
high | I | 1..1 | SimpleQuantity | A fixed quantity (no comparator) Fixed Value: {"value":3.0}
text | | 0..1 | string | Text based reference range in an observation
component | ?I | 0..* | BackboneElement | Component results
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | ? | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ? | 0..1 | Quantity, CodeableConcept, string, Range, Ratio, SampledData, Attachment, time, dateTime, Period | Actual component result
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: Observation Value Absent Reason (extensible)
interpretation | | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Example Lipid Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
```

```
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(DeviceRequest)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|
   Reference(NutritionOrder)|Reference(ProcedureRequest)|Reference(ReferralRequest) Fulfills plan, proposal or order
    --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended +  -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation  --></category>
 <code><!-- 1..1 CodeableConcept LDL Cholesterol -measured or calculated per code  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     Who and/or what this is about --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Healthcare event during which this observation is made --></context>
 <effective[x]><!-- 0..1 dateTime|Period
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this was made available -->
 <performer><!-- 0..* Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(RelatedPerson)
     Who is responsible for the observation --></performer>
 <valueQuantity><!-- 0..1 Quantity LDL Cholesterol value --></valueQuantity>
 <dataAbsentReason><!-- 0..1 CodeableConcept Why the result is missing  --></dataAbsentReason>
 <interpretation><!-- 0..1 CodeableConcept + | ++ | +++ | - | -- | ---  --></interpretation>
 <comment value="[string]"/><!-- 0..1 Comments about result -->
 <bodySite><!-- 0..1 CodeableConcept Observed body part  --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done  --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device)|Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <low><!-- 0..0 Quantity
      Low Range, if relevant --></low>
  <high><!--  1..1 Quantity
      A fixed quantity (no comparator) --></high>
  <type><!-- 0..0 CodeableConcept Reference range qualifier  --></type>
  <appliesTo><!-- 0..0 CodeableConcept Reference range population  --></appliesTo>
  <age><!-- 0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <related>  0..0 BackboneElement  <!--  0..0 Resource related to this observation -->
 </related>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type)  --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|
    Range|Ratio|SampledData|Attachment|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!-- 0..1 CodeableConcept Why the component result is missing  --></dataAbsentReason>
  <interpretation><!-- 0..1 CodeableConcept High, low, normal, etc.  --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business Identifier for observation
// value[x]: Fulfills plan, proposal or order. One of these 7:
    "basedOn" : [{ Reference(CarePlan) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(DeviceRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(MedicationRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(NutritionOrder) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ProcedureRequest) }], //Fulfills plan, proposal or order
    "basedOn" : [{ Reference(ReferralRequest) }], //Fulfills plan, proposal or order
    "status" : "<code>", // R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], //Classification of  type of observation
    "code" : { CodeableConcept }, // R! LDL Cholesterol -measured or calculated per code
// value[x]: Who and/or what this is about. One of these 4:
    "subject" : { Reference(Patient) }, //Who and/or what this is about
    "subject" : { Reference(Group) }, //Who and/or what this is about
    "subject" : { Reference(Device) }, //Who and/or what this is about
    "subject" : { Reference(Location) }, //Who and/or what this is about
// value[x]: Healthcare event during which this observation is made. One of these 2:
    "context" : { Reference(Encounter) }, //Healthcare event during which this observation is made
    "context" : { Reference(EpisodeOfCare) }, //Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 2:
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for observation
    "issued" : "<instant>", //Date/Time this was made available
// value[x]: Who is responsible for the observation. One of these 4:
```

```
    "performer" : [{ Reference(Practitioner) }] //Who is responsible for the observation
    "performer" : [{ Reference(Organization) }], //Who is responsible for the observation
    "performer" : [{ Reference(Patient) }], //Who is responsible for the observation
    "performer" : [{ Reference(RelatedPerson) }], //Who is responsible for the observation
    "valueQuantity" : { Quantity }, //LDL Cholesterol value
    "dataAbsentReason" : { CodeableConcept }, //Why the result is missing
    "interpretation" : { CodeableConcept }, //+ | ++ | +++ | - | -- | ---
    "comment" : "<string>", //Comments about result
    "bodySite" : { CodeableConcept }, //Observed body part
    "method" : { CodeableConcept }, //How it was done
    "specimen" : { Reference(Specimen) }, //Specimen used for this observation
// value[x]: (Measurement) Device. One of these 2:
    "device" : { Reference(Device) }, //(Measurement) Device
    "device" : { Reference(DeviceMetric) }, //(Measurement) Device
    "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

Other representations of profile: Schematron

10.2.12.28.2 Terminology Bindings

Path	Name	Conformance	ValueSet
Observation.language	Common Languages	extensible	Common Languages
Observation.status	ObservationStatus	required	ObservationStatus
Observation.category	Observation Category Codes	preferred	Observation Category Codes
Observation.code	LDLCodes	required	LDLCodes
Observation.dataAbsentReason	Observation Value Absent Reason	extensible	Observation Value Absent Reason
Observation.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes
Observation.bodySite	SNOMED CT Body Structures	example	SNOMED CT Body Structures
Observation.method	Observation Methods	example	Observation Methods
Observation.related.type	ObservationRelationshipType	required	ObservationRelationshipType
Observation.component.code	LOINC Codes	example	LOINC Codes
Observation.component.dataAbsentReason	Observation Value Absent Reason	extensible	Observation Value Absent Reason
Observation.component.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes

10.2.13 DiagnosticReport HL7 Extensions

URL: https://hl7.org/fhir/STU3/diagnosticreport-extensions.html

Defines common extensions used with or related to the DiagnosticReport resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.13 DiagnosticReport HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

10.2.13.1 Content

Extensions:

diagnosticReport-locationPerformed | locationPerformed : Facility location where this report was prepared.

diagnosticReport-summaryOf | summaryOf : A summary report that points to subordinate target reports.

diagnosticReport-replaces | replaces : The report replaces the target resource. For example, when a final anatomic pathology report replaces a preliminary anatomic pathology report replaces where the subsequent observation of case and report may be on more or different material (specimen). Note that this is not same concept asDiagnosticReport.status = preliminary of final, but industry definition of preliminary and final.

diagnosticReport-extends | extends : The report references related ("sibling") reports.

diagnosticReport-addendumOf | addendumOf : The supplements or provides additional information for the target report.

10.2.14 DiagnosticReport-Genetics

URL: https://hl7.org/fhir/STU3/diagnosticreport-genetic.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.14 DiagnosticReport-Genetics

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

10.2.14.1 Overview

The DiagnosticReport-genetics profile extends the DiagnosticReport resource to enable reporting of structured genetic test results by referring to Observation-genetics profile. In addition, the genetics profile denotes condition context for genetic testing, which may influence reported variants and interpretation for large genomic testing panels and provides references to the FamilyMemberHistory resource, the Condition resource.

For use cases, details on how this resource interact with other Clinical Genomics resources or profiles, please refer to implementation guidance document here .

10.2.14.2 Scope and Usage

The DiagnosticReport-genetics profile supports reporting of DNA variants at the genomic, cDNA, and protein change level. In addition, a condition context may be provided, as AssessedCondition. For large genomic tests, a condition may be used as an input into the analytic pipeline to aid in the identification of clinically relevant variants related to the test order.

Hl7 Clinical Genomics Work Group emphasizes the importance of transmitting structured genetic findings within the clinical, translational, and research environments fully integrated with other clinical data, in order to drive outcomes analysis, operational decision making, discovery research, and public health reporting.

Here is the document of HL7 Version 3 Domain Analysis Model where the examples used in genetics profile are from (Page 5).

10.2.14.3 Content

Profiles:
DiagnosticReport-Genetics | Describes how the DiagnosticReport resource is used to report structured genetic test results
Extensions:
DiagnosticReport-geneticsAssessedCondition | AssessedCondition : Used to denote condition context for genetic testing, which may influence reported variants and interpretation for large genomic testing panels e.g. lung cancer or familial breast cancer.
DiagnosticReport-geneticsFamilyMemberHistory | FamilyMemberHistory : Significant health events and conditions for a person related to the patient relevant in the context of care for the patient.
DiagnosticReport-geneticsAnalysis | Analysis : Knowledge-based comments on the effect of the sequence on patient's condition/medication reaction.
Examples:
Genetics -with family | An example of a genetics test report with Family member history
Comprehensive report | An example of a comprehensive bone marrow report
PGx Example | An example of a PGx haplotype report

10.2.14.4 Search Parameters

Search parameters defined by this package. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths | Source
assessed-condition | reference | Condition assessed by genetic test | | XML / JSON

10.2.14.5 StructureDefinition: DiagnosticReport-Genetics

URL: https://hl7.org/fhir/STU3/diagnosticreport-genetics.html
This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.14.5 StructureDefinition: DiagnosticReport-Genetics

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/diagnosticreport-genetics`

Describes how the DiagnosticReport resource is used to report structured genetic test results

This profile was published on Fri, Oct 9, 2015 00:00+1100 as a draft by Health Level Seven International (Clinical Genomics).

10.2.14.5.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from DiagnosticReport.

Summary

Prohibited: 1 element

Extensions

This structure refers to these extensions:

* http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAssessedCondition

* http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsFamilyMemberHistory

* http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAnalysis

This structure is derived from DiagnosticReport.

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | | 0..*

codedDiagnosis | | 0..0
DiagnosticReport-geneticsAssessedCondition | | 0..* | Reference(Condition) | AssessedCondition URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAssessedCondition
DiagnosticReport-geneticsFamilyMemberHistory | | 0..* | Reference(FamilyMemberHistory) | FamilyHistory URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsFamilyMemberHistory
DiagnosticReport-geneticsAnalysis | | 0..* | (Complex) | Analysis URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAnalysis
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | I | 0..* | | A Diagnostic report - a combination of request information, atomic results, images, interpretation, as well as formatted reports
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
DiagnosticReport-geneticsAssessedCondition | I | 0..* | Reference(Condition) | AssessedCondition URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAssessedCondition
DiagnosticReport-geneticsFamilyMemberHistory | I | 0..* | Reference(FamilyMemberHistory) | FamilyHistory URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsFamilyMemberHistory
DiagnosticReport-geneticsAnalysis | I | 0..* | (Complex) | Analysis URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAnalysis
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business identifier for report
basedOn | | 0..* | Reference(CarePlan), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | What was requested
status | ?!? | 1..1 | code | registered | partial | preliminary | final + Binding: DiagnosticReportStatus (required)
category | ? | 0..1 | CodeableConcept | Service category Binding: Diagnostic Service Section Codes (example)
code | ? | 1..1 | CodeableConcept | Name/Code for this diagnostic report Binding: LOINC Diagnostic Report Codes (preferred)
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | The subject of the report - usually, but not always, the patient
context | ? | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Health care event when test ordered
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for report
issued | ? | 0..1 | instant | DateTime this version was released
performer | ?I | 0..* | BackboneElement | Participants in producing the report
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
role | ? | 0..1 | CodeableConcept | Type of performer Binding: Procedure Performer Role Codes (example)
actor | ? | 1..1 | Reference(Practitioner), Reference(Organization) | Practitioner or Organization participant
specimen | | 0..* | Reference(Specimen) | Specimens this report is based on
result | | 0..* | Reference(Observation) | Observations - simple, or complex nested groups
imagingStudy | | 0..* | Reference(ImagingStudy), Reference(ImagingManifest) | Reference to full details of imaging associated with the diagnostic report
image | ?I | 0..* | BackboneElement | Key images associated with this report
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
comment | | 0..1 | string | Comment about the image (e.g. explanation)
link | ? | 1..1 | Reference(Media) | Reference to the image source
conclusion | | 0..1 | string | Clinical Interpretation of test results
presentedForm | | 0..* | Attachment | Entire report as issued
Documentation for this format

```
<!-- DiagnosticReport-Genetics -->

<DiagnosticReport xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business identifier for report --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|Reference(NutritionOrder)|
   Reference(ProcedureRequest)|Reference(ReferralRequest) What was requested --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | partial | preliminary | final +  -->
 <category><!-- 0..1 CodeableConcept Service category  --></category>
 <code><!-- 1..1 CodeableConcept Name/Code for this diagnostic report  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     The subject of the report - usually, but not always, the patient --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
```

```
    Health care event when test ordered --></context>
 <effective[x]><!-- 0..1 dateTime|Period
    Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 DateTime this version was released -->
 <performer>  0..* BackboneElement  <!--  0..* Participants in producing the report -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <role><!-- 0..1 CodeableConcept Type of performer  --></role>
  <actor><!-- 1..1 Reference(Practitioner)|Reference(Organization)
      Practitioner or Organization  participant --></actor>
 </performer>
 <specimen><!-- 0..* Reference(Specimen)
    Specimens this report is based on --></specimen>
 <result><!-- 0..* Reference(Observation)
    Observations - simple, or complex nested groups --></result>
 <imagingStudy><!-- 0..* Reference(ImagingStudy)|Reference(ImagingManifest)
    Reference to full details of imaging associated with the diagnostic report --></imagingStudy>
 <image>  0..* BackboneElement  <!--  0..* Key images associated with this report -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <comment value="[string]"/><!-- 0..1 Comment about the image (e.g. explanation) -->
  <link><!-- 1..1 Reference(Media)
      Reference to the image source --></link>
 </image>
 <conclusion value="[string]"/><!-- 0..1 Clinical Interpretation of test results -->
 <codedDiagnosis><!-- 0..0 CodeableConcept Codes for the conclusion  --></codedDiagnosis>
 <presentedForm><!-- 0..* Attachment
    Entire report as issued --></presentedForm>
</DiagnosticReport>
```

```
{ // DiagnosticReport-Genetics
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified orderOpen
      { // AssessedCondition // C?
        // from Element: extension
      },
      { // FamilyHistory // C?
        // from Element: extension
      },
      { // Analysis // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business identifier for report
// value[x]: What was requested. One of these 6:
    "basedOn" : [{ Reference(CarePlan) }], //What was requested
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //What was requested
    "basedOn" : [{ Reference(MedicationRequest) }], //What was requested
    "basedOn" : [{ Reference(NutritionOrder) }], //What was requested
    "basedOn" : [{ Reference(ProcedureRequest) }], //What was requested
    "basedOn" : [{ Reference(ReferralRequest) }], //What was requested
    "status" : "<code>", // R! registered | partial | preliminary | final +
    "category" : { CodeableConcept }, //Service category
    "code" : { CodeableConcept }, // R! Name/Code for this diagnostic report
// value[x]: The subject of the report - usually, but not always, the patient. One of these 4:
    "subject" : { Reference(Patient) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Group) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Device) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Location) }, //The subject of the report - usually, but not always, the patient
// value[x]: Health care event when test ordered. One of these 2:
    "context" : { Reference(Encounter) }, //Health care event when test ordered
    "context" : { Reference(EpisodeOfCare) }, //Health care event when test ordered
// value[x]: Clinically relevant time/time-period for report. One of these 2:
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for report
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for report
    "issued" : "<instant>", //DateTime this version was released
    "performer" : [{ BackboneElement }] // C?Participants in producing the report
    "specimen" : [{ Reference(Specimen) }], //Specimens this report is based on
    "result" : [{ Reference(Observation) }], //Observations - simple, or complex nested groups
// value[x]: Reference to full details of imaging associated with the diagnostic report. One of these 2:
    "imagingStudy" : [{ Reference(ImagingStudy) }], //Reference to full details of imaging associated with the
    diagnostic report
    "imagingStudy" : [{ Reference(ImagingManifest) }], //Reference to full details of imaging associated with the
    diagnostic report
    "image" : [{ BackboneElement }], // C?Key images associated with this report
    "conclusion" : "<string>", //Clinical Interpretation of test results
    "presentedForm" : [{ Attachment }], //Entire report as issued
  }
```

This structure is derived from DiagnosticReport.

Summary

Prohibited: 1 element

Extensions

This structure refers to these extensions:

* http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAssessedCondition

* http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsFamilyMemberHistory

* http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAnalysis

Differential View

This structure is derived from DiagnosticReport.

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | | 0..*
codedDiagnosis | | 0..0
DiagnosticReport-geneticsAssessedCondition | | 0..* | Reference(Condition) | AssessedCondition URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAssessedCondition
DiagnosticReport-geneticsFamilyMemberHistory | | 0..* | Reference(FamilyMemberHistory) | FamilyHistory URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsFamilyMemberHistory
DiagnosticReport-geneticsAnalysis | | 0..* | (Complex) | Analysis URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAnalysis
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | I | 0..* | | A Diagnostic report - a combination of request information, atomic results, images, interpretation, as well as formatted reports
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
DiagnosticReport-geneticsAssessedCondition | I | 0..* | Reference(Condition) | AssessedCondition URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAssessedCondition
DiagnosticReport-geneticsFamilyMemberHistory | I | 0..* | Reference(FamilyMemberHistory) | FamilyHistory URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsFamilyMemberHistory
DiagnosticReport-geneticsAnalysis | I | 0..* | (Complex) | Analysis URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAnalysis
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business identifier for report
basedOn | | 0..* | Reference(CarePlan), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | What was requested
status | ?!? | 1..1 | code | registered | partial | preliminary | final + Binding: DiagnosticReportStatus (required)
category | ? | 0..1 | CodeableConcept | Service category Binding: Diagnostic Service Section Codes (example)
code | ? | 1..1 | CodeableConcept | Name/Code for this diagnostic report Binding: LOINC Diagnostic Report Codes (preferred)
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | The subject of the report - usually, but not always, the patient
context | ? | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Health care event when test ordered
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for report
issued | ? | 0..1 | instant | DateTime this version was released
performer | ?I | 0..* | BackboneElement | Participants in producing the report
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
role | ? | 0..1 | CodeableConcept | Type of performer Binding: Procedure Performer Role Codes (example)
actor | ? | 1..1 | Reference(Practitioner), Reference(Organization) | Practitioner or Organization participant
specimen | | 0..* | Reference(Specimen) | Specimens this report is based on
result | | 0..* | Reference(Observation) | Observations - simple, or complex nested groups
imagingStudy | | 0..* | Reference(ImagingStudy), Reference(ImagingManifest) | Reference to full details of imaging associated with the diagnostic report
image | ?I | 0..* | BackboneElement | Key images associated with this report
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
comment | | 0..1 | string | Comment about the image (e.g. explanation)
link | ? | 1..1 | Reference(Media) | Reference to the image source
conclusion | | 0..1 | string | Clinical Interpretation of test results
presentedForm | | 0..* | Attachment | Entire report as issued
Documentation for this format

XML Template

```
<!-- DiagnosticReport-Genetics -->

<DiagnosticReport xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
```

```
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business identifier for report --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|Reference(NutritionOrder)|
   Reference(ProcedureRequest)|Reference(ReferralRequest) What was requested --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | partial | preliminary | final +  -->
 <category><!-- 0..1 CodeableConcept Service category  --></category>
 <code><!-- 1..1 CodeableConcept Name/Code for this diagnostic report  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     The subject of the report - usually, but not always, the patient --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Health care event when test ordered --></context>
 <effective[x]><!-- 0..1 dateTime|Period
     Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 DateTime this version was released -->
 <performer>  0..* BackboneElement  <!--  0..* Participants in producing the report -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <role><!-- 0..1 CodeableConcept Type of performer  --></role>
  <actor><!-- 1..1 Reference(Practitioner)|Reference(Organization)
      Practitioner or Organization  participant --></actor>
 </performer>
 <specimen><!-- 0..* Reference(Specimen)
     Specimens this report is based on --></specimen>
 <result><!-- 0..* Reference(Observation)
     Observations - simple, or complex nested groups --></result>
 <imagingStudy><!-- 0..* Reference(ImagingStudy)|Reference(ImagingManifest)
     Reference to full details of imaging associated with the diagnostic report --></imagingStudy>
 <image>  0..* BackboneElement  <!--  0..* Key images associated with this report -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <comment value="[string]"/><!-- 0..1 Comment about the image (e.g. explanation) -->
  <link><!-- 1..1 Reference(Media)
      Reference to the image source --></link>
 </image>
 <conclusion value="[string]"/><!-- 0..1 Clinical Interpretation of test results -->
 <codedDiagnosis><!-- 0..0 CodeableConcept Codes for the conclusion  --></codedDiagnosis>
 <presentedForm><!-- 0..* Attachment
     Entire report as issued --></presentedForm>
</DiagnosticReport>
```

JSON Template

```
{ // DiagnosticReport-Genetics
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified orderOpen
      { // AssessedCondition // C?
        // from Element: extension
      },
      { // FamilyHistory // C?
        // from Element: extension
      },
      { // Analysis // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business identifier for report
// value[x]: What was requested. One of these 6:
    "basedOn" : [{ Reference(CarePlan) }], //What was requested
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //What was requested
    "basedOn" : [{ Reference(MedicationRequest) }], //What was requested
    "basedOn" : [{ Reference(NutritionOrder) }], //What was requested
    "basedOn" : [{ Reference(ProcedureRequest) }], //What was requested
    "basedOn" : [{ Reference(ReferralRequest) }], //What was requested
    "status" : "<code>", // R! registered | partial | preliminary | final +
    "category" : { CodeableConcept }, //Service category
    "code" : { CodeableConcept }, // R! Name/Code for this diagnostic report
// value[x]: The subject of the report - usually, but not always, the patient. One of these 4:
    "subject" : { Reference(Patient) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Group) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Device) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Location) }, //The subject of the report - usually, but not always, the patient
// value[x]: Health care event when test ordered. One of these 2:
    "context" : { Reference(Encounter) }, //Health care event when test ordered
    "context" : { Reference(EpisodeOfCare) }, //Health care event when test ordered
// value[x]: Clinically relevant time/time-period for report. One of these 2:
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for report
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for report
    "issued" : "<instant>", //DateTime this version was released
    "performer" : [{ BackboneElement }] // C?Participants in producing the report
    "specimen" : [{ Reference(Specimen) }], //Specimens this report is based on
```

```
    "result" : [{ Reference(Observation) }], //Observations - simple, or complex nested groups
// value[x]: Reference to full details of imaging associated with the diagnostic report. One of these 2:
    "imagingStudy" : [{ Reference(ImagingStudy) }], //Reference to full details of imaging associated with the
    diagnostic report
    "imagingStudy" : [{ Reference(ImagingManifest) }], //Reference to full details of imaging associated with the
    diagnostic report
    "image" : [{ BackboneElement }], // C?Key images associated with this report
    "conclusion" : "<string>", //Clinical Interpretation of test results
    "presentedForm" : [{ Attachment }], //Entire report as issued
  }
```

Other representations of profile: Schematron

10.2.14.5.2 Terminology Bindings

Path | Name | Conformance | ValueSet
DiagnosticReport.language | Common Languages | extensible | Common Languages
DiagnosticReport.status | DiagnosticReportStatus | required | DiagnosticReportStatus
DiagnosticReport.category | Diagnostic Service Section Codes | example | Diagnostic Service Section Codes
DiagnosticReport.code | LOINC Diagnostic Report Codes | preferred | LOINC Diagnostic Report Codes
DiagnosticReport.performer.role | Procedure Performer Role Codes | example | Procedure Performer Role Codes

10.2.14.10 Diagnosticreport-genetic-DiagnosticReport-assessed-condition

URL: https://hl7.org/fhir/STU3/diagnosticreport-genetic-DiagnosticReport-assessed-condition.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.14.10 Diagnosticreport-genetic-DiagnosticReport-assessed-condition

Search Parameter assessed-condition (type: reference) defined as part of the diagnosticreport-genetic Profile.

Condition assessed by genetic test

Other Details:

* Full URL : http://hl7.org/fhir/SearchParameter/diagnosticreport-genetic-DiagnosticReport-assessed-condition

* Paths :

* Targets :

10.2.15 Profile for HLA Genotyping Results

URL: https://hl7.org/fhir/STU3/diagnosticreport-hla-results.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.15 Profile for HLA Genotyping Results

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

10.2.15.1 Overview

The DiagnosticReport-genetics profile extends the DiagnosticReport resource to enable reporting of structured genetic test results by referring to Observation-genetics profile. In addition, the genetics profile denotes condition context for genetic testing, which may influence reported variants and interpretation for large genomic testing panels and provides references to the FamilyMemberHistory resource, the Condition resource.

For use cases, details on how this resource interact with other Clinical Genomics resources or profiles, please refer to implementation guidance document here .

10.2.15.2 Scope and Usage

The DiagnosticReport-genetics profile supports reporting of DNA variants at the genomic, cDNA, and protein change level. In addition, a condition context may be provided, as AssessedCondition. For large genomic tests, a condition may be used as an input into the analytic pipeline to aid in the identification of clinically relevant variants related to the test order.

Hl7 Clinical Genomics Work Group emphasizes the importance of transmitting structured genetic findings within the clinical, translational, and research environments fully integrated with other clinical data, in order to drive outcomes analysis, operational decision making, discovery research, and public health reporting.

Here is the document of HL7 Version 3 Domain Analysis Model where the examples used in genetics profile are from (Page 5).

10.2.15.3 Content

Profiles:
HLAResult | Describes how the HLA genotyping results
Extensions:
hla-genotyping-results-allele-database | allele-database : Allele Database.
hla-genotyping-results-glstring | glstring : glstring.

hla-genotyping-results-haploid | haploid : haploid.
hla-genotyping-results-method | method : The platform, methodology and software applied at the time of the genotyping.
Examples:
HLA genotyping | An example of a HLA genotyping results report

10.2.15.4 StructureDefinition: HLAResult

URL: https://hl7.org/fhir/STU3/hlaresult.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.15.4 StructureDefinition: HLAResult

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/hlaresult`

Describes how the HLA genotyping results

This profile was published on Fri, Oct 9, 2015 00:00+1100 as a draft by Health Level Seven International (Clinical Genomics).

10.2.15.4.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from DiagnosticReport.

Summary

Extensions

This structure refers to these extensions:

* http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database

* http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring

* http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid

* http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method

This structure is derived from DiagnosticReport.

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | | 0..*
hla-genotyping-results-allele-database | | 0..1 | CodeableConcept | Allele Database URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database
hla-genotyping-results-glstring | | 0..1 | (Complex) | Glstring URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring
hla-genotyping-results-haploid | | 0..* | (Complex) | Haploid URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid
hla-genotyping-results-method | | 0..1 | CodeableConcept | The platform, methodology and software applied at the time of the genotyping URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | I | 0..* | | A Diagnostic report - a combination of request information, atomic results, images, interpretation, as well as formatted reports
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
hla-genotyping-results-allele-database | I | 0..1 | CodeableConcept | Allele Database URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database
hla-genotyping-results-glstring | I | 0..1 | (Complex) | Glstring URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring
hla-genotyping-results-haploid | I | 0..* | (Complex) | Haploid URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid
hla-genotyping-results-method | I | 0..1 | CodeableConcept | The platform, methodology and software applied at the time of the genotyping URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business identifier for report
basedOn | | 0..* | Reference(CarePlan), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | What was requested
status | ?!? | 1..1 | code | registered | partial | preliminary | final + Binding: DiagnosticReportStatus (required)

category | ? | 0..1 | CodeableConcept | Service category Binding: Diagnostic Service Section Codes (example)
code | ? | 1..1 | CodeableConcept | Name/Code for this diagnostic report Binding: LOINC Diagnostic Report Codes (preferred)
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | The subject of the report - usually, but not always, the patient
context | ? | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Health care event when test ordered
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for report
issued | ? | 0..1 | instant | DateTime this version was released
performer | ?I | 0..* | BackboneElement | Participants in producing the report
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
role | ? | 0..1 | CodeableConcept | Type of performer Binding: Procedure Performer Role Codes (example)
actor | ? | 1..1 | Reference(Practitioner), Reference(Organization) | Practitioner or Organization participant
specimen | | 0..* | Reference(Specimen) | Specimens this report is based on
result | | 0..* | Reference(Observation) | Observations - simple, or complex nested groups
imagingStudy | | 0..* | Reference(ImagingStudy), Reference(ImagingManifest) | Reference to full details of imaging associated with the diagnostic report
image | ?I | 0..* | BackboneElement | Key images associated with this report
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
comment | | 0..1 | string | Comment about the image (e.g. explanation)
link | ? | 1..1 | Reference(Media) | Reference to the image source
conclusion | | 0..1 | string | Clinical Interpretation of test results
codedDiagnosis | | 0..* | CodeableConcept | Codes for the conclusion Binding: SNOMED CT Clinical Findings (example)
presentedForm | | 0..* | Attachment | Entire report as issued
Documentation for this format

```
<!-- Profile for HLA Genotyping Results -->

<DiagnosticReport xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business identifier for report --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|Reference(NutritionOrder)|
   Reference(ProcedureRequest)|Reference(ReferralRequest) What was requested --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | partial | preliminary | final +  -->
 <category><!-- 0..1 CodeableConcept Service category  --></category>
 <code><!-- 1..1 CodeableConcept Name/Code for this diagnostic report  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     The subject of the report - usually, but not always, the patient --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Health care event when test ordered --></context>
 <effective[x]><!-- 0..1 dateTime|Period
     Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 DateTime this version was released -->
 <performer>  0..* BackboneElement  <!--  0..* Participants in producing the report -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <role><!-- 0..1 CodeableConcept Type of performer  --></role>
  <actor><!-- 1..1 Reference(Practitioner)|Reference(Organization)
      Practitioner or Organization  participant --></actor>
 </performer>
 <specimen><!-- 0..* Reference(Specimen)
     Specimens this report is based on --></specimen>
 <result><!-- 0..* Reference(Observation)
     Observations - simple, or complex nested groups --></result>
 <imagingStudy><!-- 0..* Reference(ImagingStudy)|Reference(ImagingManifest)
     Reference to full details of imaging associated with the diagnostic report --></imagingStudy>
 <image>  0..* BackboneElement  <!--  0..* Key images associated with this report -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <comment value="[string]"/><!-- 0..1 Comment about the image (e.g. explanation) -->
  <link><!-- 1..1 Reference(Media)
      Reference to the image source --></link>
 </image>
 <conclusion value="[string]"/><!-- 0..1 Clinical Interpretation of test results -->
 <codedDiagnosis><!-- 0..* CodeableConcept Codes for the conclusion  --></codedDiagnosis>
 <presentedForm><!-- 0..* Attachment
     Entire report as issued --></presentedForm>
</DiagnosticReport>
```

```
{ // Profile for HLA Genotyping Results
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
```

```
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified orderOpen
      { // Allele Database // C?
        // from Element: extension
      },
      { // Glstring // C?
        // from Element: extension
      },
      { // Haploid // C?
        // from Element: extension
      },
      { // The platform, methodology and software applied at the time of the■genotyping // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business identifier for report
// value[x]: What was requested. One of these 6:
    "basedOn" : [{ Reference(CarePlan) }], //What was requested
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //What was requested
    "basedOn" : [{ Reference(MedicationRequest) }], //What was requested
    "basedOn" : [{ Reference(NutritionOrder) }], //What was requested
    "basedOn" : [{ Reference(ProcedureRequest) }], //What was requested
    "basedOn" : [{ Reference(ReferralRequest) }], //What was requested
    "status" : "<code>", // R! registered | partial | preliminary | final +
    "category" : { CodeableConcept }, //Service category
    "code" : { CodeableConcept }, // R! Name/Code for this diagnostic report
// value[x]: The subject of the report - usually, but not always, the patient. One of these 4:
    "subject" : { Reference(Patient) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Group) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Device) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Location) }, //The subject of the report - usually, but not always, the patient
// value[x]: Health care event when test ordered. One of these 2:
    "context" : { Reference(Encounter) }, //Health care event when test ordered
    "context" : { Reference(EpisodeOfCare) }, //Health care event when test ordered
// value[x]: Clinically relevant time/time-period for report. One of these 2:
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for report
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for report
    "issued" : "<instant>", //DateTime this version was released
    "performer" : [{ BackboneElement }], // C?Participants in producing the report
    "specimen" : [{ Reference(Specimen) }] //Specimens this report is based on
    "result" : [{ Reference(Observation) }], //Observations - simple, or complex nested groups
// value[x]: Reference to full details of imaging associated with the diagnostic report. One of these 2:
    "imagingStudy" : [{ Reference(ImagingStudy) }], //Reference to full details of imaging associated with the
    diagnostic report
    "imagingStudy" : [{ Reference(ImagingManifest) }], //Reference to full details of imaging associated with the
    diagnostic report
    "image" : [{ BackboneElement }], // C?Key images associated with this report
    "conclusion" : "<string>", //Clinical Interpretation of test results
    "codedDiagnosis" : [{ CodeableConcept }], //Codes for the conclusion
    "presentedForm" : [{ Attachment }], //Entire report as issued
  }
```

This structure is derived from DiagnosticReport.

Summary

Extensions

This structure refers to these extensions:

* http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database

* http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring

* http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid

* http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method

Differential View

This structure is derived from DiagnosticReport.

Name | Flags | Card. | Type | Description & Constraints

DiagnosticReport | | 0..*

hla-genotyping-results-allele-database | | 0..1 | CodeableConcept | Allele Database URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database

hla-genotyping-results-glstring | | 0..1 | (Complex) | Glstring URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring

hla-genotyping-results-haploid | | 0..* | (Complex) | Haploid URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid

hla-genotyping-results-method | | 0..1 | CodeableConcept | The platform, methodology and software applied at the time of the genotyping URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method

Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints

DiagnosticReport | I | 0..* | | A Diagnostic report - a combination of request information, atomic results, images, interpretation, as well as formatted reports

id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
hla-genotyping-results-allele-database | I | 0..1 | CodeableConcept | Allele Database URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database
hla-genotyping-results-glstring | I | 0..1 | (Complex) | Glstring URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring
hla-genotyping-results-haploid | I | 0..* | (Complex) | Haploid URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid
hla-genotyping-results-method | I | 0..1 | CodeableConcept | The platform, methodology and software applied at the time of the genotyping URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Business identifier for report
basedOn | | 0..* | Reference(CarePlan), Reference(ImmunizationRecommendation), Reference(MedicationRequest), Reference(NutritionOrder), Reference(ProcedureRequest), Reference(ReferralRequest) | What was requested
status | ?!? | 1..1 | code | registered | partial | preliminary | final + Binding: DiagnosticReportStatus (required)
category | ? | 0..1 | CodeableConcept | Service category Binding: Diagnostic Service Section Codes (example)
code | ? | 1..1 | CodeableConcept | Name/Code for this diagnostic report Binding: LOINC Diagnostic Report Codes (preferred)
subject | ? | 0..1 | Reference(Patient), Reference(Group), Reference(Device), Reference(Location) | The subject of the report - usually, but not always, the patient
context | ? | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Health care event when test ordered
effective[x] | ? | 0..1 | dateTime, Period | Clinically relevant time/time-period for report
issued | ? | 0..1 | instant | DateTime this version was released
performer | ?I | 0..* | BackboneElement | Participants in producing the report
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
role | ? | 0..1 | CodeableConcept | Type of performer Binding: Procedure Performer Role Codes (example)
actor | ? | 1..1 | Reference(Practitioner), Reference(Organization) | Practitioner or Organization participant
specimen | | 0..* | Reference(Specimen) | Specimens this report is based on
result | | 0..* | Reference(Observation) | Observations - simple, or complex nested groups
imagingStudy | | 0..* | Reference(ImagingStudy), Reference(ImagingManifest) | Reference to full details of imaging associated with the diagnostic report
image | ?I | 0..* | BackboneElement | Key images associated with this report
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
comment | | 0..1 | string | Comment about the image (e.g. explanation)
link | ? | 1..1 | Reference(Media) | Reference to the image source
conclusion | | 0..1 | string | Clinical Interpretation of test results
codedDiagnosis | | 0..* | CodeableConcept | Codes for the conclusion Binding: SNOMED CT Clinical Findings (example)
presentedForm | | 0..* | Attachment | Entire report as issued
Documentation for this format

XML Template

```
<!-- Profile for HLA Genotyping Results -->

<DiagnosticReport xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Business identifier for report --></identifier>
 <basedOn><!-- 0..*
    Reference(CarePlan)|Reference(ImmunizationRecommendation)|Reference(MedicationRequest)|Reference(NutritionOrder)|
   Reference(ProcedureRequest)|Reference(ReferralRequest) What was requested --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | partial | preliminary | final +  -->
 <category><!-- 0..1 CodeableConcept Service category  --></category>
 <code><!-- 1..1 CodeableConcept Name/Code for this diagnostic report  --></code>
 <subject><!-- 0..1 Reference(Patient)|Reference(Group)|Reference(Device)|Reference(Location)
     The subject of the report - usually, but not always, the patient --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Health care event when test ordered --></context>
 <effective[x]><!-- 0..1 dateTime|Period
     Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 DateTime this version was released -->
 <performer>  0..* BackboneElement  <!--  0..* Participants in producing the report -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <role><!-- 0..1 CodeableConcept Type of performer  --></role>
  <actor><!-- 1..1 Reference(Practitioner)|Reference(Organization)
```

```
     Practitioner or Organization  participant --></actor>
 </performer>
 <specimen><!-- 0..* Reference(Specimen)
     Specimens this report is based on --></specimen>
 <result><!-- 0..* Reference(Observation)
     Observations - simple, or complex nested groups --></result>
 <imagingStudy><!-- 0..* Reference(ImagingStudy)|Reference(ImagingManifest)
     Reference to full details of imaging associated with the diagnostic report --></imagingStudy>
 <image>  0..* BackboneElement  <!--  0..* Key images associated with this report -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <comment value="[string]"/><!-- 0..1 Comment about the image (e.g. explanation) -->
  <link><!-- 1..1 Reference(Media)
      Reference to the image source --></link>
 </image>
 <conclusion value="[string]"/><!-- 0..1 Clinical Interpretation of test results -->
 <codedDiagnosis><!-- 0..* CodeableConcept Codes for the conclusion  --></codedDiagnosis>
 <presentedForm><!-- 0..* Attachment
     Entire report as issued --></presentedForm>
</DiagnosticReport>
```

JSON Template

```
{ // Profile for HLA Genotyping Results
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified orderOpen
      { // Allele Database // C?
        // from Element: extension
      },
      { // Glstring // C?
        // from Element: extension
      },
      { // Haploid // C?
        // from Element: extension
      },
      { // The platform, methodology and software applied at the time of the■genotyping // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Business identifier for report
// value[x]: What was requested. One of these 6:
    "basedOn" : [{ Reference(CarePlan) }], //What was requested
    "basedOn" : [{ Reference(ImmunizationRecommendation) }], //What was requested
    "basedOn" : [{ Reference(MedicationRequest) }], //What was requested
    "basedOn" : [{ Reference(NutritionOrder) }], //What was requested
    "basedOn" : [{ Reference(ProcedureRequest) }], //What was requested
    "basedOn" : [{ Reference(ReferralRequest) }], //What was requested
    "status" : "<code>", // R! registered | partial | preliminary | final +
    "category" : { CodeableConcept }, //Service category
    "code" : { CodeableConcept }, // R! Name/Code for this diagnostic report
// value[x]: The subject of the report - usually, but not always, the patient. One of these 4:
    "subject" : { Reference(Patient) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Group) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Device) }, //The subject of the report - usually, but not always, the patient
    "subject" : { Reference(Location) }, //The subject of the report - usually, but not always, the patient
// value[x]: Health care event when test ordered. One of these 2:
    "context" : { Reference(Encounter) }, //Health care event when test ordered
    "context" : { Reference(EpisodeOfCare) }, //Health care event when test ordered
// value[x]: Clinically relevant time/time-period for report. One of these 2:
    "effectiveDateTime" : "<dateTime>", //Clinically relevant time/time-period for report
    "effectivePeriod" : { Period }, //Clinically relevant time/time-period for report
    "issued" : "<instant>", //DateTime this version was released
    "performer" : [{ BackboneElement }], // C?Participants in producing the report
    "specimen" : [{ Reference(Specimen) }] //Specimens this report is based on
    "result" : [{ Reference(Observation) }], //Observations - simple, or complex nested groups
// value[x]: Reference to full details of imaging associated with the diagnostic report. One of these 2:
    "imagingStudy" : [{ Reference(ImagingStudy) }], //Reference to full details of imaging associated with the
    diagnostic report
    "imagingStudy" : [{ Reference(ImagingManifest) }], //Reference to full details of imaging associated with the
    diagnostic report
    "image" : [{ BackboneElement }], // C?Key images associated with this report
    "conclusion" : "<string>", //Clinical Interpretation of test results
    "codedDiagnosis" : [{ CodeableConcept }], //Codes for the conclusion
    "presentedForm" : [{ Attachment }], //Entire report as issued
  }
```

Other representations of profile: Schematron

10.2.15.4.2 Terminology Bindings

Path	Name	Conformance	ValueSet
DiagnosticReport.language	Common Languages	extensible	Common Languages
DiagnosticReport.status	DiagnosticReportStatus	required	DiagnosticReportStatus
DiagnosticReport.category	Diagnostic Service Section Codes	example	Diagnostic Service Section Codes
DiagnosticReport.code	LOINC Diagnostic Report Codes	preferred	LOINC Diagnostic Report Codes

DiagnosticReport.performer.role | Procedure Performer Role Codes | example | Procedure Performer Role Codes
DiagnosticReport.codedDiagnosis | SNOMED CT Clinical Findings | example | SNOMED CT Clinical Findings

10.2.16 Resource DiagnosticReport - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/diagnosticreport-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.2.16 Resource DiagnosticReport - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner

R2 : R3 Converson maps for DiagnosticReport.

Functional status for this map: 11 tests of which 3 fail to execute. 8 fail round-trip testing and 3 r3 resources are invalid (22 errors). (see documentation)

10.2.16.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/DiagnosticReport2to3" = "R2 to R3 Conversions for DiagnosticReport"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DiagnosticReport" alias DiagnosticReportR2 as source
uses "http://hl7.org/fhir/StructureDefinition/DiagnosticReport" alias DiagnosticReport as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  DiagnosticReport extends DomainResource
  input src : DiagnosticReportR2 as source
  input tgt : DiagnosticReport as target

  "DiagnosticReport-identifier" : for src.identifier make tgt.identifier
  "DiagnosticReport-status" : for src.status make tgt.status
  "DiagnosticReport-category" : for src.category make tgt.category
  "DiagnosticReport-code" : for src.code make tgt.code
  "DiagnosticReport-subject" : for src.subject make tgt.subject
  "DiagnosticReport-encounter" : for src.encounter make tgt.context
  "DiagnosticReport-effective" : for src.effective make tgt.effective
  "DiagnosticReport-issued" : for src.issued make tgt.issued
  "DiagnosticReport-performer" : for src.performer as v make tgt.performer as p, p.actor as v
  "DiagnosticReport-request" : for src.request make tgt.basedOn
  "DiagnosticReport-specimen" : for src.specimen make tgt.specimen
  "DiagnosticReport-result" : for src.result make tgt.result
  "DiagnosticReport-imagingStudy" : for src.imagingStudy make tgt.imagingStudy
  "DiagnosticReport-image" : for src.image as vs0 make tgt.image as vt0 then drpImage(vs0, vt0)
  "DiagnosticReport-conclusion" : for src.conclusion make tgt.conclusion
  "DiagnosticReport-codedDiagnosis" : for src.codedDiagnosis make tgt.codedDiagnosis
  "DiagnosticReport-presentedForm" : for src.presentedForm make tgt.presentedForm
endgroup

group drpImage extends BackboneElement
  input src as source
  input tgt as target

  "DiagnosticReport.image-comment" : for src.comment make tgt.comment
  "DiagnosticReport.image-link" : for src.link make tgt.link
endgroup
```

10.2.16.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/DiagnosticReport3to2" = "R3 to R2 Conversion for DiagnosticReport"

uses "http://hl7.org/fhir/StructureDefinition/DiagnosticReport" alias DiagnosticReport as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DiagnosticReport" alias DiagnosticReportR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  DiagnosticReport extends DomainResource
  input src : DiagnosticReport as source
  input tgt : DiagnosticReportR2 as target

  "DiagnosticReport-identifier" : for src.identifier make tgt.identifier
  "DiagnosticReport-status" : for src.status make tgt.status
  "DiagnosticReport-category" : for src.category make tgt.category
  "DiagnosticReport-code" : for src.code make tgt.code
  "DiagnosticReport-subject" : for src.subject make tgt.subject
  "DiagnosticReport-encounter" : for src.context make tgt.encounter
  "DiagnosticReport-effective" : for src.effective make tgt.effective
  "DiagnosticReport-issued" : for src.issued make tgt.issued
  "DiagnosticReport-performer" : for src.performer as v then drpPerformer(v, tgt)
  "DiagnosticReport-request" : for src.basedOn make tgt.request
  "DiagnosticReport-specimen" : for src.specimen make tgt.specimen
  "DiagnosticReport-result" : for src.result make tgt.result
  "DiagnosticReport-imagingStudy" : for src.imagingStudy make tgt.imagingStudy
  "DiagnosticReport-image" : for src.image as vs0 make tgt.image as vt0 then drpImage(vs0, vt0)
  "DiagnosticReport-conclusion" : for src.conclusion make tgt.conclusion
  "DiagnosticReport-codedDiagnosis" : for src.codedDiagnosis make tgt.codedDiagnosis
```

```
  "DiagnosticReport-presentedForm" : for src.presentedForm make tgt.presentedForm
endgroup

group drpPerformer extends BackboneElement
  input src as source
  input tgt as target

  "DiagnosticReport.performer-actor" : for src.actor make tgt.performer

endgroup

group drpImage extends BackboneElement
  input src as source
  input tgt as target

  "DiagnosticReport.image-comment" : for src.comment make tgt.comment
  "DiagnosticReport.image-link" : for src.link make tgt.link
endgroup
```

10.2.16.3 R3 Validation Errors

f001 | ERROR @ DiagnosticReport.contained: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.contained.status': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.contained: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.contained.intent': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.contained.extension: The extension http://hl7.org/fhir/diagnosticorder-r2-marker is unknown, and not allowed here (src = InstanceValidator) ERROR @ DiagnosticReport.basedOn.contained: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.basedOn.contained.status': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.basedOn.contained: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.basedOn.contained.intent': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.basedOn.contained.extension: The extension http://hl7.org/fhir/diagnosticorder-r2-marker is unknown, and not allowed here (src = InstanceValidator)

f202 | ERROR @ DiagnosticReport.contained: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.contained.status': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.contained: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.contained.intent': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.contained: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.contained.code': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.contained.extension: The extension http://hl7.org/fhir/diagnosticorder-r2-marker is unknown, and not allowed here (src = InstanceValidator) ERROR @ DiagnosticReport.basedOn.contained: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.basedOn.contained.status': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.basedOn.contained: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.basedOn.contained.intent': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.basedOn.contained: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.basedOn.contained.code': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.basedOn.contained.extension: The extension http://hl7.org/fhir/diagnosticorder-r2-marker is unknown, and not allowed here (src = InstanceValidator)

micro | ERROR @ DiagnosticReport.contained[26]: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.contained[26].status': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.contained[26]: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.contained[26].intent': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.contained[26]: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.contained[26].code': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.contained[26].extension: The extension http://hl7.org/fhir/diagnosticorder-r2-marker is unknown, and not allowed here (src = InstanceValidator) ERROR @ DiagnosticReport.basedOn.contained: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.basedOn.contained.status': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.basedOn.contained: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.basedOn.contained.intent': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.basedOn.contained: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'DiagnosticReport.basedOn.contained.code': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ DiagnosticReport.basedOn.contained.extension: The extension http://hl7.org/fhir/diagnosticorder-r2-marker is unknown, and not allowed here (src = InstanceValidator)

10.3 Resource ImagingStudy - Content

URL: https://hl7.org/fhir/STU3/imagingstudy.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Imaging Integration Work Group | Maturity Level: 3 | Trial Use | Compartments: Patient

Representation of the content produced in a DICOM imaging study. A study comprises a set of series, each of which includes a set of Service-Object Pair Instances (SOP Instances - images or other data) acquired or produced in a common context. A series is of only one modality (e.g. X-ray, CT, MR, ultrasound), but a study may have multiple series of different modalities.

10.3.1 Scope and Usage

ImagingStudy provides information on a DICOM imaging study, and the series and imaging objects in that study. It also provides information on how to retrieve that information (in a native DICOM format, or in a rendered format, such as JPEG). ImagingStudy is used to make available information about all parts of a single DICOM study.

This resource provides mappings of its elements to DICOM attributes. DICOM attributes are identified by a 32-bit tag, presented in canonical form as two four-digit hexadecimal values within parentheses and separated by a comma, e.g. (0008,103E). The name and value representation (data type) of each attribute can be found in DICOM Part 6 Data Dictionary

. The use of the attributes in the context of information objects, including detailed description of use, can be found in DICOM Part 3 Information Object Definitions . Attributes used in the DICOM query information models, such as "Number of Instances in Study", can be found in DICOM Part 4 Annex C .

ImagingStudy provides access to significant DICOM information, but will only eliminate the need for DICOM query (e.g., QIDO-RS) in the simplest cases. The DICOM instances are not stored in the ImagingStudy resource; use of a DICOM WADO-RS server or other storage mechanism is needed.

Only a single DICOM study may be referenced from one ImagingStudy. In many cases, only one ImagingStudy will reference a particular DICOM study, but this is not required.

10.3.2 Boundaries and Relationships

In contrast to ImagingManifest, this resource represents all of the known imaging objects from a single study. Imaging Manifest represents selected instances from multiple studies for one patient.

ImagingStudy is used for DICOM imaging and associated information. Use Media to track non-DICOM images, video, or audio. Binary can be used to store arbitrary content. DocumentReference allow indexing and retrieval of clinical “documents” with relevant metadata.

This resource is referenced by ChargeItem, ClinicalImpression, DiagnosticReport and ImagingManifest

10.3.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
ImagingStudy | | | DomainResource | A set of images produced in single study (one or more series of references images) Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
uid | ? | 1..1 | oid | Formal DICOM identifier for the study
accession | ? | 0..1 | Identifier | Related workflow identifier ("Accession Number")
identifier | ? | 0..* | Identifier | Other identifiers for the study
availability | ? | 0..1 | code | ONLINE | OFFLINE | NEARLINE | UNAVAILABLE InstanceAvailability (Required)
modalityList | ? | 0..* | Coding | All series modality if actual acquisition modalities Acquisition Modality Codes (Extensible)
patient | ? | 1..1 | Reference(Patient) | Who the images are of
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Originating context
started | ? | 0..1 | dateTime | When the study was started
basedOn | ? | 0..* | Reference(ReferralRequest | CarePlan | ProcedureRequest) | Request fulfilled
referrer | ? | 0..1 | Reference(Practitioner) | Referring physician
interpreter | ? | 0..* | Reference(Practitioner) | Who interpreted images
endpoint | ? | 0..* | Reference(Endpoint) | Study access endpoint
numberOfSeries | ? | 0..1 | unsignedInt | Number of Study Related Series
numberOfInstances | ? | 0..1 | unsignedInt | Number of Study Related Instances
procedureReference | ? | 0..* | Reference(Procedure) | The performed Procedure reference
procedureCode | ? | 0..* | CodeableConcept | The performed procedure code Procedure Codes (SNOMED CT) (Example)
reason | ? | 0..1 | CodeableConcept | Why the study was requested Procedure Reason Codes (Example)
description | ? | 0..1 | string | Institution-generated description
series | ? | 0..* | BackboneElement | Each study has one or more series of instances
uid | ? | 1..1 | oid | Formal DICOM identifier for this series
number | ? | 0..1 | unsignedInt | Numeric identifier of this series
modality | ? | 1..1 | Coding | The modality of the instances in the series Acquisition Modality Codes (Extensible)
description | ? | 0..1 | string | A short human readable summary of the series
numberOfInstances | ? | 0..1 | unsignedInt | Number of Series Related Instances
availability | ? | 0..1 | code | ONLINE | OFFLINE | NEARLINE | UNAVAILABLE InstanceAvailability (Required)
endpoint | ? | 0..* | Reference(Endpoint) | Series access endpoint
bodySite | ? | 0..1 | Coding | Body part examined SNOMED CT Body Structures (Example)
laterality | ? | 0..1 | Coding | Body part laterality Laterality (Example)
started | ? | 0..1 | dateTime | When the series started
performer | ? | 0..* | Reference(Practitioner) | Who performed the series
instance | | 0..* | BackboneElement | A single SOP instance from the series
uid | | 1..1 | oid | Formal DICOM identifier for this instance
number | | 0..1 | unsignedInt | The number of this instance in the series
sopClass | | 1..1 | oid | DICOM class type
title | | 0..1 | string | Description of instance
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImagingStudy xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <uid value="[oid]"/><!-- 1..1 Formal DICOM identifier for the study -->
 <accession><!-- 0..1 Identifier Related workflow identifier ("Accession Number") --></accession>
 <identifier><!-- 0..* Identifier Other identifiers for the study --></identifier>
 <availability value="[code]"/><!-- 0..1 ONLINE | OFFLINE | NEARLINE | UNAVAILABLE -->
 <modalityList><!-- 0..* Coding All series modality if actual acquisition modalities --></modalityList>
 <patient><!-- 1..1 Reference(Patient) Who the images are of --></patient>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Originating context --></context>
 <started value="[dateTime]"/><!-- 0..1 When the study was started -->
 <basedOn><!-- 0..* Reference(ReferralRequest|CarePlan|ProcedureRequest) Request fulfilled --></basedOn>
 <referrer><!-- 0..1 Reference(Practitioner) Referring physician --></referrer>
 <interpreter><!-- 0..* Reference(Practitioner) Who interpreted images --></interpreter>
 <endpoint><!-- 0..* Reference(Endpoint) Study access endpoint --></endpoint>
 <numberOfSeries value="[unsignedInt]"/><!-- 0..1 Number of Study Related Series -->
 <numberOfInstances value="[unsignedInt]"/><!-- 0..1 Number of Study Related Instances -->
 <procedureReference><!-- 0..* Reference(Procedure) The performed Procedure reference --></procedureReference>
 <procedureCode><!-- 0..* CodeableConcept The performed procedure code --></procedureCode>
 <reason><!-- 0..1 CodeableConcept Why the study was requested --></reason>
 <description value="[string]"/><!-- 0..1 Institution-generated description -->
 <series>  <!-- 0..* Each study has one or more series of instances -->
  <uid value="[oid]"/><!-- 1..1 Formal DICOM identifier for this series -->
  <number value="[unsignedInt]"/><!-- 0..1 Numeric identifier of this series -->
  <modality><!-- 1..1 Coding The modality of the instances in the series --></modality>
  <description value="[string]"/><!-- 0..1 A short human readable summary of the series -->
  <numberOfInstances value="[unsignedInt]"/><!-- 0..1 Number of Series Related Instances -->
  <availability value="[code]"/><!-- 0..1 ONLINE | OFFLINE | NEARLINE | UNAVAILABLE -->
  <endpoint><!-- 0..* Reference(Endpoint) Series access endpoint --></endpoint>
  <bodySite><!-- 0..1 Coding Body part examined --></bodySite>
  <laterality><!-- 0..1 Coding Body part laterality --></laterality>
  <started value="[dateTime]"/><!-- 0..1 When the series started -->
  <performer><!-- 0..* Reference(Practitioner) Who performed the series --></performer>
  <instance>  <!-- 0..* A single SOP instance from the series -->
   <uid value="[oid]"/><!-- 1..1 Formal DICOM identifier for this instance -->
   <number value="[unsignedInt]"/><!-- 0..1 The number of this instance in the series -->
   <sopClass value="[oid]"/><!-- 1..1 DICOM class type -->
   <title value="[string]"/><!-- 0..1 Description of instance -->
  </instance>
 </series>
</ImagingStudy>
```

JSON Template

```
{
  "resourceType" : "ImagingStudy",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "uid" : "<oid>", // R!  Formal DICOM identifier for the study
  "accession" : { Identifier }, // Related workflow identifier ("Accession Number")
  "identifier" : [{ Identifier }], // Other identifiers for the study
  "availability" : "<code>", // ONLINE | OFFLINE | NEARLINE | UNAVAILABLE
  "modalityList" : [{ Coding }], // All series modality if actual acquisition modalities
  "patient" : { Reference(Patient) }, // R!  Who the images are of
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Originating context
  "started" : "<dateTime>", // When the study was started
  "basedOn" : [{ Reference(ReferralRequest|CarePlan|ProcedureRequest) }], // Request fulfilled
  "referrer" : { Reference(Practitioner) }, // Referring physician
  "interpreter" : [{ Reference(Practitioner) }], // Who interpreted images
  "endpoint" : [{ Reference(Endpoint) }], // Study access endpoint
  "numberOfSeries" : "<unsignedInt>", // Number of Study Related Series
  "numberOfInstances" : "<unsignedInt>", // Number of Study Related Instances
  "procedureReference" : [{ Reference(Procedure) }], // The performed Procedure reference
  "procedureCode" : [{ CodeableConcept }], // The performed procedure code
  "reason" : { CodeableConcept }, // Why the study was requested
  "description" : "<string>", // Institution-generated description
  "series" : [{ // Each study has one or more series of instances
    "uid" : "<oid>", // R!  Formal DICOM identifier for this series
    "number" : "<unsignedInt>", // Numeric identifier of this series
    "modality" : { Coding }, // R!  The modality of the instances in the series
    "description" : "<string>", // A short human readable summary of the series
    "numberOfInstances" : "<unsignedInt>", // Number of Series Related Instances
    "availability" : "<code>", // ONLINE | OFFLINE | NEARLINE | UNAVAILABLE
    "endpoint" : [{ Reference(Endpoint) }], // Series access endpoint
    "bodySite" : { Coding }, // Body part examined
    "laterality" : { Coding }, // Body part laterality
    "started" : "<dateTime>", // When the series started
    "performer" : [{ Reference(Practitioner) }], // Who performed the series
    "instance" : [{ // A single SOP instance from the series
      "uid" : "<oid>", // R!  Formal DICOM identifier for this instance
      "number" : "<unsignedInt>", // The number of this instance in the series
      "sopClass" : "<oid>", // R!  DICOM class type
      "title" : "<string>" // Description of instance
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ImagingStudy;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImagingStudy.uid [ oid ]; # 1..1 Formal DICOM identifier for the study
  fhir:ImagingStudy.accession [ Identifier ]; # 0..1 Related workflow identifier ("Accession Number")
  fhir:ImagingStudy.identifier [ Identifier ], ... ; # 0..* Other identifiers for the study
  fhir:ImagingStudy.availability [ code ]; # 0..1 ONLINE | OFFLINE | NEARLINE | UNAVAILABLE
  fhir:ImagingStudy.modalityList [ Coding ], ... ; # 0..* All series modality if actual acquisition modalities
  fhir:ImagingStudy.patient [ Reference(Patient) ]; # 1..1 Who the images are of
  fhir:ImagingStudy.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Originating context
  fhir:ImagingStudy.started [ dateTime ]; # 0..1 When the study was started
  fhir:ImagingStudy.basedOn [ Reference(ReferralRequest|CarePlan|ProcedureRequest) ], ... ; # 0..* Request fulfilled
  fhir:ImagingStudy.referrer [ Reference(Practitioner) ]; # 0..1 Referring physician
  fhir:ImagingStudy.interpreter [ Reference(Practitioner) ], ... ; # 0..* Who interpreted images
  fhir:ImagingStudy.endpoint [ Reference(Endpoint) ], ... ; # 0..* Study access endpoint
  fhir:ImagingStudy.numberOfSeries [ unsignedInt ]; # 0..1 Number of Study Related Series
  fhir:ImagingStudy.numberOfInstances [ unsignedInt ]; # 0..1 Number of Study Related Instances
  fhir:ImagingStudy.procedureReference [ Reference(Procedure) ], ... ; # 0..* The performed Procedure reference
  fhir:ImagingStudy.procedureCode [ CodeableConcept ], ... ; # 0..* The performed procedure code
  fhir:ImagingStudy.reason [ CodeableConcept ]; # 0..1 Why the study was requested
  fhir:ImagingStudy.description [ string ]; # 0..1 Institution-generated description
  fhir:ImagingStudy.series [ # 0..* Each study has one or more series of instances
    fhir:ImagingStudy.series.uid [ oid ]; # 1..1 Formal DICOM identifier for this series
    fhir:ImagingStudy.series.number [ unsignedInt ]; # 0..1 Numeric identifier of this series
    fhir:ImagingStudy.series.modality [ Coding ]; # 1..1 The modality of the instances in the series
    fhir:ImagingStudy.series.description [ string ]; # 0..1 A short human readable summary of the series
    fhir:ImagingStudy.series.numberOfInstances [ unsignedInt ]; # 0..1 Number of Series Related Instances
    fhir:ImagingStudy.series.availability [ code ]; # 0..1 ONLINE | OFFLINE | NEARLINE | UNAVAILABLE
    fhir:ImagingStudy.series.endpoint [ Reference(Endpoint) ], ... ; # 0..* Series access endpoint
    fhir:ImagingStudy.series.bodySite [ Coding ]; # 0..1 Body part examined
    fhir:ImagingStudy.series.laterality [ Coding ]; # 0..1 Body part laterality
    fhir:ImagingStudy.series.started [ dateTime ]; # 0..1 When the series started
    fhir:ImagingStudy.series.performer [ Reference(Practitioner) ], ... ; # 0..* Who performed the series
    fhir:ImagingStudy.series.instance [ # 0..* A single SOP instance from the series
      fhir:ImagingStudy.series.instance.uid [ oid ]; # 1..1 Formal DICOM identifier for this instance
      fhir:ImagingStudy.series.instance.number [ unsignedInt ]; # 0..1 The number of this instance in the series
      fhir:ImagingStudy.series.instance.sopClass [ oid ]; # 1..1 DICOM class type
      fhir:ImagingStudy.series.instance.title [ string ]; # 0..1 Description of instance
    ], ...;
  ], ...;
]
```

Changes since DSTU2

ImagingStudy
ImagingStudy.context | Renamed from order to context Max Cardinality changed from * to 1 Remove Reference(DiagnosticOrder), Add Reference(Encounter), Add Reference(EpisodeOfCare)
ImagingStudy.basedOn | Added Element
ImagingStudy.interpreter | Max Cardinality changed from 1 to *
ImagingStudy.endpoint | Added Element
ImagingStudy.numberOfSeries | Min Cardinality changed from 1 to 0
ImagingStudy.numberOfInstances | Min Cardinality changed from 1 to 0
ImagingStudy.procedureReference | Renamed from procedure to procedureReference
ImagingStudy.procedureCode | Added Element
ImagingStudy.reason | Added Element
ImagingStudy.series.numberOfInstances | Min Cardinality changed from 1 to 0
ImagingStudy.series.endpoint | Added Element
ImagingStudy.series.performer | Added Element
ImagingStudy.url | deleted
ImagingStudy.series.url | deleted
ImagingStudy.series.instance.type | deleted
ImagingStudy.series.instance.content | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. 2 fail round-trip testing and 1 r3 resources are invalid (2 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints
ImagingStudy | | | DomainResource | A set of images produced in single study (one or more series of references images) Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
uid | ? | 1..1 | oid | Formal DICOM identifier for the study
accession | ? | 0..1 | Identifier | Related workflow identifier ("Accession Number")
identifier | ? | 0..* | Identifier | Other identifiers for the study
availability | ? | 0..1 | code | ONLINE | OFFLINE | NEARLINE | UNAVAILABLE InstanceAvailability (Required)

modalityList | ? | 0..* | Coding | All series modality if actual acquisition modalities Acquisition Modality Codes (Extensible)
patient | ? | 1..1 | Reference(Patient) | Who the images are of
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Originating context
started | ? | 0..1 | dateTime | When the study was started
basedOn | ? | 0..* | Reference(ReferralRequest | CarePlan | ProcedureRequest) | Request fulfilled
referrer | ? | 0..1 | Reference(Practitioner) | Referring physician
interpreter | ? | 0..* | Reference(Practitioner) | Who interpreted images
endpoint | ? | 0..* | Reference(Endpoint) | Study access endpoint
numberOfSeries | ? | 0..1 | unsignedInt | Number of Study Related Series
numberOfInstances | ? | 0..1 | unsignedInt | Number of Study Related Instances
procedureReference | ? | 0..* | Reference(Procedure) | The performed Procedure reference
procedureCode | ? | 0..* | CodeableConcept | The performed procedure code Procedure Codes (SNOMED CT) (Example)
reason | ? | 0..1 | CodeableConcept | Why the study was requested Procedure Reason Codes (Example)
description | ? | 0..1 | string | Institution-generated description
series | ? | 0..* | BackboneElement | Each study has one or more series of instances
uid | ? | 1..1 | oid | Formal DICOM identifier for this series
number | ? | 0..1 | unsignedInt | Numeric identifier of this series
modality | ? | 1..1 | Coding | The modality of the instances in the series Acquisition Modality Codes (Extensible)
description | ? | 0..1 | string | A short human readable summary of the series
numberOfInstances | ? | 0..1 | unsignedInt | Number of Series Related Instances
availability | ? | 0..1 | code | ONLINE | OFFLINE | NEARLINE | UNAVAILABLE InstanceAvailability (Required)
endpoint | ? | 0..* | Reference(Endpoint) | Series access endpoint
bodySite | ? | 0..1 | Coding | Body part examined SNOMED CT Body Structures (Example)
laterality | ? | 0..1 | Coding | Body part laterality Laterality (Example)
started | ? | 0..1 | dateTime | When the series started
performer | ? | 0..* | Reference(Practitioner) | Who performed the series
instance | | 0..* | BackboneElement | A single SOP instance from the series
uid | | 1..1 | oid | Formal DICOM identifier for this instance
number | | 0..1 | unsignedInt | The number of this instance in the series
sopClass | | 1..1 | oid | DICOM class type
title | | 0..1 | string | Description of instance
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImagingStudy xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <uid value="[oid]"/><!-- 1..1 Formal DICOM identifier for the study -->
 <accession><!-- 0..1 Identifier Related workflow identifier ("Accession Number") --></accession>
 <identifier><!-- 0..* Identifier Other identifiers for the study --></identifier>
 <availability value="[code]"/><!-- 0..1 ONLINE | OFFLINE | NEARLINE | UNAVAILABLE -->
 <modalityList><!-- 0..* Coding All series modality if actual acquisition modalities --></modalityList>
 <patient><!-- 1..1 Reference(Patient) Who the images are of --></patient>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Originating context --></context>
 <started value="[dateTime]"/><!-- 0..1 When the study was started -->
 <basedOn><!-- 0..* Reference(ReferralRequest|CarePlan|ProcedureRequest) Request fulfilled --></basedOn>
 <referrer><!-- 0..1 Reference(Practitioner) Referring physician --></referrer>
 <interpreter><!-- 0..* Reference(Practitioner) Who interpreted images --></interpreter>
 <endpoint><!-- 0..* Reference(Endpoint) Study access endpoint --></endpoint>
 <numberOfSeries value="[unsignedInt]"/><!-- 0..1 Number of Study Related Series -->
 <numberOfInstances value="[unsignedInt]"/><!-- 0..1 Number of Study Related Instances -->
 <procedureReference><!-- 0..* Reference(Procedure) The performed Procedure reference --></procedureReference>
 <procedureCode><!-- 0..* CodeableConcept The performed procedure code --></procedureCode>
 <reason><!-- 0..1 CodeableConcept Why the study was requested --></reason>
 <description value="[string]"/><!-- 0..1 Institution-generated description -->
 <series>  <!-- 0..* Each study has one or more series of instances -->
  <uid value="[oid]"/><!-- 1..1 Formal DICOM identifier for this series -->
  <number value="[unsignedInt]"/><!-- 0..1 Numeric identifier of this series -->
  <modality><!-- 1..1 Coding The modality of the instances in the series --></modality>
  <description value="[string]"/><!-- 0..1 A short human readable summary of the series -->
  <numberOfInstances value="[unsignedInt]"/><!-- 0..1 Number of Series Related Instances -->
  <availability value="[code]"/><!-- 0..1 ONLINE | OFFLINE | NEARLINE | UNAVAILABLE -->
  <endpoint><!-- 0..* Reference(Endpoint) Series access endpoint --></endpoint>
  <bodySite><!-- 0..1 Coding Body part examined --></bodySite>
  <laterality><!-- 0..1 Coding Body part laterality --></laterality>
  <started value="[dateTime]"/><!-- 0..1 When the series started -->
  <performer><!-- 0..* Reference(Practitioner) Who performed the series --></performer>
  <instance>  <!-- 0..* A single SOP instance from the series -->
   <uid value="[oid]"/><!-- 1..1 Formal DICOM identifier for this instance -->
   <number value="[unsignedInt]"/><!-- 0..1 The number of this instance in the series -->
   <sopClass value="[oid]"/><!-- 1..1 DICOM class type -->
   <title value="[string]"/><!-- 0..1 Description of instance -->
  </instance>
 </series>
</ImagingStudy>
```

JSON Template

```
{
  "resourceType" : "ImagingStudy",
  // from Resource: id, meta, implicitRules, and language
```

```
  // from DomainResource: text, contained, extension, and modifierExtension
  "uid" : "<oid>", // R!  Formal DICOM identifier for the study
  "accession" : { Identifier }, // Related workflow identifier ("Accession Number")
  "identifier" : [{ Identifier }], // Other identifiers for the study
  "availability" : "<code>", // ONLINE | OFFLINE | NEARLINE | UNAVAILABLE
  "modalityList" : [{ Coding }], // All series modality if actual acquisition modalities
  "patient" : { Reference(Patient) }, // R!  Who the images are of
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Originating context
  "started" : "<dateTime>", // When the study was started
  "basedOn" : [{ Reference(ReferralRequest|CarePlan|ProcedureRequest) }], // Request fulfilled
  "referrer" : { Reference(Practitioner) }, // Referring physician
  "interpreter" : [{ Reference(Practitioner) }], // Who interpreted images
  "endpoint" : [{ Reference(Endpoint) }], // Study access endpoint
  "numberOfSeries" : "<unsignedInt>", // Number of Study Related Series
  "numberOfInstances" : "<unsignedInt>", // Number of Study Related Instances
  "procedureReference" : [{ Reference(Procedure) }], // The performed Procedure reference
  "procedureCode" : [{ CodeableConcept }], // The performed procedure code
  "reason" : { CodeableConcept }, // Why the study was requested
  "description" : "<string>", // Institution-generated description
  "series" : [{ // Each study has one or more series of instances
    "uid" : "<oid>", // R!  Formal DICOM identifier for this series
    "number" : "<unsignedInt>", // Numeric identifier of this series
    "modality" : { Coding }, // R!  The modality of the instances in the series
    "description" : "<string>", // A short human readable summary of the series
    "numberOfInstances" : "<unsignedInt>", // Number of Series Related Instances
    "availability" : "<code>", // ONLINE | OFFLINE | NEARLINE | UNAVAILABLE
    "endpoint" : [{ Reference(Endpoint) }], // Series access endpoint
    "bodySite" : { Coding }, // Body part examined
    "laterality" : { Coding }, // Body part laterality
    "started" : "<dateTime>", // When the series started
    "performer" : [{ Reference(Practitioner) }], // Who performed the series
    "instance" : [{ // A single SOP instance from the series
      "uid" : "<oid>", // R!  Formal DICOM identifier for this instance
      "number" : "<unsignedInt>", // The number of this instance in the series
      "sopClass" : "<oid>", // R!  DICOM class type
      "title" : "<string>" // Description of instance
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ImagingStudy;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImagingStudy.uid [ oid ]; # 1..1 Formal DICOM identifier for the study
  fhir:ImagingStudy.accession [ Identifier ]; # 0..1 Related workflow identifier ("Accession Number")
  fhir:ImagingStudy.identifier [ Identifier ], ... ; # 0..* Other identifiers for the study
  fhir:ImagingStudy.availability [ code ]; # 0..1 ONLINE | OFFLINE | NEARLINE | UNAVAILABLE
  fhir:ImagingStudy.modalityList [ Coding ], ... ; # 0..* All series modality if actual acquisition modalities
  fhir:ImagingStudy.patient [ Reference(Patient) ]; # 1..1 Who the images are of
  fhir:ImagingStudy.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Originating context
  fhir:ImagingStudy.started [ dateTime ]; # 0..1 When the study was started
  fhir:ImagingStudy.basedOn [ Reference(ReferralRequest|CarePlan|ProcedureRequest) ], ... ; # 0..* Request fulfilled
  fhir:ImagingStudy.referrer [ Reference(Practitioner) ]; # 0..1 Referring physician
  fhir:ImagingStudy.interpreter [ Reference(Practitioner) ], ... ; # 0..* Who interpreted images
  fhir:ImagingStudy.endpoint [ Reference(Endpoint) ], ... ; # 0..* Study access endpoint
  fhir:ImagingStudy.numberOfSeries [ unsignedInt ]; # 0..1 Number of Study Related Series
  fhir:ImagingStudy.numberOfInstances [ unsignedInt ]; # 0..1 Number of Study Related Instances
  fhir:ImagingStudy.procedureReference [ Reference(Procedure) ], ... ; # 0..* The performed Procedure reference
  fhir:ImagingStudy.procedureCode [ CodeableConcept ], ... ; # 0..* The performed procedure code
  fhir:ImagingStudy.reason [ CodeableConcept ]; # 0..1 Why the study was requested
  fhir:ImagingStudy.description [ string ]; # 0..1 Institution-generated description
  fhir:ImagingStudy.series [ # 0..* Each study has one or more series of instances
    fhir:ImagingStudy.series.uid [ oid ]; # 1..1 Formal DICOM identifier for this series
    fhir:ImagingStudy.series.number [ unsignedInt ]; # 0..1 Numeric identifier of this series
    fhir:ImagingStudy.series.modality [ Coding ]; # 1..1 The modality of the instances in the series
    fhir:ImagingStudy.series.description [ string ]; # 0..1 A short human readable summary of the series
    fhir:ImagingStudy.series.numberOfInstances [ unsignedInt ]; # 0..1 Number of Series Related Instances
    fhir:ImagingStudy.series.availability [ code ]; # 0..1 ONLINE | OFFLINE | NEARLINE | UNAVAILABLE
    fhir:ImagingStudy.series.endpoint [ Reference(Endpoint) ], ... ; # 0..* Series access endpoint
    fhir:ImagingStudy.series.bodySite [ Coding ]; # 0..1 Body part examined
    fhir:ImagingStudy.series.laterality [ Coding ]; # 0..1 Body part laterality
    fhir:ImagingStudy.series.started [ dateTime ]; # 0..1 When the series started
    fhir:ImagingStudy.series.performer [ Reference(Practitioner) ], ... ; # 0..* Who performed the series
    fhir:ImagingStudy.series.instance [ # 0..* A single SOP instance from the series
      fhir:ImagingStudy.series.instance.uid [ oid ]; # 1..1 Formal DICOM identifier for this instance
      fhir:ImagingStudy.series.instance.number [ unsignedInt ]; # 0..1 The number of this instance in the series
      fhir:ImagingStudy.series.instance.sopClass [ oid ]; # 1..1 DICOM class type
      fhir:ImagingStudy.series.instance.title [ string ]; # 0..1 Description of instance
    ], ...;
  ], ...;
]
```

Changes since DSTU2

ImagingStudy
ImagingStudy.context | Renamed from order to context Max Cardinality changed from * to 1 Remove Reference(DiagnosticOrder), Add Reference(Encounter), Add Reference(EpisodeOfCare)
ImagingStudy.basedOn | Added Element
ImagingStudy.interpreter | Max Cardinality changed from 1 to *
ImagingStudy.endpoint | Added Element
ImagingStudy.numberOfSeries | Min Cardinality changed from 1 to 0
ImagingStudy.numberOfInstances | Min Cardinality changed from 1 to 0
ImagingStudy.procedureReference | Renamed from procedure to procedureReference
ImagingStudy.procedureCode | Added Element
ImagingStudy.reason | Added Element
ImagingStudy.series.numberOfInstances | Min Cardinality changed from 1 to 0
ImagingStudy.series.endpoint | Added Element
ImagingStudy.series.performer | Added Element
ImagingStudy.url | deleted
ImagingStudy.series.url | deleted
ImagingStudy.series.instance.type | deleted
ImagingStudy.series.instance.content | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. 2 fail round-trip testing and 1 r3 resources are invalid (2 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

10.3.3.1 Terminology Bindings

Path | Definition | Type | Reference
ImagingStudy.availability ImagingStudy.series.availability | Availability of the resource | Required | InstanceAvailability
ImagingStudy.modalityList ImagingStudy.series.modality | Type of acquired data in the instance | Extensible | Acquisition Modality Codes
ImagingStudy.procedureCode | The performed procedure type | Example | Procedure Codes (SNOMED CT)
ImagingStudy.reason | The reason for the study | Example | Procedure Reason Codes
ImagingStudy.series.bodySite | Codes describing anatomical locations. May include laterality. | Example | SNOMED CT Body Structures
ImagingStudy.series.laterality | Codes describing body site laterality (left, right, etc.). | Example | Laterality

10.3.3.2 Implementation Notes

A referenced DICOM SOP instance could be:

* A single- or multi-frame, still or video image captured by a variety of imaging modalities, such as X-ray, MR, and ultrasound;

* A set of various presentation parameters, including annotation and markup;

* A set of measurements or a report, including radiation dose report and CAD analysis;

* An encapsulated PDF or CDA document;

* A list of instances, such as key “of interest” images, or instances to be “deleted”; or

* Other DICOM content.

UID values follow the FHIR convention of expressing UIDs as URNs. For example, the DICOM Study Instance UID of 1.2.250.1.59.40211.12345678.678910 is expressed as “urn:oid:1.2.250.1.59.40211.12345678.678910”.

The ImagingManifest.study.endpoint elements and ImagingManifest.study.series.endpoint elements indicate network services that can be used to access the studies, series, or instances; for example, a DICOM WADO-RS server. An ImagingManifest.study.series.endpoint of a particular Endpoint.connectionType provides that service for that series, and all contained instances. An ImagingManifest.study.endpoint of a particularconnection type provides that service for all series in that study that do not have a specified Endpoint of that type, and their contained instances. That is, an ImagingManifest.study.series.endpoint overrides a ImagingManifest.study.endpoint of the same connection type. (Since each study, or individual series of a study can be stored on different imaging archive servers, per-series endpoints are required. For the identified services and use cases, all instances within a series would be stored together, and thus instance-level endpoints are not defined.)

Different Endpoint connection types may have different capabilities, protocols or requirements; and the specified Endpoint.url may require manipulation. For the details on use of imaging-related Endpoint connection types, See below for details.

10.3.3.2.1 WADO-RS

An Endpoint.connectionType of code dicom-wado-rsi, system http://hl7.org/fhir/endpoint-connection-type, identifies a DICOM WADO-RS service. The Endpoint.address identifies the HTTP(S) service base url. That is, only the scheme, authority and path are included. Sub-services, such as study, shall not be specified. The path shall not contain a trailing slash.

The DICOM WADO-RS (Web Access to DICOM Objects, RESTful mode) service uses a RESTful approach to instance retrieval. This service allows for retrieval of native DICOM SOP instances, or instances “rendered” into other formats, including JPEG and MPEG. The media type of a response is specified by the request Accept header (preferred); or, by the accept query parameters. Supported media types depend on the capabilities of the WADO-RS server and the classification of the instance as “single frame,” “multi-frame,” “video,” “text,” or “other.” The WADO-RS service also allows retrieval of study or

series level information.

The path to retrieve a DICOM instance is constructed by appending the appropriate sub-resource paths to the Endpoint.address value.

For example, a native DICOM PS3.10 instance file can be retrieved (if consistent with the Accept header) by performing a GET on a URL constructed from a Endpoint.address of “https://pacs.hospital.org/wado-rs”, the study.uid value of “urn:oid:1.2.250.1.59.40211.12345678.678910”, study.series.uid value of “urn:oid:1.2.250.1.59.40211.789001276.14556172.67789”, and study.series.instance.uid value of “urn:oid:1.2.250.1.59.40211.2678810.87991027.899772.2”:

```
https://pacs.hospital.org/wado-rs/studies/1.2.250.1.59.40211.12345678.678910/series/1.2.250.1.59.40211.789001276.1455617
    2.67789/instances/1.2.250.1.59.40211.2678810.87991027.899772.2
```

Query parameters on the "rendered" sub-resource can control other aspects of the rendering including: the rendered dimensions, the quality (compression ratio), the region of interest to render, the brightness/contrast (window center/width) adjustments, and whether to “burn” patient or study demographics into the rendered result. Specific frames of a multi-frame instance may be retrieved using the frames sub-resource.

For example, provided the Accept header indicates a preference for image/jpeg, the example above can be extended with parameters that cause a JPEG thumbnail (100 columns by 100 rows) of a region extending from the top-left corner of the original image, across 1000 and down 3000 pixels, to be retrieved (additional sub-resource and parameters emphasized):

```
https://pacs.hospital.org/wado-rs/studies/1.2.250.1.59.40211.12345678.678910/series/1.2.250.1.59.40211.789001276.1455617
    2.67789/instances/1.2.250.1.59.40211.2678810.87991027.899772.2/rendered?viewport=100,100,0,0,1000,3000
```

For further details on DICOM WADO-RS capabilities including additional rendering parameters, see DICOM PS 3.18 .

10.3.3.2.2 WADO-URI

An Endpoint.connectionType of code dicom-wado-uri, system http://hl7.org/fhir/endpoint-connection-type, identifies a DICOM WADO-URI service. The Endpoint.address identifies the HTTP(S) service base url. That is, only the scheme, authority and path are included. Neither a quetstion mark (?) nor any query parameters shall be included.

The DICOM WADO-URI (Web Access to DICOM Objects, URI mode) service uses HTTP query parameter syntax. This service allows for retrieval of native DICOM instances, or instances “rendered” into other formats, including JPEG and MPEG. The media type of a response is specified by the request Accept header (preferred); or, by the contentType query parameter. Supported media types depend on the classification of the instance as “single frame,” “multi-frame,” “video,” “text,” or “other.”

The query to retrieve a DICOM instance is constructed by appending the appropriate query parameters to the Endpoint.address.url.

For example, a native DICOM PS3.10 instance file can be retrieved (if consistent with the Accept header) by performing a GET on a URL constructed from a Endpoint.address.url of “https://pacs.hospital.org/wado-uri”, the study.uid value of “urn:oid:1.2.250.1.59.40211.12345678.678910”, study.series.uid value of “urn:oid:1.2.250.1.59.40211.789001276.14556172.67789”, and study.series.instance.uid value of “urn:oid:1.2.250.1.59.40211.2678810.87991027.899772.2”:

```
https://pacs.hospital.org/wado-uri?requestType=WADO&studyUID=1.2.250.1.59.40211.12345678.678910&seriesUID=1.2.250.1.59.4
    0211.789001276.14556172.67789&objectUID=1.2.250.1.59.40211.2678810.87991027.899772.2
```

Additional query parameters can control other aspects of the rendering including rendered dimensions, quality (compression ratio), the region of interest within the image to render, brightness/contrast (window center/width) adjustments, whether to “burn” patient or study demographics into the rendered result, and which frame of a multi-frame instance to retrieve.

For example, provided the Accept header indicates a preference for image/jpeg, the example above can be extended with parameters that cause a JPEG thumbnail (100 columns by 100 rows) of the left half of the image to be retrieved (additional parameters emphasized):

```
https://pacs.hospital.org/wado-uri?requestType=WADO&studyUID=1.2.250.1.59.40211.12345678.678910&seriesUID=1.2.250.1.59.4
    0211.789001276.14556172.67789&objectUID=1.2.250.1.59.40211.2678810.87991027.899772.2&rows=100&columns=100&region=0,0
    ,0.5,1
```

For further details on DICOM WADO-URI capabilities including additional rendering parameters, see DICOM PS 3.18 .

10.3.3.2.3 IID

An Endpoint.connectionType of code ihe-iid, system http://hl7.org/fhir/endpoint-connection-type, identifies an IHE Invoke Image Display (IID) service. The Endpoint.address identifies the HTTP(S) service base url. That is, only the scheme, authority and path are included. Neither the question mark (“?”) nor any query parameters shall be included.

The IHE Invoke Image Display (IID) service provides a standardized mechanism to launch a viewer in a particular study context. (IID also supports invoking a particular patient context, but that is not profiled here.) An IID-type Endpoint should be used only at the study level. As well as invoking the viewer on a particular study, query parameters can request particular viewer capabilities, image quality, and more.

To launch a viewer, append the appropriate query parameters to Endpoint.address value.

For example, given an Endpoint.address of https://pacs.hospital.org/IHEInvokeImageDisplay, to invoke a diagnostic quality viewer on the study with study.uid value of “urn:uri:1.2.250.1.59.40211.12345678.678910”, the following URL would be constructed:

```
https://pacs.hospital.org/IHEInvokeImageDisplay?requestType=STUDY&studyUID=1.2.250.1.59.40211.12345678.678910&diagnostic
    Quality=true
```

For further details on IHE Invoke Image Display capabilities including additional parameters, see the IHE Technical Frameworks , or the introduction on the IHE IID Profile Wiki .

10.3.3.3 Use Case

10.3.3.3.1 EHR access to imaging studies

Amy, a family physician, would like to see a list of available studies for her patient, Alex. Her EHR client makes a FHIR call for all ImagingStudy objects available for Alex. In the response, she is able to see the study date, procedure, modality, and accession number, for each study returned. There is enough information provided in the response to obtain a thumbnail via a WADO-RS call, or to launch a viewer using an IHE Radiology - Invoke Image Display (IID) profile call using the url elements found in the ImagingStudy.

10.3.3.3.2 Comprehensive Imaging Scheduled Workflow

Joe Angina complains of shortness of breath and occasional chest pain to his primary care physician, Dr. Pat Down at Local MultiClinic, who orders a stress echocardiogram; the order is created as a FHIR Task resource to manage the workflow, with a link to a ProcedureRequest resource with the details of the request. The order is scheduled and assigned to cardiologist Dr. Art Skann, also at Local MultiClinic.

On the scheduled day of the exam, Joe arrives at the echo lab to meet with Dr. Skann and have the study done. Dr. Skann's workstation shows the daily list of Task, and he follows the link to retrieve the ProcedureRequest. (He may follow the links through the referenced Patient resource to access Joe's electronic medical record, but that is not the concern of this storyboard.)

The Task and ProcedureRequest has been transcoded to a DICOM Modality Worklist Scheduled Procedure Step, and in the echo lab the equipment has downloaded the Modality Worklist. The study is performed, and the acquired images and sonographer's preliminary measurements are stored in the Local MultiClinic Picture Archiving and Communication System (PACS). The PACS creates an ImagingStudy resource for each study it manages.

Dr. Skann interprets the study on a PACS workstation, and he selects two key image frames to be included in the diagnostic report; this selection is stored back to the PACS as a DICOM Key Object Selection with the title "For Report Attachment", and the PACS makes it available (transcodes it) as a FHIR ImagingManifest resource. Dr. Skann dictates the report using a structured data entry report writing program, including a recommendation for a cardiac catheterization procedure, and signs it. The report writing program formats the report as a CDA document, retrieves the ImagingManifest resource, and inserts the referenced key images into the report.

Dr. Down meets again with Joe, and they review the results of the stress test. Joe has a question about the findings that the key images in the report do not show, so Dr. Down uses the Local MultiClinic EMR to query the PACS for the full ImagingStudy resource, and uses the references there to open an image display for the full study. Joe agrees to proceed to catheterization, and Dr. Down sends a referral to the Ginormous University Hospital cath department, and triggers the PACS to share the echo study through the Metropolitan Health Information Exchange.

The PACS creates a manifest of the study as an ImagingManifest resource, which includes all the images but excludes the sonographer's preliminary measurements (which as a matter of policy are not shared outside the Local MultiClinic). The manifest is published to the Metro HIE. (In accordance with IHE XDS-I , the images themselves are not directly published to the HIE, but available for on-demand retrieval from the PACS.)

At Ginormous Hospital, Dr. Cora Plummer receives the cath referral, and looks up the study in the Metro HIE registry. She retrieves the study manifest ImagingManifest, and uses it to access the shared images, which she uses to prepare for the cath procedure.

10.3.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
accession | token | The accession identifier for the study | ImagingStudy.accession
basedon | reference | The order for the image | ImagingStudy.basedOn (ReferralRequest, CarePlan, ProcedureRequest)
bodysite | token | The body site studied | ImagingStudy.series.bodySite
context | reference | The context of the study | ImagingStudy.context (EpisodeOfCare, Encounter)
dicom-class | uri | The type of the instance | ImagingStudy.series.instance.sopClass
endpoint | reference | The endpoint for te study or series | ImagingStudy.endpoint | ImagingStudy.series.endpoint (Endpoint)
identifier | token | Other identifiers for the Study | ImagingStudy.identifier | 26 Resources
modality | token | The modality of the series | ImagingStudy.series.modality
patient | reference | Who the study is about | ImagingStudy.patient (Patient) | 31 Resources
performer | reference | The person who performed the study | ImagingStudy.series.performer (Practitioner)
reason | token | The reason for the study | ImagingStudy.reason
series | uri | The identifier of the series of images | ImagingStudy.series.uid
started | date | When the study was started | ImagingStudy.started
study | uri | The study identifier for the image | ImagingStudy.uid
uid | uri | The instance unique identifier | ImagingStudy.series.instance.uid

10.3.5 Resource ImagingStudy - Examples

URL: https://hl7.org/fhir/STU3/imagingstudy-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.3.5 Resource ImagingStudy - Examples

Imaging Integration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Example List:

Example of imagingstudy | example | XML | JSON | Turtle
Example XR imaging study with full elements | example-xr | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.3.6 Resource ImagingStudy - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/imagingstudy-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.3.6 Resource ImagingStudy - Detailed Descriptions

Imaging Integration Work Group | Maturity Level: 3 | Trial Use | Compartments: Patient

Detailed Descriptions for the elements in the ImagingStudy resource.

ImagingStudy
Definition | Representation of the content produced in a DICOM imaging study. A study comprises a set of series, each of which includes a set of Service-Object Pair Instances (SOP Instances - images or other data) acquired or produced in a common context. A series is of only one modality (e.g. X-ray, CT, MR, ultrasound), but a study may have multiple series of different modalities.
Control | 1..1
ImagingStudy.uid
Definition | Formal identifier for the study.
Control | 1..1
Type | oid
Alternate Names | StudyInstanceUID
Summary | true
ImagingStudy.accession
Definition | Accession Number is an identifier related to some aspect of imaging workflow and data management. Usage may vary across different institutions. See for instance IHE Radiology Technical Framework Volume 1 Appendix A .
Control | 0..1
Type | Identifier
Alternate Names | AccessionNumber
Summary | true
ImagingStudy.identifier
Definition | Other identifiers for the study.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Alternate Names | StudyID
Summary | true
ImagingStudy.availability
Definition | Availability of study (online, offline, or nearline).
Control | 0..1
Terminology Binding | InstanceAvailability (Required)
Type | code
Alternate Names | InstanceAvailability
Summary | true
ImagingStudy.modalityList
Definition | A list of all the Series.ImageModality values that are actual acquisition modalities, i.e. those in the DICOM Context Group 29 (value set OID 1.2.840.10008.6.1.19).
Control | 0..*
Terminology Binding | Acquisition Modality Codes (Extensible)
Type | Coding
Alternate Names | ModalitiesInStudy
Summary | true
ImagingStudy.patient
Definition | The patient imaged in the study.
Control | 1..1
Type | Reference(Patient)
Summary | true
ImagingStudy.context
Definition | The encounter or episode at which the request is initiated.
Control | 0..1

Type | Reference(Encounter | EpisodeOfCare)
Summary | true
ImagingStudy.started
Definition | Date and time the study started.
Control | 0..1
Type | dateTime
Alternate Names | StudyDate; StudyTime
Summary | true
ImagingStudy.basedOn
Definition | A list of the diagnostic requests that resulted in this imaging study being performed.
Control | 0..*
Type | Reference(ReferralRequest | CarePlan | ProcedureRequest)
Requirements | To support grouped procedures (one imaging study supporting multiple ordered procedures, e.g. chest/abdomen/pelvis CT).
Summary | true
ImagingStudy.referrer
Definition | The requesting/referring physician.
Control | 0..1
Type | Reference(Practitioner)
Alternate Names | ReferringPhysiciansName
Summary | true
ImagingStudy.interpreter
Definition | Who read the study and interpreted the images or other content.
Control | 0..*
Type | Reference(Practitioner)
Alternate Names | Name of Physician(s) Reading Study
Summary | true
ImagingStudy.endpoint
Definition | The network service providing access (e.g., query, view, or retrieval) for the study. See implementation notes for information about using DICOM endpoints. A study-level endpoint applies to each series in the study, unless overridden by a series-level endpoint with the same Endpoint.type.
Control | 0..*
Type | Reference(Endpoint)
Requirements | Access methods for viewing (e.g., IHE's IID profile) or retrieving (e.g., DICOM's WADO-URI and WADO-RS) the study or the study's series or instances. The study-level baseLocation applies to each series in the study, unless overridden in series by a series-level baseLocation of the same type.
Summary | true
Comments | Typical endpoint types include DICOM WADO-RS, which is used to retrieve DICOM instances in native or rendered (e.g., JPG, PNG), formats using a RESTful API; DICOM WADO-URI, which can similarly retrieve native or rendered instances, except using an HTTP query-based approach; DICOM QIDO-RS, which allows RESTful query for DICOM information without retrieving the actual instances; or IHE Invoke Image Display (IID), which provides standard invocation of an imaging web viewer.
ImagingStudy.numberOfSeries
Definition | Number of Series in the Study. This value given may be larger than the number of series elements this Resource contains due to resource availability, security, or other factors. This element should be present if any series elements are present.
Control | 0..1
Type | unsignedInt
Alternate Names | NumberOfStudyRelatedSeries
Summary | true
ImagingStudy.numberOfInstances
Definition | Number of SOP Instances in Study. This value given may be larger than the number of instance elements this resource contains due to resource availability, security, or other factors. This element should be present if any instance elements are present.
Control | 0..1
Type | unsignedInt
Alternate Names | NumberOfStudyRelatedInstances
Summary | true
ImagingStudy.procedureReference
Definition | A reference to the performed Procedure.
Control | 0..*
Type | Reference(Procedure)
Summary | true
ImagingStudy.procedureCode
Definition | The code for the performed procedure type.
Control | 0..*
Terminology Binding | Procedure Codes (SNOMED CT) (Example)
Type | CodeableConcept
Alternate Names | ProcedureCodeSequence
Summary | true
ImagingStudy.reason
Definition | Description of clinical condition indicating why the ImagingStudy was requested.
Control | 0..1
Terminology Binding | Procedure Reason Codes (Example)
Type | CodeableConcept
Summary | true
ImagingStudy.description

Definition | Institution-generated description or classification of the Study performed.
Control | 0..1
Type | string
Alternate Names | StudyDescription
Summary | true
ImagingStudy.series
Definition | Each study has one or more series of images or other content.
Control | 0..*
Summary | true
ImagingStudy.series.uid
Definition | Formal identifier for this series.
Control | 1..1
Type | oid
Alternate Names | SeriesInstanceUID
Summary | true
ImagingStudy.series.number
Definition | The numeric identifier of this series in the study.
Control | 0..1
Type | unsignedInt
Alternate Names | SeriesNumber
Summary | true
ImagingStudy.series.modality
Definition | The modality of this series sequence.
Control | 1..1
Terminology Binding | Acquisition Modality Codes (Extensible)
Type | Coding
Alternate Names | Modality
Summary | true
ImagingStudy.series.description
Definition | A description of the series.
Control | 0..1
Type | string
Alternate Names | SeriesDescription
Summary | true
ImagingStudy.series.numberOfInstances
Definition | Number of SOP Instances in the Study. The value given may be larger than the number of instance elements this resource contains due to resource availability, security, or other factors. This element should be present if any instance elements are present.
Control | 0..1
Type | unsignedInt
Alternate Names | NumberOfSeriesRelatedInstances
Summary | true
ImagingStudy.series.availability
Definition | Availability of series (online, offline or nearline).
Control | 0..1
Terminology Binding | InstanceAvailability (Required)
Type | code
Alternate Names | InstanceAvailability
Summary | true
ImagingStudy.series.endpoint
Definition | The network service providing access (e.g., query, view, or retrieval) for this series. See implementation notes for information about using DICOM endpoints. A series-level endpoint, if present, has precedence over a study-level endpoint with the same Endpoint.type.
Control | 0..*
Type | Reference(Endpoint)
Requirements | Access methods for retrieving (e.g., DICOM's WADO-URI and WADO-RS) the series or the series' instances. A baseLocation specified at the series level has precedence over a baseLocation of the same type specified at the study level.
Summary | true
Comments | Typical endpoint types include DICOM WADO-RS, which is used to retrieve DICOM instances in native or rendered (e.g., JPG, PNG) formats using a RESTful API; DICOM WADO-URI, which can similarly retrieve native or rendered instances, except using an HTTP query-based approach; and DICOM QIDO-RS, which allows RESTful query for DICOM information without retrieving the actual instances.
ImagingStudy.series.bodySite
Definition | The anatomic structures examined. See DICOM Part 16 Annex L (http://dicom.nema.org/medical/dicom/current/output/chtml/part16/chapter_L.html) for DICOM to SNOMED-CT mappings. The bodySite may indicate the laterality of body part imaged; if so, it shall be consistent with any content of ImagingStudy.series.laterality.
Control | 0..1
Terminology Binding | SNOMED CT Body Structures (Example)
Type | Coding
Alternate Names | BodyPartExamined
Summary | true
ImagingStudy.series.laterality
Definition | The laterality of the (possibly paired) anatomic structures examined. E.g., the left knee, both lungs, or unpaired abdomen. If present, shall be consistent with any laterality information indicated in ImagingStudy.series.bodySite.
Control | 0..1
Terminology Binding | Laterality (Example)

Type | Coding
Summary | true
ImagingStudy.series.started
Definition | The date and time the series was started.
Control | 0..1
Type | dateTime
Summary | true
ImagingStudy.series.performer
Definition | The physician or operator (often the radiology technician) who performed the series. The performer is recorded at the series level, since each series in a study may be performed by a different practitioner, at different times, and using different devices. A series may be performed by multiple practitioners.
Control | 0..*
Type | Reference(Practitioner)
Alternate Names | PerformingPhysicianName; OperatorName
Summary | true
ImagingStudy.series.instance
Definition | A single SOP instance within the series, e.g. an image, or presentation state.
Control | 0..*
ImagingStudy.series.instance.uid
Definition | Formal identifier for this image or other content.
Control | 1..1
Type | oid
Alternate Names | SOPInstanceUID
ImagingStudy.series.instance.number
Definition | The number of instance in the series.
Control | 0..1
Type | unsignedInt
Alternate Names | InstanceNumber
ImagingStudy.series.instance.sopClass
Definition | DICOM instance type.
Control | 1..1
Type | oid
Alternate Names | SOPClassUID
ImagingStudy.series.instance.title
Definition | The description of the instance.
Control | 0..1
Type | string
Comments | Particularly for post-acquisition analytic objects, such as SR, presentation states, value mapping, etc.

10.3.7 Resource ImagingStudy - Mappings

URL: https://hl7.org/fhir/STU3/imagingstudy-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.3.7 Resource ImagingStudy - Mappings

Imaging Integration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Mappings:

RIM Mapping

DICOM Tag Mapping

Mappings for the imagingstudy resource.

10.3.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

ImagingStudy | Act[classCode=COMPOSITION, moodCode=EVN]
uid | .id
accession | .inboundRelationship[typeCode=COMP].source[classCode=ACSN, moodCode=EVN].id
identifier | .id (no means to distinguish from uid)
availability | Not supported
modalityList | .code
patient | .participation[typeCode=SBJ].role
context | inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
started | .effectiveTime
basedOn | .outboundRelationship[typeCode=FLFS].target[classCode=DGIMG, moodCode=RQO]
referrer | .outboundRelationship[typeCode=FLFS].target[classCode=DGIMG, moodCode=RQO].participation[typeCode=AUT].role
interpreter | .participation[typeCode=PRF].role
endpoint | Not supported
numberOfSeries | .outboundRelationship[typeCode=COMP, subsetCode=SUMM].target[classCode=OBSSER, moodCode=EVN].repeatNumber
numberOfInstances | .outboundRelationship[typeCode=COMP, subsetCode=SUMM].target[classCode=DGIMG, moodCode=EVN].repeatNumber
procedureReference | .outboundRelationship[typeCode=COMP].target[classCode=PROC, moodCode=EVN]
procedureCode | .outboundRelationship[typeCode=COMP].target[classCode=PROC, moodCode=EVN]

reason | .reasonCode
description | .text
series | .outboundRelationship[typeCode=COMP].target[classCode=OBSSER, moodCode=EVN]
uid | .id
number | .inboundRelationship[type=COMP, source[classCode=COMPOSITION, moodCode=EVN]].sequenceNumber
modality | .code
description | .text
numberOfInstances | .outboundRelationship[typeCode=COMP, subsetCode=SUMM].target[classCode=DGIMG, moodCode=EVN].repeatNumber
availability | .inboundRelationship[typeCode=COMP]. source[classCode=OBS, moodCode=EVN, code="availability"].value
endpoint | Not supported
bodySite | .targetSiteCode
laterality | .targetSiteCode
started | .effectiveTime
performer | .participation[typeCode=PRF].role
instance | .outboundRelationship[typeCode=COMP].target[classCode=DGIMG, moodCode=EVN]
uid | .id
number | .inboundRelationship[typeCode=COMP, source[classCode=OBSSER, moodCode=EVN]].sequenceNumber
sopClass | .inboundRelationship[typeCode=COMP]. source[classCode=OBS, moodCode=EVN, code="sop class"].value
title | .title

10.3.7.2 Mappings for DICOM Tag Mapping (http://nema.org/dicom)

ImagingStudy | Reference IHE radiology TF vol 2 table 4.14-1
uid | (0020,000D)
accession | (0008,0050)
identifier | (0020,0010)
availability
modalityList | (0008,0061)
patient | (0010/*)
context
started | (0008,0020)+(0008,0030)
basedOn
referrer
interpreter | (0008,1060)
endpoint
numberOfSeries | (0020,1206)
numberOfInstances | (0020,1208)
procedureReference | (0008,1032)
procedureCode
reason
description | (0008,1030)
series
uid | (0020,000E)
number | (0020,0011)
modality | (0008,0060)
description | (0008,103E)
numberOfInstances | (0020,1209)
availability | (0008,0056)
endpoint
bodySite | (0018,0015)
laterality | (0020,0060)
started | (0008,0021) + (0008,0031)
performer | (0008,1050) | (0008,1072)
instance
uid | (0008,0018)
number | (0020,0013)
sopClass | (0008,0016)
title | (0070,0080) | (0040,A043) > (0008,0104) | (0042,0010) | (0008,0008)

10.3.8 Resource ImagingStudy - Design Notes

URL: https://hl7.org/fhir/STU3/imagingstudy-explanations.html

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10.3.8 Resource ImagingStudy - Design Notes

to do

10.3.9 Resource ImagingStudy - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/imagingstudy-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.3.9 Resource ImagingStudy - Extensions & Profiles

Imaging Integration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the ImagingStudy resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

10.3.9.1 Profiles

No Profiles defined for this resource

10.3.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

auditevent-SOPClass | SOPClass | for AuditEvent HL7 Extensions
procedure-causedBy | causedBy | for Procedure HL7 Extensions

10.3.9.3 Search Extensions

No Search Extensions defined for this resource

10.3.10 Resource ImagingStudy - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/imagingstudy-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.3.10 Resource ImagingStudy - Detailed Descriptions

Imaging Integration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

R2 : R3 Converson maps for ImagingStudy.

Functional status for this map: 2 tests that all execute ok. 2 fail round-trip testing and 1 r3 resources are invalid (2 errors). (see documentation)

10.3.10.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/ImagingStudy2to3" = "R2 to R3 Conversions for ImagingStudy"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/ImagingStudy" alias ImagingStudyR2 as source
uses "http://hl7.org/fhir/StructureDefinition/ImagingStudy" alias ImagingStudy as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  ImagingStudy extends DomainResource
  input src : ImagingStudyR2 as source
  input tgt : ImagingStudy as target

  "ImagingStudy-started" : for src.started make tgt.started
  "ImagingStudy-patient" : for src.patient make tgt.patient
  "ImagingStudy-uid" : for src.uid make tgt.uid
  "ImagingStudy-accession" : for src.accession make tgt.accession
  "ImagingStudy-identifier" : for src.identifier make tgt.identifier
  // no order
  "ImagingStudy-modalityList" : for src.modalityList make tgt.modalityList
  "ImagingStudy-referrer" : for src.referrer make tgt.referrer
  "ImagingStudy-availability" : for src.availability make tgt.availability
  "ImagingStudy-url" : for src.url as v make tgt.extension as t, t.url = 'http://hl7.org/fhir/ImagingStudy-r2-url',
    t.value = v
  "ImagingStudy-numberOfSeries" : for src.numberOfSeries make tgt.numberOfSeries
  "ImagingStudy-numberOfInstances" : for src.numberOfInstances make tgt.numberOfInstances
  "ImagingStudy-procedure" : for src.procedure make tgt.procedureReference
  "ImagingStudy-interpreter" : for src.interpreter make tgt.interpreter
  "ImagingStudy-description" : for src.description make tgt.description
  "ImagingStudy-series" : for src.series as vs0 make tgt.series as vt0 then imgSeries(vs0, vt0)
endgroup

group imgSeries extends BackboneElement
  input src as source
  input tgt as target

  "ImagingStudy.series-uid" : for src.uid make tgt.uid
  "ImagingStudy.series-number" : for src.number make tgt.number
  "ImagingStudy.series-modality" : for src.modality make tgt.modality
  "ImagingStudy.series-description" : for src.description make tgt.description
  "ImagingStudy.series-numberOfInstances" : for src.numberOfInstances make tgt.numberOfInstances
  "ImagingStudy.series-availability" : for src.availability make tgt.availability
  "ImagingStudy.series-url" : for src.url as v make tgt.extension as t, t.url =
```

```
    'http://hl7.org/fhir/ImagingStudy-r2-url', t.value = v
  "ImagingStudy.series-bodySite" : for src.bodySite make tgt.bodySite
  "ImagingStudy.series-laterality" : for src.laterality make tgt.laterality
  "ImagingStudy.series-started" : for src.started make tgt.started
  "ImagingStudy.series-instance" : for src.instance as vs0 make tgt.instance as vt0 then imgInstance(vs0, vt0)
endgroup

group imgInstance extends BackboneElement
  input src as source
  input tgt as target

  "ImagingStudy.series.instance-number" : for src.number make tgt.number
  "ImagingStudy.series.instance-uid" : for src.uid make tgt.uid
  "ImagingStudy.series.instance-sopClass" : for src.sopClass make tgt.sopClass
  "ImagingStudy.series.instance-title" : for src.title make tgt.title
endgroup
```

10.3.10.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/ImagingStudy3to2" = "R3 to R2 Conversion for ImagingStudy"

uses "http://hl7.org/fhir/StructureDefinition/ImagingStudy" alias ImagingStudy as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/ImagingStudy" alias ImagingStudyR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  ImagingStudy extends DomainResource
  input src : ImagingStudy as source
  input tgt : ImagingStudyR2 as target

  "ImagingStudy-started" : for src.started make tgt.started
  "ImagingStudy-patient" : for src.patient make tgt.patient
  "ImagingStudy-uid" : for src.uid make tgt.uid
  "ImagingStudy-accession" : for src.accession make tgt.accession
  "ImagingStudy-identifier" : for src.identifier make tgt.identifier
  "ImagingStudy-modalityList" : for src.modalityList make tgt.modalityList
  "ImagingStudy-referrer" : for src.referrer make tgt.referrer
  "ImagingStudy-availability" : for src.availability make tgt.availability
  "ImagingStudy-url" : for src.extension as e where e.url = 'http://hl7.org/fhir/ImagingStudy-r2-url' make tgt.modified
    = (e.value)
  "ImagingStudy-numberOfSeries" : for src.numberOfSeries make tgt.numberOfSeries
  "ImagingStudy-numberOfInstances" : for src.numberOfInstances make tgt.numberOfInstances
  "ImagingStudy-procedure" : for src.procedureReference make tgt.procedure
  "ImagingStudy-interpreter" : for src.interpreter make tgt.interpreter
  "ImagingStudy-description" : for src.description make tgt.description
  "ImagingStudy-series" : for src.series as vs0 make tgt.series as vt0 then imgSeries(vs0, vt0)
endgroup

group imgSeries extends BackboneElement
  input src as source
  input tgt as target

  "ImagingStudy.series-uid" : for src.uid make tgt.uid
  "ImagingStudy.series-number" : for src.number make tgt.number
  "ImagingStudy.series-modality" : for src.modality make tgt.modality
  "ImagingStudy.series-description" : for src.description make tgt.description
  "ImagingStudy.series-numberOfInstances" : for src.numberOfInstances make tgt.numberOfInstances
  "ImagingStudy.series-availability" : for src.availability make tgt.availability
  "ImagingStudy.series-url" : for src.extension as e where e.url = 'http://hl7.org/fhir/ImagingStudy-r2-url' make
    tgt.modified = (e.value)
  "ImagingStudy.series-bodySite" : for src.bodySite make tgt.bodySite
  "ImagingStudy.series-laterality" : for src.laterality make tgt.laterality
  "ImagingStudy.series-started" : for src.started make tgt.started
  "ImagingStudy.series-instance" : for src.instance as vs0 make tgt.instance as vt0 then imgInstance(vs0, vt0)
endgroup

group imgInstance extends BackboneElement
  input src as source
  input tgt as target

  "ImagingStudy.series.instance-number" : for src.number make tgt.number
  "ImagingStudy.series.instance-uid" : for src.uid make tgt.uid
  "ImagingStudy.series.instance-sopClass" : for src.sopClass make tgt.sopClass
  "ImagingStudy.series.instance-title" : for src.title make tgt.title
endgroup


group imgSrc
  input src as source
  input tgt as target

  "imgSrc" : for src.url make tgt.url
endgroup
```

10.3.10.3 R3 Validation Errors

imagingstudy-example | ERROR @ ImagingStudy.extension[1]: The extension http://hl7.org/fhir/StructureDefinition/imagingstudy-radiationDose is unknown, and not allowed here (src = InstanceValidator) ERROR @ ImagingStudy.extension[2]: The extension http://hl7.org/fhir/StructureDefinition/imagingstudy-radiationDuration is unknown, and not allowed here (src = InstanceValidator)

10.4 Resource ImagingManifest - Content

URL: https://hl7.org/fhir/STU3/imagingmanifest.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Imaging Integration Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

A text description of the DICOM SOP instances selected in the ImagingManifest; or the reason for, or significance of, the selection.

10.4.1 Scope and Usage

This resource provides information on a selected set of imaging objects, along with information on how to retrieve those instances (either in native DICOM format, or in a rendered format, such as JPEG), or launch an image viewer. The ImagingManifest is used to make available information concerning images etc. that are intended to be exchanged into other clinical contexts such as diagnostic reports, Care Plans, etc.

ImagingManifest provides a FHIR transformation of a DICOM Key Object Selection file as profiled by the IHE's XDS-I profile. Although ImagingManifest can address certain uses outside XDS-I (such as launching a viewer), it does not provide the full capabilities of a general DICOM Key Object Selection.

More than one ImagingManifest may reference instances from a particular DICOM study (and ImagingStudy). A particular ImagingManifest may reference instances from more than one DICOM study (and ImagingStudy). An ImagingManifest may reference all instances, or only selected instances from a study.

10.4.2 Boundaries and Relationships

In distinction to ImagingStudy, this resource is a set of specifically selected objects, potentially from multiple studies on the same patient. ImagingStudy is intended as the resource that identifies a single complete study in itself.

10.4.3 Background and Context

This resource corresponds to a subset of the DICOM Key Object Selection (KOS) SOP Class, and provides a FHIR access to the content of KOS SOP Instances. The content is closely based on the definitions of the equivalent DICOM constructs, and informed by usage patterns already established through DICOM implementation practices, including IHE KIN, TCE, and XDS-I profiles.

The DICOM access methods provide access using the rich controls of the DICOM access methodology indicated. A DICOM capable client may use these access methods to gain full access to the DICOM objects and header.

This resource is referenced by diagnosticreport

10.4.4 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name	Flags	Card.	Type	Description & Constraints
ImagingManifest			DomainResource	Key Object Selection Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier	?	0..1	Identifier	SOP Instance UID
patient	?	1..1	Reference(Patient)	Patient of the selected objects
authoringTime	?	0..1	dateTime	Time when the selection of instances was made
author	?	0..1	Reference(Practitioner | Device | Organization | Patient | RelatedPerson)	Author (human or machine)
description	?	0..1	string	Description text
study	?	1..*	BackboneElement	Study identity of the selected instances
uid	?	1..1	oid	Study instance UID
imagingStudy	?	0..1	Reference(ImagingStudy)	Reference to ImagingStudy
endpoint	?	0..*	Reference(Endpoint)	Study access service endpoint
series	?	1..*	BackboneElement	Series identity of the selected instances
uid	?	1..1	oid	Series instance UID
endpoint	?	0..*	Reference(Endpoint)	Series access endpoint
instance	?	1..*	BackboneElement	The selected instance
sopClass	?	1..1	oid	SOP class UID of instance
uid	?	1..1	oid	Selected instance UID

Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImagingManifest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier SOP Instance UID --></identifier>
 <patient><!-- 1..1 Reference(Patient) Patient of the selected objects --></patient>
 <authoringTime value="[dateTime]"/><!-- 0..1 Time when the selection of instances was made -->
 <author><!-- 0..1 Reference(Practitioner|Device|Organization|Patient|
   RelatedPerson) Author (human or machine) --></author>
 <description value="[string]"/><!-- 0..1 Description text -->
 <study>  <!-- 1..* Study identity of the selected instances -->
  <uid value="[oid]"/><!-- 1..1 Study instance UID -->
  <imagingStudy><!-- 0..1 Reference(ImagingStudy) Reference to ImagingStudy --></imagingStudy>
  <endpoint><!-- 0..* Reference(Endpoint) Study access service endpoint --></endpoint>
  <series>  <!-- 1..* Series identity of the selected instances -->
   <uid value="[oid]"/><!-- 1..1 Series instance UID -->
   <endpoint><!-- 0..* Reference(Endpoint) Series access endpoint --></endpoint>
   <instance>  <!-- 1..* The selected instance -->
    <sopClass value="[oid]"/><!-- 1..1 SOP class UID of instance -->
    <uid value="[oid]"/><!-- 1..1 Selected instance UID -->
   </instance>
  </series>
 </study>
</ImagingManifest>
```

JSON Template

```
{
  "resourceType" : "ImagingManifest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // SOP Instance UID
  "patient" : { Reference(Patient) }, // R!  Patient of the selected objects
  "authoringTime" : "<dateTime>", // Time when the selection of instances was made
  "author" : { Reference(Practitioner|Device|Organization|Patient|
   RelatedPerson) }, // Author (human or machine)
  "description" : "<string>", // Description text
  "study" : [{ // R!  Study identity of the selected instances
    "uid" : "<oid>", // R!  Study instance UID
    "imagingStudy" : { Reference(ImagingStudy) }, // Reference to ImagingStudy
    "endpoint" : [{ Reference(Endpoint) }], // Study access service endpoint
    "series" : [{ // R!  Series identity of the selected instances
      "uid" : "<oid>", // R!  Series instance UID
      "endpoint" : [{ Reference(Endpoint) }], // Series access endpoint
      "instance" : [{ // R!  The selected instance
        "sopClass" : "<oid>", // R!  SOP class UID of instance
        "uid" : "<oid>" // R!  Selected instance UID
      }]
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ImagingManifest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImagingManifest.identifier [ Identifier ]; # 0..1 SOP Instance UID
  fhir:ImagingManifest.patient [ Reference(Patient) ]; # 1..1 Patient of the selected objects
  fhir:ImagingManifest.authoringTime [ dateTime ]; # 0..1 Time when the selection of instances was made
  fhir:ImagingManifest.author [ Reference(Practitioner|Device|Organization|Patient|RelatedPerson) ]; # 0..1 Author
    (human or machine)
  fhir:ImagingManifest.description [ string ]; # 0..1 Description text
  fhir:ImagingManifest.study [ # 1..* Study identity of the selected instances
    fhir:ImagingManifest.study.uid [ oid ]; # 1..1 Study instance UID
    fhir:ImagingManifest.study.imagingStudy [ Reference(ImagingStudy) ]; # 0..1 Reference to ImagingStudy
    fhir:ImagingManifest.study.endpoint [ Reference(Endpoint) ], ... ; # 0..* Study access service endpoint
    fhir:ImagingManifest.study.series [ # 1..* Series identity of the selected instances
      fhir:ImagingManifest.study.series.uid [ oid ]; # 1..1 Series instance UID
      fhir:ImagingManifest.study.series.endpoint [ Reference(Endpoint) ], ... ; # 0..* Series access endpoint
      fhir:ImagingManifest.study.series.instance [ # 1..* The selected instance
        fhir:ImagingManifest.study.series.instance.sopClass [ oid ]; # 1..1 SOP class UID of instance
        fhir:ImagingManifest.study.series.instance.uid [ oid ]; # 1..1 Selected instance UID
      ], ...;
    ], ...;
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

ImagingManifest | | | DomainResource | Key Object Selection Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..1 | Identifier | SOP Instance UID
patient | ? | 1..1 | Reference(Patient) | Patient of the selected objects
authoringTime | ? | 0..1 | dateTime | Time when the selection of instances was made
author | ? | 0..1 | Reference(Practitioner | Device | Organization | Patient | RelatedPerson) | Author (human or machine)
description | ? | 0..1 | string | Description text
study | ? | 1..* | BackboneElement | Study identity of the selected instances
uid | ? | 1..1 | oid | Study instance UID
imagingStudy | ? | 0..1 | Reference(ImagingStudy) | Reference to ImagingStudy
endpoint | ? | 0..* | Reference(Endpoint) | Study access service endpoint
series | ? | 1..* | BackboneElement | Series identity of the selected instances
uid | ? | 1..1 | oid | Series instance UID
endpoint | ? | 0..* | Reference(Endpoint) | Series access endpoint
instance | ? | 1..* | BackboneElement | The selected instance
sopClass | ? | 1..1 | oid | SOP class UID of instance
uid | ? | 1..1 | oid | Selected instance UID
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImagingManifest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier SOP Instance UID --></identifier>
 <patient><!-- 1..1 Reference(Patient) Patient of the selected objects --></patient>
 <authoringTime value="[dateTime]"/><!-- 0..1 Time when the selection of instances was made -->
 <author><!-- 0..1 Reference(Practitioner|Device|Organization|Patient|
   RelatedPerson) Author (human or machine) --></author>
 <description value="[string]"/><!-- 0..1 Description text -->
 <study>  <!-- 1..* Study identity of the selected instances -->
  <uid value="[oid]"/><!-- 1..1 Study instance UID -->
  <imagingStudy><!-- 0..1 Reference(ImagingStudy) Reference to ImagingStudy --></imagingStudy>
  <endpoint><!-- 0..* Reference(Endpoint) Study access service endpoint --></endpoint>
  <series>  <!-- 1..* Series identity of the selected instances -->
   <uid value="[oid]"/><!-- 1..1 Series instance UID -->
   <endpoint><!-- 0..* Reference(Endpoint) Series access endpoint --></endpoint>
   <instance>  <!-- 1..* The selected instance -->
    <sopClass value="[oid]"/><!-- 1..1 SOP class UID of instance -->
    <uid value="[oid]"/><!-- 1..1 Selected instance UID -->
   </instance>
  </series>
 </study>
</ImagingManifest>
```

JSON Template

```
{
  "resourceType" : "ImagingManifest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // SOP Instance UID
  "patient" : { Reference(Patient) }, // R!  Patient of the selected objects
  "authoringTime" : "<dateTime>", // Time when the selection of instances was made
  "author" : { Reference(Practitioner|Device|Organization|Patient|
   RelatedPerson) }, // Author (human or machine)
  "description" : "<string>", // Description text
  "study" : [{ // R!  Study identity of the selected instances
    "uid" : "<oid>", // R!  Study instance UID
    "imagingStudy" : { Reference(ImagingStudy) }, // Reference to ImagingStudy
    "endpoint" : [{ Reference(Endpoint) }], // Study access service endpoint
    "series" : [{ // R!  Series identity of the selected instances
      "uid" : "<oid>", // R!  Series instance UID
      "endpoint" : [{ Reference(Endpoint) }], // Series access endpoint
      "instance" : [{ // R!  The selected instance
        "sopClass" : "<oid>", // R!  SOP class UID of instance
        "uid" : "<oid>" // R!  Selected instance UID
      }]
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ImagingManifest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImagingManifest.identifier [ Identifier ]; # 0..1 SOP Instance UID
  fhir:ImagingManifest.patient [ Reference(Patient) ]; # 1..1 Patient of the selected objects
  fhir:ImagingManifest.authoringTime [ dateTime ]; # 0..1 Time when the selection of instances was made
  fhir:ImagingManifest.author [ Reference(Practitioner|Device|Organization|Patient|RelatedPerson) ]; # 0..1 Author
```

```
    (human or machine)
  fhir:ImagingManifest.description [ string ]; # 0..1 Description text
  fhir:ImagingManifest.study [ # 1..* Study identity of the selected instances
    fhir:ImagingManifest.study.uid [ oid ]; # 1..1 Study instance UID
    fhir:ImagingManifest.study.imagingStudy [ Reference(ImagingStudy) ]; # 0..1 Reference to ImagingStudy
    fhir:ImagingManifest.study.endpoint [ Reference(Endpoint) ], ... ; # 0..* Study access service endpoint
    fhir:ImagingManifest.study.series [ # 1..* Series identity of the selected instances
      fhir:ImagingManifest.study.series.uid [ oid ]; # 1..1 Series instance UID
      fhir:ImagingManifest.study.series.endpoint [ Reference(Endpoint) ], ... ; # 0..* Series access endpoint
      fhir:ImagingManifest.study.series.instance [ # 1..* The selected instance
        fhir:ImagingManifest.study.series.instance.sopClass [ oid ]; # 1..1 SOP class UID of instance
        fhir:ImagingManifest.study.series.instance.uid [ oid ]; # 1..1 Selected instance UID
      ], ...;
    ], ...;
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

10.4.4.1 Implementation Notes

A referenced DICOM SOP instance could be:

* A single- or multi-frame, still or video image captured by a variety of imaging modalities, such as X-ray, MR, and ultrasound;

* A set of various presentation parameters, including annotation and markup;

* A set of measurements or a report, including radiation dose report and CAD analysis;

* An encapsulated PDF or CDA document;

* A list of instances, such as key “of interest” images, or instances to be “deleted”; or

* Other DICOM content.

UID values follow the FHIR convention of expressing UIDs as URNs. For example, the DICOM Study Instance UID of 1.2.250.1.59.40211.12345678.678910 is expressed as “urn:oid:1.2.250.1.59.40211.12345678.678910”.

The ImagingManifest.study.endpoint elements and ImagingManifest.study.series.endpoint elements indicate network services that can be used to access the studies, series, or instances; for example, a DICOM WADO-RS server. An ImagingManifest.study.series.endpoint of a particular Endpoint.connectionType provides that service for that series, and all contained instances. An ImagingManifest.study.endpoint of a particularconnection type provides that service for all series in that study that do not have a specified Endpoint of that type, and their contained instances. That is, an ImagingManifest.study.series.endpoint overrides a ImagingManifest.study.endpoint of the same connection type. (Since each study, or individual series of a study can be stored on different imaging archive servers, per-series endpoints are required. For the identified services and use cases, all instances within a series would be stored together, and thus instance-level endpoints are not defined.)

Different Endpoint connection types may have different capabilities, protocols or requirements; and the specified Endpoint.url may require manipulation. For the details on use of imaging-related Endpoint connection types, See ImagingStudy Implementation Notes for details.

10.4.4.2 Use Case

10.4.4.2.1 EHR access to imaging studies

Amy, a family physician, is accessing a cross-enterprise document registry that contains radiology objects (IHE Radiology XDS-I), to discover studies for her patient, Alex. Her EHR client makes a FHIR call for all ImagingManifest objects available for Alex. In the response, she is able to get study identifiers for each study that has been published to the registry. There is enough information provided in the response to obtain a thumbnail via a WADO-RS call, or to launch a viewer using an IHE Radiology - Invoke Image Display (IID) profile call using the url elements found in the ImagingManifest. In each result, there is a reference to the ImagingStudy FHIR object which can provide more information about each study.

10.4.4.2.2 Comprehensive Imaging Scheduled Workflow

Joe Angina complains of shortness of breath and occasional chest pain to his primary care physician, Dr. Pat Down at Local MultiClinic, who orders a stress echocardiogram; the order is created as a FHIR Task resource to manage the workflow, with a link to a ProcedureRequest resource with the details of the request. The order is scheduled and assigned to cardiologist Dr. Art Skann, also at Local MultiClinic.

On the scheduled day of the exam, Joe arrives at the echo lab to meet with Dr. Skann and have the study done. Dr. Skann’s workstation shows the daily list of Task, and he follows the link to retrieve the ProcedureRequest. (He may follow the links through the referenced Patient resource to access Joe’s electronic medical record, but that is not the concern of this storyboard.)

The Task and ProcedureRequest has been transcoded to a DICOM Modality Worklist Scheduled Procedure Step, and in the echo lab the equipment has downloaded the Modality Worklist. The study is performed, and the acquired images and sonographer’s preliminary measurements are stored in the Local MultiClinic Picture Archiving and Communication System (PACS). The PACS creates an ImagingStudy resource for each study it manages.

Dr. Skann interprets the study on a PACS workstation, and he selects two key image frames to be included in the diagnostic report; this selection is stored back to the PACS as a DICOM Key Object Selection with the title "For Report Attachment", and the PACS makes it available (transcodes it) as a FHIR ImagingManifest resource. Dr. Skann dictates the report using a structured data entry report writing program, including a recommendation for a cardiac catheterization procedure, and signs it. The report writing program formats the report as a CDA document, retrieves the ImagingManifest resource, and inserts the referenced key images into the report.

Dr. Down meets again with Joe, and they review the results of the stress test. Joe has a question about the findings that the key images in the report do not show, so Dr. Down uses the Local MultiClinic EMR to query the PACS for the full ImagingStudy resource, and uses the references there to open an image display for the full study. Joe agrees to proceed to catheterization, and Dr. Down sends a referral to the Ginormous University Hospital cath department, and triggers the PACS to share the echo study through the Metropolitan Health Information Exchange.

The PACS creates a manifest of the study as an ImagingManifest resource, which includes all the images but excludes the sonographer's preliminary measurements (which as a matter of policy are not shared outside the Local MultiClinic). The manifest is published to the Metro HIE. (In accordance with IHE XDS-I , the images themselves are not directly published to the HIE, but available for on-demand retrieval from the PACS.)

At Ginormous Hospital, Dr. Cora Plummer receives the cath referral, and looks up the study in the Metro HIE registry. She retrieves the study manifest ImagingManifest, and uses it to access the shared images, which she uses to prepare for the cath procedure.

10.4.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
author | reference | Author of the ImagingManifest (or a DICOM Key Object Selection which it represents) | ImagingManifest.author (Practitioner, Organization, Device, Patient, RelatedPerson)
authoring-time | date | Time of the ImagingManifest (or a DICOM Key Object Selection which it represents) authoring | ImagingManifest.authoringTime
endpoint | reference | The endpoint for the study or series | ImagingManifest.study.endpoint | ImagingManifest.study.series.endpoint (Endpoint)
identifier | token | UID of the ImagingManifest (or a DICOM Key Object Selection which it represents) | ImagingManifest.identifier
imaging-study | reference | ImagingStudy resource selected in the ImagingManifest (or a DICOM Key Object Selection which it represents) | ImagingManifest.study.imagingStudy (ImagingStudy)
patient | reference | Subject of the ImagingManifest (or a DICOM Key Object Selection which it represents) | ImagingManifest.patient (Patient) | 31 Resources
selected-study | uri | Study selected in the ImagingManifest (or a DICOM Key Object Selection which it represents) | ImagingManifest.study.uid

10.4.6 Resource ImagingManifest - Examples

URL: https://hl7.org/fhir/STU3/imagingmanifest-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.4.6 Resource ImagingManifest - Examples

Imaging Integration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

Example List:

Example of imagingmanifest | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.4.7 Resource ImagingManifest - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/imagingmanifest-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.4.7 Resource ImagingManifest - Detailed Descriptions

Imaging Integration Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the ImagingManifest resource.

ImagingManifest
Definition | A text description of the DICOM SOP instances selected in the ImagingManifest; or the reason for, or significance of, the selection.
Control | 1..1
Alternate Names | Manifest; XDS-I summary; Key Images; (in limited use) Key Object Selection (KOS)
ImagingManifest.identifier
Definition | Unique identifier of the DICOM Key Object Selection (KOS) that this resource represents.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1

Type | Identifier
Requirements | DICOM SOP Instance is always assigned with a unique identifier (UID).
Alternate Names | SOPInstanceUID
Summary | true
ImagingManifest.patient
Definition | A patient resource reference which is the patient subject of all DICOM SOP Instances in this ImagingManifest.
Control | 1..1
Type | Reference(Patient)
Requirements | SOP Instances in ImagingManifest must be from the same patient.
Summary | true
Comments | SOP instances selected in the ImagingManifest can be from different studies, but must be of the same patient.
ImagingManifest.authoringTime
Definition | Date and time when the selection of the referenced instances were made. It is (typically) different from the creation date of the selection resource, and from dates associated with the referenced instances (e.g. capture time of the referenced image).
Control | 0..1
Type | dateTime
Requirements | Date and time when the selection was made can be important to understand the content of selection.
Alternate Names | ObservationDateTime
Summary | true
ImagingManifest.author
Definition | Author of ImagingManifest. It can be a human author or a device which made the decision of the SOP instances selected. For example, a radiologist selected a set of imaging SOP instances to attach in a diagnostic report, and a CAD application may author a selection to describe SOP instances it used to generate a detection conclusion.
Control | 0..1
Type | Reference(Practitioner | Device | Organization | Patient | RelatedPerson)
Requirements | Track the selection decision maker.
Summary | true
ImagingManifest.description
Definition | Free text narrative description of the ImagingManifest. The value may be derived from the DICOM Standard Part 16, CID-7010 descriptions (e.g. Best in Set, Complete Study Content). Note that those values cover the wide range of uses of the DICOM Key Object Selection object, several of which are not supported by ImagingManifest. Specifically, there is no expected behavior associated with descriptions that suggest referenced images be removed or not used.
Control | 0..1
Type | string
Requirements | Need to provide a narrative description of the SOP instances in the selection.
Alternate Names | TextValue
Summary | true
Comments | Free text narrative description of the ImagingManifest.
ImagingManifest.study
Definition | Study identity and locating information of the DICOM SOP instances in the selection.
Control | 1..*
Requirements | DICOM SOP instances are identified through the UIDs of the study - series - instance hierarchy. In addition, the locator at the study level provide a means of retrieving the entire study.
Summary | true
Comments | Study component represents the study level identity and locator information of the DICOM SOP instances in the selection. It is the top level identity of the hierarchical identification of the instances.
ImagingManifest.study.uid
Definition | Study instance UID of the SOP instances in the selection.
Control | 1..1
Type | oid
Requirements | Study instance UID is required to fully identify the DICOM SOP instances in the selection.
Alternate Names | StudyInstanceUID; Referenced study instance UID
Summary | true
ImagingManifest.study.imagingStudy
Definition | Reference to the Imaging Study in FHIR form.
Control | 0..1
Type | Reference(ImagingStudy)
Requirements | Used to retrieve the ImagingStudy that contain the images referenced in the Imaging ObjectSelection.study.
Summary | true
ImagingManifest.study.endpoint
Definition | The network service providing access (e.g., query, view, or retrieval) for the study. See implementation notes for information about using DICOM endpoints. A study-level endpoint applies to each series in the study, unless overridden by a series-level endpoint with the same Endpoint.type.
Control | 0..*
Type | Reference(Endpoint)
Requirements | Access methods for viewing (e.g. IHE's IID profile) or retrieving (e.g., DICOM's WADO-URI and WADO-RS) the study or the study's series or instances. The study-level baseLocation applies to each series in the study, unless overridden in series by a series-level baseLocation of the same type.
Summary | true
ImagingManifest.study.series
Definition | Series identity and locating information of the DICOM SOP instances in the selection.
Control | 1..*

Requirements | DICOM SOP instances are identified through the UIDs of the study - series - instance hierarchy. In addition, the locator at the series level provides a means of retrieving the entire series.
Summary | true
Comments | Series component represents the series level identity and locator information of the DICOM SOP instances in the selection.
ImagingManifest.study.series.uid
Definition | Series instance UID of the SOP instances in the selection.
Control | 1..1
Type | oid
Requirements | Series instance UID is required to fully identify the DICOM SOP instances in the selection.
Alternate Names | SeriesInstanceUID; Referenced series instance UID
Summary | true
ImagingManifest.study.series.endpoint
Definition | The network service providing access (e.g., query, view, or retrieval) for this series. See implementation notes for information about using DICOM endpoints. A series-level endpoint, if present, has precedence over a study-level endpoint with the same Endpoint.type.
Control | 0..*
Type | Reference(Endpoint)
Requirements | Access methods for retrieving (e.g., DICOM's WADO-URI and WADO-RS) the series or the series' instances. A baseLocation specified at the series level has precedence over a baseLocation of the same type specified at the study level.
Summary | true
ImagingManifest.study.series.instance
Definition | Identity and locating information of the selected DICOM SOP instances.
Control | 1..*
Requirements | DICOM SOP instances are identified through the UIDs of the study - series - instance hierarchy. This is the instance level identity of the instance in the selection.
Summary | true
Comments | SOP Instance component identify the instance selected, along with the study and series identities to form the DICOM identity hierarchy.
ImagingManifest.study.series.instance.sopClass
Definition | SOP class UID of the selected instance.
Control | 1..1
Type | oid
Requirements | DICOM SOP instances can be an image or other data object. The SOP class UID provide the accurate information about what type the instance is.
Alternate Names | SOPClassUID
Summary | true
Comments | SOP class UID identifies the type of the selected instances, e.g. CT image, Gray scale softcopy presentation state, ECG waveform, etc.
ImagingManifest.study.series.instance.uid
Definition | SOP Instance UID of the selected instance.
Control | 1..1
Type | oid
Requirements | SOP instance UID is required to fully identify the DICOM SOP instance in the selection.
Alternate Names | SOPInstanceUID; Referenced SOP instance UID
Summary | true
Comments | SOP instance UID identifies the instance.

10.4.8 Resource ImagingManifest - Mappings

URL: https://hl7.org/fhir/STU3/imagingmanifest-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.4.8 Resource ImagingManifest - Mappings

Imaging Integration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

DICOM Tag Mapping

Mappings for the imagingmanifest resource.

10.4.8.1 Mappings for RIM Mapping (http://hl7.org/v3)

ImagingManifest | Act[classCode=COMPOSITION, moodCode=EVN]
identifier | .id
patient | .participation[typeCode=SBJ].role
authoringTime | .effectiveTime
author | .participation[typeCode=PRF].role
description | .text
study | Act[classCode=COMPOSITION, moodCode=EVN]
uid | .id
imagingStudy | n/a
endpoint | n/a
series | n/a
uid | .id

endpoint | n/a
instance | n/a
sopClass | .inboundRelationship[typeCode=COMP]. source[classCode=OBS, moodCode=EVN, code="sop class"].value
uid | .id

10.4.8.2 Mappings for DICOM Tag Mapping (http://nema.org/dicom)

ImagingManifest | Key Object Selection SOP Class (1.2.840.10008.5.1.4.1.1.88.59)
identifier | (0008,0018) - SOP Instance UID (of the Key Object Selection SOP Class)
patient | (0010,0010) - Patient Name (0010,0020) - Patient ID (0010,0021) - Issuer of Patient ID
authoringTime | (0040,A032) - Observation Date Time
author | (0040,A730) - Content Sequence TID 1002 - Observer Context
description | (0040,A160) - Text Value SR content item immediately under the root item, concept name code = (113012, DCM, "Key Object Description")
study
uid | (0020,000D) - Study Instance UID
imagingStudy | imagingStudy (0020,000D) - Study Instance UID
endpoint
series
uid | (0020,000E) - Series Instance UID
endpoint
instance
sopClass | (0008,0016) SOP Class UID
uid | (0008,0018) - SOP Instance UID

10.4.9 Resource ImagingManifest - Design Notes

URL: https://hl7.org/fhir/STU3/imagingmanifest-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.4.9 Resource ImagingManifest - Design Notes

to do

10.4.10 Resource ImagingManifest - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/imagingmanifest-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.4.10 Resource ImagingManifest - Extensions & Profiles

Imaging Integration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the ImagingManifest resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

10.4.10.1 Profiles

No Profiles defined for this resource

10.4.10.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

10.4.10.3 Search Extensions

No Search Extensions defined for this resource

10.5 Resource Sequence - Content

URL: https://hl7.org/fhir/STU3/sequence.html

while the amino acid string should be constrained within the range:

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Clinical Genomics Work Group | Maturity Level: 1 | Trial Use | Compartments: Not linked to any defined compartments

Raw data describing a biological sequence.

10.5.1 Scope and Usage

The Sequence resource is designed to describe an atomic sequence which contains the alignment sequencing test result and multiple variations. Atomic sequences can be connected by link element and they will lead to sequence graph. By this method, a sequence can be reported. Complete genetic sequence information, of which specific genetic variations are a part, is reported by reference to the GA4GH repository. Thus, the FHIR Sequence resource avoids large genomic payloads in a manner analogous to how the FHIR ImagingStudy resource references large images maintained in other systems. For use cases, details on how this resource interact with other Clinical Genomics resources or profiles, please refer to implementation guidance document here .

10.5.1.1 Genetic Standards and Resources include:

* Variant Databases: dbSNP , ClinVar , and COSMIC

* Reference Sequences: RefSeq and ENSEMBL

This resource is designed to describe sequence variations with clinical significance with information such as:

* Name of the variation represented

* Type of the variation

* Gene region occupied by the variation

* Tissue source used to determine genotype of the variation

* Quality of the result

It is strongly encouraged to provide all available information in this resource for any reported variants, because receiving systems (e.g. discovery research, outcomes analysis, and public health reporting) may use this information to normalize variants over time or across sources. However, these data should not be used to dynamically correct/change variant representations for clinical use outside of the laboratory, due to insufficient information.

Implementers should be aware that semantic equivalency of results of genetic variants cannot be guaranteed unless there is an agreed upon standard between sending and receiving systems.

10.5.2 Boundaries and Relationships

Focus of the resource is to provide sequencing alignment data immediately relevant to what the interpretation on clinical decision-making originates from. Hence data such as precise read of DNA sequences and sequence alignment are not included; such data are nonetheless accessible through references to GA4GH (Global Alliance for Genomics and Health) API. The Sequence resource will be referenced by Observation to provide variant information. As clinical assessments/diagnosis of a patient are typically captured in the Condition resource or the ClinicalImpression resource, the Sequence resource can be referenced by the Condition resource to provide specific genetic data to support an assertions. This is analogous to how Condition references other resources, such as AllergyIntolerance, Procedure, and Questionnaire resources.

This resource is referenced by observation

10.5.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Sequence | ?I | | DomainResource | Information about a biological sequence + Only 0 and 1 are valid for coordinateSystem Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | Unique ID for this particular sequence. This is a FHIR-defined id

type | ? | 0..1 | code | aa | dna | rna sequenceType (Example)

coordinateSystem | ? | 1..1 | integer | Base number of coordinate system (0 for 0-based numbering or coordinates, inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end)

patient | ? | 0..1 | Reference(Patient) | Who and/or what this is about

specimen | ? | 0..1 | Reference(Specimen) | Specimen used for sequencing

device | ? | 0..1 | Reference(Device) | The method for sequencing

performer | ? | 0..1 | Reference(Organization) | Who should be responsible for test result

quantity | ? | 0..1 | Quantity | The number of copies of the seqeunce of interest. (RNASeq)

referenceSeq | ?I | 0..1 | BackboneElement | A sequence used as reference + Only +1 and -1 are valid for strand + GenomeBuild and chromosome must be both contained if either one of them is contained + Have and only have one of the following elements in referenceSeq : 1. genomeBuild ; 2 referenceSeqId; 3. referenceSeqPointer; 4. referenceSeqString;

chromosome | ? | 0..1 | CodeableConcept | Chromosome containing genetic finding chromosome-human (Example)

genomeBuild | ? | 0..1 | string | The Genome Build used for reference, following GRCh build versions e.g. 'GRCh 37'

referenceSeqId | ? | 0..1 | CodeableConcept | Reference identifier ENSEMBL (Example)

referenceSeqPointer | ? | 0..1 | Reference(Sequence) | A Pointer to another Sequence entity as reference sequence

referenceSeqString | ? | 0..1 | string | A string to represent reference sequence
strand | ? | 0..1 | integer | Directionality of DNA (+1/-1)
windowStart | ? | 1..1 | integer | Start position of the window on the reference sequence
windowEnd | ? | 1..1 | integer | End position of the window on the reference sequence
variant | ? | 0..* | BackboneElement | Variant in sequence
start | ? | 0..1 | integer | Start position of the variant on the reference sequence
end | ? | 0..1 | integer | End position of the variant on the reference sequence
observedAllele | ? | 0..1 | string | Allele that was observed
referenceAllele | ? | 0..1 | string | Allele in the reference sequence
cigar | ? | 0..1 | string | Extended CIGAR string for aligning the sequence with reference bases
variantPointer | ? | 0..1 | Reference(Observation) | Pointer to observed variant information
observedSeq | ? | 0..1 | string | Sequence that was observed
quality | ? | 0..* | BackboneElement | An set of value as quality of sequence
type | ? | 1..1 | code | indel | snp | unknown qualityType (Required)
standardSequence | ? | 0..1 | CodeableConcept | Standard sequence for comparison FDA-StandardSequence (Example)
start | ? | 0..1 | integer | Start position of the sequence
end | ? | 0..1 | integer | End position of the sequence
score | ? | 0..1 | Quantity | Quality score for the comparison
method | ? | 0..1 | CodeableConcept | Method to get quality FDA-Method (Example)
truthTP | ? | 0..1 | decimal | True positives from the perspective of the truth data
queryTP | ? | 0..1 | decimal | True positives from the perspective of the query data
truthFN | ? | 0..1 | decimal | False negatives
queryFP | ? | 0..1 | decimal | False positives
gtFP | ? | 0..1 | decimal | False positives where the non-REF alleles in the Truth and Query Call Sets match
precision | ? | 0..1 | decimal | Precision of comparison
recall | ? | 0..1 | decimal | Recall of comparison
fScore | ? | 0..1 | decimal | F-score
readCoverage | ? | 0..1 | integer | Average number of reads representing a given nucleotide in the reconstructed sequence
repository | ? | 0..* | BackboneElement | External repository which contains detailed report related with observedSeq in this resource
type | ? | 1..1 | code | directlink | openapi | login | oauth | other repositoryType (Required)
url | ? | 0..1 | uri | URI of the repository
name | ? | 0..1 | string | Repository's name
datasetId | ? | 0..1 | string | Id of the dataset that used to call for dataset in repository
variantsetId | ? | 0..1 | string | Id of the variantset that used to call for variantset in repository
readsetId | ? | 0..1 | string | Id of the read
pointer | ? | 0..* | Reference(Sequence) | Pointer to next atomic sequence
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Sequence xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique ID for this particular sequence. This is a FHIR-defined id --></identifier>
 <type value="[code]"/><!-- 0..1 aa | dna | rna -->
 <coordinateSystem value="[integer]"/><!-- 1..1 Base number of coordinate system (0 for 0-based numbering or
    coordinates, inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end) -->
 <patient><!-- 0..1 Reference(Patient) Who and/or what this is about --></patient>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for sequencing --></specimen>
 <device><!-- 0..1 Reference(Device) The method for sequencing --></device>
 <performer><!-- 0..1 Reference(Organization) Who should be responsible for test result --></performer>
 <quantity><!-- 0..1 Quantity The number of copies of the seqeunce of interest.  (RNASeq) --></quantity>
 <referenceSeq>  <!-- 0..1 A sequence used as reference -->
  <chromosome><!-- 0..1 CodeableConcept Chromosome containing genetic finding --></chromosome>
  <genomeBuild value="[string]"/><!-- 0..1 The Genome Build used for reference, following GRCh build versions e.g. 'GRCh
     37' -->
  <referenceSeqId><!-- 0..1 CodeableConcept Reference identifier --></referenceSeqId>
  <referenceSeqPointer><!-- 0..1 Reference(Sequence) A Pointer to another Sequence entity as reference sequence
    --></referenceSeqPointer>
  <referenceSeqString value="[string]"/><!-- 0..1 A string to represent reference sequence -->
  <strand value="[integer]"/><!-- 0..1 Directionality of DNA ( +1/-1) -->
  <windowStart value="[integer]"/><!-- 1..1 Start position of the window on the  reference sequence -->
  <windowEnd value="[integer]"/><!-- 1..1 End position of the window on the reference sequence -->
 </referenceSeq>
 <variant>  <!-- 0..* Variant in sequence -->
  <start value="[integer]"/><!-- 0..1 Start position of the variant on the  reference sequence -->
  <end value="[integer]"/><!-- 0..1 End position of the variant on the reference sequence -->
  <observedAllele value="[string]"/><!-- 0..1 Allele that was observed -->
  <referenceAllele value="[string]"/><!-- 0..1 Allele in the reference sequence -->
  <cigar value="[string]"/><!-- 0..1 Extended CIGAR string for aligning the sequence with reference bases -->
  <variantPointer><!-- 0..1 Reference(Observation) Pointer to observed variant information --></variantPointer>
 </variant>
 <observedSeq value="[string]"/><!-- 0..1 Sequence that was observed -->
 <quality>  <!-- 0..* An set of value as quality of sequence -->
  <type value="[code]"/><!-- 1..1 indel | snp | unknown -->
  <standardSequence><!-- 0..1 CodeableConcept Standard sequence for comparison --></standardSequence>
  <start value="[integer]"/><!-- 0..1 Start position of the sequence -->
  <end value="[integer]"/><!-- 0..1 End position of the sequence -->
  <score><!-- 0..1 Quantity Quality score for the comparison --></score>
  <method><!-- 0..1 CodeableConcept Method to get quality --></method>
```

```
  <truthTP value="[decimal]"/><!-- 0..1 True positives from the perspective of the truth data -->
  <queryTP value="[decimal]"/><!-- 0..1 True positives from the perspective of the query data -->
  <truthFN value="[decimal]"/><!-- 0..1 False negatives -->
  <queryFP value="[decimal]"/><!-- 0..1 False positives -->
  <gtFP value="[decimal]"/><!-- 0..1 False positives where the non-REF alleles in the Truth and Query Call Sets match
    -->
  <precision value="[decimal]"/><!-- 0..1 Precision of comparison -->
  <recall value="[decimal]"/><!-- 0..1 Recall of comparison -->
  <fScore value="[decimal]"/><!-- 0..1 F-score -->
 </quality>
 <readCoverage value="[integer]"/><!-- 0..1 Average number of reads representing a given nucleotide in the reconstructed
     sequence -->
 <repository>  <!-- 0..* External repository which contains detailed report related with observedSeq in this resource
    -->
  <type value="[code]"/><!-- 1..1 directlink | openapi | login | oauth | other -->
  <url value="[uri]"/><!-- 0..1 URI of the repository -->
  <name value="[string]"/><!-- 0..1 Repository's name -->
  <datasetId value="[string]"/><!-- 0..1 Id of the dataset that used to call for dataset in repository -->
  <variantsetId value="[string]"/><!-- 0..1 Id of the variantset that used to call for variantset in repository -->
  <readsetId value="[string]"/><!-- 0..1 Id of the read -->
 </repository>
 <pointer><!-- 0..* Reference(Sequence) Pointer to next atomic sequence --></pointer>
</Sequence>
```

JSON Template

```
{
  "resourceType" : "Sequence",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique ID for this particular sequence. This is a FHIR-defined id
  "type" : "<code>", // aa | dna | rna
  "coordinateSystem" : <integer>, // R!  Base number of coordinate system (0 for 0-based numbering or coordinates,
    inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end)
  "patient" : { Reference(Patient) }, // Who and/or what this is about
  "specimen" : { Reference(Specimen) }, // Specimen used for sequencing
  "device" : { Reference(Device) }, // The method for sequencing
  "performer" : { Reference(Organization) }, // Who should be responsible for test result
  "quantity" : { Quantity }, // The number of copies of the seqeunce of interest.  (RNASeq)
  "referenceSeq" : { // A sequence used as reference
    "chromosome" : { CodeableConcept }, // Chromosome containing genetic finding
    "genomeBuild" : "<string>", // The Genome Build used for reference, following GRCh build versions e.g. 'GRCh 37'
    "referenceSeqId" : { CodeableConcept }, // Reference identifier
    "referenceSeqPointer" : { Reference(Sequence) }, // A Pointer to another Sequence entity as reference sequence
    "referenceSeqString" : "<string>", // A string to represent reference sequence
    "strand" : <integer>, // Directionality of DNA ( +1/-1)
    "windowStart" : <integer>, // R!  Start position of the window on the  reference sequence
    "windowEnd" : <integer> // R!  End position of the window on the reference sequence
  },
  "variant" : [{ // Variant in sequence
    "start" : <integer>, // Start position of the variant on the  reference sequence
    "end" : <integer>, // End position of the variant on the reference sequence
    "observedAllele" : "<string>", // Allele that was observed
    "referenceAllele" : "<string>", // Allele in the reference sequence
    "cigar" : "<string>", // Extended CIGAR string for aligning the sequence with reference bases
    "variantPointer" : { Reference(Observation) } // Pointer to observed variant information
  }],
  "observedSeq" : "<string>", // Sequence that was observed
  "quality" : [{ // An set of value as quality of sequence
    "type" : "<code>", // R!  indel | snp | unknown
    "standardSequence" : { CodeableConcept }, // Standard sequence for comparison
    "start" : <integer>, // Start position of the sequence
    "end" : <integer>, // End position of the sequence
    "score" : { Quantity }, // Quality score for the comparison
    "method" : { CodeableConcept }, // Method to get quality
    "truthTP" : <decimal>, // True positives from the perspective of the truth data
    "queryTP" : <decimal>, // True positives from the perspective of the query data
    "truthFN" : <decimal>, // False negatives
    "queryFP" : <decimal>, // False positives
    "gtFP" : <decimal>, // False positives where the non-REF alleles in the Truth and Query Call Sets match
    "precision" : <decimal>, // Precision of comparison
    "recall" : <decimal>, // Recall of comparison
    "fScore" : <decimal> // F-score
  }],
  "readCoverage" : <integer>, // Average number of reads representing a given nucleotide in the reconstructed sequence
  "repository" : [{ // External repository which contains detailed report related with observedSeq in this resource
    "type" : "<code>", // R!  directlink | openapi | login | oauth | other
    "url" : "<uri>", // URI of the repository
    "name" : "<string>", // Repository's name
    "datasetId" : "<string>", // Id of the dataset that used to call for dataset in repository
    "variantsetId" : "<string>", // Id of the variantset that used to call for variantset in repository
    "readsetId" : "<string>" // Id of the read
  }],
  "pointer" : [{ Reference(Sequence) }] // Pointer to next atomic sequence
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[ a fhir:Sequence;
```

```
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Sequence.identifier [ Identifier ], ... ; # 0..* Unique ID for this particular sequence. This is a FHIR-defined
    id
  fhir:Sequence.type [ code ]; # 0..1 aa | dna | rna
  fhir:Sequence.coordinateSystem [ integer ]; # 1..1 Base number of coordinate system (0 for 0-based numbering or
    coordinates, inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end)
  fhir:Sequence.patient [ Reference(Patient) ]; # 0..1 Who and/or what this is about
  fhir:Sequence.specimen [ Reference(Specimen) ]; # 0..1 Specimen used for sequencing
  fhir:Sequence.device [ Reference(Device) ]; # 0..1 The method for sequencing
  fhir:Sequence.performer [ Reference(Organization) ]; # 0..1 Who should be responsible for test result
  fhir:Sequence.quantity [ Quantity ]; # 0..1 The number of copies of the seqeunce of interest.  (RNASeq)
  fhir:Sequence.referenceSeq [ # 0..1 A sequence used as reference
    fhir:Sequence.referenceSeq.chromosome [ CodeableConcept ]; # 0..1 Chromosome containing genetic finding
    fhir:Sequence.referenceSeq.genomeBuild [ string ]; # 0..1 The Genome Build used for reference, following GRCh build
    versions e.g. 'GRCh 37'
    fhir:Sequence.referenceSeq.referenceSeqId [ CodeableConcept ]; # 0..1 Reference identifier
    fhir:Sequence.referenceSeq.referenceSeqPointer [ Reference(Sequence) ]; # 0..1 A Pointer to another Sequence entity
    as reference sequence
    fhir:Sequence.referenceSeq.referenceSeqString [ string ]; # 0..1 A string to represent reference sequence
    fhir:Sequence.referenceSeq.strand [ integer ]; # 0..1 Directionality of DNA ( +1/-1)
    fhir:Sequence.referenceSeq.windowStart [ integer ]; # 1..1 Start position of the window on the  reference sequence
    fhir:Sequence.referenceSeq.windowEnd [ integer ]; # 1..1 End position of the window on the reference sequence
  ];
  fhir:Sequence.variant [ # 0..* Variant in sequence
    fhir:Sequence.variant.start [ integer ]; # 0..1 Start position of the variant on the  reference sequence
    fhir:Sequence.variant.end [ integer ]; # 0..1 End position of the variant on the reference sequence
    fhir:Sequence.variant.observedAllele [ string ]; # 0..1 Allele that was observed
    fhir:Sequence.variant.referenceAllele [ string ]; # 0..1 Allele in the reference sequence
    fhir:Sequence.variant.cigar [ string ]; # 0..1 Extended CIGAR string for aligning the sequence with reference bases
    fhir:Sequence.variant.variantPointer [ Reference(Observation) ]; # 0..1 Pointer to observed variant information
  ], ...;
  fhir:Sequence.observedSeq [ string ]; # 0..1 Sequence that was observed
  fhir:Sequence.quality [ # 0..* An set of value as quality of sequence
    fhir:Sequence.quality.type [ code ]; # 1..1 indel | snp | unknown
    fhir:Sequence.quality.standardSequence [ CodeableConcept ]; # 0..1 Standard sequence for comparison
    fhir:Sequence.quality.start [ integer ]; # 0..1 Start position of the sequence
    fhir:Sequence.quality.end [ integer ]; # 0..1 End position of the sequence
    fhir:Sequence.quality.score [ Quantity ]; # 0..1 Quality score for the comparison
    fhir:Sequence.quality.method [ CodeableConcept ]; # 0..1 Method to get quality
    fhir:Sequence.quality.truthTP [ decimal ]; # 0..1 True positives from the perspective of the truth data
    fhir:Sequence.quality.queryTP [ decimal ]; # 0..1 True positives from the perspective of the query data
    fhir:Sequence.quality.truthFN [ decimal ]; # 0..1 False negatives
    fhir:Sequence.quality.queryFP [ decimal ]; # 0..1 False positives
    fhir:Sequence.quality.gtFP [ decimal ]; # 0..1 False positives where the non-REF alleles in the Truth and Query Call
     Sets match
    fhir:Sequence.quality.precision [ decimal ]; # 0..1 Precision of comparison
    fhir:Sequence.quality.recall [ decimal ]; # 0..1 Recall of comparison
    fhir:Sequence.quality.fScore [ decimal ]; # 0..1 F-score
  ], ...;
  fhir:Sequence.readCoverage [ integer ]; # 0..1 Average number of reads representing a given nucleotide in the
    reconstructed sequence
  fhir:Sequence.repository [ # 0..* External repository which contains detailed report related with observedSeq in this
    resource
    fhir:Sequence.repository.type [ code ]; # 1..1 directlink | openapi | login | oauth | other
    fhir:Sequence.repository.url [ uri ]; # 0..1 URI of the repository
    fhir:Sequence.repository.name [ string ]; # 0..1 Repository's name
    fhir:Sequence.repository.datasetId [ string ]; # 0..1 Id of the dataset that used to call for dataset in repository
    fhir:Sequence.repository.variantsetId [ string ]; # 0..1 Id of the variantset that used to call for variantset in
    repository
    fhir:Sequence.repository.readsetId [ string ]; # 0..1 Id of the read
  ], ...;
  fhir:Sequence.pointer [ Reference(Sequence) ], ... ; # 0..* Pointer to next atomic sequence
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

Sequence | ?I | | DomainResource | Information about a biological sequence + Only 0 and 1 are valid for coordinateSystem Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | Unique ID for this particular sequence. This is a FHIR-defined id

type | ? | 0..1 | code | aa | dna | rna sequenceType (Example)

coordinateSystem | ? | 1..1 | integer | Base number of coordinate system (0 for 0-based numbering or coordinates, inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end)

patient | ? | 0..1 | Reference(Patient) | Who and/or what this is about

specimen | ? | 0..1 | Reference(Specimen) | Specimen used for sequencing

device | ? | 0..1 | Reference(Device) | The method for sequencing

performer | ? | 0..1 | Reference(Organization) | Who should be responsible for test result

quantity | ? | 0..1 | Quantity | The number of copies of the seqeunce of interest. (RNASeq)

referenceSeq | ?I | 0..1 | BackboneElement | A sequence used as reference + Only +1 and -1 are valid for strand + GenomeBuild and chromosome must be both contained if either one of them is contained + Have and only have one of the following elements in referenceSeq : 1. genomeBuild ; 2 referenceSeqId; 3. referenceSeqPointer; 4. referenceSeqString;
chromosome | ? | 0..1 | CodeableConcept | Chromosome containing genetic finding chromosome-human (Example)
genomeBuild | ? | 0..1 | string | The Genome Build used for reference, following GRCh build versions e.g. 'GRCh 37'
referenceSeqId | ? | 0..1 | CodeableConcept | Reference identifier ENSEMBL (Example)
referenceSeqPointer | ? | 0..1 | Reference(Sequence) | A Pointer to another Sequence entity as reference sequence
referenceSeqString | ? | 0..1 | string | A string to represent reference sequence
strand | ? | 0..1 | integer | Directionality of DNA (+1/-1)
windowStart | ? | 1..1 | integer | Start position of the window on the reference sequence
windowEnd | ? | 1..1 | integer | End position of the window on the reference sequence
variant | ? | 0..* | BackboneElement | Variant in sequence
start | ? | 0..1 | integer | Start position of the variant on the reference sequence
end | ? | 0..1 | integer | End position of the variant on the reference sequence
observedAllele | ? | 0..1 | string | Allele that was observed
referenceAllele | ? | 0..1 | string | Allele in the reference sequence
cigar | ? | 0..1 | string | Extended CIGAR string for aligning the sequence with reference bases
variantPointer | ? | 0..1 | Reference(Observation) | Pointer to observed variant information
observedSeq | ? | 0..1 | string | Sequence that was observed
quality | ? | 0..* | BackboneElement | An set of value as quality of sequence
type | ? | 1..1 | code | indel | snp | unknown qualityType (Required)
standardSequence | ? | 0..1 | CodeableConcept | Standard sequence for comparison FDA-StandardSequence (Example)
start | ? | 0..1 | integer | Start position of the sequence
end | ? | 0..1 | integer | End position of the sequence
score | ? | 0..1 | Quantity | Quality score for the comparison
method | ? | 0..1 | CodeableConcept | Method to get quality FDA-Method (Example)
truthTP | ? | 0..1 | decimal | True positives from the perspective of the truth data
queryTP | ? | 0..1 | decimal | True positives from the perspective of the query data
truthFN | ? | 0..1 | decimal | False negatives
queryFP | ? | 0..1 | decimal | False positives
gtFP | ? | 0..1 | decimal | False positives where the non-REF alleles in the Truth and Query Call Sets match
precision | ? | 0..1 | decimal | Precision of comparison
recall | ? | 0..1 | decimal | Recall of comparison
fScore | ? | 0..1 | decimal | F-score
readCoverage | ? | 0..1 | integer | Average number of reads representing a given nucleotide in the reconstructed sequence
repository | ? | 0..* | BackboneElement | External repository which contains detailed report related with observedSeq in this resource
type | ? | 1..1 | code | directlink | openapi | login | oauth | other repositoryType (Required)
url | ? | 0..1 | uri | URI of the repository
name | ? | 0..1 | string | Repository's name
datasetId | ? | 0..1 | string | Id of the dataset that used to call for dataset in repository
variantsetId | ? | 0..1 | string | Id of the variantset that used to call for variantset in repository
readsetId | ? | 0..1 | string | Id of the read
pointer | ? | 0..* | Reference(Sequence) | Pointer to next atomic sequence
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Sequence xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique ID for this particular sequence. This is a FHIR-defined id --></identifier>
 <type value="[code]"/><!-- 0..1 aa | dna | rna -->
 <coordinateSystem value="[integer]"/><!-- 1..1 Base number of coordinate system (0 for 0-based numbering or
    coordinates, inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end) -->
 <patient><!-- 0..1 Reference(Patient) Who and/or what this is about --></patient>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for sequencing --></specimen>
 <device><!-- 0..1 Reference(Device) The method for sequencing --></device>
 <performer><!-- 0..1 Reference(Organization) Who should be responsible for test result --></performer>
 <quantity><!-- 0..1 Quantity The number of copies of the seqeunce of interest.  (RNASeq) --></quantity>
 <referenceSeq>  <!-- 0..1 A sequence used as reference -->
  <chromosome><!-- 0..1 CodeableConcept Chromosome containing genetic finding --></chromosome>
  <genomeBuild value="[string]"/><!-- 0..1 The Genome Build used for reference, following GRCh build versions e.g. 'GRCh
     37' -->
  <referenceSeqId><!-- 0..1 CodeableConcept Reference identifier --></referenceSeqId>
  <referenceSeqPointer><!-- 0..1 Reference(Sequence) A Pointer to another Sequence entity as reference sequence
    --></referenceSeqPointer>
  <referenceSeqString value="[string]"/><!-- 0..1 A string to represent reference sequence -->
  <strand value="[integer]"/><!-- 0..1 Directionality of DNA ( +1/-1) -->
  <windowStart value="[integer]"/><!-- 1..1 Start position of the window on the  reference sequence -->
  <windowEnd value="[integer]"/><!-- 1..1 End position of the window on the reference sequence -->
 </referenceSeq>
 <variant>  <!-- 0..* Variant in sequence -->
  <start value="[integer]"/><!-- 0..1 Start position of the variant on the  reference sequence -->
  <end value="[integer]"/><!-- 0..1 End position of the variant on the reference sequence -->
  <observedAllele value="[string]"/><!-- 0..1 Allele that was observed -->
  <referenceAllele value="[string]"/><!-- 0..1 Allele in the reference sequence -->
  <cigar value="[string]"/><!-- 0..1 Extended CIGAR string for aligning the sequence with reference bases -->
  <variantPointer><!-- 0..1 Reference(Observation) Pointer to observed variant information --></variantPointer>
```

```
 </variant>
 <observedSeq value="[string]"/><!-- 0..1 Sequence that was observed -->
 <quality>  <!-- 0..* An set of value as quality of sequence -->
  <type value="[code]"/><!-- 1..1 indel | snp | unknown -->
  <standardSequence><!-- 0..1 CodeableConcept Standard sequence for comparison --></standardSequence>
  <start value="[integer]"/><!-- 0..1 Start position of the sequence -->
  <end value="[integer]"/><!-- 0..1 End position of the sequence -->
  <score><!-- 0..1 Quantity Quality score for the comparison --></score>
  <method><!-- 0..1 CodeableConcept Method to get quality --></method>
  <truthTP value="[decimal]"/><!-- 0..1 True positives from the perspective of the truth data -->
  <queryTP value="[decimal]"/><!-- 0..1 True positives from the perspective of the query data -->
  <truthFN value="[decimal]"/><!-- 0..1 False negatives -->
  <queryFP value="[decimal]"/><!-- 0..1 False positives -->
  <gtFP value="[decimal]"/><!-- 0..1 False positives where the non-REF alleles in the Truth and Query Call Sets match
    -->
  <precision value="[decimal]"/><!-- 0..1 Precision of comparison -->
  <recall value="[decimal]"/><!-- 0..1 Recall of comparison -->
  <fScore value="[decimal]"/><!-- 0..1 F-score -->
 </quality>
 <readCoverage value="[integer]"/><!-- 0..1 Average number of reads representing a given nucleotide in the reconstructed
     sequence -->
 <repository>  <!-- 0..* External repository which contains detailed report related with observedSeq in this resource
    -->
  <type value="[code]"/><!-- 1..1 directlink | openapi | login | oauth | other -->
  <url value="[uri]"/><!-- 0..1 URI of the repository -->
  <name value="[string]"/><!-- 0..1 Repository's name -->
  <datasetId value="[string]"/><!-- 0..1 Id of the dataset that used to call for dataset in repository -->
  <variantsetId value="[string]"/><!-- 0..1 Id of the variantset that used to call for variantset in repository -->
  <readsetId value="[string]"/><!-- 0..1 Id of the read -->
 </repository>
 <pointer><!-- 0..* Reference(Sequence) Pointer to next atomic sequence --></pointer>
</Sequence>
```

JSON Template

```
{
  "resourceType" : "Sequence",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique ID for this particular sequence. This is a FHIR-defined id
  "type" : "<code>", // aa | dna | rna
  "coordinateSystem" : <integer>, // R!  Base number of coordinate system (0 for 0-based numbering or coordinates,
    inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end)
  "patient" : { Reference(Patient) }, // Who and/or what this is about
  "specimen" : { Reference(Specimen) }, // Specimen used for sequencing
  "device" : { Reference(Device) }, // The method for sequencing
  "performer" : { Reference(Organization) }, // Who should be responsible for test result
  "quantity" : { Quantity }, // The number of copies of the seqeunce of interest.  (RNASeq)
  "referenceSeq" : { // A sequence used as reference
    "chromosome" : { CodeableConcept }, // Chromosome containing genetic finding
    "genomeBuild" : "<string>", // The Genome Build used for reference, following GRCh build versions e.g. 'GRCh 37'
    "referenceSeqId" : { CodeableConcept }, // Reference identifier
    "referenceSeqPointer" : { Reference(Sequence) }, // A Pointer to another Sequence entity as reference sequence
    "referenceSeqString" : "<string>", // A string to represent reference sequence
    "strand" : <integer>, // Directionality of DNA ( +1/-1)
    "windowStart" : <integer>, // R!  Start position of the window on the  reference sequence
    "windowEnd" : <integer> // R!  End position of the window on the reference sequence
  },
  "variant" : [{ // Variant in sequence
    "start" : <integer>, // Start position of the variant on the  reference sequence
    "end" : <integer>, // End position of the variant on the reference sequence
    "observedAllele" : "<string>", // Allele that was observed
    "referenceAllele" : "<string>", // Allele in the reference sequence
    "cigar" : "<string>", // Extended CIGAR string for aligning the sequence with reference bases
    "variantPointer" : { Reference(Observation) } // Pointer to observed variant information
  }],
  "observedSeq" : "<string>", // Sequence that was observed
  "quality" : [{ // An set of value as quality of sequence
    "type" : "<code>", // R!  indel | snp | unknown
    "standardSequence" : { CodeableConcept }, // Standard sequence for comparison
    "start" : <integer>, // Start position of the sequence
    "end" : <integer>, // End position of the sequence
    "score" : { Quantity }, // Quality score for the comparison
    "method" : { CodeableConcept }, // Method to get quality
    "truthTP" : <decimal>, // True positives from the perspective of the truth data
    "queryTP" : <decimal>, // True positives from the perspective of the query data
    "truthFN" : <decimal>, // False negatives
    "queryFP" : <decimal>, // False positives
    "gtFP" : <decimal>, // False positives where the non-REF alleles in the Truth and Query Call Sets match
    "precision" : <decimal>, // Precision of comparison
    "recall" : <decimal>, // Recall of comparison
    "fScore" : <decimal> // F-score
  }],
  "readCoverage" : <integer>, // Average number of reads representing a given nucleotide in the reconstructed sequence
  "repository" : [{ // External repository which contains detailed report related with observedSeq in this resource
    "type" : "<code>", // R!  directlink | openapi | login | oauth | other
    "url" : "<uri>", // URI of the repository
    "name" : "<string>", // Repository's name
    "datasetId" : "<string>", // Id of the dataset that used to call for dataset in repository
    "variantsetId" : "<string>", // Id of the variantset that used to call for variantset in repository
```

```
    "readsetId" : "<string>" // Id of the read
  }],
  "pointer" : [{ Reference(Sequence) }] // Pointer to next atomic sequence
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Sequence;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Sequence.identifier [ Identifier ], ... ; # 0..* Unique ID for this particular sequence. This is a FHIR-defined
    id
  fhir:Sequence.type [ code ]; # 0..1 aa | dna | rna
  fhir:Sequence.coordinateSystem [ integer ]; # 1..1 Base number of coordinate system (0 for 0-based numbering or
    coordinates, inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end)
  fhir:Sequence.patient [ Reference(Patient) ]; # 0..1 Who and/or what this is about
  fhir:Sequence.specimen [ Reference(Specimen) ]; # 0..1 Specimen used for sequencing
  fhir:Sequence.device [ Reference(Device) ]; # 0..1 The method for sequencing
  fhir:Sequence.performer [ Reference(Organization) ]; # 0..1 Who should be responsible for test result
  fhir:Sequence.quantity [ Quantity ]; # 0..1 The number of copies of the seqeunce of interest.  (RNASeq)
  fhir:Sequence.referenceSeq [ # 0..1 A sequence used as reference
    fhir:Sequence.referenceSeq.chromosome [ CodeableConcept ]; # 0..1 Chromosome containing genetic finding
    fhir:Sequence.referenceSeq.genomeBuild [ string ]; # 0..1 The Genome Build used for reference, following GRCh build
    versions e.g. 'GRCh 37'
    fhir:Sequence.referenceSeq.referenceSeqId [ CodeableConcept ]; # 0..1 Reference identifier
    fhir:Sequence.referenceSeq.referenceSeqPointer [ Reference(Sequence) ]; # 0..1 A Pointer to another Sequence entity
    as reference sequence
    fhir:Sequence.referenceSeq.referenceSeqString [ string ]; # 0..1 A string to represent reference sequence
    fhir:Sequence.referenceSeq.strand [ integer ]; # 0..1 Directionality of DNA ( +1/-1)
    fhir:Sequence.referenceSeq.windowStart [ integer ]; # 1..1 Start position of the window on the  reference sequence
    fhir:Sequence.referenceSeq.windowEnd [ integer ]; # 1..1 End position of the window on the reference sequence
  ];
  fhir:Sequence.variant [ # 0..* Variant in sequence
    fhir:Sequence.variant.start [ integer ]; # 0..1 Start position of the variant on the  reference sequence
    fhir:Sequence.variant.end [ integer ]; # 0..1 End position of the variant on the reference sequence
    fhir:Sequence.variant.observedAllele [ string ]; # 0..1 Allele that was observed
    fhir:Sequence.variant.referenceAllele [ string ]; # 0..1 Allele in the reference sequence
    fhir:Sequence.variant.cigar [ string ]; # 0..1 Extended CIGAR string for aligning the sequence with reference bases
    fhir:Sequence.variant.variantPointer [ Reference(Observation) ]; # 0..1 Pointer to observed variant information
  ], ...;
  fhir:Sequence.observedSeq [ string ]; # 0..1 Sequence that was observed
  fhir:Sequence.quality [ # 0..* An set of value as quality of sequence
    fhir:Sequence.quality.type [ code ]; # 1..1 indel | snp | unknown
    fhir:Sequence.quality.standardSequence [ CodeableConcept ]; # 0..1 Standard sequence for comparison
    fhir:Sequence.quality.start [ integer ]; # 0..1 Start position of the sequence
    fhir:Sequence.quality.end [ integer ]; # 0..1 End position of the sequence
    fhir:Sequence.quality.score [ Quantity ]; # 0..1 Quality score for the comparison
    fhir:Sequence.quality.method [ CodeableConcept ]; # 0..1 Method to get quality
    fhir:Sequence.quality.truthTP [ decimal ]; # 0..1 True positives from the perspective of the truth data
    fhir:Sequence.quality.queryTP [ decimal ]; # 0..1 True positives from the perspective of the query data
    fhir:Sequence.quality.truthFN [ decimal ]; # 0..1 False negatives
    fhir:Sequence.quality.queryFP [ decimal ]; # 0..1 False positives
    fhir:Sequence.quality.gtFP [ decimal ]; # 0..1 False positives where the non-REF alleles in the Truth and Query Call
     Sets match
    fhir:Sequence.quality.precision [ decimal ]; # 0..1 Precision of comparison
    fhir:Sequence.quality.recall [ decimal ]; # 0..1 Recall of comparison
    fhir:Sequence.quality.fScore [ decimal ]; # 0..1 F-score
  ], ...;
  fhir:Sequence.readCoverage [ integer ]; # 0..1 Average number of reads representing a given nucleotide in the
    reconstructed sequence
  fhir:Sequence.repository [ # 0..* External repository which contains detailed report related with observedSeq in this
    resource
    fhir:Sequence.repository.type [ code ]; # 1..1 directlink | openapi | login | oauth | other
    fhir:Sequence.repository.url [ uri ]; # 0..1 URI of the repository
    fhir:Sequence.repository.name [ string ]; # 0..1 Repository's name
    fhir:Sequence.repository.datasetId [ string ]; # 0..1 Id of the dataset that used to call for dataset in repository
    fhir:Sequence.repository.variantsetId [ string ]; # 0..1 Id of the variantset that used to call for variantset in
    repository
    fhir:Sequence.repository.readsetId [ string ]; # 0..1 Id of the read
  ], ...;
  fhir:Sequence.pointer [ Reference(Sequence) ], ... ; # 0..* Pointer to next atomic sequence
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

10.5.3.1 Terminology Bindings

Path	Definition	Type	Reference
Sequence.type	Type if a sequence -- DNA, RNA, or amino acid sequence	Example	sequenceType
Sequence.referenceSeq.chromosome	Chromosome number for human	Example	chromosome-human

Sequence.referenceSeq.referenceSeqId | Reference identifier | Example | ENSEMBL
Sequence.quality.type | Type for quality report | Required | qualityType
Sequence.quality.standardSequence | Reference identifier of the sequence that used to mark the quality of tested samples. | Example | FDA-StandardSequence
Sequence.quality.method | The method used to evaluate the numerical quality of the observed sequence. | Example | FDA-Method
Sequence.repository.type | Type for access of external URI | Required | repositoryType

10.5.3.2 Constraints

* seq-3: Only 0 and 1 are valid for coordinateSystem (expression : coordinateSystem = 1 or coordinateSystem = 0)

* seq-4: On Sequence.referenceSeq: Only +1 and -1 are valid for strand (expression on Sequence.referenceSeq: strand.empty() or strand = 1 or strand = -1)

* seq-5: On Sequence.referenceSeq: GenomeBuild and chromosome must be both contained if either one of them is contained (expression on Sequence.referenceSeq: (chromosome.empty() and genomeBuild.empty()) or (chromosome.exists() and genomeBuild.exists()))

* seq-6: On Sequence.referenceSeq: Have and only have one of the following elements in referenceSeq : 1. genomeBuild ; 2 referenceSeqId; 3. referenceSeqPointer; 4. referenceSeqString; (expression on Sequence.referenceSeq: (genomeBuild.count()+referenceSeqId.count()+ referenceSeqPointer.count()+ referenceSeqString.count()) = 1)

10.5.4 Notes:

10.5.4.1 Sequence Coordinate System

When saving the variant information, the nucleic acid will be numbered with order. Some files are using 0-based coordiantes (e.g. BCD file format) while some files are using 1-based coordinates (e.g. VCF file format) . The element coordinateSystem in Sequence Resource contains this information.

Sequence.coordinateSystem constraints within two possible values: 0 for 0-based system, which will mark the sequence from number 0, while 1 for 1-based system, which will begin marking the first position with number 1. The significant difference between two system is the end position. In 0-based system, the end position is exclusive , which means the last position will not be contained in the sequence window while in 1-based system, the end position is inclusive , which means the last position is included in the sequence window. Note both systems has an inclusive start position.

For example, ACGTGCAT will be numbered from 1 to 8 in 1-based system and will be numbered from 0 to 8 in 0-based system to mark flanks (i.e. place between two Nucleotide). So the interval [3,5] in 1-based system is GTG while interval [2,5) in 0-based system is same segment GTG.

10.5.4.2 Choice of Strand

There are lots of definition concerning with the Directionality of DNA or RNA. Here strand element in Sequence resource is defined to have constraints with value +1 and -1 , to indicate what is the direction for nucleotide series. In order to avoid confusion, we use +1 to express the "plus" strand (5' to 3') and -1 to express the "minus" strand (3' to 5'). Here is a very simple mapping that indicates which number the expression will be represented.

+1 | -1
Watson | Crick
Sense | Antisense
positive | negative

Here is a small-scale example: if 5' GCGATATCGCAAA 3' is the data, then GC..AAA is plus strand while AAA..CG is the minus strand.

10.5.4.3 String usage for Reference Sequence and Observed Sequence

We hope that string of observedSeq can be constrained more than just any normal string but with notation tables. Here we present what the nucleotide acid stirng should be constrianed within the range:

A --> adenosine | M --> A C (amino) | U --> uridine | H --> A C T | V --> G C A
C --> cytidine | S --> G C (strong) | D --> G A T | K --> G T (keto)
G --> guanine | W --> A T (weak) | R --> G A (purine) | N --> A G C T (any)
T --> thymidine | B --> G T C | Y --> T C (pyrimidine) | - --> gap of indeterminate length

A alanine | P proline | B aspartate or asparagine | Q glutamine
C cystine | R arginine | D aspartate | S serine
E glutamate | T threonine | F phenylalanine | U selenocysteine
G glycine | V valine | H histidine | W tryptophan
I isoleucine | Y tyrosine | K lysine | Z glutamate or glutamine
L leucine | X any | M methionine | * translation stop
N asparagine | - gap of indeterminate length

10.5.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
chromosome | token | Chromosome number of the reference sequence | Sequence.referenceSeq.chromosome

coordinate | composite | Search parameter for region of the reference DNA sequence string. This will refer to part of a locus or part of a gene where search region will be represented in 1-based system. Since the coordinateSystem can either be 0-based or 1-based, this search query will include the result of both coordinateSystem that contains the equivalent segment of the gene or whole genome sequence. For example, a search for sequence can be represented as `coordinate=1$lt345$gt123`, this means it will search for the Sequence resource on chromosome 1 and with position >123 and <345, where in 1-based system resource, all strings within region 1:124-344 will be revealed, while in 0-based system resource, all strings within region 1:123-344 will be revealed. You may want to check detail about 0-based v.s. 1-based above. | Sequence.variant
end | number | End position (0-based exclusive, which menas the acid at this position will not be included, 1-based inclusive, which means the acid at this position will be included) of the reference sequence. | Sequence.referenceSeq.windowEnd
identifier | token | The unique identity for a particular sequence | Sequence.identifier
patient | reference | The subject that the observation is about | Sequence.patient (Patient)
start | number | Start position (0-based inclusive, 1-based inclusive, that means the nucleic acid or amino acid at this position will be included) of the reference sequence. | Sequence.referenceSeq.windowStart
type | token | Amino Acid Sequence/ DNA Sequence / RNA Sequence | Sequence.type

10.5.6 Resource Sequence - Examples

URL: https://hl7.org/fhir/STU3/sequence-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.5.6 Resource Sequence - Examples

Clinical Genomics Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

Simple sequence example | example | XML | JSON | Turtle
Sequence example from precisionFDA | fda-example | XML | JSON | Turtle
Example of representing complex variants in Sequence Resource | sequence-complex-variant | XML | JSON | Turtle
Example of a single varaint on a reference Sequence related to a haplotype observation in PGx example | example-pgx-1 | XML | JSON | Turtle
Example of another single varaint on a reference Sequence related to a haplotype observation in PGx example, which will form with the other haplotype data to be a diplotype observation. | example-pgx-2 | XML | JSON | Turtle
Example of a TPMT SNP data that support a haplotype observation | example-TPMT-one | XML | JSON | Turtle
Example of another TPMT SNP data that support a haplotype observation | example-TPMT-two | XML | JSON | Turtle
Sequence Graphic example-1 | graphic-example-1 | XML | JSON | Turtle
Sequence Graphic example-2 | graphic-example-2 | XML | JSON | Turtle
Sequence Graphic example-3 | graphic-example-3 | XML | JSON | Turtle
Sequence Graphic example-4 | graphic-example-4 | XML | JSON | Turtle
Sequence Graphic example-5 | graphic-example-5 | XML | JSON | Turtle
Example of a record with 0-based coordinatesystem | coord-0-base | XML | JSON | Turtle
Example of a record with 1-based coordinatesystem | coord-1-base | XML | JSON | Turtle
Sequence example for vcf comparison on precisionFDA | fda-vcf-comparison | XML | JSON | Turtle
Sequence example for vcf and Hap.py comparison on precisionFDA | fda-vcfeval-comparison | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.5.7 Resource Sequence - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/sequence-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.5.7 Resource Sequence - Detailed Descriptions

Clinical Genomics Work Group | Maturity Level: 1 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Sequence resource.

Sequence
Definition | Raw data describing a biological sequence.
Control | 1..1
Summary | true
Invariants | Defined on this element seq-3: Only 0 and 1 are valid for coordinateSystem (expression : coordinateSystem = 1 or coordinateSystem = 0, xpath: count(f:coordinateSystem[@value=0 and @value=1]) = 1)
Sequence.identifier
Definition | A unique identifier for this particular sequence instance. This is a FHIR-defined id.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Allows sequences to be distinguished and referenced.
Summary | true
Sequence.type
Definition | Amino Acid Sequence/ DNA Sequence / RNA Sequence.
Control | 0..1
Terminology Binding | sequenceType (Example)

Type | code
Summary | true
Sequence.coordinateSystem
Definition | Whether the sequence is numbered starting at 0 (0-based numbering or coordinates, inclusive start, exclusive end) or starting at 1 (1-based numbering, inclusive start and inclusive end).
Control | 1..1
Type | integer
Summary | true
Sequence.patient
Definition | The patient whose sequencing results are described by this resource.
Control | 0..1
Type | Reference(Patient)
Summary | true
Sequence.specimen
Definition | Specimen used for sequencing.
Control | 0..1
Type | Reference(Specimen)
Summary | true
Sequence.device
Definition | The method for sequencing, for example, chip information.
Control | 0..1
Type | Reference(Device)
Summary | true
Sequence.performer
Definition | The organization or lab that should be responsible for this result.
Control | 0..1
Type | Reference(Organization)
Summary | true
Sequence.quantity
Definition | The number of copies of the seqeunce of interest. (RNASeq).
Control | 0..1
Type | Quantity
Summary | true
Sequence.referenceSeq
Definition | A sequence that is used as a reference to describe variants that are present in a sequence analyzed.
Control | 0..1
Summary | true
Invariants | Defined on this element seq-4: Only +1 and -1 are valid for strand (expression : strand.empty() or strand = 1 or strand = -1, xpath: not(exists(f:strand)) or count(f:strand[@value=-1 and @value=1]) = 1) seq-5: GenomeBuild and chromosome must be both contained if either one of them is contained (expression : (chromosome.empty() and genomeBuild.empty()) or (chromosome.exists() and genomeBuild.exists()), xpath: (exists(f:chromosome) and exists(f:genomeBuild)) or (not(exists(f:chromosome)) and not(exists(f:genomeBuild)))) seq-6: Have and only have one of the following elements in referenceSeq : 1. genomeBuild ; 2 referenceSeqId; 3. referenceSeqPointer; 4. referenceSeqString; (expression : (genomeBuild.count()+referenceSeqId.count()+ referenceSeqPointer.count()+ referenceSeqString.count()) = 1, xpath: count(f:genomeBuild)+count(f:referenceSeqId)+count(f:referenceSeqPointer)+count(f:referenceSeqString)=1)
Sequence.referenceSeq.chromosome
Definition | Structural unit composed of a nucleic acid molecule which controls its own replication through the interaction of specific proteins at one or more origins of replication (SO:0000340).
Control | 0..1
Terminology Binding | chromosome-human (Example)
Type | CodeableConcept
Summary | true
Sequence.referenceSeq.genomeBuild
Definition | The Genome Build used for reference, following GRCh build versions e.g. 'GRCh 37'. Version number must be included if a versioned release of a primary build was used.
Control | 0..1
Type | string
Summary | true
Sequence.referenceSeq.referenceSeqId
Definition | Reference identifier of reference sequence submitted to NCBI. It must match the type in the Sequence.type field. For example, the prefix, “NG” identifies reference sequence for genes, “NM” for messenger RNA transcripts, and “NP_” for amino acid sequences.
Control | 0..1
Terminology Binding | ENSEMBL (Example)
Type | CodeableConcept
Summary | true
Sequence.referenceSeq.referenceSeqPointer
Definition | A Pointer to another Sequence entity as reference sequence.
Control | 0..1
Type | Reference(Sequence)
Summary | true
Sequence.referenceSeq.referenceSeqString
Definition | A string like "ACGT".
Control | 0..1
Type | string

Summary | true
Sequence.referenceSeq.strand
Definition | Directionality of DNA sequence. Available values are "1" for the plus strand (5' to 3')/Watson/Sense/positive and "-1" for the minus strand(3' to 5')/Crick/Antisense/negative.
Control | 0..1
Type | integer
Summary | true
Sequence.referenceSeq.windowStart
Definition | Start position of the window on the reference sequence. If the coordinate system is either 0-based or 1-based, then start position is inclusive.
Control | 1..1
Type | integer
Summary | true
Sequence.referenceSeq.windowEnd
Definition | End position of the window on the reference sequence. If the coordinate system is 0-based then end is is exclusive and does not include the last position. If the coordinate system is 1-base, then end is inclusive and includes the last position.
Control | 1..1
Type | integer
Summary | true
Sequence.variant
Definition | The definition of variant here originates from Sequence ontology (variant_of). This element can represent amino acid or nucleic sequence change(including insertion,deletion,SNP,etc.) It can represent some complex mutation or segment variation with the assist of CIGAR string.
Control | 0..*
Summary | true
Sequence.variant.start
Definition | Start position of the variant on the reference sequence.If the coordinate system is either 0-based or 1-based, then start position is inclusive.
Control | 0..1
Type | integer
Summary | true
Sequence.variant.end
Definition | End position of the variant on the reference sequence.If the coordinate system is 0-based then end is is exclusive and does not include the last position. If the coordinate system is 1-base, then end is inclusive and includes the last position.
Control | 0..1
Type | integer
Summary | true
Sequence.variant.observedAllele
Definition | An allele is one of a set of coexisting sequence variants of a gene (SO:0001023). Nucleotide(s)/amino acids from start position of sequence to stop position of sequence on the positive (+) strand of the observed sequence. When the sequence type is DNA, it should be the sequence on the positive (+) strand. This will lay in the range between variant.start and variant.end.
Control | 0..1
Type | string
Summary | true
Sequence.variant.referenceAllele
Definition | An allele is one of a set of coexisting sequence variants of a gene (SO:0001023). Nucleotide(s)/amino acids from start position of sequence to stop position of sequence on the positive (+) strand of the reference sequence. When the sequence type is DNA, it should be the sequence on the positive (+) strand. This will lay in the range between variant.start and variant.end.
Control | 0..1
Type | string
Summary | true
Sequence.variant.cigar
Definition | Extended CIGAR string for aligning the sequence with reference bases. See detailed documentation here .
Control | 0..1
Type | string
Summary | true
Sequence.variant.variantPointer
Definition | A pointer to an Observation containing variant information.
Control | 0..1
Type | Reference(Observation)
Summary | true
Sequence.observedSeq
Definition | Sequence that was observed. It is the result marked by referenceSeq along with variant records on referenceSeq. This shall starts from referenceSeq.windowStart and end by referenceSeq.windowEnd.
Control | 0..1
Type | string
Summary | true
Sequence.quality
Definition | An experimental feature attribute that defines the quality of the feature in a quantitative way, such as a phred quality score (SO:0001686).
Control | 0..*
Summary | true
Sequence.quality.type
Definition | INDEL / SNP / Undefined variant.

Control | 1..1
Terminology Binding | qualityType (Required)
Type | code
Summary | true
Sequence.quality.standardSequence
Definition | Gold standard sequence used for comparing against.
Control | 0..1
Terminology Binding | FDA-StandardSequence (Example)
Type | CodeableConcept
Summary | true
Sequence.quality.start
Definition | Start position of the sequence. If the coordinate system is either 0-based or 1-based, then start position is inclusive.
Control | 0..1
Type | integer
Summary | true
Sequence.quality.end
Definition | End position of the sequence.If the coordinate system is 0-based then end is is exclusive and does not include the last position. If the coordinate system is 1-base, then end is inclusive and includes the last position.
Control | 0..1
Type | integer
Summary | true
Sequence.quality.score
Definition | The score of an experimentally derived feature such as a p-value (SO:0001685).
Control | 0..1
Type | Quantity
Summary | true
Sequence.quality.method
Definition | Which method is used to get sequence quality.
Control | 0..1
Terminology Binding | FDA-Method (Example)
Type | CodeableConcept
Summary | true
Sequence.quality.truthTP
Definition | True positives, from the perspective of the truth data, i.e. the number of sites in the Truth Call Set for which there are paths through the Query Call Set that are consistent with all of the alleles at this site, and for which there is an accurate genotype call for the event.
Control | 0..1
Type | decimal
Summary | true
Sequence.quality.queryTP
Definition | True positives, from the perspective of the query data, i.e. the number of sites in the Query Call Set for which there are paths through the Truth Call Set that are consistent with all of the alleles at this site, and for which there is an accurate genotype call for the event.
Control | 0..1
Type | decimal
Summary | true
Sequence.quality.truthFN
Definition | False negatives, i.e. the number of sites in the Truth Call Set for which there is no path through the Query Call Set that is consistent with all of the alleles at this site, or sites for which there is an inaccurate genotype call for the event. Sites with correct variant but incorrect genotype are counted here.
Control | 0..1
Type | decimal
Summary | true
Sequence.quality.queryFP
Definition | False positives, i.e. the number of sites in the Query Call Set for which there is no path through the Truth Call Set that is consistent with this site. Sites with correct variant but incorrect genotype are counted here.
Control | 0..1
Type | decimal
Summary | true
Sequence.quality.gtFP
Definition | The number of false positives where the non-REF alleles in the Truth and Query Call Sets match (i.e. cases where the truth is 1/1 and the query is 0/1 or similar).
Control | 0..1
Type | decimal
Summary | true
Sequence.quality.precision
Definition | QUERY.TP / (QUERY.TP + QUERY.FP).
Control | 0..1
Type | decimal
Summary | true
Sequence.quality.recall
Definition | TRUTH.TP / (TRUTH.TP + TRUTH.FN).
Control | 0..1
Type | decimal
Summary | true

Sequence.quality.fScore
Definition | Harmonic mean of Recall and Precision, computed as: 2 * precision * recall / (precision + recall).
Control | 0..1
Type | decimal
Summary | true
Sequence.readCoverage
Definition | Coverage (read depth or depth) is the average number of reads representing a given nucleotide in the reconstructed sequence.
Control | 0..1
Type | integer
Summary | true
Sequence.repository
Definition | Configurations of the external repository. The repository shall store target's observedSeq or records related with target's observedSeq.
Control | 0..*
Summary | true
Sequence.repository.type
Definition | Click and see / RESTful API / Need login to see / RESTful API with authentication / Other ways to see resource.
Control | 1..1
Terminology Binding | repositoryType (Required)
Type | code
Summary | true
Sequence.repository.url
Definition | URI of an external repository which contains further details about the genetics data.
Control | 0..1
Type | uri
Summary | true
Sequence.repository.name
Definition | URI of an external repository which contains further details about the genetics data.
Control | 0..1
Type | string
Summary | true
Sequence.repository.datasetId
Definition | Id of the variant in this external repository. The server will understand how to use this id to call for more info about datasets in external repository.
Control | 0..1
Type | string
Summary | true
Sequence.repository.variantsetId
Definition | Id of the variantset in this external repository. The server will understand how to use this id to call for more info about variantsets in external repository.
Control | 0..1
Type | string
Summary | true
Sequence.repository.readsetId
Definition | Id of the read in this external repository.
Control | 0..1
Type | string
Summary | true
Sequence.pointer
Definition | Pointer to next atomic sequence which at most contains one variant.
Control | 0..*
Type | Reference(Sequence)
Summary | true

10.5.8 Resource Sequence - Mappings

URL: https://hl7.org/fhir/STU3/sequence-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.5.8 Resource Sequence - Mappings

Clinical Genomics Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings for the sequence resource.

10.5.9 Resource Sequence - Design Notes

URL: https://hl7.org/fhir/STU3/sequence-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.5.9 Resource Sequence - Design Notes

to do

10.5.10 Resource Sequence - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/sequence-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.5.10 Resource Sequence - Extensions & Profiles

Clinical Genomics Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Sequence resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

10.5.10.1 Profiles

No Profiles defined for this resource

10.5.10.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

observation-geneticsSequence | Sequence | for Observation-genetics

10.5.10.3 Search Extensions

No Search Extensions defined for this resource

10.6 Resource Specimen - Content

URL: https://hl7.org/fhir/STU3/specimen.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Patient, Practitioner

A sample to be used for analysis.

10.6.1 Scope and Usage

Any material sample:

* taken from a biological entity, living or dead

* taken from a physical object or the environment

Some specimens are biological and can contain one or more components including but not limited to cellular molecules, cells, tissues, organs, body fluids, embryos, and body excretory products (source: NCI Thesaurus , modified).

The specimen resource covers substances used for diagnostic and environmental testing. The focus of the specimen resource is the process for gathering, maintaining and processing the specimen as well as where the specimen originated. This is distinct from the use of Substance which is only used when these other aspects are not relevant.

10.6.2 Background and Context

The current definition of the specimen resource contains only basic information about specimen containers. It does not address the recursive nature of containers or the tracking of the location of a container within its parent container (for instance: a tube in a tray in a rack in a freezer). The frequency with which these elements are tracked may depend on the context of use; general lab, biobanking, etc. Comments from reviewers on the appropriate scope for this resource, and the need for tracking related specimen management attributes, are welcomed.

This resource is referenced by DiagnosticReport, Media, Observation, ProcedureRequest and Sequence

10.6.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Specimen | | | DomainResource | Sample for analysis Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Identifier
accessionIdentifier | ? | 0..1 | Identifier | Identifier assigned by the lab
status | ?!? | 0..1 | code | available | unavailable | unsatisfactory | entered-in-error SpecimenStatus (Required)
type | ? | 0..1 | CodeableConcept | Kind of material that forms the specimen v2 Specimen Type (Example)
subject | ? | 1..1 | Reference(Patient | Group | Device | Substance) | Where the specimen came from. This may be from the patient(s) or from the environment or a device
receivedTime | ? | 0..1 | dateTime | The time when specimen was received for processing
parent | | 0..* | Reference(Specimen) | Specimen from which this specimen originated
request | | 0..* | Reference(ProcedureRequest) | Why the specimen was collected
collection | | 0..1 | BackboneElement | Collection details
collector | ? | 0..1 | Reference(Practitioner) | Who collected the specimen
collected[x] | ? | 0..1 | | Collection time
collectedDateTime | | | dateTime
collectedPeriod | | | Period
quantity | | 0..1 | SimpleQuantity | The quantity of specimen collected
method | | 0..1 | CodeableConcept | Technique used to perform collection FHIR Specimen Collection Method (Example)
bodySite | | 0..1 | CodeableConcept | Anatomical collection site SNOMED CT Body Structures (Example)
processing | | 0..* | BackboneElement | Processing and processing step details
description | | 0..1 | string | Textual description of procedure
procedure | | 0..1 | CodeableConcept | Indicates the treatment step applied to the specimen SpecimenProcessingProcedure (Example)
additive | | 0..* | Reference(Substance) | Material used in the processing step
time[x] | | 0..1 | | Date and time of specimen processing
timeDateTime | | | dateTime
timePeriod | | | Period
container | | 0..* | BackboneElement | Direct container of specimen (tube/slide, etc.)
identifier | ? | 0..* | Identifier | Id for the container
description | | 0..1 | string | Textual description of the container
type | | 0..1 | CodeableConcept | Kind of container directly associated with specimen SpecimenContainer (Example)
capacity | | 0..1 | SimpleQuantity | Container volume or size
specimenQuantity | | 0..1 | SimpleQuantity | Quantity of specimen within container
additive[x] | | 0..1 | | Additive associated with container v2 Additive/Preservative (Example)
additiveCodeableConcept | | | CodeableConcept
additiveReference | | | Reference(Substance)
note | | 0..* | Annotation | Comments
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Specimen xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Identifier --></identifier>
 <accessionIdentifier><!-- 0..1 Identifier Identifier assigned by the lab --></accessionIdentifier>
 <status value="[code]"/><!-- 0..1 available | unavailable | unsatisfactory | entered-in-error -->
 <type><!-- 0..1 CodeableConcept Kind of material that forms the specimen --></type>
 <subject><!-- 1..1 Reference(Patient|Group|Device|Substance) Where the specimen came from. This may be from the
    patient(s) or from the environment or a device --></subject>
 <receivedTime value="[dateTime]"/><!-- 0..1 The time when specimen was received for processing -->
 <parent><!-- 0..* Reference(Specimen) Specimen from which this specimen originated --></parent>
 <request><!-- 0..* Reference(ProcedureRequest) Why the specimen was collected --></request>
 <collection>  <!-- 0..1 Collection details -->
  <collector><!-- 0..1 Reference(Practitioner) Who collected the specimen --></collector>
  <collected[x]><!-- 0..1 dateTime|Period Collection time --></collected[x]>
  <quantity><!-- 0..1 Quantity(SimpleQuantity) The quantity of specimen collected --></quantity>
  <method><!-- 0..1 CodeableConcept Technique used to perform collection --></method>
  <bodySite><!-- 0..1 CodeableConcept Anatomical collection site --></bodySite>
 </collection>
 <processing>  <!-- 0..* Processing and processing step details -->
  <description value="[string]"/><!-- 0..1 Textual description of procedure -->
  <procedure><!-- 0..1 CodeableConcept Indicates the treatment step  applied to the specimen --></procedure>
  <additive><!-- 0..* Reference(Substance) Material used in the processing step --></additive>
  <time[x]><!-- 0..1 dateTime|Period Date and time of specimen processing --></time[x]>
 </processing>
 <container>  <!-- 0..* Direct container of specimen (tube/slide, etc.) -->
  <identifier><!-- 0..* Identifier Id for the container --></identifier>
  <description value="[string]"/><!-- 0..1 Textual description of the container -->
  <type><!-- 0..1 CodeableConcept Kind of container directly associated with specimen --></type>
  <capacity><!-- 0..1 Quantity(SimpleQuantity) Container volume or size --></capacity>
  <specimenQuantity><!-- 0..1 Quantity(SimpleQuantity) Quantity of specimen within container --></specimenQuantity>
  <additive[x]><!-- 0..1 CodeableConcept|Reference(Substance) Additive associated with container --></additive[x]>
 </container>
 <note><!-- 0..* Annotation Comments --></note>
</Specimen>
```

JSON Template

```
{
  "resourceType" : "Specimen",
  // from Resource: id, meta, implicitRules, and language
```

```
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Identifier
  "accessionIdentifier" : { Identifier }, // Identifier assigned by the lab
  "status" : "<code>", // available | unavailable | unsatisfactory | entered-in-error
  "type" : { CodeableConcept }, // Kind of material that forms the specimen
  "subject" : { Reference(Patient|Group|Device|Substance) }, // R!  Where the specimen came from. This may be from the
    patient(s) or from the environment or a device
  "receivedTime" : "<dateTime>", // The time when specimen was received for processing
  "parent" : [{ Reference(Specimen) }], // Specimen from which this specimen originated
  "request" : [{ Reference(ProcedureRequest) }], // Why the specimen was collected
  "collection" : { // Collection details
    "collector" : { Reference(Practitioner) }, // Who collected the specimen
    // collected[x]: Collection time. One of these 2:
    "collectedDateTime" : "<dateTime>",
    "collectedPeriod" : { Period },
    "quantity" : { Quantity(SimpleQuantity) }, // The quantity of specimen collected
    "method" : { CodeableConcept }, // Technique used to perform collection
    "bodySite" : { CodeableConcept } // Anatomical collection site
  },
  "processing" : [{ // Processing and processing step details
    "description" : "<string>", // Textual description of procedure
    "procedure" : { CodeableConcept }, // Indicates the treatment step  applied to the specimen
    "additive" : [{ Reference(Substance) }], // Material used in the processing step
    // time[x]: Date and time of specimen processing. One of these 2:
    "timeDateTime" : "<dateTime>"
    "timePeriod" : { Period }
  }],
  "container" : [{ // Direct container of specimen (tube/slide, etc.)
    "identifier" : [{ Identifier }], // Id for the container
    "description" : "<string>", // Textual description of the container
    "type" : { CodeableConcept }, // Kind of container directly associated with specimen
    "capacity" : { Quantity(SimpleQuantity) }, // Container volume or size
    "specimenQuantity" : { Quantity(SimpleQuantity) }, // Quantity of specimen within container
    // additive[x]: Additive associated with container. One of these 2:
    "additiveCodeableConcept" : { CodeableConcept }
    "additiveReference" : { Reference(Substance) }
  }],
  "note" : [{ Annotation }] // Comments
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Specimen;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Specimen.identifier [ Identifier ], ... ; # 0..* External Identifier
  fhir:Specimen.accessionIdentifier [ Identifier ]; # 0..1 Identifier assigned by the lab
  fhir:Specimen.status [ code ]; # 0..1 available | unavailable | unsatisfactory | entered-in-error
  fhir:Specimen.type [ CodeableConcept ]; # 0..1 Kind of material that forms the specimen
  fhir:Specimen.subject [ Reference(Patient|Group|Device|Substance) ]; # 1..1 Where the specimen came from. This may be
    from the patient(s) or from the environment or a device
  fhir:Specimen.receivedTime [ dateTime ]; # 0..1 The time when specimen was received for processing
  fhir:Specimen.parent [ Reference(Specimen) ], ... ; # 0..* Specimen from which this specimen originated
  fhir:Specimen.request [ Reference(ProcedureRequest) ], ... ; # 0..* Why the specimen was collected
  fhir:Specimen.collection [ # 0..1 Collection details
    fhir:Specimen.collection.collector [ Reference(Practitioner) ]; # 0..1 Who collected the specimen
    # Specimen.collection.collected[x] : 0..1 Collection time. One of these 2
      fhir:Specimen.collection.collectedDateTime [ dateTime ]
      fhir:Specimen.collection.collectedPeriod [ Period ]
    fhir:Specimen.collection.quantity [ Quantity(SimpleQuantity) ]; # 0..1 The quantity of specimen collected
    fhir:Specimen.collection.method [ CodeableConcept ]; # 0..1 Technique used to perform collection
    fhir:Specimen.collection.bodySite [ CodeableConcept ]; # 0..1 Anatomical collection site
  ];
  fhir:Specimen.processing [ # 0..* Processing and processing step details
    fhir:Specimen.processing.description [ string ]; # 0..1 Textual description of procedure
    fhir:Specimen.processing.procedure [ CodeableConcept ]; # 0..1 Indicates the treatment step  applied to the specimen
    fhir:Specimen.processing.additive [ Reference(Substance) ], ... ; # 0..* Material used in the processing step
    # Specimen.processing.time[x] : 0..1 Date and time of specimen processing. One of these 2
      fhir:Specimen.processing.timeDateTime [ dateTime ]
      fhir:Specimen.processing.timePeriod [ Period ]
  ], ...;
  fhir:Specimen.container [ # 0..* Direct container of specimen (tube/slide, etc.)
    fhir:Specimen.container.identifier [ Identifier ], ... ; # 0..* Id for the container
    fhir:Specimen.container.description [ string ]; # 0..1 Textual description of the container
    fhir:Specimen.container.type [ CodeableConcept ]; # 0..1 Kind of container directly associated with specimen
    fhir:Specimen.container.capacity [ Quantity(SimpleQuantity) ]; # 0..1 Container volume or size
    fhir:Specimen.container.specimenQuantity [ Quantity(SimpleQuantity) ]; # 0..1 Quantity of specimen within container
    # Specimen.container.additive[x] : 0..1 Additive associated with container. One of these 2
      fhir:Specimen.container.additiveCodeableConcept [ CodeableConcept ]
      fhir:Specimen.container.additiveReference [ Reference(Substance) ]
  ], ...;
  fhir:Specimen.note [ Annotation ], ... ; # 0..* Comments
]
```

Changes since DSTU2

Specimen
Specimen.request | Added Element
Specimen.processing | Added Element
Specimen.processing.description | Added Element
Specimen.processing.procedure | Added Element
Specimen.processing.additive | Added Element
Specimen.processing.time[x] | Added Element
Specimen.note | Added Element
Specimen.collection.comment | deleted
Specimen.treatment | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (1 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints
Specimen | | | DomainResource | Sample for analysis Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Identifier
accessionIdentifier | ? | 0..1 | Identifier | Identifier assigned by the lab
status | ?!? | 0..1 | code | available | unavailable | unsatisfactory | entered-in-error SpecimenStatus (Required)
type | ? | 0..1 | CodeableConcept | Kind of material that forms the specimen v2 Specimen Type (Example)
subject | ? | 1..1 | Reference(Patient | Group | Device | Substance) | Where the specimen came from. This may be from the patient(s) or from the environment or a device
receivedTime | ? | 0..1 | dateTime | The time when specimen was received for processing
parent | | 0..* | Reference(Specimen) | Specimen from which this specimen originated
request | | 0..* | Reference(ProcedureRequest) | Why the specimen was collected
collection | | 0..1 | BackboneElement | Collection details
collector | ? | 0..1 | Reference(Practitioner) | Who collected the specimen
collected[x] | ? | 0..1 | | Collection time
collectedDateTime | | | dateTime
collectedPeriod | | | Period
quantity | | 0..1 | SimpleQuantity | The quantity of specimen collected
method | | 0..1 | CodeableConcept | Technique used to perform collection FHIR Specimen Collection Method (Example)
bodySite | | 0..1 | CodeableConcept | Anatomical collection site SNOMED CT Body Structures (Example)
processing | | 0..* | BackboneElement | Processing and processing step details
description | | 0..1 | string | Textual description of procedure
procedure | | 0..1 | CodeableConcept | Indicates the treatment step applied to the specimen SpecimenProcessingProcedure (Example)
additive | | 0..* | Reference(Substance) | Material used in the processing step
time[x] | | 0..1 | | Date and time of specimen processing
timeDateTime | | | dateTime
timePeriod | | | Period
container | | 0..* | BackboneElement | Direct container of specimen (tube/slide, etc.)
identifier | ? | 0..* | Identifier | Id for the container
description | | 0..1 | string | Textual description of the container
type | | 0..1 | CodeableConcept | Kind of container directly associated with specimen SpecimenContainer (Example)
capacity | | 0..1 | SimpleQuantity | Container volume or size
specimenQuantity | | 0..1 | SimpleQuantity | Quantity of specimen within container
additive[x] | | 0..1 | | Additive associated with container v2 Additive/Preservative (Example)
additiveCodeableConcept | | | CodeableConcept
additiveReference | | | Reference(Substance)
note | | 0..* | Annotation | Comments
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Specimen xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Identifier --></identifier>
 <accessionIdentifier><!-- 0..1 Identifier Identifier assigned by the lab --></accessionIdentifier>
 <status value="[code]"/><!-- 0..1 available | unavailable | unsatisfactory | entered-in-error -->
 <type><!-- 0..1 CodeableConcept Kind of material that forms the specimen --></type>
 <subject><!-- 1..1 Reference(Patient|Group|Device|Substance) Where the specimen came from. This may be from the
    patient(s) or from the environment or a device --></subject>
 <receivedTime value="[dateTime]"/><!-- 0..1 The time when specimen was received for processing -->
 <parent><!-- 0..* Reference(Specimen) Specimen from which this specimen originated --></parent>
 <request><!-- 0..* Reference(ProcedureRequest) Why the specimen was collected --></request>
 <collection>  <!-- 0..1 Collection details -->
  <collector><!-- 0..1 Reference(Practitioner) Who collected the specimen --></collector>
  <collected[x]><!-- 0..1 dateTime|Period Collection time --></collected[x]>
  <quantity><!-- 0..1 Quantity(SimpleQuantity) The quantity of specimen collected --></quantity>
  <method><!-- 0..1 CodeableConcept Technique used to perform collection --></method>
  <bodySite><!-- 0..1 CodeableConcept Anatomical collection site --></bodySite>
```

```
 </collection>
 <processing>  <!-- 0..* Processing and processing step details -->
  <description value="[string]"/><!-- 0..1 Textual description of procedure -->
  <procedure><!-- 0..1 CodeableConcept Indicates the treatment step  applied to the specimen --></procedure>
  <additive><!-- 0..* Reference(Substance) Material used in the processing step --></additive>
  <time[x]><!-- 0..1 dateTime|Period Date and time of specimen processing --></time[x]>
 </processing>
 <container>  <!-- 0..* Direct container of specimen (tube/slide, etc.) -->
  <identifier><!-- 0..* Identifier Id for the container --></identifier>
  <description value="[string]"/><!-- 0..1 Textual description of the container -->
  <type><!-- 0..1 CodeableConcept Kind of container directly associated with specimen --></type>
  <capacity><!-- 0..1 Quantity(SimpleQuantity) Container volume or size --></capacity>
  <specimenQuantity><!-- 0..1 Quantity(SimpleQuantity) Quantity of specimen within container --></specimenQuantity>
  <additive[x]><!-- 0..1 CodeableConcept|Reference(Substance) Additive associated with container --></additive[x]>
 </container>
 <note><!-- 0..* Annotation Comments --></note>
</Specimen>
```

JSON Template

```
{
  "resourceType" : "Specimen",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Identifier
  "accessionIdentifier" : { Identifier }, // Identifier assigned by the lab
  "status" : "<code>", // available | unavailable | unsatisfactory | entered-in-error
  "type" : { CodeableConcept }, // Kind of material that forms the specimen
  "subject" : { Reference(Patient|Group|Device|Substance) }, // R!  Where the specimen came from. This may be from the
    patient(s) or from the environment or a device
  "receivedTime" : "<dateTime>", // The time when specimen was received for processing
  "parent" : [{ Reference(Specimen) }], // Specimen from which this specimen originated
  "request" : [{ Reference(ProcedureRequest) }], // Why the specimen was collected
  "collection" : { // Collection details
    "collector" : { Reference(Practitioner) }, // Who collected the specimen
    // collected[x]: Collection time. One of these 2:
    "collectedDateTime" : "<dateTime>",
    "collectedPeriod" : { Period },
    "quantity" : { Quantity(SimpleQuantity) }, // The quantity of specimen collected
    "method" : { CodeableConcept }, // Technique used to perform collection
    "bodySite" : { CodeableConcept } // Anatomical collection site
  },
  "processing" : [{ // Processing and processing step details
    "description" : "<string>", // Textual description of procedure
    "procedure" : { CodeableConcept }, // Indicates the treatment step  applied to the specimen
    "additive" : [{ Reference(Substance) }], // Material used in the processing step
    // time[x]: Date and time of specimen processing. One of these 2:
    "timeDateTime" : "<dateTime>"
    "timePeriod" : { Period }
  }],
  "container" : [{ // Direct container of specimen (tube/slide, etc.)
    "identifier" : [{ Identifier }], // Id for the container
    "description" : "<string>", // Textual description of the container
    "type" : { CodeableConcept }, // Kind of container directly associated with specimen
    "capacity" : { Quantity(SimpleQuantity) }, // Container volume or size
    "specimenQuantity" : { Quantity(SimpleQuantity) }, // Quantity of specimen within container
    // additive[x]: Additive associated with container. One of these 2:
    "additiveCodeableConcept" : { CodeableConcept }
    "additiveReference" : { Reference(Substance) }
  }],
  "note" : [{ Annotation }] // Comments
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Specimen;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Specimen.identifier [ Identifier ], ... ; # 0..* External Identifier
  fhir:Specimen.accessionIdentifier [ Identifier ]; # 0..1 Identifier assigned by the lab
  fhir:Specimen.status [ code ]; # 0..1 available | unavailable | unsatisfactory | entered-in-error
  fhir:Specimen.type [ CodeableConcept ]; # 0..1 Kind of material that forms the specimen
  fhir:Specimen.subject [ Reference(Patient|Group|Device|Substance) ]; # 1..1 Where the specimen came from. This may be
    from the patient(s) or from the environment or a device
  fhir:Specimen.receivedTime [ dateTime ]; # 0..1 The time when specimen was received for processing
  fhir:Specimen.parent [ Reference(Specimen) ], ... ; # 0..* Specimen from which this specimen originated
  fhir:Specimen.request [ Reference(ProcedureRequest) ], ... ; # 0..* Why the specimen was collected
  fhir:Specimen.collection [ # 0..1 Collection details
    fhir:Specimen.collection.collector [ Reference(Practitioner) ]; # 0..1 Who collected the specimen
    # Specimen.collection.collected[x] : 0..1 Collection time. One of these 2
      fhir:Specimen.collection.collectedDateTime [ dateTime ]
      fhir:Specimen.collection.collectedPeriod [ Period ]
    fhir:Specimen.collection.quantity [ Quantity(SimpleQuantity) ]; # 0..1 The quantity of specimen collected
    fhir:Specimen.collection.method [ CodeableConcept ]; # 0..1 Technique used to perform collection
    fhir:Specimen.collection.bodySite [ CodeableConcept ]; # 0..1 Anatomical collection site
  ];
```

```
  fhir:Specimen.processing [ # 0..* Processing and processing step details
    fhir:Specimen.processing.description [ string ]; # 0..1 Textual description of procedure
    fhir:Specimen.processing.procedure [ CodeableConcept ]; # 0..1 Indicates the treatment step  applied to the specimen
    fhir:Specimen.processing.additive [ Reference(Substance) ], ... ; # 0..* Material used in the processing step
    # Specimen.processing.time[x] : 0..1 Date and time of specimen processing. One of these 2
      fhir:Specimen.processing.timeDateTime [ dateTime ]
      fhir:Specimen.processing.timePeriod [ Period ]
  ], ...;
  fhir:Specimen.container [ # 0..* Direct container of specimen (tube/slide, etc.)
    fhir:Specimen.container.identifier [ Identifier ], ... ; # 0..* Id for the container
    fhir:Specimen.container.description [ string ]; # 0..1 Textual description of the container
    fhir:Specimen.container.type [ CodeableConcept ]; # 0..1 Kind of container directly associated with specimen
    fhir:Specimen.container.capacity [ Quantity(SimpleQuantity) ]; # 0..1 Container volume or size
    fhir:Specimen.container.specimenQuantity [ Quantity(SimpleQuantity) ]; # 0..1 Quantity of specimen within container
    # Specimen.container.additive[x] : 0..1 Additive associated with container. One of these 2
      fhir:Specimen.container.additiveCodeableConcept [ CodeableConcept ]
      fhir:Specimen.container.additiveReference [ Reference(Substance) ]
  ], ...;
  fhir:Specimen.note [ Annotation ], ... ; # 0..* Comments
]
```

Changes since DSTU2

Specimen
Specimen.request | Added Element
Specimen.processing | Added Element
Specimen.processing.description | Added Element
Specimen.processing.procedure | Added Element
Specimen.processing.additive | Added Element
Specimen.processing.time[x] | Added Element
Specimen.note | Added Element
Specimen.collection.comment | deleted
Specimen.treatment | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (1 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

10.6.3.1 Terminology Bindings

Path | Definition | Type | Reference
Specimen.status | Codes providing the status/availability of a specimen. | Required | SpecimenStatus
Specimen.type | The type of the specimen. | Example | v2 Specimen Type
Specimen.collection.method | The technique that is used to perform the procedure | Example | FHIR Specimen Collection Method
Specimen.collection.bodySite | Codes describing anatomical locations. May include laterality. | Example | SNOMED CT Body Structures
Specimen.processing.procedure | Type indicating the technique used to process the specimen | Example | SpecimenProcessingProcedure
Specimen.container.type | Type of specimen container | Example | SpecimenContainer
Specimen.container.additive[x] | Substance added to specimen container | Example | v2 Additive/Preservative

10.6.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
accession | token | The accession number associated with the specimen | Specimen.accessionIdentifier
bodysite | token | The code for the body site from where the specimen originated | Specimen.collection.bodySite
collected | date | The date the specimen was collected | Specimen.collection.collected
collector | reference | Who collected the specimen | Specimen.collection.collector (Practitioner)
container | token | The kind of specimen container | Specimen.container.type
container-id | token | The unique identifier associated with the specimen container | Specimen.container.identifier
identifier | token | The unique identifier associated with the specimen | Specimen.identifier
parent | reference | The parent of the specimen | Specimen.parent (Specimen)
patient | reference | The patient the specimen comes from | Specimen.subject (Patient)
status | token | available | unavailable | unsatisfactory | entered-in-error | Specimen.status
subject | reference | The subject of the specimen | Specimen.subject (Group, Device, Patient, Substance)
type | token | The specimen type | Specimen.type

10.6.5 Resource Specimen - Examples

URL: https://hl7.org/fhir/STU3/specimen-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.6.5 Resource Specimen - Examples

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

Example List:

General Specimen Example | 101 | XML | JSON | Turtle
Bacterial Isolate from Stool sample | isolate | XML | JSON | Turtle
VMA, Random Urine Sample | vma-urine | XML | JSON | Turtle
Serum Sample | sst | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.6.6 Resource Specimen - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/specimen-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.6.6 Resource Specimen - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Patient, Practitioner

Detailed Descriptions for the elements in the Specimen resource.

Specimen
Definition | A sample to be used for analysis.
Control | 1..1
Specimen.identifier
Definition | Id for specimen.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
Specimen.accessionIdentifier
Definition | The identifier assigned by the lab when accessioning specimen(s). This is not necessarily the same as the specimen identifier, depending on local lab procedures.
Control | 0..1
Type | Identifier
Summary | true
Specimen.status
Definition | The availability of the specimen.
Control | 0..1
Terminology Binding | SpecimenStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
Specimen.type
Definition | The kind of material that forms the specimen.
Control | 0..1
Terminology Binding | v2 Specimen Type (Example)
Type | CodeableConcept
Summary | true
Comments | The type can change the way that a specimen is handled, and drives what kind of analyses can properly be performed on the specimen. It is frequently used in diagnostic work flow decision making systems.
Specimen.subject
Definition | Where the specimen came from. This may be from the patient(s) or from the environment or a device.
Control | 1..1
Type | Reference(Patient | Group | Device | Substance)
Requirements | Must know the subject context.
Summary | true
Specimen.receivedTime
Definition | Time when specimen was received for processing or testing.
Control | 0..1
Type | dateTime
Summary | true
Specimen.parent
Definition | Reference to the parent (source) specimen which is used when the specimen was either derived from or a component of another specimen.
Control | 0..*
Type | Reference(Specimen)
Comments | The parent specimen could be the source from which the current specimen is derived by some processing step (e.g. an aliquot or isolate or extracted nucleic acids from clinical samples) or one of many specimens that were combined to create a pooled sample.
Specimen.request
Definition | Details concerning a test or procedure request that required a specimen to be collected.
Control | 0..*

Type | Reference(ProcedureRequest)
Comments | The request may be explicit or implied such with a ProcedureRequest that requires a blood draw.
Specimen.collection
Definition | Details concerning the specimen collection.
Control | 0..1
Specimen.collection.collector
Definition | Person who collected the specimen.
Control | 0..1
Type | Reference(Practitioner)
Summary | true
Specimen.collection.collected[x]
Definition | Time when specimen was collected from subject - the physiologically relevant time.
Control | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Specimen.collection.quantity
Definition | The quantity of specimen collected; for instance the volume of a blood sample, or the physical measurement of an anatomic pathology sample.
Control | 0..1
Type | SimpleQuantity
Specimen.collection.method
Definition | A coded value specifying the technique that is used to perform the procedure.
Control | 0..1
Terminology Binding | FHIR Specimen Collection Method (Example)
Type | CodeableConcept
Specimen.collection.bodySite
Definition | Anatomical location from which the specimen was collected (if subject is a patient). This is the target site. This element is not used for environmental specimens.
Control | 0..1
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Comments | If the use case requires BodySite to be handled as a separate resource instead of an inline coded element (e.g. to identify and track separately) then use the standard extension body-site-instance.
Specimen.processing
Definition | Details concerning processing and processing steps for the specimen.
Control | 0..*
Specimen.processing.description
Definition | Textual description of procedure.
Control | 0..1
Type | string
Specimen.processing.procedure
Definition | A coded value specifying the procedure used to process the specimen.
Control | 0..1
Terminology Binding | SpecimenProcessingProcedure (Example)
Type | CodeableConcept
Specimen.processing.additive
Definition | Material used in the processing step.
Control | 0..*
Type | Reference(Substance)
Specimen.processing.time[x]
Definition | A record of the time or period when the specimen processing occurred. For example the time of sample fixation or the period of time the sample was in formalin.
Control | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Specimen.container
Definition | The container holding the specimen. The recursive nature of containers; i.e. blood in tube in tray in rack is not addressed here.
Control | 0..*
Specimen.container.identifier
Definition | Id for container. There may be multiple; a manufacturer's bar code, lab assigned identifier, etc. The container ID may differ from the specimen id in some circumstances.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
Specimen.container.description
Definition | Textual description of the container.
Control | 0..1
Type | string
Specimen.container.type
Definition | The type of container associated with the specimen (e.g. slide, aliquot, etc.).
Control | 0..1

Terminology Binding | SpecimenContainer (Example)
Type | CodeableConcept
Specimen.container.capacity
Definition | The capacity (volume or other measure) the container may contain.
Control | 0..1
Type | SimpleQuantity
Specimen.container.specimenQuantity
Definition | The quantity of specimen in the container; may be volume, dimensions, or other appropriate measurements, depending on the specimen type.
Control | 0..1
Type | SimpleQuantity
Specimen.container.additive[x]
Definition | Introduced substance to preserve, maintain or enhance the specimen. Examples: Formalin, Citrate, EDTA.
Control | 0..1
Terminology Binding | v2 Additive/Preservative (Example)
Type | CodeableConcept|Reference(Substance)
[x] Note | See Choice of Data Types for further information about how to use [x]
Specimen.note
Definition | To communicate any details or issues about the specimen or during the specimen collection. (for example: broken vial, sent with patient, frozen).
Control | 0..*
Type | Annotation

10.6.7 Resource Specimen - Mappings

URL: https://hl7.org/fhir/STU3/specimen-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.6.7 Resource Specimen - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

Mappings:

RIM Mapping

HL7 v2 Mapping

Mappings for the specimen resource.

10.6.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

Specimen | Role[classCode=SPEC]
identifier | .id
accessionIdentifier | .participation[typeCode=SBJ].act[classCode=ACSN, moodCode=EVN].id
status | status
type | .code
subject | .scoper or .scoper.playsRole
receivedTime | .participation[typeCode=SBJ].act[code=SPCREC, moodCode=EVN].effectiveTime
parent | .scoper (if parent) .player.scopesRole[classCode=SPEC].player (if child)
request | outboundRelationship[typeCode=FLFS].target
collection | .participation[typeCode=SBJ].act[classCode=SPECCOLLECT, moodCode=EVN]
collector | .participation[typeCode=PFM].role
collected[x] | .effectiveTime
quantity | .participation[typeCode=SBJ].role[classCode=SPEC].player.quantity
method | .methodCode
bodySite | .targetSiteCode
processing | .participation[typeCode=SBJ].act[code=SPCTRT, moodCode=EVN]
description | .text
procedure | .code
additive | .participation[typeCode=CSM].role[classCode=ADTV].code
time[x] | .effectiveTime
container | .player.scopingRole[classCode=CONT].scoper
identifier | .id
description | .desc
type | .code
capacity | .quantity
specimenQuantity | .playedRole[classCode=CONT].quantity
additive[x] | .scopesRole[classCode=ADTV].player
note | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="annotation"].value

10.6.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Specimen
identifier | SPM-2

accessionIdentifier | SPM-30 (v2.7+)
status | SPM-20
type | SPM-4 and possibly SPM-5
subject
receivedTime | SPM-18
parent
request | ORC? OBR-2/3?
collection | SPM-14
collector | PRT
collected[x] | SPM-17
quantity | SPM-12
method | SPM-7
bodySite | SPM-8 and SPM-9
processing
description
procedure
additive | SPM-6
time[x]
container
identifier | SAC-3
description
type | SPM-27
capacity | one of: SAC 16, SAC17 and SAC 20, or SAC 21 and SAC 20
specimenQuantity | SAC-23 (value) SAC-24 (units
additive[x] | SAC-27
note | OBX

10.6.8 Resource Specimen - Design Notes

URL: https://hl7.org/fhir/STU3/specimen-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.6.8 Resource Specimen - Design Notes

to do

10.6.9 Resource Specimen - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/specimen-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.6.9 Resource Specimen - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Specimen resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

10.6.9.1 Profiles

No Profiles defined for this resource

10.6.9.2 Extensions

specimen-collectionPriority | collectionPriority | for Specimen HL7 Extensions
specimen-isDryWeight | isDryWeight | for Specimen HL7 Extensions
specimen-processingTime | processingTime | for Specimen HL7 Extensions
specimen-sequenceNumber | sequenceNumber | for Specimen HL7 Extensions
specimen-specialHandling | specialHandling | for Specimen HL7 Extensions

Extensions that reference this resource:

procedurerequest-geneticsItem | Item | for ProcedureRequest-Genetics

10.6.9.3 Search Extensions

No Search Extensions defined for this resource

10.6.10 Specimen HL7 Extensions

URL: https://hl7.org/fhir/STU3/specimen-extensions.html

Defines common extensions used with or related to the Specimen resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.6.10 Specimen HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

10.6.10.1 Content

Extensions:

specimen-sequenceNumber | sequenceNumber : An assigned number on the specimen denoting the order of collection.

specimen-collectionPriority | collectionPriority : The urgency of sample collection, such as STAT, ASAP, ASAP-ED, ROUTINE, ROUTINE-AM, etc.…

specimen-specialHandling | specialHandling : Special handling during the collection, transport, or storage of the specimen.

specimen-isDryWeight | isDryWeight : If the recorded quantity of the specimen is reported as a weight, if it is a dry weight.

specimen-processingTime | processingTime : Period or duration of processing.

10.6.11 Resource Specimen - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/specimen-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.6.11 Resource Specimen - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

R2 : R3 Converson maps for Specimen.

Functional status for this map: 6 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (1 errors). (see documentation)

10.6.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Specimen2to3" = "R2 to R3 Conversions for Specimen"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Specimen" alias SpecimenR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Specimen" alias Specimen as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Specimen extends DomainResource
  input src : SpecimenR2 as source
  input tgt : Specimen as target

  "Specimen-identifier" : for src.identifier make tgt.identifier
  "Specimen-accessionIdentifier" : for src.accessionIdentifier make tgt.accessionIdentifier
  "Specimen-status" : for src.status make tgt.status
  "Specimen-type" : for src.type make tgt.type
  "Specimen-subject" : for src.subject make tgt.subject
  "Specimen-receivedTime" : for src.receivedTime make tgt.receivedTime
  "Specimen-parent" : for src.parent make tgt.parent
  "Specimen-collection" : for src.collection as vs0 make tgt.collection as vt0 then spmCollection(vs0, vt0, tgt)
  "Specimen-treatment" : for src.treatment as vs0 make tgt.processing as vt0 then spmTreatment(vs0, vt0)
  "Specimen-container" : for src.container as vs0 make tgt.container as vt0 then spmContainer(vs0, vt0)
endgroup

group spmCollection extends BackboneElement
  input src as source
  input tgt as target
  input tgts as target

  "Specimen.collection-collector" : for src.collector make tgt.collector
  "Specimen.collection-collected" : for src.collected make tgt.collected
  "Specimen.collection-comment" : for src.comment make tgts.note
  "Specimen.collection-quantity" : for src.quantity make tgt.quantity
  "Specimen.collection-method" : for src.method make tgt.method
  "Specimen.collection-bodySite" : for src.bodySite make tgt.bodySite
endgroup

group spmTreatment extends BackboneElement
  input src as source
  input tgt as target

  "Specimen.treatment-description" : for src.description make tgt.description
  "Specimen.treatment-procedure" : for src.procedure make tgt.procedure
  "Specimen.treatment-additive" : for src.additive make tgt.additive
endgroup

group spmContainer extends BackboneElement
  input src as source
  input tgt as target

  "Specimen.container-identifier" : for src.identifier make tgt.identifier
  "Specimen.container-description" : for src.description make tgt.description
  "Specimen.container-type" : for src.type make tgt.type
```

```
  "Specimen.container-capacity" : for src.capacity make tgt.capacity
  "Specimen.container-specimenQuantity" : for src.specimenQuantity make tgt.specimenQuantity
  "Specimen.container-additive" : for src.additive make tgt.additive
endgroup
```

10.6.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Specimen3to2" = "R3 to R2 Conversion for Specimen"

uses "http://hl7.org/fhir/StructureDefinition/Specimen" alias Specimen as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Specimen" alias SpecimenR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Specimen extends DomainResource
  input src : Specimen as source
  input tgt : SpecimenR2 as target

  "Specimen-identifier" : for src.identifier make tgt.identifier
  "Specimen-accessionIdentifier" : for src.accessionIdentifier make tgt.accessionIdentifier
  "Specimen-status" : for src.status make tgt.status
  "Specimen-type" : for src.type make tgt.type
  "Specimen-subject" : for src.subject make tgt.subject
  "Specimen-receivedTime" : for src.receivedTime make tgt.receivedTime
  "Specimen-parent" : for src.parent make tgt.parent
  "Specimen-collection" : for src.collection as vs0 make tgt.collection as vt0 then spmCollection(vs0, vt0, src)
  "Specimen-treatment" : for src.processing as vs0 make tgt.treatment as vt0 then spmTreatment(vs0, vt0)
  "Specimen-container" : for src.container as vs0 make tgt.container as vt0 then spmContainer(vs0, vt0)
endgroup

group spmCollection extends BackboneElement
  input src as source
  input tgt as target
  input ssrc as source

  "Specimen.collection-collector" : for src.collector make tgt.collector
  "Specimen.collection-collected" : for src.collected make tgt.collected
  "Specimen.collection-quantity" : for src.quantity make tgt.quantity
  "Specimen.collection-comment" : for ssrc.note make tgt.comment
  "Specimen.collection-method" : for src.method make tgt.method
  "Specimen.collection-bodySite" : for src.bodySite make tgt.bodySite
endgroup

group spmTreatment extends BackboneElement
  input src as source
  input tgt as target

  "Specimen.treatment-description" : for src.description make tgt.description
  "Specimen.treatment-procedure" : for src.procedure make tgt.procedure
  "Specimen.treatment-additive" : for src.additive make tgt.additive
endgroup

group spmContainer extends BackboneElement
  input src as source
  input tgt as target

  "Specimen.container-identifier" : for src.identifier make tgt.identifier
  "Specimen.container-description" : for src.description make tgt.description
  "Specimen.container-type" : for src.type make tgt.type
  "Specimen.container-capacity" : for src.capacity make tgt.capacity
  "Specimen.container-specimenQuantity" : for src.specimenQuantity make tgt.specimenQuantity
  "Specimen.container-additive" : for src.additive make tgt.additive
endgroup
```

10.6.11.3 R3 Validation Errors

specimen-example | ERROR @ Specimen.processing.extension: The extension http://hl7.org/fhir/StructureDefinition/specimen-treatmentTime is unknown, and not allowed here (src = InstanceValidator)

10.7 Resource BodySite - Content

URL: https://hl7.org/fhir/STU3/bodysite.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient

Record details about the anatomical location of a specimen or body part. This resource may be used when a coded concept does not provide the necessary detail needed for the use case.

10.7.1 Scope and Usage

The BodySite resource contains details about the anatomical location of a specimen or body part, including patient information, identifiers, as well as text descriptons and images. It provides for the addition of qualifiers such as laterality and directionality to the anatomic location for those use cases where precoordination of codes is not possible. The BodySite resource supports recording and tracking of an anatomic location or structure on a patient outside the context of another

resource. For example it can be the target of a Procedure resource or Observation resource.

10.7.2 Boundaries and Relationships

The BodySite resource is not intended to substitute for precoordination of codes. If precoordination of codes is supported by an implementation the codeableConcept should be used. This resource is not intended for describing the type of anatomical location but rather a specific body site on a specific patient.

10.7.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
BodySite | | | DomainResource | Specific and identified anatomical location Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Bodysite identifier
active | ?!? | 0..1 | boolean | Whether this body site record is in active use
code | ? | 0..1 | CodeableConcept | Named anatomical location SNOMED CT Body Structures (Example)
qualifier | | 0..* | CodeableConcept | Modification to location code Bodysite Location Qualifier (Example)
description | ? | 0..1 | string | Anatomical location description
image | | 0..* | Attachment | Attached images
patient | ? | 1..1 | Reference(Patient) | Who this is about
Documentation for this format

UML Diagram (Legend)

XML Template

```
<BodySite xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Bodysite identifier --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this body site record is in active use -->
 <code><!-- 0..1 CodeableConcept Named anatomical location --></code>
 <qualifier><!-- 0..* CodeableConcept Modification to location code --></qualifier>
 <description value="[string]"/><!-- 0..1 Anatomical location description -->
 <image><!-- 0..* Attachment Attached images --></image>
 <patient><!-- 1..1 Reference(Patient) Who this is about --></patient>
</BodySite>
```

JSON Template

```
{
  "resourceType" : "BodySite",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Bodysite identifier
  "active" : <boolean>, // Whether this body site record is in active use
  "code" : { CodeableConcept }, // Named anatomical location
  "qualifier" : [{ CodeableConcept }], // Modification to location code
  "description" : "<string>", // Anatomical location description
  "image" : [{ Attachment }], // Attached images
  "patient" : { Reference(Patient) } // R!  Who this is about
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:BodySite;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:BodySite.identifier [ Identifier ], ... ; # 0..* Bodysite identifier
  fhir:BodySite.active [ boolean ]; # 0..1 Whether this body site record is in active use
  fhir:BodySite.code [ CodeableConcept ]; # 0..1 Named anatomical location
  fhir:BodySite.qualifier [ CodeableConcept ], ... ; # 0..* Modification to location code
  fhir:BodySite.description [ string ]; # 0..1 Anatomical location description
  fhir:BodySite.image [ Attachment ], ... ; # 0..* Attached images
  fhir:BodySite.patient [ Reference(Patient) ]; # 1..1 Who this is about
]
```

Changes since DSTU2

BodySite

BodySite.active | Added Element
BodySite.qualifier | Renamed from modifier to qualifier

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
BodySite | | | DomainResource | Specific and identified anatomical location Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Bodysite identifier
active | ?!? | 0..1 | boolean | Whether this body site record is in active use
code | ? | 0..1 | CodeableConcept | Named anatomical location SNOMED CT Body Structures (Example)
qualifier | | 0..* | CodeableConcept | Modification to location code Bodysite Location Qualifier (Example)
description | ? | 0..1 | string | Anatomical location description
image | | 0..* | Attachment | Attached images
patient | ? | 1..1 | Reference(Patient) | Who this is about
Documentation for this format

UML Diagram (Legend)

XML Template

```
<BodySite xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Bodysite identifier --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this body site record is in active use -->
 <code><!-- 0..1 CodeableConcept Named anatomical location --></code>
 <qualifier><!-- 0..* CodeableConcept Modification to location code --></qualifier>
 <description value="[string]"/><!-- 0..1 Anatomical location description -->
 <image><!-- 0..* Attachment Attached images --></image>
 <patient><!-- 1..1 Reference(Patient) Who this is about --></patient>
</BodySite>
```

JSON Template

```
{
  "resourceType" : "BodySite",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Bodysite identifier
  "active" : <boolean>, // Whether this body site record is in active use
  "code" : { CodeableConcept }, // Named anatomical location
  "qualifier" : [{ CodeableConcept }], // Modification to location code
  "description" : "<string>", // Anatomical location description
  "image" : [{ Attachment }], // Attached images
  "patient" : { Reference(Patient) } // R!  Who this is about
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:BodySite;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:BodySite.identifier [ Identifier ], ... ; # 0..* Bodysite identifier
  fhir:BodySite.active [ boolean ]; # 0..1 Whether this body site record is in active use
  fhir:BodySite.code [ CodeableConcept ]; # 0..1 Named anatomical location
  fhir:BodySite.qualifier [ CodeableConcept ], ... ; # 0..* Modification to location code
  fhir:BodySite.description [ string ]; # 0..1 Anatomical location description
  fhir:BodySite.image [ Attachment ], ... ; # 0..* Attached images
  fhir:BodySite.patient [ Reference(Patient) ]; # 1..1 Who this is about
]
```

Changes since DSTU2

BodySite
BodySite.active | Added Element
BodySite.qualifier | Renamed from modifier to qualifier

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

10.7.3.1 Terminology Bindings

Path | Definition | Type | Reference

BodySite.code | Codes describing anatomical locations. May include laterality | Example | SNOMED CT Body Structures
BodySite.qualifier | Concepts modifying the anatomic location | Example | Bodysite Location Qualifier

10.7.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
code | token | Named anatomical location | BodySite.code
identifier | token | Identifier for this instance of the anatomical location | BodySite.identifier
patient | reference | Patient to whom bodysite belongs | BodySite.patient (Patient)

10.7.5 Resource BodySite - Examples

URL: https://hl7.org/fhir/STU3/bodysite-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.7.5 Resource BodySite - Examples

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Example List:

This example demonstrates using the bodysite resource to identify a fetus within systems that choose not to treat a fetus as a Patient | fetus | XML | JSON | Turtle
This example demonstrates using the bodysite resource to identify a specific tumor or lesion that will have multiple observations and/or procedures performed on it over time | tumor | XML | JSON | Turtle
This example demonstrates using the bodysite resource to identify skin patches or other portions of a person or animal that are a focus of a clinical trial and that are subject to repeated observations and/or procedures over time | skin-patch | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.7.6 Resource BodySite - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/bodysite-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.7.6 Resource BodySite - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient

Detailed Descriptions for the elements in the BodySite resource.

BodySite
Definition | Record details about the anatomical location of a specimen or body part. This resource may be used when a coded concept does not provide the necessary detail needed for the use case.
Control | 1..1
Alternate Names | anatomical location
BodySite.identifier
Definition | Identifier for this instance of the anatomical location.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
BodySite.active
Definition | Whether this body site is in active use.
Control | 0..1
Type | boolean
Is Modifier | true
Default Value | true
Requirements | Need to be able to mark a body site entry as not to be used because it was created in error.
Summary | true
Comments | This element is labeled as a modifier because it may be used to mark that the resource was created in error.
BodySite.code
Definition | Named anatomical location - ideally coded where possible.
Control | 0..1
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Summary | true
BodySite.qualifier
Definition | Qualifier to refine the anatomical location. These include qualifiers for laterality, relative location, directionality, number, and plane.
Control | 0..*
Terminology Binding | Bodysite Location Qualifier (Example)

Type | CodeableConcept
BodySite.description
Definition | A summary, charactarization or explanation of the anatomic location.
Control | 0..1
Type | string
Summary | true
Comments | This description could include any visual markings used to orientate the viewer e.g. external reference points, special sutures, ink markings.
BodySite.image
Definition | Image or images used to identify a location.
Control | 0..*
Type | Attachment
BodySite.patient
Definition | The person to which the body site belongs.
Control | 1..1
Type | Reference(Patient)
Summary | true

10.7.7 Resource BodySite - Mappings

URL: https://hl7.org/fhir/STU3/bodysite-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.7.7 Resource BodySite - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Mappings:

RIM Mapping

HL7 v2 Mapping

Open EHR Archetype Mapping

Mappings for the bodysite resource.

10.7.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

BodySite | ActSIte
identifier | .id
active | statusCode
code | .targetSiteCode
qualifier | .targetSiteCode
description | .text
image | outboundRelationship[typeCode=COMP].target
patient | .scoper or .scoper.playsRole

10.7.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

BodySite
identifier
active
code | OBX-20
qualifier | No distinct element defined in V2.x. The IHE Radiology Technical Framework directs the use of OBR-15-specimen source to contain laterality for imaging studies.
description
image
patient

10.7.7.3 Mappings for Open EHR Archetype Mapping (http://openehr.org)

BodySite | Archetype ID openEHR-EHR-CLUSTER.anatomical_location.v1 Concept name Anatomical Location Concept description Record details about an anatomical location. Keywords location, site, anatomical Copyright © openEHR Foundation Purpose To record details about anatomical location. Misuse Not for specifiying unilateral/bilateral occurrence - this is related to an evaluation which perhaps includes multiple locations. Author Author name: Heather Leslie Organization: Ocean Informatics Email: heather.leslie@oceaninformatics.com Date of Origination: 2008-11-10 Translator Norwegian Bokmål: Lars Bitsch-Larsen, Haukeland University Hospital of Bergen, Norway, MD, DEAA, MBA, spec in anesthesia, spec in tropical medicine. Arabic (Syria): Mona Saleh Other contributors Heather Leslie, Ocean Informatics, Australia (Editor) Ian McNicoll, freshEHR Clinical Informatics, United Kingdom (Editor) Dunmail Hodkinson, Black Pear Software Ltd, UK References Use cases: - 5LICS, midclavicular line - L forearm, 2cm anterior to the cubital fossa - Right iliac fossa - 5 cm below L tibial tuberosity - 2 cm medial to R nipple - 4 cm below R costal margin in midclavicular line - R femur - midline nose - medial aspect of R great toe nail.
identifier | Specific location Cluster Optional (0..1) Cardinality: Mandatory, repeating, unordered (1..*)
active
code | Name of location
qualifier | Combines Side, Numerical identifier, Anatomical plane,and Aspect
description | Description of anatomical location/Visual markings/orientation
image | Image

patient

10.7.8 Resource BodySite - Design Notes

URL: https://hl7.org/fhir/STU3/bodysite-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.7.8 Resource BodySite - Design Notes

to do

10.7.9 Resource BodySite - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/bodysite-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.7.9 Resource BodySite - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the BodySite resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

10.7.9.1 Profiles

No Profiles defined for this resource

10.7.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

body-site-instance | Body Site Instance | for General Extensions for use by FHIR Implementers
condition-targetBodySite | targetBodySite | for Condition HL7 Extensions
procedure-approachBodySite | approachBodySite | for Procedure HL7 Extensions
procedure-targetBodySite | targetBodySite | for Procedure HL7 Extensions
procedurerequest-approachBodySite | approachBodySite | for Procedure Request HL7 Extensions
procedurerequest-targetBodySite | targetBodySite | for Procedure Request HL7 Extensions

10.7.9.3 Search Extensions

No Search Extensions defined for this resource

10.7.10 Resource BodySite - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/bodysite-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.7.10 Resource BodySite - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

R2 : R3 Converson maps for BodySite.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

10.7.10.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/BodySite2to3" = "R2 to R3 Conversions for BodySite"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/BodySite" alias BodySiteR2 as source
uses "http://hl7.org/fhir/StructureDefinition/BodySite" alias BodySite as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/DomainResource2to3"
imports "http://hl7.org/fhir/StructureMap/Attachment2to3"
imports "http://hl7.org/fhir/StructureMap/Identifier2to3"
imports "http://hl7.org/fhir/StructureMap/Reference2to3"
imports "http://hl7.org/fhir/StructureMap/CodeableConcept2to3"

group for type+types  BodySite extends DomainResource
  input src : BodySiteR2 as source
  input tgt : BodySite as target

  "BodySite-patient" : for src.patient make tgt.patient
  "BodySite-identifier" : for src.identifier make tgt.identifier
```

```
  "BodySite-code" : for src.code make tgt.code
  "BodySite-modifier" : for src.modifier make tgt.qualifier
  "BodySite-description" : for src.description make tgt.description
  "BodySite-image" : for src.image make tgt.image

endgroup
```

10.7.10.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/BodySite3to2" = "R3 to R2 Conversion for BodySite"

uses "http://hl7.org/fhir/StructureDefinition/BodySite" alias BodySite as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/BodySite" alias BodySiteR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/DomainResource3to2"
imports "http://hl7.org/fhir/StructureMap/Attachment3to2"
imports "http://hl7.org/fhir/StructureMap/Identifier3to2"
imports "http://hl7.org/fhir/StructureMap/Reference3to2"
imports "http://hl7.org/fhir/StructureMap/CodeableConcept3to2"

group for type+types  BodySite extends DomainResource
  input src : BodySite as source
  input tgt : BodySiteR2 as target

  "BodySite-patient" : for src.patient make tgt.patient
  "BodySite-identifier" : for src.identifier make tgt.identifier
  "BodySite-code" : for src.code make tgt.code
  "BodySite-modifier" : for src.qualifier make tgt.modifier
  "BodySite-description" : for src.description make tgt.description
  "BodySite-image" : for src.image make tgt.image

endgroup
```

10.7.10.3 R3 Validation Errors

10.8 Genomics Implementation Guidance

URL: https://hl7.org/fhir/STU3/genomics.html

(see the publication above for details about each)

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

10.8 Genomics Implementation Guidance

Clinical Genomics Work Group | Maturity Level: 1 | Ballot Status: Trial Use

Table of Contents

* Background
* Overview
* Sequence Resource
* Observation-genetics Profile
* DiagnosticReport-genetics Profile
* ProcedureRequest-genetics Profile
* HLA genotyping results Profile
* Relationship among Sequence resource and genetics profiles
* HL7 Domain Analysis Model Use Cases
* Other Genomics Use Cases
* Clinical Genomics Apps
* Appendix 1: From DSTU2 Genomics to STU3

10.8 Background

Rapid advances in sequencing technologies, clinical genetics tests for whole genome and exome sequencing are allowing sophisticated genetics testing to be used by providers and patients in clinical decisions. Results from such tests are used to identify distinct genetic variants that may contribute to syndromes, conditions and/or predictive responses to treatments. The implementation of precision medicine will depend upon having such data to diagnose patients, choose medications, and predict the course of disease and care, but will require standards and effective user interfaces.

A current technical challenge exists in interoperability, the ability to access and share clinical and genetics data. The challenges of interoperability includes collection, coding, and retrieval to scale. An individual's genetic data set is large, complex and requires curation. Unfortunately, incompatible systems and nomenclatures are already in use. A standards-based ontology that could be adopted to integrate both genetic data and clinical information systems will be crucial to accelerating the integration of precision medicine and to make sense of genetic testing results in a complete clinical context.

One approach for collecting, coding, and retrieving genetics data comes from the Global Alliance for Genomics and Health (GA4GH). The GA4GH organization has built and is refining an API and data model for the exchange of full sequence genomic information across multiple research organizations and platforms. The GA4GH focuses on the needs of researchers.

A second approach is evolving from HL7 through FHIR. FHIR is attractive because it is relatively easy to implement because it is comprised of a set of modular components called resources, which can be easily and incrementally assembled into working systems. The clinical requirements for genetics data is, relative to genomics research needs, utilitarian and reductive because it is about distilling and extracting particular genetics data produced by ever more sophisticated testing for use at the point-of-care. This has made FHIR a very functional framework to initiate an interoperable clinical genetics data standardization to which multiple stakeholders have contributed to this guide.

FHIR DSTU2 introduced a standard genetics profile that applies to the FHIR Observation resource. Using this profile, Observation payloads can return genetic testing results in a standardized manner.

Extending Observations rather than creating a new, dedicated FHIR genetics data resource is consistent with the FHIR community mandate because it only adds new resources to an existing resource library when there is a compelling case to do so. We provide the case to add a Sequence resource in this specification.

10.8 Overview

STU3 moves beyond FHIR DSTU2 Standard Genetics profile on Observation allowing increased granularity and less ambiguity by creating a new resource to be called Sequence. This resource will be used to hold clinically relevant sequence data in a manner that is both efficient and versatile integrating new and as yet undefined types of genomic and other -omics data that will soon be commonly entered into health records for clinical use. Sequence will be leveraged by other FHIR resources, including Observation. This is consistent with how all FHIR resources are designed and used.

The September 2014 Informative Ballot (“HL7 Clinical Genomics, Domain Analysis Model: Clinical Sequencing Release 1”) provided guiding use cases, which initially informed development of the initial Standard Genetics profile that is found in FHIR DSTU2. The same use cases also led to a second Project to develop a Sequence resource (“Develop FHIR sequence resource for Clinical Genomics”). A preliminary effort to address these issues has been explored and published in context of the Substitutable Medical Applications and Reusable Technologies (SMART) Platforms Project and described in an article (“SMART on FHIR Genomics: Facilitating standardized clinico-genomic apps ”).

Sequence is designed to hold genetic sequences in blocks relevant to actionable clinical decision-making. Extensions to Sequence address complex cases and can associate it with repositories for retrieving a patient’s full sequence data, such as those defined by GA4GH. Other changes include a suite of genetics profiles for other FHIR resources. In addition, the Observation-genetics profile adds new references so that an Observation can report genetics test results to be integrated into the EHR. There are also new genetics-extension profiles for DiagnosticReport, ProcedureRequest and FamilyMemberHistory, respectively, to extend them to report genetics results. We have given all of these FHIR genetics profiles the suffix “-genetics” (e.g. “DiagnosticReport-genetics profile”). New profiles on top of DiagnosticReport have been created for reporting HLA genotyping results.

On the following pages, we elaborate upon the rationale for the proposed design, introducing in some detail the following resource and profiles:

* A Sequence resource

* An Observation-genetics profile on Observation

* A DiagnosticReport-genetics profile on DiagnosticReport

* A ProcedureRequest-genetics profile on ProcedureRequest

* An HLA-genotyping-results profile on DiagnosticReport

With these resource and profiles, FHIR can support a large set of clinical use cases (see Section 9, 10 , and 11) and is thus positioned to address all emergent -omics use cases, including Next-Generation Sequencing (NGS). These tools are simple to implement, will optimize payload sizes, and help developers avoid redundant retrieval of data. Appendix 1 of this document shows how DSTU 2.0 can be mapped to the new additions to the resource.

10.8 Sequence Resource

10.8.1 Structure Diagram

Name | Flags | Card. | Type | Description & Constraints

Sequence | ?I | | DomainResource | Information about a biological sequence + Only 0 and 1 are valid for coordinateSystem Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | Unique ID for this particular sequence. This is a FHIR-defined id

type | ? | 0..1 | code | aa | dna | rna sequenceType (Example)

coordinateSystem | ? | 1..1 | integer | Base number of coordinate system (0 for 0-based numbering or coordinates, inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end)

patient | ? | 0..1 | Reference(Patient) | Who and/or what this is about

specimen | ? | 0..1 | Reference(Specimen) | Specimen used for sequencing

device | ? | 0..1 | Reference(Device) | The method for sequencing

performer | ? | 0..1 | Reference(Organization) | Who should be responsible for test result

quantity | ? | 0..1 | Quantity | The number of copies of the seqeunce of interest. (RNASeq)

referenceSeq | ?I | 0..1 | BackboneElement | A sequence used as reference + Only +1 and -1 are valid for strand + GenomeBuild and chromosome must be both contained if either one of them is contained + Have and only have one of the following elements in referenceSeq : 1. genomeBuild ; 2

referenceSeqId; 3. referenceSeqPointer; 4. referenceSeqString;
chromosome | ? | 0..1 | CodeableConcept | Chromosome containing genetic finding chromosome-human (Example)
genomeBuild | ? | 0..1 | string | The Genome Build used for reference, following GRCh build versions e.g. 'GRCh 37'
referenceSeqId | ? | 0..1 | CodeableConcept | Reference identifier ENSEMBL (Example)
referenceSeqPointer | ? | 0..1 | Reference(Sequence) | A Pointer to another Sequence entity as reference sequence
referenceSeqString | ? | 0..1 | string | A string to represent reference sequence
strand | ? | 0..1 | integer | Directionality of DNA (+1/-1)
windowStart | ? | 1..1 | integer | Start position of the window on the reference sequence
windowEnd | ? | 1..1 | integer | End position of the window on the reference sequence
variant | ? | 0..* | BackboneElement | Variant in sequence
start | ? | 0..1 | integer | Start position of the variant on the reference sequence
end | ? | 0..1 | integer | End position of the variant on the reference sequence
observedAllele | ? | 0..1 | string | Allele that was observed
referenceAllele | ? | 0..1 | string | Allele in the reference sequence
cigar | ? | 0..1 | string | Extended CIGAR string for aligning the sequence with reference bases
variantPointer | ? | 0..1 | Reference(Observation) | Pointer to observed variant information
observedSeq | ? | 0..1 | string | Sequence that was observed
quality | ? | 0..* | BackboneElement | An set of value as quality of sequence
type | ? | 1..1 | code | indel | snp | unknown qualityType (Required)
standardSequence | ? | 0..1 | CodeableConcept | Standard sequence for comparison FDA-StandardSequence (Example)
start | ? | 0..1 | integer | Start position of the sequence
end | ? | 0..1 | integer | End position of the sequence
score | ? | 0..1 | Quantity | Quality score for the comparison
method | ? | 0..1 | CodeableConcept | Method to get quality FDA-Method (Example)
truthTP | ? | 0..1 | decimal | True positives from the perspective of the truth data
queryTP | ? | 0..1 | decimal | True positives from the perspective of the query data
truthFN | ? | 0..1 | decimal | False negatives
queryFP | ? | 0..1 | decimal | False positives
gtFP | ? | 0..1 | decimal | False positives where the non-REF alleles in the Truth and Query Call Sets match
precision | ? | 0..1 | decimal | Precision of comparison
recall | ? | 0..1 | decimal | Recall of comparison
fScore | ? | 0..1 | decimal | F-score
readCoverage | ? | 0..1 | integer | Average number of reads representing a given nucleotide in the reconstructed sequence
repository | ? | 0..* | BackboneElement | External repository which contains detailed report related with observedSeq in this resource
type | ? | 1..1 | code | directlink | openapi | login | oauth | other repositoryType (Required)
url | ? | 0..1 | uri | URI of the repository
name | ? | 0..1 | string | Repository's name
datasetId | ? | 0..1 | string | Id of the dataset that used to call for dataset in repository
variantsetId | ? | 0..1 | string | Id of the variantset that used to call for variantset in repository
readsetId | ? | 0..1 | string | Id of the read
pointer | ? | 0..* | Reference(Sequence) | Pointer to next atomic sequence
Documentation for this format

10.8.2 Description

The Sequence resource is designed for next-generation sequencing data. Patients' observed sequences should be represented by recording reference sequence id/string and detected variants. To specify how it proceed, here is a picture below:

Sequence.coordinateSystem: This element shall be constrained into only two possible values: 0 for 0-based system and 1 for 1-based system. Below is the picture that could explain what's the difference between these two systems:

Here are two examples that clarify the usage in both cases (they represent same segment part):

* 0-based example: here Source:XML,JSON

* 1-based example: here Source:XML,JSON

Sequence.referenceSeq: Four optional ways are provided to represent reference sequence in Sequence resource:

* Sequence.referenceSeq.referenceSeqId: Reference sequence id in public database;

* Sequence.referenceSeq.referenceSeqString: Reference sequence string;

* Sequence.referenceSeq.referenceSeqPointer: Reference to observed sequence stored in another sequence entity;

* Sequence.referenceSeq.genomeBuild, Sequence.reference.chromosome: The combination of genome build and chromosome.

The window selects a range from the reference sequence (or genome) that is used to define building block of a current sequence (e.g. Sequence resource instance 1).

Sequence.referenceSeq.strand: Only two possible values can be made by strand, +1 for plus strand while -1 for minus strand. Since the directionality of the sequence string might be represented in different word in different omics scenario, below are simple example of how to map other expressions into its correlated value:

Map to +1 | Map to -1
5?-to-3? direction | 3?-to-5? direction
Watson | Crick
Sense | Antisense

Positive | Negative

Sequence.quality: Quality scores for bases in the sequence. Is is intended to be compliant with emerging regulatory needs needs (eg: those found at PrecisionFDA).

Sequence.variant: This complex element is used for encoding sequence. When the information of reference sequence and variants are provided, the observed sequence will be derived.

10.8.2.1 Internal Pointers

Sequence.patient: This element points to a Patient identifier to show that this sequence is related to the same patient.

Sequence.specimen: A pointer to specimen identifier, if needed.

Sequence.device: A pointer to Device identifier which is used for describing sequencing method (such as chip id, chip manufacturer etc.)

Sequence.pointer: A pointer to a Sequence instance for the next sequence block to build a sequence graph.

10.8.2.2 External Pointers

Sequence.repository: This complex element is used to provide a clarifying structure, a base URL, and/or relevant IDs when referring to an external repository.

GA4GH Repository Example. If the Sequence resource refers to a GA4GH repository for read info, references to a GA4GH full sequence dataset should conform to GA4GH data models and accessed via the GA4GH API. The URL of a GA4GH repository, ids of a GA4GH variant and read group are contained in the Sequence resource. The URL of a GA4GH repository is an api_base of a GA4GH server that could be called for sequence data. The GA4GH variant set is a collection of call sets and the GA4GH call set is a collection of variant calls, typically for one sample. A variant call represents a determination of genotype with respect to that variant.

VariantSet definition: A VariantSet is a collection of variants and variant calls intended to be analyzed together.

CallSet definition: A CallSet is a collection of calls that were generated by the same analysis of the same sample.

A read group is a collection of reads produced by a sequencer. A read group set typically models reads corresponding to one sample, sequenced one way, and aligned one way. The API reference of Google Genomics is a GA4GH repository built by Google and provides details of the data models, such as the resource representations.

10.8.3 Usage Examples

10.8.3.1 Different way to represent sequence record with variations

We provide a detailed example to show how sequence resource can be used to represent record of observed sequence by different method.

The diagram above describes 4 optional ways provided in the Sequence resource to encode sequencing data. Here are the corresponding examples

* By referenceSeqId Source:XML,JSON,TURTLE

* By referenceSeqString Source:XML,JSON,TURTLE

* By referenceSeqPointer Source:XML,JSON,TURTLE

* By Genomebuild+chromosome Source:XML,JSON,TURTLE

10.8.3.2 Example from precision FDA

We provide one example to show how precision FDA vcf data can be upload and comform FHIR specification, and how quality in sequence resource is represented by comparison between a reference sequence and the observed sequence.

Source: XML, JSON, Turtle.

10.8.3.3 Complex Variation example

We provide one example to show how a complex variant can be represented with the help of cigar. The deletion, insertion and mutation is represented in characters along with the number of repetition.

Source: XML, JSON, Turtle.

10.8 Observation-genetics Profile

10.8.1 Structure Diagram

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..*
observation-geneticsDNASequenceVariantName | | 0..1 | CodeableConcept | DNA sequence variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNASequenceVariantName Binding: ???? (preferred)
observation-geneticsDNAVariantId | | 0..1 | CodeableConcept | DNA sequence variant ID URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNAVariantId
observation-geneticsDNASequenceVariantType | | 0..1 | CodeableConcept | DNA sequence variant type URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNASequenceVariantType
observation-geneticsAminoAcidChangeName | | 0..1 | CodeableConcept | Amino acid change URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeName Binding: ???? (preferred)

observation-geneticsAminoAcidChangeType | | 0..1 | CodeableConcept | Amino Acid Change Type URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeType

observation-geneticsGene | | 0..1 | CodeableConcept | HGNC gene symbol URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGene Binding: ???? (preferred)

observation-geneticsDNARegionName | | 0..1 | string | DNA region name URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNARegionName

observation-geneticsAlleleName | | 0..1 | CodeableConcept | Name of allele URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAlleleName Binding: ???? (preferred)

observation-geneticsAllelicState | | 0..1 | CodeableConcept | The level of occurrence of a single DNA sequence variant within a set of chromosomes: Heteroplasmic / Homoplasmic / Homozygous / Heterozygous / Hemizygous URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAllelicState

observation-geneticsAllelicFrequency | | 0..1 | decimal | Allele frequency URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAllelicFrequency

observation-geneticsCopyNumberEvent | | 0..1 | CodeableConcept | Copy number variation URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsCopyNumberEvent

observation-geneticsGenomicSourceClass | | 0..1 | CodeableConcept | Genomic source class URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGenomicSourceClass

observation-geneticsPhaseSet | | 0..1 | uri | Phase set identifier (UUID/OID) URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsPhaseSet

observation-geneticsSequence | | 0..* | Reference(Sequence) | Sequence URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsSequence

observation-geneticsInterpretation | | 0..1 | Reference(Observation) | Clinical interpretations for variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsInterpretation

Documentation for this format

10.8.2 Description

The Observation-genetics profile Observation-genetics profile is used to interpret variants from sequence resource. Clinical usage may need more specific representation of variant at locus or structural variant in whole genome.

Some of the attributes of the profile follow:

* The observation-geneticsSequence extension will refer to the Sequence resource for sequence information related to this variant.

* The observation-geneticsInterpretation extension will refer to an Observation instance which contains clinical interpretations for the variant described.

* The code, effective[x], issued, performer, method, specimen elements can be used to describe how the genetic observation (variant and sequence data) is obtained.

* Other extensions are used to describe attributes of this variant such as Genomics Source Class, Amino Acid Change Type, etc. These are mappings from v2 and lonic code reference with details can be found in this list .

Additional Observations instance will be created for variant's further analysis. For example, Observation.component element will be used for knowledge-based interpretations of the sequence variant. Here are some examples for the component.code.

LOINC Code | LOINC Element Name | Comments

51963-7 | Medication Assessed | A coded medication accessed in a pharmacogenetic test (recommend RxNorm).

51967-8 | Genetic disease assessed | A coded disease that is associated with the region of DNA covered by the genetic test (recommend SNOMED).

53037-8 | Genetic Disease Sequence Variant Interpretation | Interpretation of the pathogenicity of the DNA Sequence Variant in the context of the assessed genetic disease.

53040-2 | Drug Metabolism Sequence Variant Interpretation | Predicted phenotype for drug efficacy. A sequence variant interpretation value known to allow (responsive) or prevent (resistant) the drug to perform.

51961-1 | Drug Efficacy Sequence Variant Interpretation | Predicted phenotype for ability of drug to bind to intended site in order to deliver intended effect. A Sequence Variant interpretation value known to allow (responsive) or prevent (resistant) the drug to perform.

In the meantime, the related element in this Observation instance will point to Observaiton-genetics profile to show these clinical interpretations are further analysis for the variant. (For example, sequence variant has its pointer back to the observation, see PGx example)

10.8.3 Usage Examples

10.8.3.1 An Observation-genetics instance and its two related Observation instances for interpretations

We provide an example of an Observation-genetics instance which records a variant detected in the patient. (we call this example A)

Source: XML, JSON, Turtle.

An example of an Observation instance which records knowledge-based clinical interpretations for the variant represented in A.

Source: XML, JSON, Turtle.

An example of an Observation instance which records PCR validation test for the variant in A.

Source: XML, JSON, Turtle.

10.8.3.2 Phenotype, Diplotype and Haplotye

We provide the following examples to reveal how PGx data can be harmonized within the FHIR specification. Detailed discussion will be put here.

* Phenotype Source:XML,JSON,TURTLE

* Diplotype Source:XML,JSON,TURTLE

* Haplotype Source:XML,JSON,TURTLE

10.8 DiagnosticReport-genetics Profile

10.8.1 Structure Diagram

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | | 0..*
codedDiagnosis | | 0..0
DiagnosticReport-geneticsAssessedCondition | | 0..* | Reference(Condition) | AssessedCondition URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAssessedCondition
DiagnosticReport-geneticsFamilyMemberHistory | | 0..* | Reference(FamilyMemberHistory) | FamilyHistory URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsFamilyMemberHistory
DiagnosticReport-geneticsAnalysis | | 0..* | (Complex) | Analysis URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAnalysis
Documentation for this format

This DiagnosticReport-genetics is built on top of DiagnosticReport. The new profile is used to describe a genetics test report. The result element in DiagnosticReport will refer to the Observation resource that can lead to a bundle of genetic observations. And the element of code, effective[x], issued, performer, request, specimen are be used to describe the details of the genetic test. Extensions about AssessedCondition and FamilyMemberHistory are added. Overall, this profile extends the DiagnosticReport resource to enable reporting of structured genetic test results. In addition, it denotes condition context for genetic testing, which may influence reported variants and interpretations for large genomic testing panels.

* The DiagnosticReport-genetics profile contains two extensions referring to other resources/profiles and one complex extension.

* AssessedCondition is used to denote condition context for genetic testing, which may influence reported variants and interpretation for large genomic testing panels. It refers to the Condition resource.

* FamilyMemberHistory are significant health events and conditions for one or more persons related to the patient. It refers to the FamilyMemberHistory-Genetic profile.

* Analysis provides knowledge-based interpretation of overall results of tests performed on patients.

Examples for analysis.code:

LOINC Code | LOINC Element Name | Comments
51968-6 | Genetic Disease Analysis Overall Interpretation | Interpretation of all identified DNA Sequence variants along with any known clinical information for the benefit of aiding clinicians in understanding the results overall in either the context of diagnosis or increased risk of disease.
51964-5 | Drug Efficacy Analysis Overall Interpretation | Overall predicted phenotype for drug efficacy for all DNA Sequence Variants identified in a single case.
51971-0 | Drug metabolism analysis overall interpretation | Overall predicted phenotype for drug metabolism for all DNA Sequence Variants identified in a single case.

10.8.2 Usage Examples

An example of a genetic test report for a patient with FamilyHistory.

Source: XML, JSON, Turtle.

An example of a comprehensive bone marrow report.

Source: XML, JSON, Turtle.

10.8 ProcedureRequest-genetics profile

10.8.1 Structure Diagram

A complex extension is added on top of the ProcedureRequest resource. Here is the structure of the extension:

Name | Flags | Card. | Type | Description & Constraints
ProcedureRequest | | 0..*
procedurerequest-geneticsItem | | 0..* | (Complex) | The items the orderer requested URL: http://hl7.org/fhir/StructureDefinition/procedurerequest-geneticsItem
Documentation for this format

10.8.2 Description

To describe an order requested sequence variants detection. User must set up the code for the request and they can also refer to the corresponding sequence instance for that variant.

10.8.3 Usage Example

10.8.3.1 An order for genetics test

Here is a diagnostic request for testing 185delAG variant. The mother of the patient received results from a mutation panel (eg. MyRisk from Myriad) and she has a BRCA1 185delAG mutation. The clinician the would like to request to test the patient only for an 185delAG mutation. In this case, the diagnostic request for the patient will specify the sequence variant - 185delAG.

10.8 HLA genotyping results Profile

Human leukocyte antigen (HLA) genotyping is fundamental for research and clinical practice in immunogenetics and histocompatibility. Pointers to external locations refer to registered methods, raw NGS reads, and reference standards can be conveyed in this profile. Information about allele assignment including ambiguous results and the allele database used for assignments is stored in extensions.

The structure of the HLA typing report in this profile attempts to follow the principles outlined in the Minimum Information for ReportIng Next-generation sequence Genotyping (MIRING). These principles were identified through a series of meetings with international group of stakeholders in the application of Next Generation Sequencing (NGS) technology for genotyping the HLA and KIR loci as well as other immune-related loci (http://igdawg.org/ngs.html). MIRING describes eight principles, described in detail in Human Immunology. 2015 Dec; 76(12):954-962 . These include detailed metadata about:

* MIRING Annotation

* Reference Context

* Full Genotype

* Consensus Sequence

* Novel Polymorphisms

* Platform Documentation

* Read Processing Documentation

* Primary Data

These principles were implemented in a technical specification by extending an existing XML based format for exchanging histocompatibility and immunogenetic genotyping data called Histoimmunogenetics Markup Language (HML) to include results from NGS methodologies (https://bioinformatics.bethematchclinical.org/hla-resources/hml/). The resulting schema may be found in https://schemas.nmdp.org/. The National Marrow Donor Program (NMDP)/Be The Match uses this format for reporting HLA genotyping from potential donors and for patients needing stem cell transplants.

Both MIRING and HML were used to inform mapping data elements to FHIR resources such as Patient, Specimen, Sequence, Observation-Genetic Profile, and DiagnosticReport where possible. Several additional data elements were needed specifically for this use case, resulting in the development of a specific profile for reporting HLA genotyping results (Diagnostic Report Profile for HLA Genotyping Results).

* Minimum information for reporting next generation sequence genotyping (MIRING): Guidelines for reporting HLA and KIR genotyping via next generation sequencing (see here)

* Histoimmunogenetics Markup Language 1.0: Reporting Next Generation Sequencing-based HLA and KIR Genotyping (see here)

10.8.1 Structure Diagram of HLA genotyping results Profile

Name | Flags | Card. | Type | Description & Constraints
extension | | 0..* | Extension | URL = http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid haploid: haploid. Use on element: DiagnosticReport
locus | | 0..1 | CodeableConcept | gene region.
type | | 0..1 | CodeableConcept | haploid type.
method | | 0..1 | CodeableConcept | haploid method.
Documentation for this format

Both hla-genotyping-results-glstring and hla-genotyping-results-haploid are complex extensions.

The development of these extensions were informed from the allele-assignment structure found in HML. While allele assignment for individual loci can be reported in an Observation, here they are used to summarize the assignments at a report level. Two methods may be used for reporting HLA allele-assignments: Haploid and GL String.

Structure of hla-genotyping-results-haploid:

Name | Flags | Card. | Type | Description & Constraints
extension | | 0..* | Extension | URL = http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid haploid: haploid. Use on element: DiagnosticReport
locus | | 0..1 | CodeableConcept | gene region.
type | | 0..1 | CodeableConcept | haploid type.
method | | 0..1 | CodeableConcept | haploid method.
Documentation for this format

Haploid reporting is a method that is used to report an individual allele, or a list of possible alleles if the results are ambiguous, using NMDP Multiple Allele Codes (MAC) which is a shorthand for describing allele ambiguity (https://bioinformatics.bethematchclinical.org/hla-resources/allele-codes/). MACs are widely used for reporting HLA typing results because they are able to report long allele lists in a small amount of space, but the system has a number of shortcomings (Tissue Antigens. 2013 Aug;82(2):106-12). When reporting data using haploid, typical use is one or two haploid elements for a particular locus, but possibly more if multiple loci are covered (ex: two HLA-DRB1 haploids + one HLA-DRB3 haploid).

Within Haploid, Method indicates whether the general methodology is DNA based typing (e.g., Sequence Specific Primers (SSP), Sequence Specific Oligonucleotide Probes (SSOP), or Sequence Based Typing (SBT)), or SER which indicates serology based methods.

Example of using two hla-genotyping-results-haploid extensions to report a HLA-A genotype of HLA-A*01:AB and HLA-A*02:MN which expands to HLA-A*01:01/HLA-A*01:02 and HLA-A:02:01/HLA-A:02:02/HLA-A:02:03

```
<extension url="http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid" >
  <extension url=”locus”>
      <valueCodeableConcept>
          <coding>
              <system value="http://www.genenames.org"/>
              <code value="4931"/>
              <display value="HLA-A"/>
          </coding>
          <text value="HLA-A"/>
      </valueCodeableConcept>
  </extension>
  <extension url=”type”>
      <valueCodeableConcept>
           <coding>
              <system value="https://bioinformatics.bethematchclinical.org/hla-resources/allele-codes/"/>
              <code value="AB"/>
              <display value="01/02"/>
          </coding>
          <text value="HLA-A*01:AB"/>
      </valueCodeableConcept>
  </extension>
  <extension url=”method”>
      <valueCodeableConcept>
          <text value=”DNA”/>
      </valueCodeableConcept>
  </extension>
</extension>
<extension url="http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid" >
  <extension url=”locus”>
      <valueCodeableConcept>
          <coding>
              <system value="http://www.genenames.org"/>
              <code value="4931"/>
              <display value="HLA-A"/>
          </coding>
          <text value="HLA-A"/>
      </valueCodeableConcept>
  </extension>
  <extension url=”type”>
      <valueCodeableConcept>
           <coding>
              <system value="https://bioinformatics.bethematchclinical.org/hla-resources/allele-codes/"/>
              <code value="AB"/>
              <display value="01/02/03"/>
          </coding>
          <text value="HLA-A*02:MN"/>
      </valueCodeableConcept>
  </extension>
  <extension url=”method”>
      <valueCodeableConcept>
          <text value=”DNA”/>
      </valueCodeableConcept>
  </extension>
</extension>
```

Because of limitations with MAC, another method called GL Strings was developed that encodes the results in a text string with hierarchical set of operators to describe the relationships between alleles, lists of possible alleles, phased alleles, genotypes, lists of possible genotypes, and multilocus unphased genotypes, without losing typing information or increasing typing ambiguity. (Tissue Antigens. 2013 Aug;82(2):106-12).

The structure of HLA-genotyping-results-glstring:

Name | Flags | Card. | Type | Description & Constraints

extension | | 0..1 | Extension | URL = http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring glstring: glstring. Use on element: DiagnosticReport

url | | 0..1 | uri | glstring using a URI reference.

text | | 0..1 | string | glstring using inline data.

Documentation for this format

While the string format is easily parsed into separate components to be rendered for user viewing, GL Strings by themselves are potentially quite long and difficult to read. It often advantageous to point to a URI which may return the GL String on demand to avoid manual data entry. This is available through the URI element in this extension.

Example of using the hla-genotyping-results-glstring to report a GL String in both text and URI formats

```
<extension url="http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring">
  <extension url=”text”>
      <valueString value="HLA-A*01:01:01:01/HLA-A*01:02+HLA-A*02:01:02/HLA-A*02:02:02/HLA-A*02:03:01"/>
  </extension>
  <extension url=”uri”>
      <valueUri value="https://gl.nmdp.org/imgt-hla/3.23.0/genotype/1h"/>
  </extension>
</extension>
```

Whether reporting in Haploid or GL String formats, it is important to identify the version of the IMGT/HLA allele database, as new HLA alleles are constantly being discovered and allele assignment is based on the known alleles at the time, and so results may need to be reinterpreted later. This is done through the Allele Database element.

Structure of hla-genotyping-results-allele-database

Name | Flags | Card. | Type | Description & Constraints

extension | | 0..1 | CodeableConcept | URL = http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database allele-database: Allele Database. Use on element: DiagnosticReport

Documentation for this format

Example of using hla-genotyping-results-allele-database to report using the IMGT/HLA database, version 3.23

```
<extension url="http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database">
  <valueCodeableConcept>
      <coding>
          <system value="http://www.ebi.ac.uk/ipd/imgt/hla/"/>
          <version value="3.23"/>
      </coding>
      <text value="IMGT/HLA 3.23"/>
  </valueCodeableConcept>
</extension>
```

Overall methodology may be reported using hla-genotyping-results-method. Here a codeable concept may be used to refer to method entered into a public registry, such as the NCBI Genetic Test Registry, or a local private registry.

Structure of hla-genotyping-results-method

Name | Flags | Card. | Type | Description & Constraints

extension | | 0..1 | CodeableConcept | URL = http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method method: The platform, methodology and software applied at the time of the genotyping. Use on element: DiagnosticReport

Documentation for this format

Example of hla-genotyping-results-method to report a lab test registered in the NCBI Genetic Test Registry

```
<extension  uri="http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method">
  <valueCodeableConcept>
        <coding>
            <system value="http://www.ncbi.nlm.nih.gov/gtr/"/>
            <code value="GTR000000000.0"/>
        </coding>
        <text value=”NGS based HLA-A genotyping”
  </valueCodeableConcept>
</extension>
```

10.8.2 Usage Example

Example of a HLA genotyping report

* that was requested in a Diagnostic Request,

* with results for HLA-A, HLA-B, HLA-C loci,

* by Sequence Based Typing (SBT) of exons 2 and 3,

* using Next Generation Sequencing technologies,

* allele assignment based on the IMGT/HLA database version 3.23,

* reported as a multilocus unphased genotype using GL Strings,

* and with genotyping results for individual loci available in separate Observations

Source: XML, JSON, Turtle.

In the above example, three Observations are referenced, each containing the details surrounding the genotyping of each gene/locus. Those observations would further refer to other observations (using Observation.related) that point to phased exon sequences which are the basis for individual allele assignment. This strategy is illustrated in the following figure for one gene, and sequencing two exons for each allele. Note that references from DiagnosticReport and Observation back to Patient/Donor and Specimen are not shown for clarity, although we would include them in the resource instances.

10.8 Relationship among Sequence resource and genetics profiles

10.8.1 Pointers to Sequence resource

Sequence resource is used to transmit and represent sequencing data. In FHIR Genomics, there are several genetics profiles containing internal pointers to Sequence resource for genetic data reporting. Here is the summary:

Sequence.pointer: A bundle of Sequence instances can be connected by Sequence.pointer to build sequence graph described in Section 3. Sequence.pointer will point to the next sequence block.

Sequence extension in Observation-genetics profile: Observation-genetics profile is used to report a genetic variant found in patients. Sequence extension contains a pointer to Sequence identifier which has related sequencing read info. It will provide reference/observed allele information and quality scores for each base/sequence block.

10.8.2 Pointer to Observation resource (based on genetics profile)

Observation-genetics based Observation resource is used for interpretative genetic data. Sequence resource and genetics profiles will use internal pointers to Observation-genetics-profile based Observation instance for variant report.

Sequence.observation: A pointer to genetics Observation instance which is used for interpretations of this sequence block. Interpretations are generally about genetic variant found in this sequence block.

DiagnosticReport.result: A pointer to Observation instance. This element is used for reporting genetic result.

Item extension in ProcedureRequest-genetics profile: It is used to describe the genetic test order.

FamilyMemberHistory: A pointer to genetics Observation instance to report genetic test results of family member

10.8.3 Observation vs. Sequence

Sequence is used for raw sequencing data while genetic-profile-based Observation resource is used for reporting interpretative genetic information (eg. DNA/amino acid variant detected in the patient). Although both Sequence and Observation-genetics profiles contain variant info, they are used differently: Sequence.variant is used for encoding the sequence block; the Observation-genetics profile is mainly used for a variants test result or interpretations of raw data in Sequence resource.

10.8.4 Usage example

This picture depicts the logical relationship among these resources. And the definition of pointers are described above:

Here are the details and examples mentioned above.

Resource | Details of data structures and pointers | Links of resources example
Sequence | Sequence Data Structure | Example Here
Observation | Observation Data Structure | Example Here
ProcedureRequest | ProcedureRequest Data Structure | Example Here
DiagnosticReport | DiagnosticReport Data Structure | Example Here

Here is a specific use case to depict the whole story of FHIR in the clinical genomics setting: a ProcedureRequest is requested, then a Diagnosticreport are reported. Two Sequence instance, two Observation-haplotype instance, a Observation-diplotype and a Observation-phenotype for further interpretation are also created. (You can see the xml code and json code if you click the link.)

* Haplotype: A haplotype is a set of DNA variations, or polymorphisms, that tend to be inherited together. A haplotype can refer to a combination of alleles or to a set of single nucleotide polymorphisms (SNPs) found on the same chromosome.

* Diplotype: A diplotype is defined as a specific combination of two haplotypes.

* Genotype: A genotype is an individual’s collection of genes.

* Phenotype: A phenotype is the composite of an organism’s observable characteristics or traits, such as its morphology, development, biochemical or physiological properties, behavior and products of behavior.

You can see the relationship among these resources in following picture:

10.8 HL7 Domain Analysis Use Cases

Simulated use of the proposed Sequence resource and profiles on other FHIR resources are shown for many of the examples in Section 5 – Use Case Scenarios in the HL7 Domain Analysis Model (DAM): Clinical Genomics, Release 1, September 2014 Informative Ballot .

We show use of the FHIR design we are proposing for these cases:

* Specimen Identification

* Get references to all variants obtained from germline analysis.

* Explore antibacterial drug resistance over time (non-patient DNA)

* Clinical Sequencing – Germline Testing

* Cancer Profiling

* Decision Making Tools – Family History and Drug Dosage Calculators

* Today clinicians translate (i.e. manually re-enter) genetic data into tools for decision making. This includes family history tools and drug dosage calculators. In the future, this data will automatically be incorporated into clinical decision making tools.

* Get pharmacogenomic reporting and disease risk or diagnosis associated with variant

* Public Health

* Clinical and Research Data Warehouses

10.8.1 Specimen Identification

* Germline testing for biomarkers/mutations (usually inherited)

* Tumor testing for biomarkers/mutations (somatic / tumor specific)

* Pediatric Testing

* Infectious Disease Testing

10.8.1.1 Get references to all variants obtained from germline analysis

```
GET   /Observation?
    _profile=http://hl7.org/fhir/StructureDefinition/observation-genetics&
    GenomicSourceClass=http://loinc.org|LA6683-2
```

*Return a bundle of genetics-profile-based Observation instances

10.8.2 Clinical Sequencing – Germline Testing

For a chart review, get references to all DNA sequences related to mutations with an interpretation “Unknown Significance”

```
GET /Observation?
    subject=123&
    component-code-value-[x]= http://loinc.org|53037-8$LA6682-4
```

53037-8: LOINC code for “Genetic disease sequence variant interpretation”

LA6682-4: LOINC answer code for “Unknown Significance”.

10.8.3 Cancer Profiling

The goal of this profile methodology is to get references from all variants obtained from somatic analysis. Changes in the population of cells with particular mutations will change overtime as well as in conjunction with events such as therapy. For instance, targeted chemotherapy may kill a specific population of cancer cells with specific mutations and other cancer cell populations may survive and continue to divide. Therefore, clearly annotating these specimens as somatic variants and capturing annotations related to a time relevant to a treatment timeline may be critical for analysis.

```
GET /Observation?
    _profile=http://hl7.org/fhir/StructureDefinition/observation-genetics&
    GenomicsSourceClass=http://loinc.org|LA6684-0&
    date=2015-07-04&
    subject=123
```

10.8.4 Decision Making Tools – Family History and Drug Dosage Calculators

Today clinicians translate (i.e. manually re-enter) genetic data into tools for decision making. This includes family history tools and drug dosage calculators. In the future, this data will automatically be incorporated into clinical decision making tools.

Get family history related to one observation:

```
GET /DiagnosticReport?
    _profile=http://hl7.org/fhir/StructureDefinition/diagnosticreport-genetic&
    subject=123
```

10.8.4.1 Get pharmacogenomic reporting and disease risk or diagnosis associated with variant

```
GET /Observation?
    _profile=http://hl7.org/fhir/StructureDefinition/observation-genetics&
    subject=123&
    Interpretation.component-code= http://loinc.org|51963-7
```

51963-7 : LOINC code for “Medication Assessed ”

10.8.5 Public Health

Today Registrars manually translate clinical data into public health reporting systems. This data is used to monitor and improve public health (e.g. surveillance and clinical research). In the future, this data will be extracted from the EHR in an automated (or semi-automated) fashion.

For a breast cancer clinical genomic study, get all genetic-profile-based observations of patients with breast cancer:

```
GET /DiagnosticReport?
    _profile=http://hl7.org/fhir/StructureDefinition/diagnosticreport-genetic&
    AssessedCondition.code=http://snomed.info/sct|254837009
```

10.8.6 Clinical and Research Data Warehouses

Health data warehousing should persist data in its standardized formats, while allowing users to export subsets of the data in the warehouse into multiple ‘data marts’, optimized for specific use cases, analysis type or reporting needs.

Get all genetic-profile-based observations of patients with the variant c.181T>G

```
GET /Observation?
    _profile=http://hl7.org/fhir/StructureDefinition/observation-genetics&
    DNAVariantID=http://www.ncbi.nlm.nih.gov/projects/SNP|rs58238560
```

10.8 Other Genomics Use Cases

During the development of the FHIR Genomics design, CGWG participants have commented on the earlier use cases and/or proposed new use cases. In this section, we list describe of these new cases and demonstrate how the proposed design will address them.

10.8.1 Get a patient’s family members genetics report

This example is proposed by Kevin Hughes. Family history is useful for clinicians to know more about the condition of the patient.

```
Get /FamilyMemberHistory?
    _profile=http://hl7.org/fhir/StructureDefinition/familymemberhistory-genetic&
    patient=123
```

10.8.2 Panel

Search for results from nephrotic syndrome panel

```
Get /DiagnosticReport&
    _profile=http://hl7.org/fhir/StructureDefinition/diagnosticreport-genetic&
```

```
code=N0336&
patient=123
```

10.8.3 Sequence Test Metrics

10.8 Clinical Genomic Apps

The idea for a Sequence resource grew out, in part, the SMART Platforms Project, which explored creating clinical genomic apps to integrated traditional EMR clinical data and genomic data to show data visualization and analysis, including CDS that depended upon both types of data. Below are a couple of examples. Several apps have already been designed including Genomics Advisor, SMART Precision Cancer Medicine, and Diabetes Bear EMR. Below, one of these apps will be described. To include other apps in this section, please feel free to add a note on it and how it uses FHIR/Genomics calls.

10.8.1 Genomics Advisor

The SMART on FHIR Genomics Advisor was an app incorporating genomics data to show risk of disease, drug susceptibility, and related conditions based upon genotype. Technically, this app was architected (see below) by combining data from independent data services, a SMART on FHIR clinical server for clinical information and one for a SMART on FHIR Genomics data server for genomic data. The set of FHIR API calls that are necessary to support this app are shown below:

The presentation of the app looks like this:

10.8 Appendix 1: From DSTU2 Genomics to STU3

The table shows how the data elements in the DSTU2 Observation resource as extended by the Standard Genetics Profile would be mapped to the new Sequence resource.

DSTU2 Standard Genetics profile on Observation | Observation-genetics profile | Maps to Sequence (Proposed)
geneticsGenomeBuild | Sequence extension | referenceSeq.genomicsBuild
geneticsChromosome | Sequence extension | referenceSeq.chromosome
geneticsGenomicsStart | Sequence extension | variant.start
geneticsGenomicsStop | Sequence extension | variant.end
geneticsReferenceAllele | Sequence extension | variant.referenceAllele
geneticsObservedAllele | Sequence extension | variant.observedAllele
geneticsSpecies | Sequence extension | specimen
geneticsAllelicState | Allelic State extension
geneticsAllelicFrequency | Allelic Frequency extension
geneticsReadCoverage | Sequence extension | readCoverage
geneticsCopyNumberEvent | Copy number event extension
geneticsTranscriptReferenceSequenceId | Transcript reference sequence identifier extension
geneticsProteinReferenceSequenceId | Protein reference sequence identifier
geneticsCIGAR | Sequence extension | variant.cigar
geneticsVariationId | DNA variation id extension
geneticsVariationType | DNA sequence variation type extension
geneticsAminoAcidChange | Amino acid change extension
geneticsAminoAcidChangeType | Amino acid change type extension
geneticsGene | Gene extension
geneticsDNARegionName | DNA region name extension
geneticsGenomicSourceClass | Genomic source class extension